

Obstetrics & Gynaecology

World of Revision

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GYNAECOLOGY REVISION : PART 1

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Anatomy of Uterus

00:00:42

Normal position of uterus :

- Ante**flexed** : B/w uterus & cervix (120°).
- Ante**verted** : B/w **vagina** & cervix (90°).

Ligament which helps to keep uterus in anteverted position :
Round ligament (RL).

Ligament which prevents retroversion :
Uterosacral ligament > RL.

Ligaments which support uterus :

1. Cardinal ligament (Transverse cervical/mackenrodt's ligament).
2. Uterosacral ligament.
3. Pubocervical ligament.

Ligament which does not support uterus :
Broad ligament.

Muscle which mainly supports uterus :
Levator ani muscle :
Pubococcygeus, Iliococcygeus.

At fundus of the uterus, three structures are present :

- R : **R**ound ligament.
- T : Fallo**p**ian **t**ube.
- O : **O**varian ligament (OL).

Ligament which connects :

- Ovary to uterus : Ovarian ligament.
- Ovary to pelvic side wall : Infundibulopelvic ligament/Suspensory ligament of ovary (Contains ovarian vessels).
- Blood supply of uterus :
 - uterine artery (Branch of anterior division of internal iliac artery).
 - Utero-ovarian anastomosis.
- Nerve supply : T₁₀ - L₁.

Uterus

Cervix to Corpus Ratio :

At birth → 1 : 1.
Before puberty → 2 : 1.
At puberty → 1 : 2.
At reproductive age → 1 : 3.
After menopause → 1 : 1 (Both atrophy).

Uterus develops from : **mullerian duct**.
Round ligament develops from : **Distal end of gubernaculum**.
Ovarian ligament develops from : **Proximal end of gubernaculum**.

To visualize uterus from inside :

Hysteroscopy :

Distension media → Gas : Only diagnostic.
→ **Liquid** : Diagnosis & Rx.

unipolar cautery is used with :

Electrolyte deficient media :

- **glycine (m/c)**.
- mannitol.
- Sorbitol.

With electrolyte deficient media, procedure is stopped at : 1000 mL fluid deficit.

With electrolyte rich media, procedure is stopped at : 2500 mL fluid deficit.

Electrolyte rich media :

- NS.
- Ringer lactate. } **m/c used**
(Best media)

Hysteroscopy indications :

- Ashermann syndrome.
- Septate uterus.
- Endometrial polypectomy.
- Type 0 & I (Submucous) fibroid.

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Uterine Vasculature :

Branches of uterine artery :

- Uterine artery.
- Arcuate artery.
- Radial artery — { Basal artery.
Spiral artery.

Stepwise devascularization in PPH mx :

Ligation of :

1. Uterine artery : U/L → B/L.
2. Utero-ovarian anastomosis.
3. Anterior division of internal

iliac artery



Trisulnikov's triple ligation :

3rd is Sampson's A./Artery of RL.

Note :

Living ligature : middle layer of myometrium.

Round Ligament :

Canal of Nuck :

- Fold of peritoneum with round ligament in inguinal canal.
- Leads to hernia.

Artery of round ligament :

Sampson artery : Arises from inferior epigastric artery > uterine artery.

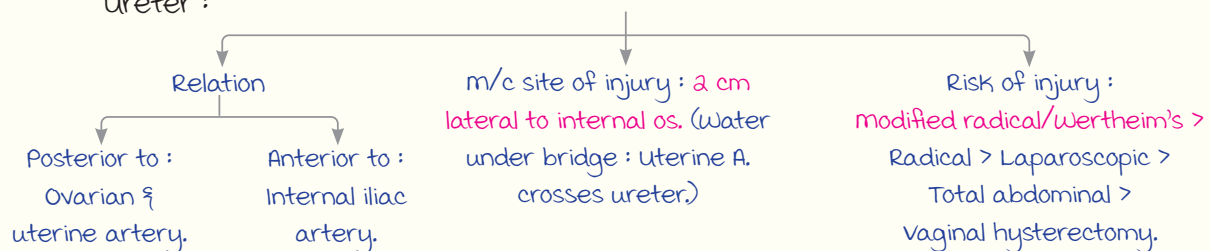
Broad Ligament :

Contents :

- Fallopian tube → 1 tube.
- RL } 2 ligaments.
- OL }
- Uterine A (UtA) } 2 arteries.
- Ovarian A }
- Epoophoron } 3 remnants.
- Paroophoron }
- Gartner's duct }

Uterine Relations :

Ureter :

Retroverted Uterus :

Retroverted/Retroflexed uterus :

- 20%.
- manually corrected.

Fixed retroverted uterus :

- Endometriosis > PID/genital TB.
- Cannot be corrected manually.

Clamping during Hysterectomies :

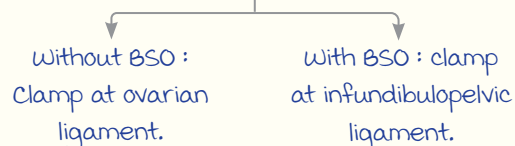
Abdominal hysterectomy :

- 1st : RTO pedicle.
- 2nd : Cardinal ligament.
- 3rd : uterine artery.
- 4th : uterosacral ligament.

vaginal hysterectomy :
Order reversed.

Total abdominal hysterectomy (TAH) :

Remove uterus + cervix



Anatomy of Cervix

00:17:56

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Cervix has Two Parts :

Endocervix :

- Supravaginal.
- Red.

Exocervix :

- Portio vaginalis.
- Pink.

Lining of :

- Endocervix : Columnar epithelium.
- Exocervix : Stratified squamous epithelium.

Junction of the endocervix & exocervix :

- Transformation zone (TZ)/ Squamocolumnar junction.
- Level : External os.
- m/c site for pre-invasive and invasive lesions of cervix.

Normal length of cervix :

- In non-pregnant female : 2.5 cm.
- In pregnancy : 4 cm.
- In pregnancy, if length of cervix is **< 2.5 cm** :
 - K/a short cervix.
 - Risk of TA abortion and preterm labour.
- ACOG recommends :

- h/o TA abortion - h/o preterm labour	TVS : Cx length b/w 16 - 24 wog
---	---------------------------------------

Cervix

- TZ moves out :
 - Puberty.
 - OCP.
 - Pregnancy.
- Ectropion :
 - When endocervix is seen moving out of exocervix.
 - Physiological in pregnancy.
 - Do not take biopsy.
 - Can cause post coital bleeding.



Ectropion

- Blood supply of cervix : Descending cervical artery (Branch of uterine artery).
- Nerve supply of cervix : S₂ - S₄.
- Lymphatic drainage of cervix :
 - H : Hypogastric/Internal iliac LN.
 - O : Obturator LN.
 - P : Paracervical LN.
 - E : External iliac LN.
- Cervix does not drain into : Superficial inguinal LN.
- Sentinel lymph node : **Paracervical LN.**
- Dührssen's incision :
 - Done in entrapped head in breech.
 - Above/below 3 o'clock & 9 o'clock (Site of artery)

Pain management & Instruments

00:23:34

Pain management :

Pain :
 → 1st stage : D/t uterine contraction (T_{10} - L_1).
 → 2nd stage : Cervical dilation (S_2 - S_4).

Analgesia	Procedure	Level / Site
Epidural analgesia	Painless labour	T_{10}
LSCS anaesthesia		T_4
Pudendal block	Instrumental delivery	Ischial spine : Pierce sacrospinous ligament
Paracervical block	Suction & evacuation	Never at 3 o'clock & 9 o'clock

Instruments :



uterine sound :

- Have calibrations.
- Olive tipped.
- Assess size and position of uterus.



Bladder sound :

No calibrations.

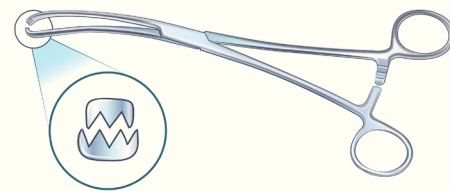


A. Anterior vaginal wall retractor :

Large loop with serrations.

B. Blunt - sharp curette :

Small loop.



vulsellum forceps :

- To hold lip of cervix (Non pregnant).
- Has ratchet lock.
- Curved blade with gap.
- Rat like teeth.

Note :

- Cervix of pregnant woman : **Sponge holding forceps.**
- Allis forceps : No curve.
- Kocher's forceps : No gap.

Anatomy of Vagina

00:29:54

----- Active space -----

- Lined by : Stratified non-keratinized squamous epithelium.
- Has inhabitant bacteria : (Lactobacilli) **Döderlein** → Convert glycogen into lactic acid.
- Döderlein bacilli appear at puberty and disappear at menopause.
- Vagina develops from :
 - Upper part : mullerian duct.
 - Lower part : urogenital sinus.
- Vagina does not have glands.

pH of vagina :

- At birth : Acidic.
- During childhood : Alkaline (6 - 8).
- At puberty : Alkaline → Acidic.
- In reproductive age : **4 - 4.5**.
- During pregnancy : < 4.
- During menstruation : 6 - 8.
- After menopause : 6 - 8.

Cells of vagina :

- Cells which predominate in estrogen predominance : Superficial cell.
- Cells which predominate in progesterone predominance : Intermediate cells.
- Cells which predominate when no hormone predominance : Parabasal cells.

Vagina

De Lancey's Levels of Support :

Level 1 :

- **uterosacral & cardinal ligament.**
- Support : upper 1/3rd vagina.
- Defect leads to :
 - Apical prolapse.
 - vault prolapse.
 - Cervical elongation.
 - **Enterocoele.**

Level 2 :

- Arcus tendinous fascia.
- Support : middle 1/3rd vagina.
- Defect leads to :
 - **Cystocoele (m/c)**
 - **Rectocoele.**

Level 3 :

- Perineal body & muscles attached to it.
- Support : Lower 1/3rd vagina.
- Defect leads to :
 - **urethrocele.**
 - Rectocoele.
 - Laxed perineum.

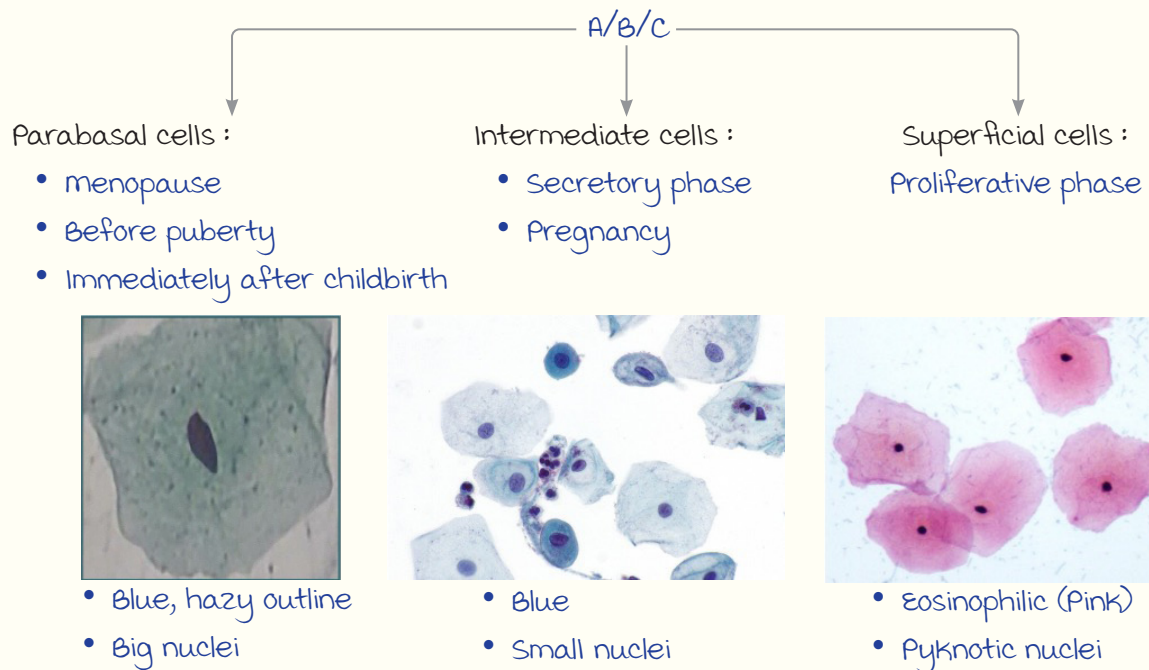
management of vaginal Prolapse :

Does not depend on age and parity.

- | | | |
|---|--|------------------------|
| <ul style="list-style-type: none"> • Cystocoele • urethrocele | } Ant. colporrhaphy

} Post. colpoperineorrhaphy | } Pelvic floor repair. |
| <ul style="list-style-type: none"> • Rectocoele • Laxed perineum | | |
| <ul style="list-style-type: none"> • Enterocoele <ul style="list-style-type: none"> - m/c : McCall culdoplasty. - Outdated : moschowitz repair, Halban's repair. | | |

----- Active space -----

Maturation Index :Vaginal pH :

- Factors responsible for acidic pH :
 - Estrogen : Presence of glycogen (For conversion to lactic acid).
 - Döderlein bacilli.
- vaginal defense lost : 10 days after birth.
- **> 4.5** : Infections.
 - Bacterial vaginosis.
 - Trichomoniasis.
- Organism tolerating acidic pH : **Candida** (≤ 4.5 pH).

Anatomy of Fallopian Tube

00:42:07

----- Active space -----

Length : 10 - 12 cm.

Lined by : Ciliated columnar epithelium.

Parts :

- From medial to lateral :
Interstitium → Isthmus → Ampulla
→ Infundibulum.
- 3 types of cells :
a. Secretory cells.
b. Ciliated cells.
c. Resting cells (Peg cells).
- Nerve supply : T₁₁, T₁₂, L₁.
- Blood supply :
- medial 1/3rd : uterine artery.
- Lateral 2/3rd : Ovarian artery.

Parts  :

- Narrowest : Interstitium.
- Second narrowest : Isthmus.
- Anatomical sphincter : Interstitium.
- Physiological sphincter : Isthmus.
- Tubal ligation is done at : Isthmus.
(uniformly thick)
- Widest part : Ampulla.
- Essure inserted at : Interstitial/
intramural part.
- m/c site of :
- Fertilisation.
- Ectopic pregnancy.
- Genital TB. } Ampulla

Fallopian Tube

Tubal Ligation :

- Permanent method of contraception.
- **Not** a reversible method/Long acting reversible contraceptive (LARC).
- Effective : FR of < 0.5%.
- Conditions :
a. Age : 22 - 49 years
b. Children : 1 child > 1 year old.
c. Ever married.
d. Spouse :
• Not undergone vasectomy.
• Spouse consent not required.

Laparoscopic tubal ligation :

- m/c method.
- use : Fallope rings (Silicone + Barium sulphate)
- Not done for : Post partum ligation.
- Can be done : Interval ligation. (After 6 weeks of delivery)
- Relative contraindication : Severe cardiac/
pulmonary disease.

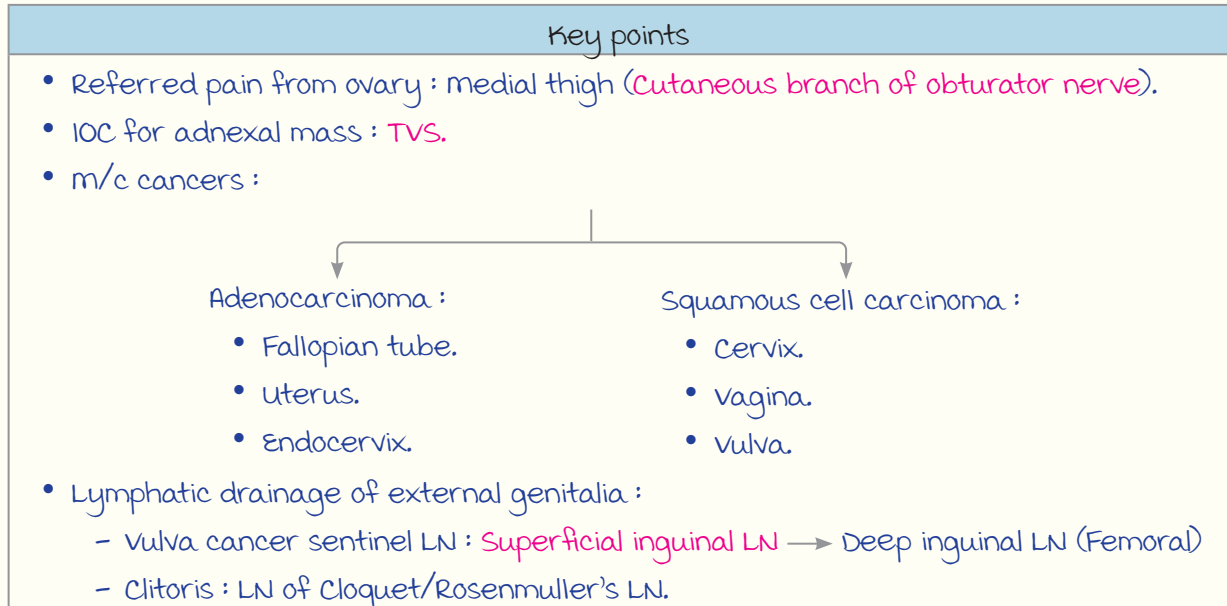
Postpartum ligation :

- Time : After 24 hrs of delivery - 1 week of delivery.
- method : mini laparotomy.
- Technique :
- Pomeroy : Single ligature. (Chromic catgut)
- modified Pomeroy : Double ligature. (Plain catgut)

----- Active space -----

Essure :

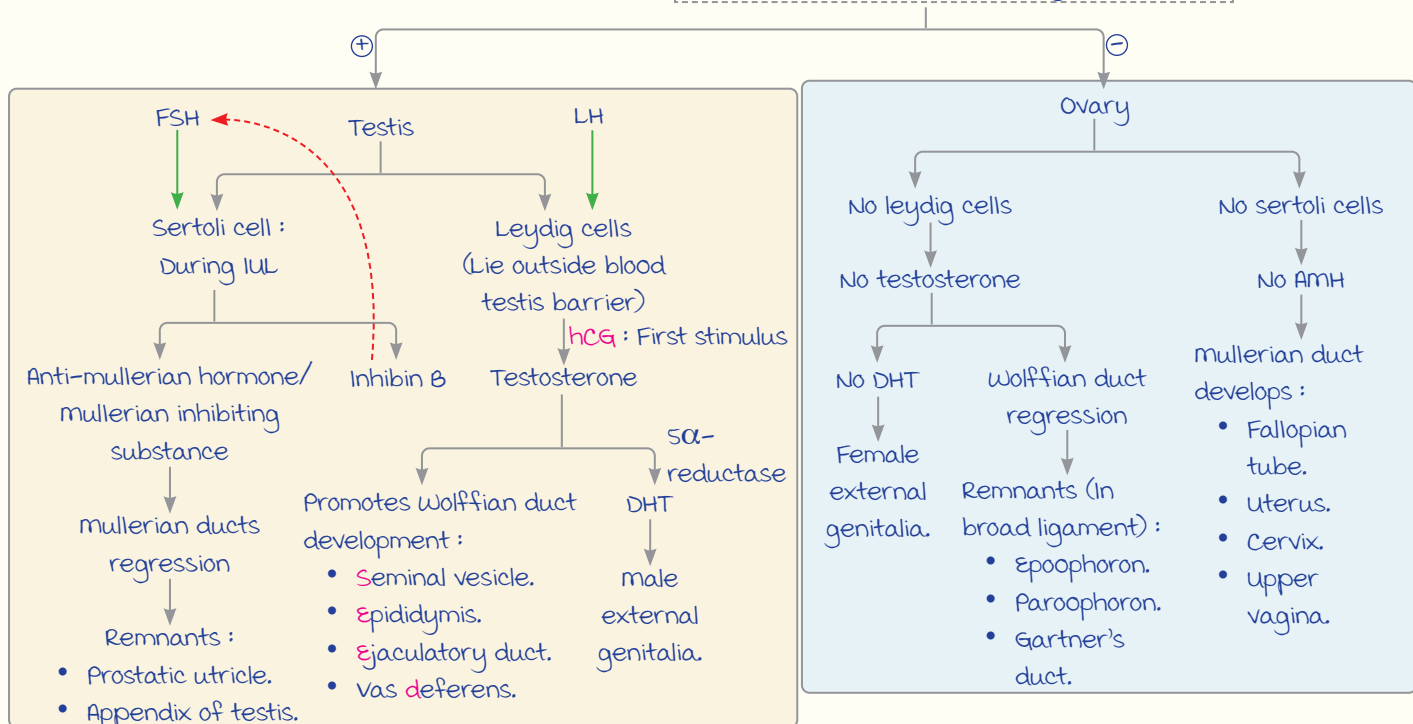
- Hysteroscopic tubal ligation.
- material : Nickel & titanium.
- effective : **After 3 months** (Check by HSG).
- Success rate : 99%.

**Gonads and External Genitalia**

00:51:20

Development of Gonads & Reproductive Tracts :Genital ridge + **SRY gene :**

- Distal, **short arm** of Y chromosome.
- Determines sex of body.



Embryological Development of External Genitalia :

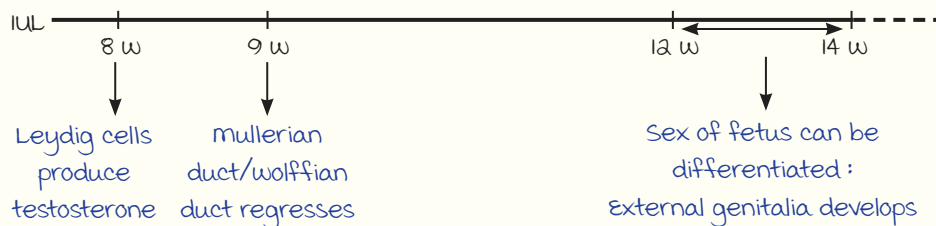
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Develop from : urogenital sinus

	Adequate testosterone (DHT (+))	Absent testosterone
External genitalia	male	Female
Homologous organs		
Genital tubercle	Penis	Clitoris
Genital fold	Scrotum	Labia majora
Genital swelling	Penile urethra	Labia minora

Inadequate testosterone
Ambiguous genitalia :
• Clitoromegaly
• Labioscrotal fold

Developmental Timeline :



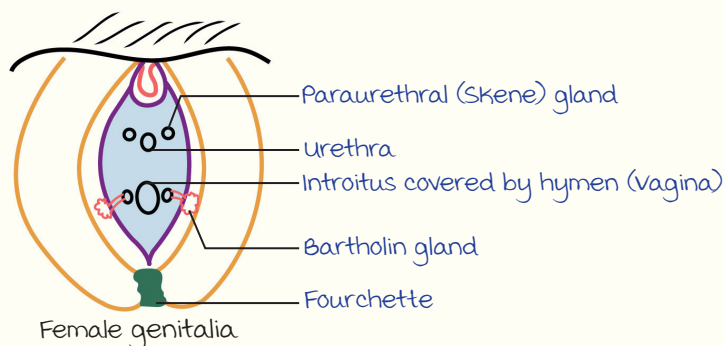
Female External Genitalia :

Bartholin's gland :

- Named criteria for cancer :
Honan's criteria.
- Location : B/w labia majora and labia minora.
- Ducts :
 - Open in vestibule outside hymen.
 - At junction of ant. 2/3rd and post. 1/3rd.
 - 4 o'clock/8 o'clock.

Vestibule :

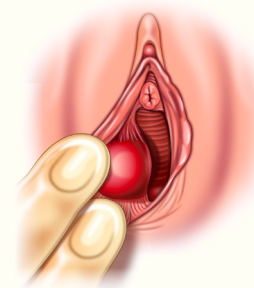
- Bounded by :
 - Ant. : Clitoris.
 - Post. : Fourchette.
 - Lateral : Labia minora.
- Openings :
 - Skene's glands.
 - Urethra.
 - Vagina.
 - Bartholin gland ducts.



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mx of Bartholin's cyst :

Bartholin cyst	mx
< 3 cm, asymptomatic	Expectant mx.
≥ 3 cm, asymptomatic	Incision and drainage f/b
Symptomatic	
Abscess	insertion of word catheter
Recurrent cyst	marsupialization



Bartholin's cyst

Vulval Cancer

01:03:18

Important One Liners :

- m/c type : Squamous cell Ca.
- 2nd m/c type : melanoma > Adeno Ca.
- Age group :
50 - 70 years (mean : 65 years).
- m/c symptom : Pruritis.
- m/c site : Labia majora > Labia minora.
- It had 2 common varieties :
a. Keratinizing (m/c).
b. Warty.

Risk Factors :

H : HPV (HPV 16).

P : → Paget's disease.
→ p53 mutation.V :

- Vulval intraepithelial neoplasia (VIN).
- Vulval atrophy.
- Vulval warts (Condyloma acuminata).

S : Smoking/Lichen sclerosis.

Staging :

Stage IA : Size < 2 cm & depth < 1 mm.

Stage IB : size ≥ 2cm or depth ≥ 1 mm

LN involved in : Stage III & IV.

vulval Cancer

management :

Stage I, 2 → Sx : Excision of tumor + LND.

Stage 3, 4 → Chemoradiation.

Surgery :

Stage IA : Simple partial vulvectomy.

IB/II : Radical vulvectomy.

Note :

- Condyloma lata
 - Herpes simplex virus
 - Vulval hamartoma
- Not risk factors for vulval cancer.

Lymphatic Dissection :

In stage IA : No LND.

If LN are palpable : B/L inguinofemoral LND.

If LN are positive : B/L inguinofemoral LND.

If LN are not palpable :

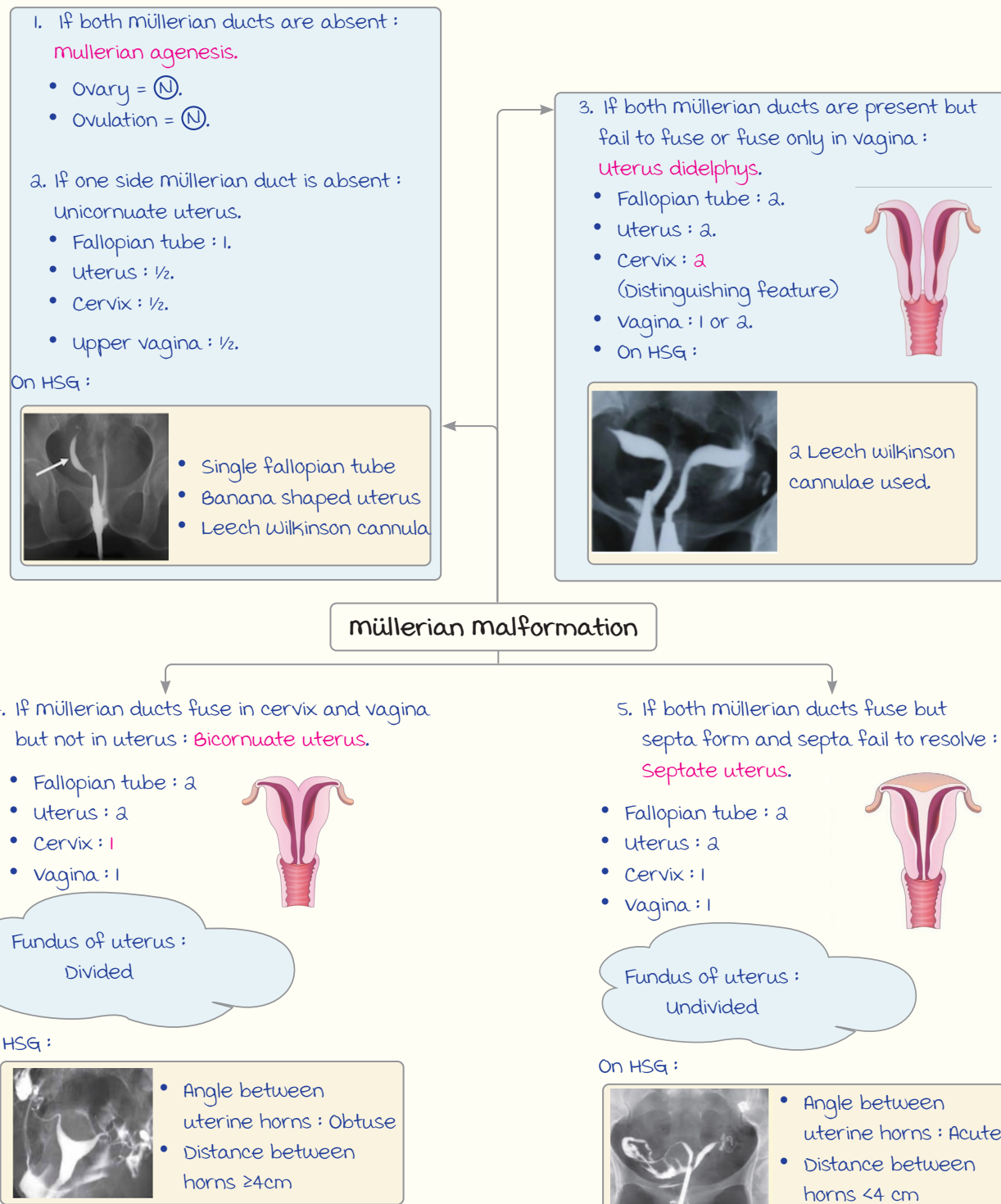
- Stage IA : No LND.
- In all other stages : B/L Inguinofemoral LND.
- Always : Do B/L inguinofemoral LN dissection.
- Except if all following conditions are fulfilled w/L :
 - Size : < 2 cms
 - +
 - Tumor should be : > 2 cms away from midline
 - +
 - Local extension : Absent.

GYNAECOLOGY REVISION : PART 2

----- Active space -----

Mullerian Malformations

00:01:43



----- Active space -----

Note :

HSG cannot differentiate between septate & bicornuate uterus.

6. Arcuate uterus :

- Fundus of uterus flat/dimpled.
- Best reproductive outcome.

7. In utero exposure to diethylstilbestrol (DES) :

- m/c uterine anomaly : Hypoplastic uterus.
- most specific uterine anomaly : T-shaped uterus.
- m/c malignancy : Clear cell cancer of vagina.



Arcuate uterus

Important Points :

m/c uterine anomaly : Septate > bicornuate uterus.

Anomaly with best reproductive outcome : Arcuate > didelphys uterus.

Anomaly with worst reproductive outcome : Septate uterus.

Associations of unicornuate uterus :

- Ectopic pregnancy.
- Ectopic ovary.
- U/L dysmenorrhea.
- U/L renal anomalies.

m/c malformation associated with renal anomalies :

Unicornuate uterus > mullerian agenesis.

m/c malformation not associated with transverse lie : uterine didelphys.

m/c complication associated with mullerian malformations :

- Obstetric : Recurrent abortions.
- Gynaecological : Endometriosis, dysmenorrhea.

Investigations for mullerian malformations :

IOC : 3D ultrasound.

Gold standard : MRI.

Last resort : Laparoscopy + hysteroscopy.

Surgery in mullerian malformations :

Indication : Recurrent abortions.

	Septate uterus	Bicornuate/didelphys uterus
Preferred Sx	Hysteroscopic resection of septum	unification Sx
Outdated Sx	Jones metroplasty	Straussman metroplasty

Prolapse

00:21:01

----- Active space -----

Supports of uterus :

1. Ligaments which support uterus :

- Cardinal/transverse cervical/
mackenrodt's ligament
- uterosacral ligament
- Pubocervical ligament

Triradiate
ligament

2. muscles which support uterus :

- Levator ani.
- Bulbospongiosus.
- Transverse perinei

Superficial
Deep

3. mechanical support :

Angle of anteversion = 90° .Angle of ante flexion = 120° .

Important Points :

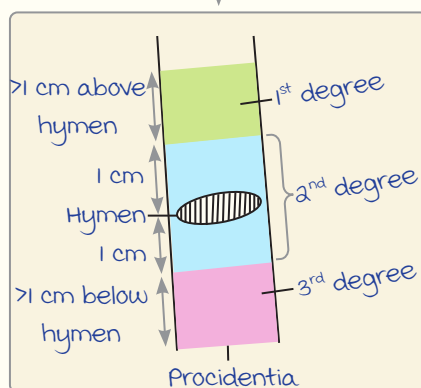
- First step for prolapse : **Retroversion**.
- Fixed retroverted uterus seen in : Endometriosis.
- Ligament which keeps uterus in anteverted position : Round ligament.
- Ligament which prevents retroversion : uterosacral ligament.
- Ligament which is secondary support : Round ligament.
- Ligament which doesn't support uterus : Broad ligament.
- Which part of Levator ani if damaged has maximum chances of causing prolapse : **Puborectalis** > pubococcygeus.

Prolapse

Classification of Prolapse :

POP-Q Classification :

Reference point : Hymen.



New Definition :

In POP-Q : Total 9 sites used

- 1st measurement : Total vaginal length.
- measurement taken while patient is performing Valsalva maneuver.

Risk Factors :

- Elderly menopausal females
- Birth trauma
- Multiparity
- Instrumental delivery

Best method to Prevent Prolapse :

Perineal/Kegel exercises.

POP-Q Grid :

Anterior vaginal wall Aa	Anterior vaginal wall Ba	Cervix or Cuff C
Genital hiatus gh	Perineal body pb	Total vaginal length tvL
Posterior vaginal wall Ap	Posterior vaginal wall Bp	Posterior fornix Dp

Procidentia :

Entire cervix & uterus lies outside the hymen.

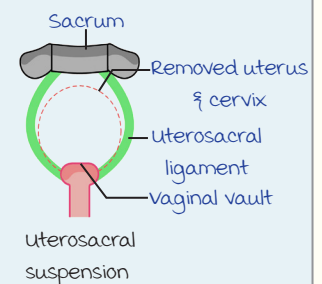
Vault Prolapse :

Prolapse of vaginal stump after hysterectomy.

management :

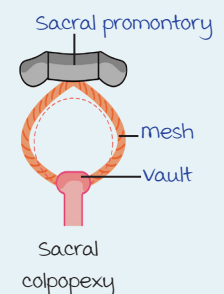
1. vaginal :

- uterosacral suspension (Prophylactic & therapeutic)
- Sacrospinous fixation.



2. Abdominal (Better outcome but more difficult) :

Sacral colpopexy : mesh attached to sacral promontory & vault.



----- Active space -----

management of uterine Prolapse (97) :

- Depends on age & parity of the female.
- Surgery done only for 3rd degree prolapse.

Case	management
Nulliparous female with prolapse (or) female wants future childbearing	Sling surgery (Colpopexy) : Prerequisite : uterocervical length should be <12.5 cm.
Reproductive age female who doesn't want future childbearing	Fothergill surgery/manchester repair
3 rd degree prolapse in a menopausal female	vaginal hysterectomy
3 rd degree prolapse in a menopausal female with cystocele & rectocele	Ward mayo's hysterectomy : vaginal hysterectomy + anterior colporrhaphy (For cystocele) + posterior colpoperineorrhaphy (For rectocele)
55-60 yr old female with prolapse & comorbidities	Le Forts colpocleisis
• Elderly female unfit for surgery • Pregnant/immediate postpartum female	Ring pessary
1 st /2 nd degree prolapse	Kegel's exercise
Decubitus ulcer (D/t venous congestion)	Acridavine + glycerine packing

SUI & Urinary Fistula

00:34:10

----- Active space -----

- Female c/o dribbling of urine on increasing intra-abdominal pressure : Stress urinary incontinence.
- Female c/o continuous dribbling of urine + normal emptying of bladder : ureterovaginal fistula.
- Female c/o continuous dribbling of urine + **no emptying of bladder** : vesicovaginal fistula.
- urine coming from urethra + vagina at the time of urination : urethrovaginal fistula.

most common type of fistula : **VVF**.

- most common cause of VVF.
 - Developed countries : Total abd. hysterectomy.
 - Developing countries : Obstructed labour.
- IOC for VVF : **methylene blue 3 swab test**.
- Gold standard for VVF : Cystoscopy.
- Surgery of choice (Sx) :
 - Vaginal : Latzko repair
 - Abdominal : Chassar moir technique/O'Connor Sx
- Post op instructions (Sx) :
 - No p/s, no P/V, no intercourse : For 3-6 months.
 - No pregnancy : For 1 year.
 - Post op catheter : 14 days.
- Time of surgical repair : identified
 - <24 hrs : Immediate repair
 - >24 hrs : After 6 wks - 3 months
- Conception after VVF repair → Deliver by cesarean section (Classical).

Other Urinary Conditions

methylene blue swab test (IOC) :

1. upper swab → Wet + unstained : ureterovaginal fistula.
2. middle swab → Blue + wet : vesicovaginal fistula (VVF).
3. Lowermost swab → Blue + wet : urethrovaginal fistula.

For SUI :

- m/c done surgery :
 - Tension free trans obturator tape (TOT)
 - Transvaginal tape
- Gold standard surgery:

Burch colposuspension (Abdominal Sx)

Proximal urethra is suspended to **Cooper's ligament**/iliopectineal ligament.

- Drugs which may be useful in SUI : Duloxetine.
- Cyclical hematuria is seen in :
 - Fistula between bladder & uterus.
 - Bladder endometriosis.

----- Active space -----

Puberty

00:44:48

Puberty :

♀ = 10.5 years

♂ = 11.5 years

1st sign of puberty in females : Growth spurt.1st sign of puberty in males : Testicular enlargement.

Sequence of puberty in females

Growth spurt

Thelarche/breast budding → Tanner stage 2
(1st visible sign)

Pubarche (Pubic & axillary hair)

Hair on mons pubis → Tanner stage 3



Peak height velocity

↓ 6 months

menarche (12.5 yrs : within 2 yrs of thelarche)

Estrogen has the following Actions**During Puberty :**

1. Breast development.
2. Development of reproductive tract : uterus development (Non pregnant).
3. Cervical mucus production.
4. ↑ Bone mass.
5. Increased clotting factors.

Puberty**Delayed Puberty :**

- Females : No thelarche by 13 yrs.
- males : No testicular enlargement by 14 yrs.
- m/c cause : Constitutional delay.
- DOC : Pulsatile GnRH.

Hormones Responsible :

- main hormone responsible for breast development : Estrogen.
- main hormone responsible for pubic hair & axillary hair development : Androgen.

Precocious Puberty :

- ♀ (m/c) : Thelarche <8 yrs.
- ♂ : Testicular enlargement <9 yrs.
- Cause : Idiopathic.
- DOC : Continuous GnRH.

Precocious Menstruation :

menarche at <10 yrs.

Central Precocious Puberty :

- D/t premature activation of HPO axis :
 - LH : ↑.
 - FSH : ↑.
 - E : ↑.
- m/c cause : Idiopathic.
- 2nd m/c cause : Brain tumors (Hamartoma).
- Ix : MRI Brain.
- DOC : Continuous GnRH.

Peripheral Precocious Puberty :

- D/t exogenous estrogen/androgen :
 - LH : ↓.
 - FSH : ↓.
 - Estrogen/androgen : ↑.
- Cause :
 - Isosexual
 - Hypothyroidism
 - Estrogen secreting ovarian tumor
 - McCune Albright syndrome
 - Heterosexual
 - CAH
 - Androgen secreting ovarian/adrenal tumors

----- Active space -----

To be done in all cases of precocious puberty (97%) :

1. Bone age estimation.
2. LH, FSH, estrogen levels :
Differentiate central vs. peripheral.
3. MRI brain : Rule out brain hamartomas.
4. TSH levels : Rule out hypothyroidism.
5. TVS : To rule out ovarian tumors.

Neurotransmitters :

- which keeps hypothalamus in dormant state : GABA.
- which is responsible for pulsatile release of GnRH : Kisspeptin.
- which has a permissive role in puberty : Leptin.

Investigation

To be done in all case of heterosexual precocious puberty :
Levels of :

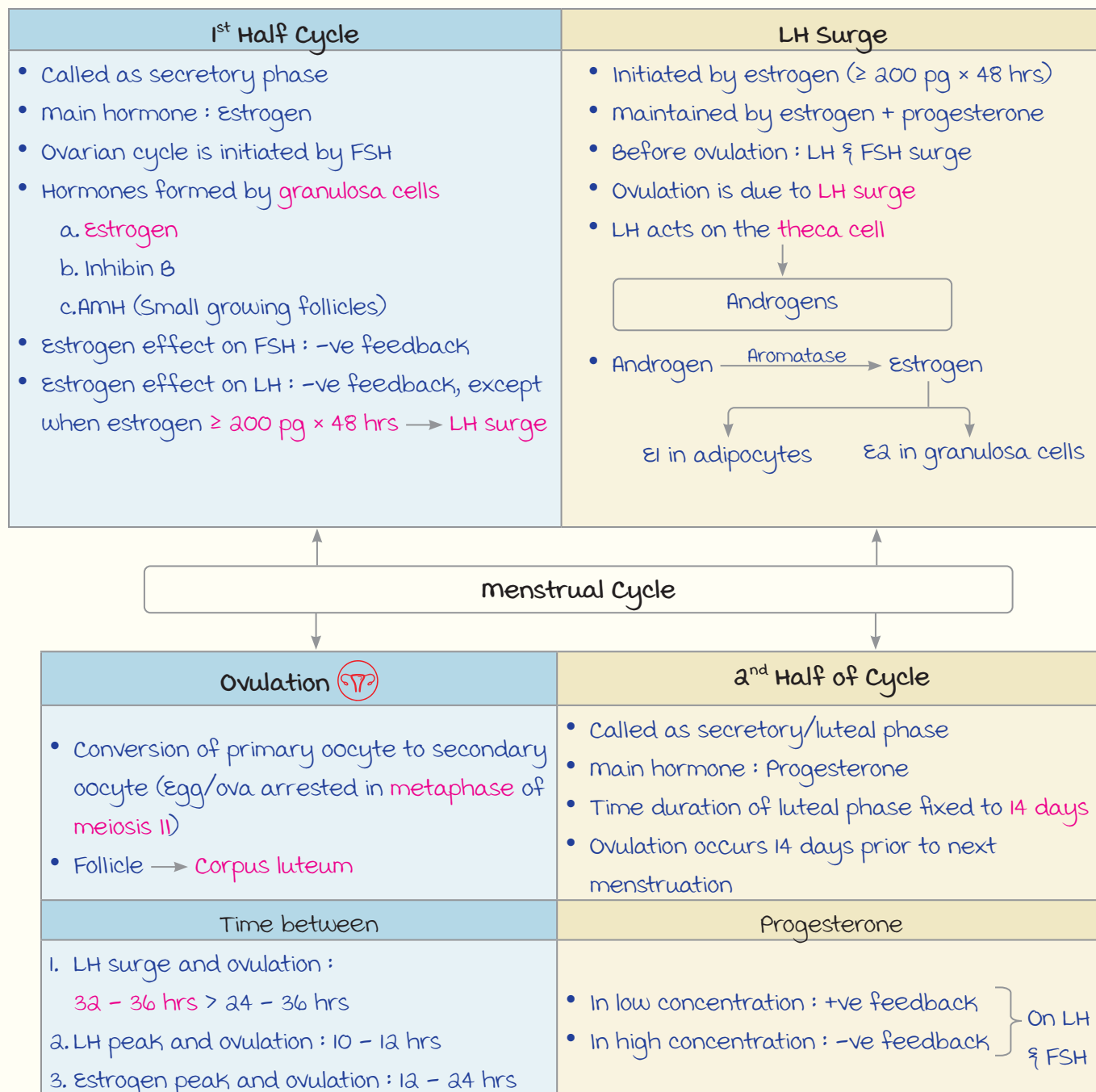
1. Testosterone.
2. DHEA-S : To rule out adrenal tumors.
3. 17-OH progesterone : To screen for CAH.

McCune-Albright syndrome
Precocious puberty
+
Polyostotic fibrous dysplasia
+
Cafe-Au-lait spots

GYNAECOLOGY REVISION : PART 3

Menstrual Cycle

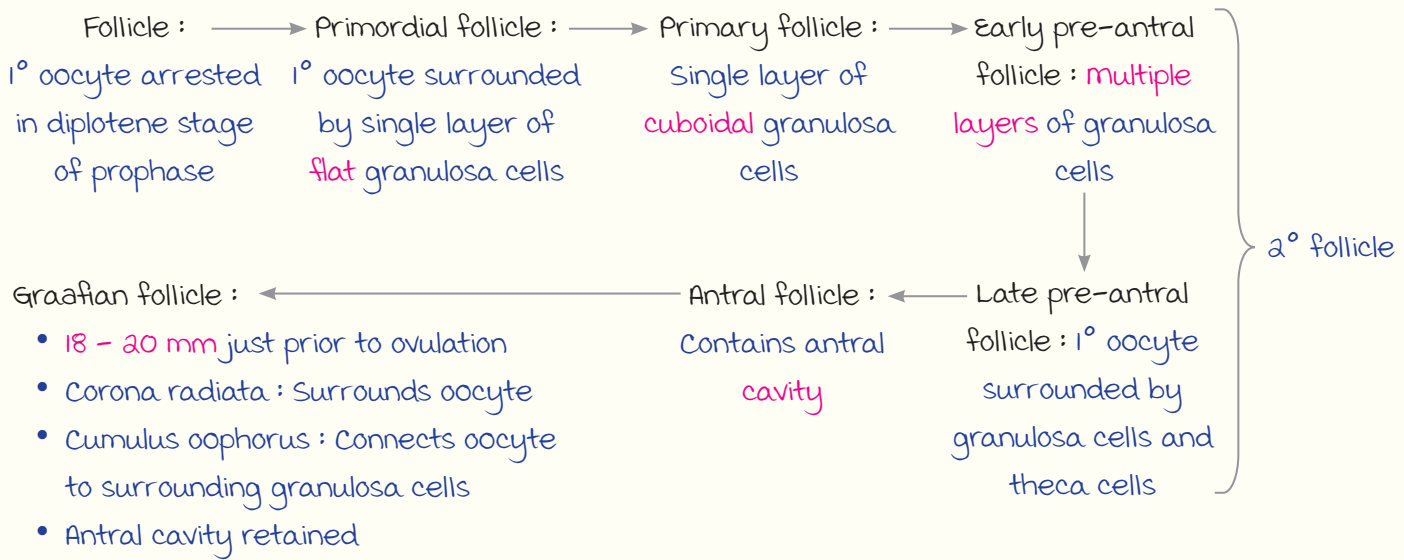
00:00:25



Note :

Hormone produced by **luteinized granulosa cells** : **Progesterone**.

----- Active space ----- Stage of Follicular Development 97 :



Note :

All follicles are 1° oocyte except : Graafian follicle → 2° oocyte.

Hormones in Gynaecology

00:16:05

----- Active space -----

- Natural GnRH is secreted by : **Arcuate nucleus** of the hypothalamus in a pulsatile manner.
- LH and FSH are secreted by : Anterior pituitary in response to GnRH.
- Natural estrogen :
 - E1 : Estrone → m/c in **menopausal females**
 - E2 : Estradiol → m/c in **reproductive age & pregnant females**
 - E3 : Estriol → **most specific** estrogen in pregnancy

most common synthetic estrogen :

- OCP : Ethinyl estradiol (EE)
- HRT :
 - Earlier : Conjugated equine estrogen (CEE)
 - New : 17 β -estradiol

Potency of estrogens (✓) : EE > CEE > E2 > E1 > E3

Third generation progestins include :

- Desogestrel
 - Norgestimate
 - Gestodene
- Least androgenic side effects

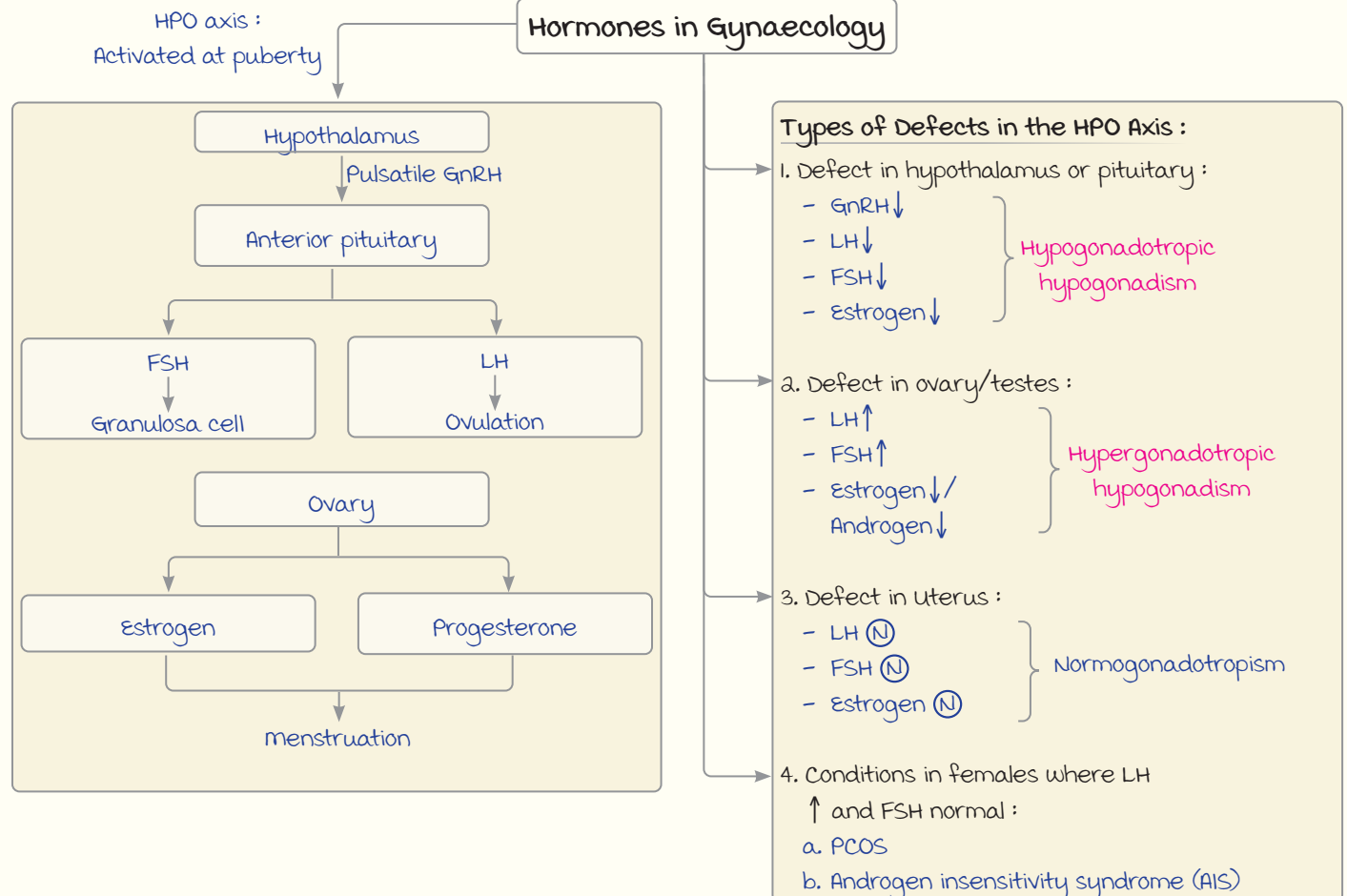
Fourth generation progestin : Anti-androgenic

Substances included :

- Cyproterone acetate
- Spironolactone
- Drospirenone
- Dienogest

m/c used in OCP : EE + 3rd gen progesterone

Ideal for PCOS patients : 3rd/4th gen progesterone



----- Active space -----

Kallmann Syndrome :

Defect in **KAL-1 gene** → Defective migration of GnRH neurons from olfactory epithelium.

Features :

- Hypogonadotropic hypogonadism :
 - GnRH ↓.
 - FSH ↓.
 - LH ↓.
 - Estrogen/Testosterone ↓.
- males > Females.
- A/w **anosmia**.

Presentation :

- Delayed puberty.
- 1° amenorrhea.
- Infertility.

DOC : Pulsatile GnRH.

Effects of Estrogen & Progesterone :

	Estrogen	Progesterone
Cervical mucus	• Thin, watery, profuse • Spinnbarkeit (Elastic) • Ferning ⊕ on microscopy (From day 7) • Favourable for sperm entry	• Thick, viscous, scanty • Non-elastic • Ferning disappears (Day 17) • Unfavourable for sperm entry (Basis of progesterone containing contraceptives)
Coagulation factors	↑ : Previous h/o thrombosis/MI → OCPs/ E + P containing contraceptives c/i	No effect on clotting factors
Salt and water	Retention	Wasting
Risk of endometrial cancer	↑	↓
Other effects	• Predominant vaginal cell : Superficial cell • Bone mineral density : ↑ • Cardioprotective : ↓LDL, ↑HDL	• ↑Risk of breast cancer • ↑Basal body temperature

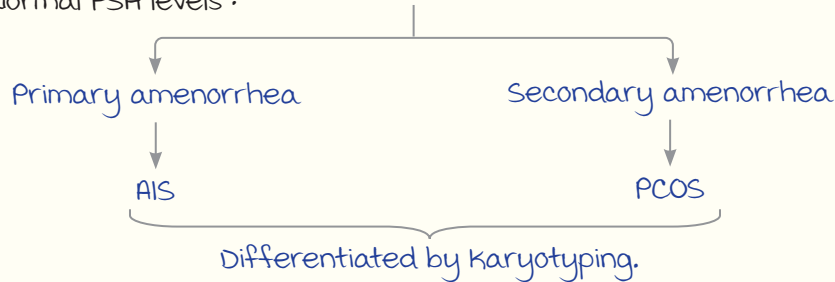
Gonadotropins :

Normal FSH : 4 - 10 IU.

Low FSH : < 4 IU (1 - 3 IU).

Best time to test for LH & FSH : Day 2 - 4 of cycle.

↑ LH + Normal FSH levels :



↑ LH & ↑ FSH + primary/secondary amenorrhea : Gonadal disorder (Ovary/Testes).

Prolactin on GnRH : -ve feedback → ↓ LH, ↓ FSH.

Hormonal Profile in Various Syndromes (97?) :

	LH	FSH	Estrogen	Other hormones
Asherman syndrome	Normal			-
Turner's syndrome	↑	↑	↓	-
Kallmann syndrome	↓	↓	↓	GnRH ↓
Sheehan syndrome	↓	↓	↓	-
PCOS/AIS	↑	Normal	⊖	Androgens : ↑↑ (Corresponds to male levels)

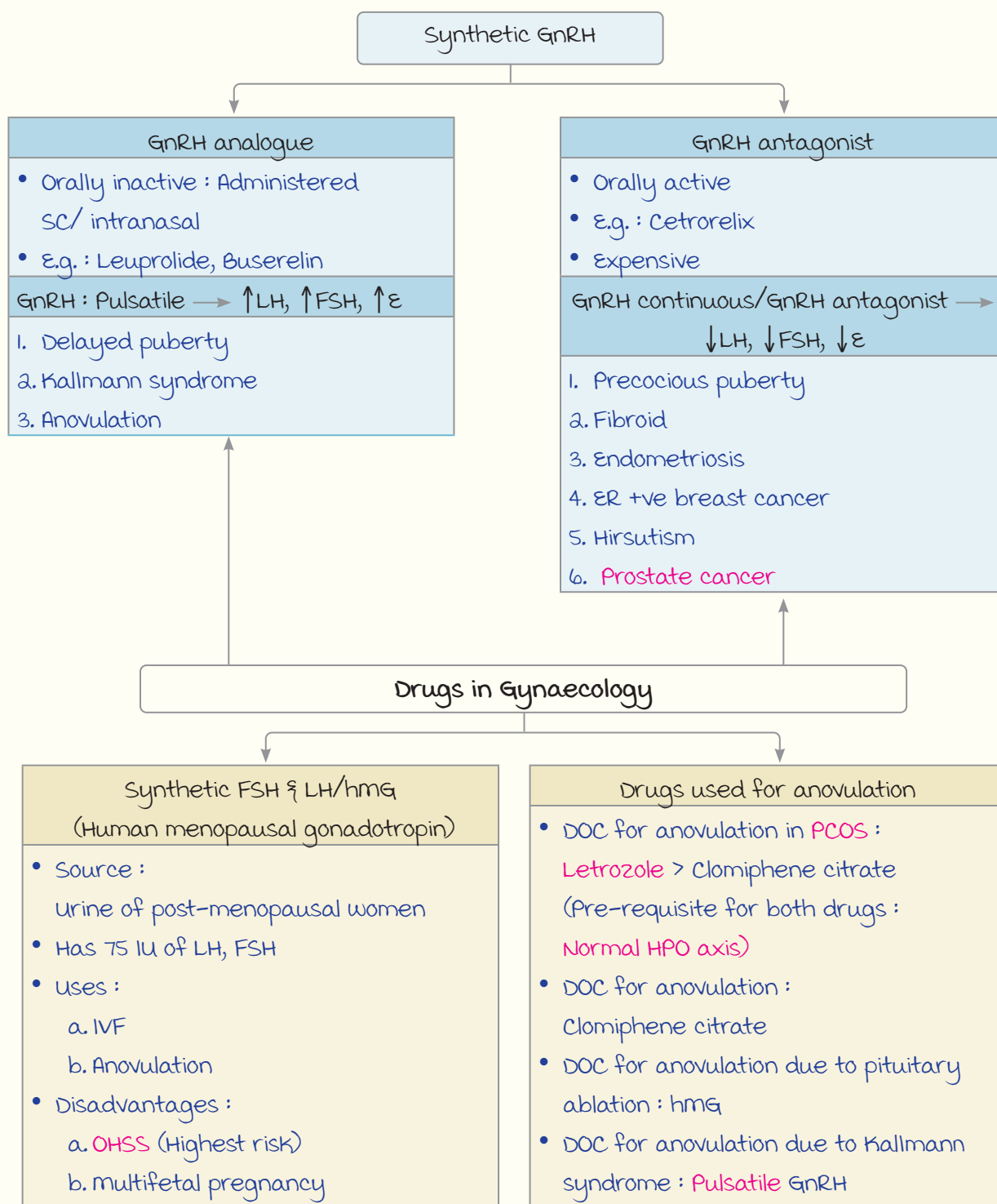
Note :

Triple layered endometrium : Late proliferative phase > Ovulation.

----- Active space -----

Drugs in Gynaecology

00:35:58



Abnormal Uterine Bleeding (AUB)

00:41:00

----- Active space -----

Normal Characteristics of Bleeding :

Length :

- Average : 28 days.
- Normal : 24 - 38 days
(Earlier : 21 - 35 days).

Blood loss :

Days :

- Days : 2 - 8 days (4.5 - 5 days).
- Volume : 80 mL (35 - 50 mL).

Types of AUB :

Hypomenorrhea :

Less blood loss : < 2 days / < 20 mL
(Asherman syndrome)

menorrhagia :

Heavy menstrual bleeding : > 80 mL / > 8 days
(Fibroid/adenomyosis)

Polymenorrhea : Short cycles

Oligomenorrhea : Long cycles / < 9 cycles in a year

metrorrhagia :

Irregular / intermenstrual bleeding (Polyp).

Causes of AUB :

- Polyp.
- Adenomyosis.
- Leiomyoma.
- malignancy / Hyperplasia.
- Coagulation disorders.
- Ovulatory dysfunction.
- Endometrial causes.
- Iatrogenic causes (IUCD insertion).
- Not classified.

Investigations :

- 1st investigation in AUB :
UPT (Except menopausal females)
 - 2nd investigation in AUB : TVS
 - If indicated :
Endometrial biopsy
- } c/i in pubertal females

AUB

1st Line Drugs in management of AUB :

1. Tranexamic acid
2. OCPs
3. Progesterone (Oral/mirena)

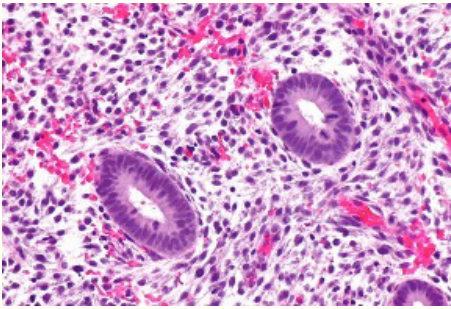
Endometrial Biopsy (🩺) :

- AKA endometrial sampling / endometrial aspiration cytology.
- Done in OPD
- Instrument :
 - India : Karman's cannula
 - worldwide : Pipelle / Vabra aspirator.
- Timing : Day 26 - 27 of cycle

Indications :

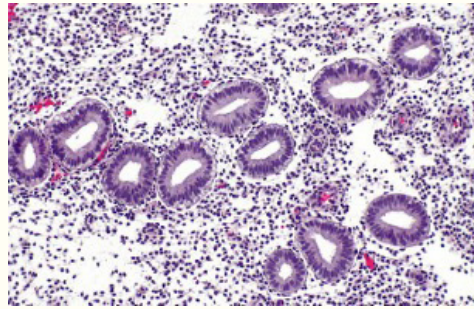
1. IOC for genital TB
 2. Can detect anovulation
 3. In case of AUB :
 - a. Reproductive female :
Endometrial thickness ≥ 11 mm / persistent bleeding
 - b. Perimenopausal female :
 - India : ≥ 40 yrs
 - worldwide : ≥ 45 yrs
 - c. Postmenopausal female : Endometrial thickness ≥ 4 mm
- } Endometrial biopsy irrespective of USG findings

----- Active space ----- Endometrial Biopsy HPE :



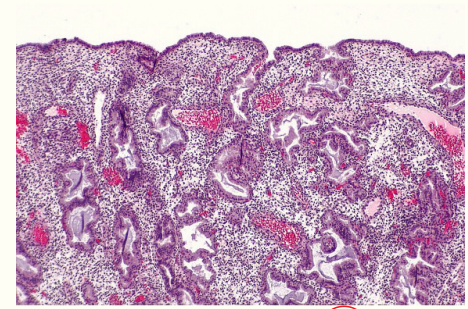
Proliferative phase :

- Simple tubular glands
- Telescoping of glands
- Pseudo-stratification
(Nuclei at different levels)



Early secretory phase > Ovulation :

- Subnuclear vacuoles :
- 1st sign of ovulation on biopsy
 - Appears on day 17 - 18



Late secretory phase (17) :

- Corkscrew/sawtooth glands
- Secretion d/t progesterone
⊕ (From corpus luteum)

Investigations for Endometrial Pathology :

IOC : Endometrial biopsy → Best done 2 - 3 days before menstruation.

Gold standard :

Hysteroscopy + Fractional curettage > Hysteroscopy + D & C > Fractional curettage
> Hysteroscopy.

Indications of fractional curettage :

- Hyperplasia with atypia.
- Insufficient sample.
- Hyperplasia without atypia + persistent bleeding despite management.
- Cervical stenosis.

AUB in Adolescent Girls :

Cause : Anovulation > Coagulopathy.

Investigation :

- UPT.
- S. Iron.
- Coagulation profile.
- C/i :
 - P/V exam.
 - Endometrial biopsy.

Doc : OCPs.

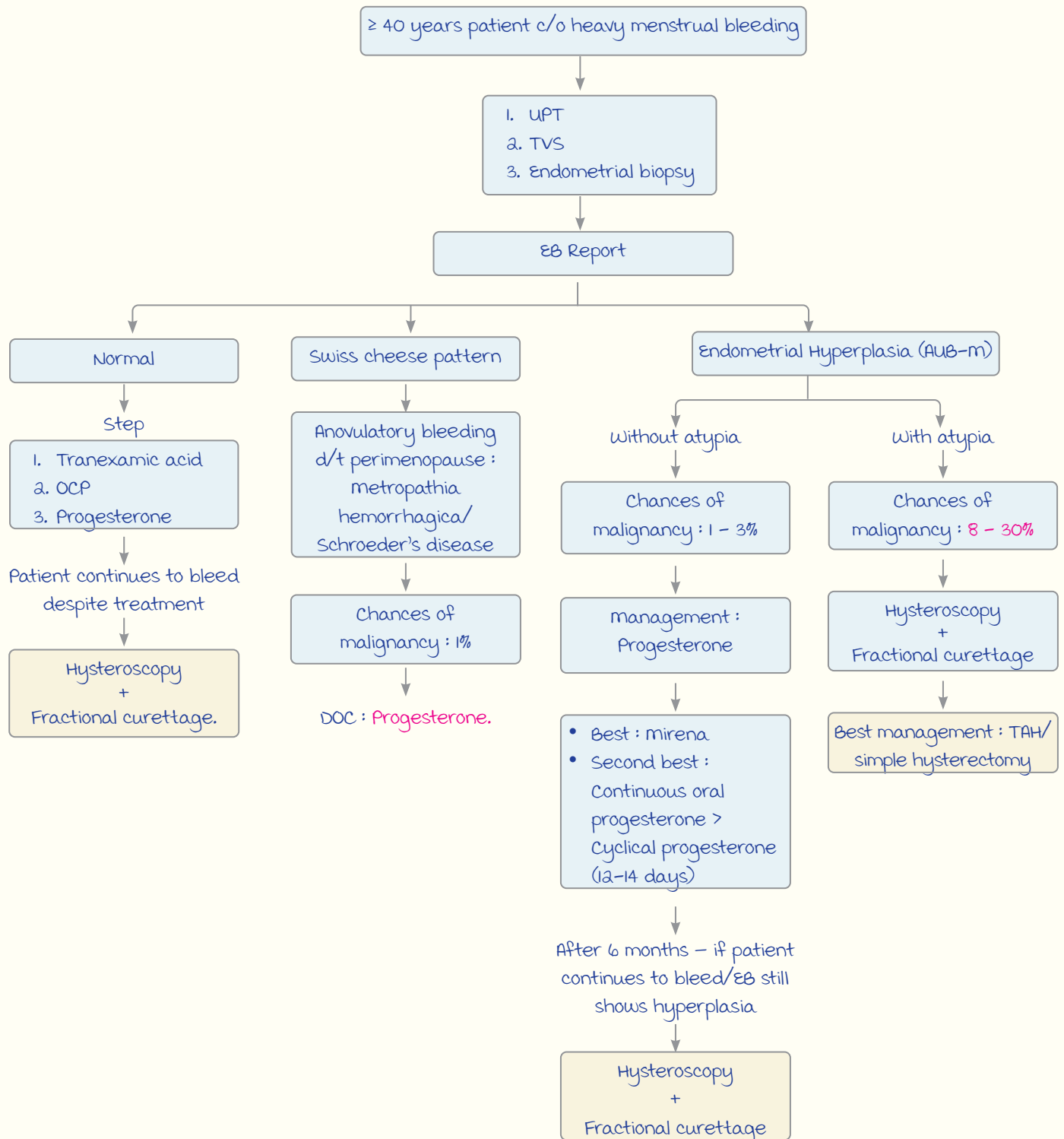
metropathia Hemorrhagica :

- Anovulatory bleeding seen in perimenopausal females.
- Presentation : 2 - 3 months of amenorrhea $\xrightarrow{F/b}$ Heavy bleeding.

Management of Perimenopausal Bleeding

00:58:55

----- Active space -----



Note :

Duration of progesterone Rx :

- Withdrawal bleeding : 4 - 6 days
- Protective effect on endometrium (Anti-proliferative) : 12 - 14 days (minimum).

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Endometrial Cancer

01:09:50

Important One Liners :most common endometrial cancer : **Adeno Ca.**

most malignant :

- Clear cell tumor
- Papillary serous tumor.

most common age : 50 – 70 yrs (mean : 60 years).

most common complaint :

Irregular bleeding (AUB-m).

most specific complaint : Postmenopausal bleeding.

most common route of spread :

Direct > lymphatic spread.

Risk Factors :

- **Family** : Familial
 - Lynch syndrome
 - BRCA 1/2 mutations
 - Cowden Syndrome (mutation of P-TEN gene on Chr 10)
- **Has** : Hypertension
- **O** : Obesity
- **L** : Late menopause/early menarche
- **D** : Diabetes
- **A** : Atypical endometrial hyperplasia
- **U** : unopposed estrogen (PCOS/HRT/Granulosa cell tumor)
- **N** : Nulliparity
- **T** : Tamoxifen

Endometrial Cancer
(Estrogen Dependent)

● : Corpus cancer syndrome
(↑Risk of endometrial Ca.)

Important Points (TV) :Gatekeeper gene for endometrial cancer : **PTEN.**

Adenocarcinoma grade 1, 2 : Type 1 endometrial Ca.

- With good prognosis.
- A/w : Excess estrogen.

Adenocarcinoma grade 3, papillary, serous, clear cell : Type 2 endometrial Ca.

- With poor prognosis.
- Not a/w ↑estrogen.

Important Stages :Stage 1 : Tumor confined to **uterus**

A. < 50% myometrial involvement.

B. ≥ 50% myometrial involvement.

Stage 2 : **Cervical** involvement.**Stage 3 :**

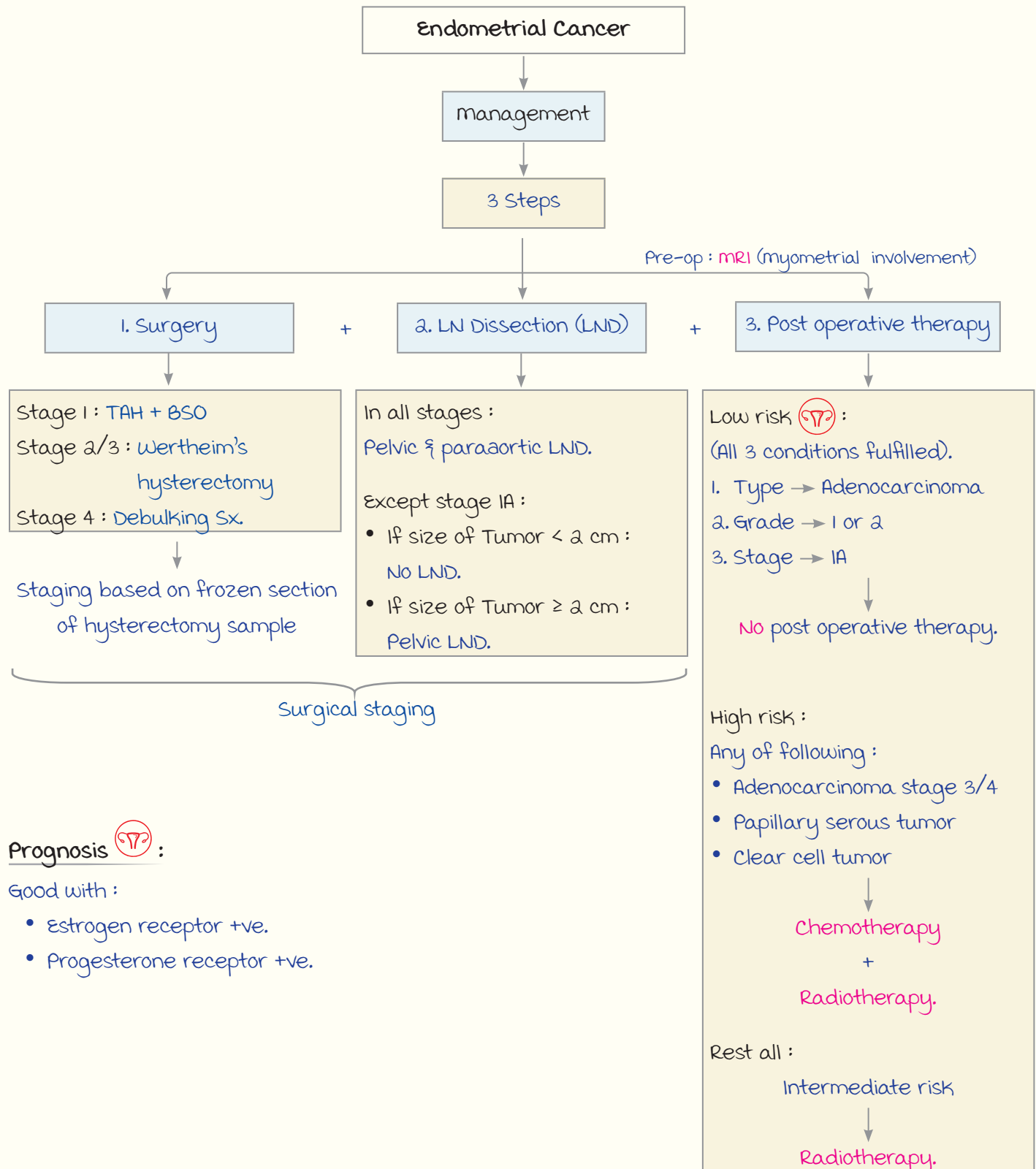
- 3A : Serosa/Adnexa involvement.
- 3B : Vagina/Parametrium involvement.
- 3C : LN involvement
 - CI : Pelvic
 - Ca : Para aortic

Note :

- Superficial inguinal LN in endometrial, ovarian & cervical Ca → **Stage 4B** (metastasis).
- Sentinel lymph node in vulval cancer : Superficial inguinal LN.

management of Endometrial Cancer :

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Prognosis ⚠️ :

Good with :

- Estrogen receptor +ve.
- Progesterone receptor +ve.

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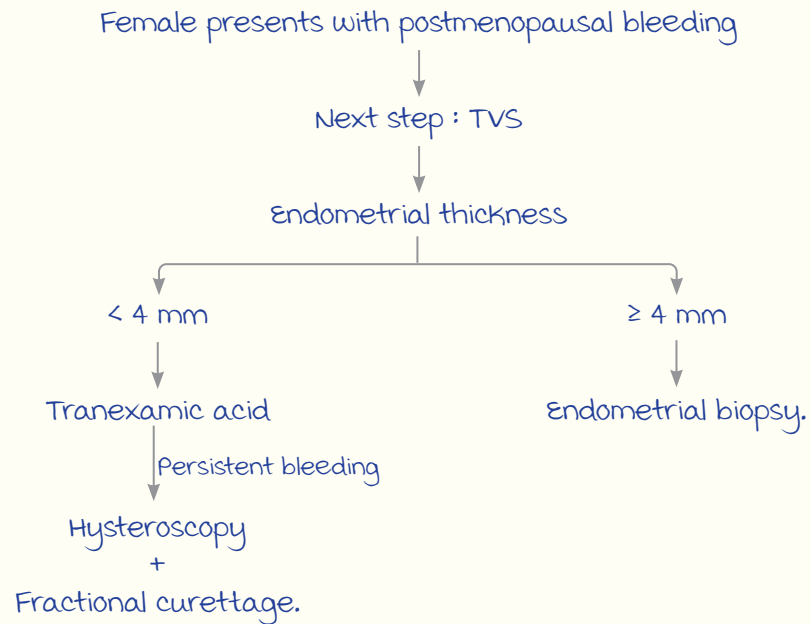
Post Menopausal Bleeding

01:31:58

Causes :

- Worldwide : Polyp > Endometrial atrophy.
- India : Cancer cervix.

management :



GYNAECOLOGY REVISION : PART 4

----- Active space -----

PCOS

00:00:20

Pathophysiology (??):

- Excessive androgen from ovary.
- (N) Androgen levels in females : < 70 ng.
- In PCOS : 70 - 200 ng.
- Due to ↑ androgen :
 - Hirsutism.
 - It is folliculotoxic.
 - No follicles mature → No Graafian follicle, no ovulation → Infertility (Anovulation)

Additional Features :

- Insulin resistance (Acanthosis nigricans).
- Obesity (m/c) :
 - In obese PCOS females :
 - $E1 : E2 = 2 : 1$ ($1 : 2 \rightarrow$ (N)).
 - menometrorrhagia.
- Abnormal gonadotropin secretion :
 - LH : ↑.
 - FSH : (N).
 - Normally LH : FSH = 1 : 1.
 - In PCOS patients : 2 : 1 or 3 : 1.
 - Test for FSH : LH is done on : D2 - D3 of cycle.

Not in Rotterdam criteria (??):

- Insulin resistance
- Necklace pattern
- Obesity
- LH : FSH
- menometrorrhagia

HAIR-AN Syndrome :

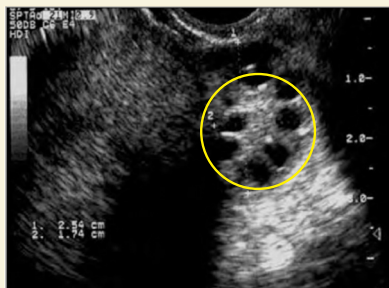
- Hyperandrogenism.
- Insulin resistance.
- Acanthosis nigricans.

PCOS

Rotterdam Criteria :

Any 2 of the 3 :

- ↑ Androgen levels :
 - Hirsutism
 - ↑ Testosterone levels.
- Ovulatory dysfunction :
 - Oligomenorrhea - Amenorrhea
- On USG :
 - Polycystic ovaries :
 - ≥ 12 follicles : - > 10 cc Volume
 - 2 - 9 mm size



Necklace pattern

Lab Findings :

Increased :

- Testosterone : ↑
- DHEA sulphate : ↑ (Adrenal glands)
- LH : ↑
- LDL : ↑
- Estrogen in obese females : ↑
- AMH ↑ (From growing follicles)

Decreased :

- Progesterone.
- SHBG (Sex hormone binding globulin) :
D/t insulin and androgens

Unchanged :

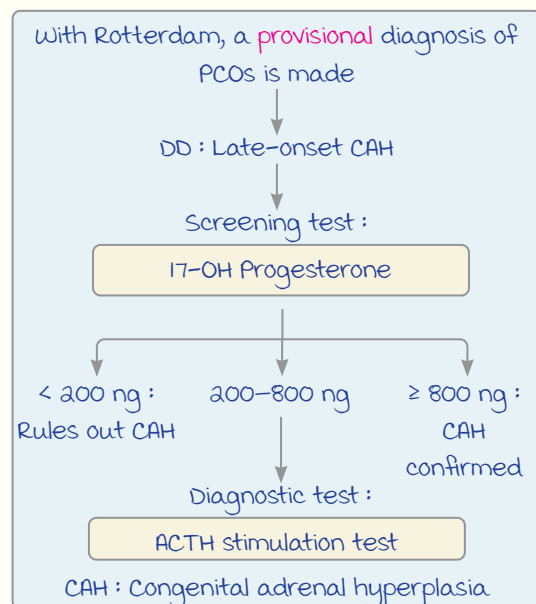
- FSH.
- Prolactin.

Progesterone challenge test (??) :
Positive (Withdrawal bleeding)

Note :

European criteria : ≥ 20 follicles.

----- Active space -----



- First step in management of PCOS : Lifestyle modification.
- Always in PCOS, treat only complaints with which the patient comes to you.
- DOC for irregular cycles : **OCP** (3rd/4th generation progesterone).
- DOC for hirsutism : **OCP**.
- DOC for insulin resistance : **metformin**.
- Drug which cannot be used in patients with PCOS who want to conceive : **Spironolactone**.

PCOS

- Chromosomes a/w PCOS : 2 & 9
- Loci a/w PCOS : THADA, DENDDIA

management of Infertility :

First step :

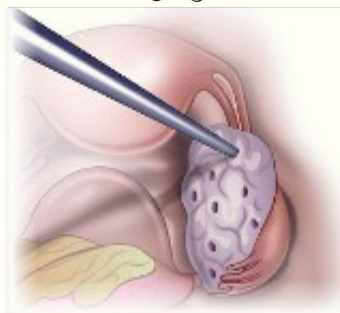
- Lifestyle modifications.
- Folic acid : 400 mcg daily (5 – 10% ovulate).

1st line :DOC : **Letrozole** > Clomiphene citrate (CC).2nd line :

- **hmg**.
- CC + metformin (Insulin resistance).
- CC + Corticosteroid (↑DHEAS).
- OCPs for 3 cycles → CC (E + P : ↓Androgens).

3rd line : Pulsatile GnRH.

Outdated surgery : Laparoscopic ovarian drilling.



- Destroy ovarian stroma

Theca cell destruction

↓ Androgen

- s/e :

Premature ovarian insufficiency
(Premature menopause).

Long Term Complications :

D/t ↑ androgen :

- Dyslipidemia : ↑LDL, ↓HDL.
- Coronary heart disease.

D/t ↑ estrogen :

- Endometrial Ca.
- Ovarian Ca +/-.
- Endometrial hyperplasia.
- NASH (Non-alcoholic steatohepatitis).

D/t obesity :

- Anxiety/depression.
- Sleep apnea syndrome.
- metabolic x syndrome.

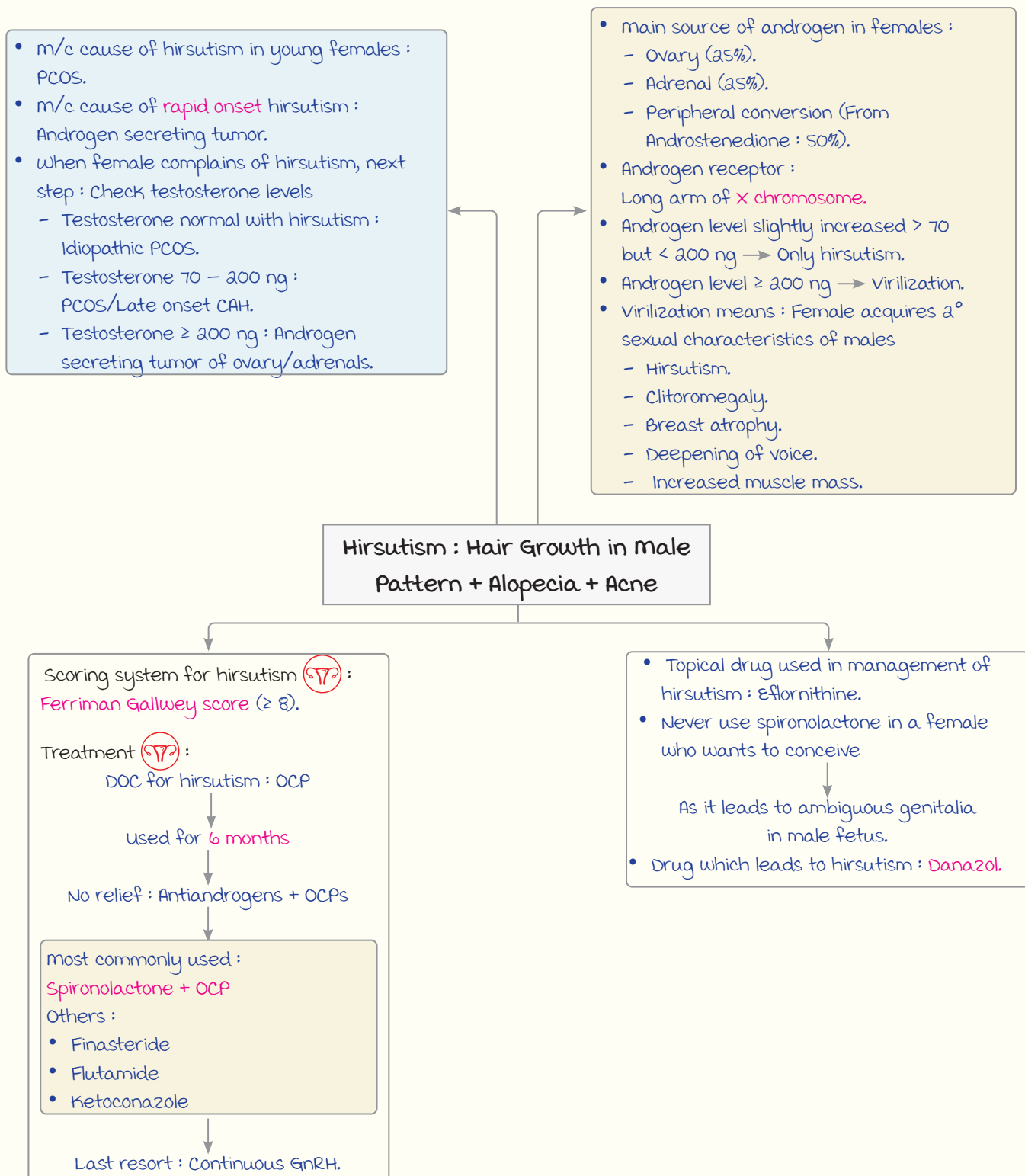
Complication not seen :

Osteoporosis (E : ⊖/↑).

Hirsutism

00:21:00

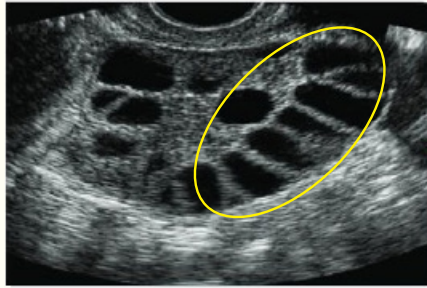
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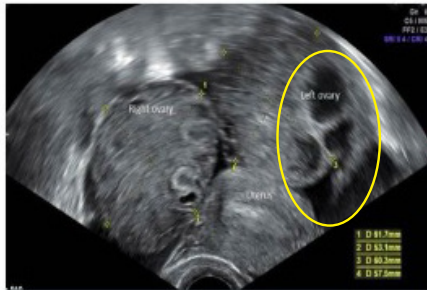
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Ovarian Hyperstimulation Syndrome

00:27:22



Small follicles : PCOS

Big follicles + ↓ stroma :
OHSS h/o IVF/Inj. hMG

OHSS

- m/c drug which leads to OHSS : **hMG**.
- Drug not causing OHSS : Letrozole.
- Triggering factor for OHSS : Inj. hCG.
- Inflammatory mediator released : VEGF (makes capillaries leaky).
- Inj. hCG :
 - Trigger ovulation (Functionally ~ LH).
 - Give : Follicles ≥ 2 & ≥ 18 mm.
 - Do not give hCG if follicles ≥ 20 and/or estradiol (E₂) > 3500 pg/mL (Each follicle $\rightarrow 200$ pg_{E₂})
- Early onset OHSS : < 9 days after Inj. hCG
- Late onset OHSS : ≥ 9 days after Inj. hCG (D/t pregnancy)

OHSS Classification :

- mild :
Abdominal discomfort,
USG $\rightarrow$ B/L enlarged ovaries.
- moderate :
USG $\rightarrow$ **Ascites** (+).
- Severe : Clinical evidence of ascites.
Critical :
End-organ damage (D/t thrombosis)
 - ARDS.
 - DIC.
 - ARF.

management :

Admit patient in case of severe/critical/if patient is pregnant.

Drugs used to decrease severity in OHSS :

- Bromocriptine (Prevents VEGF).
- IV Albumin (Volume expansion $\rightarrow$ $\uparrow$ Intravascular volume).
- Heparin (Thromboprophylaxis).

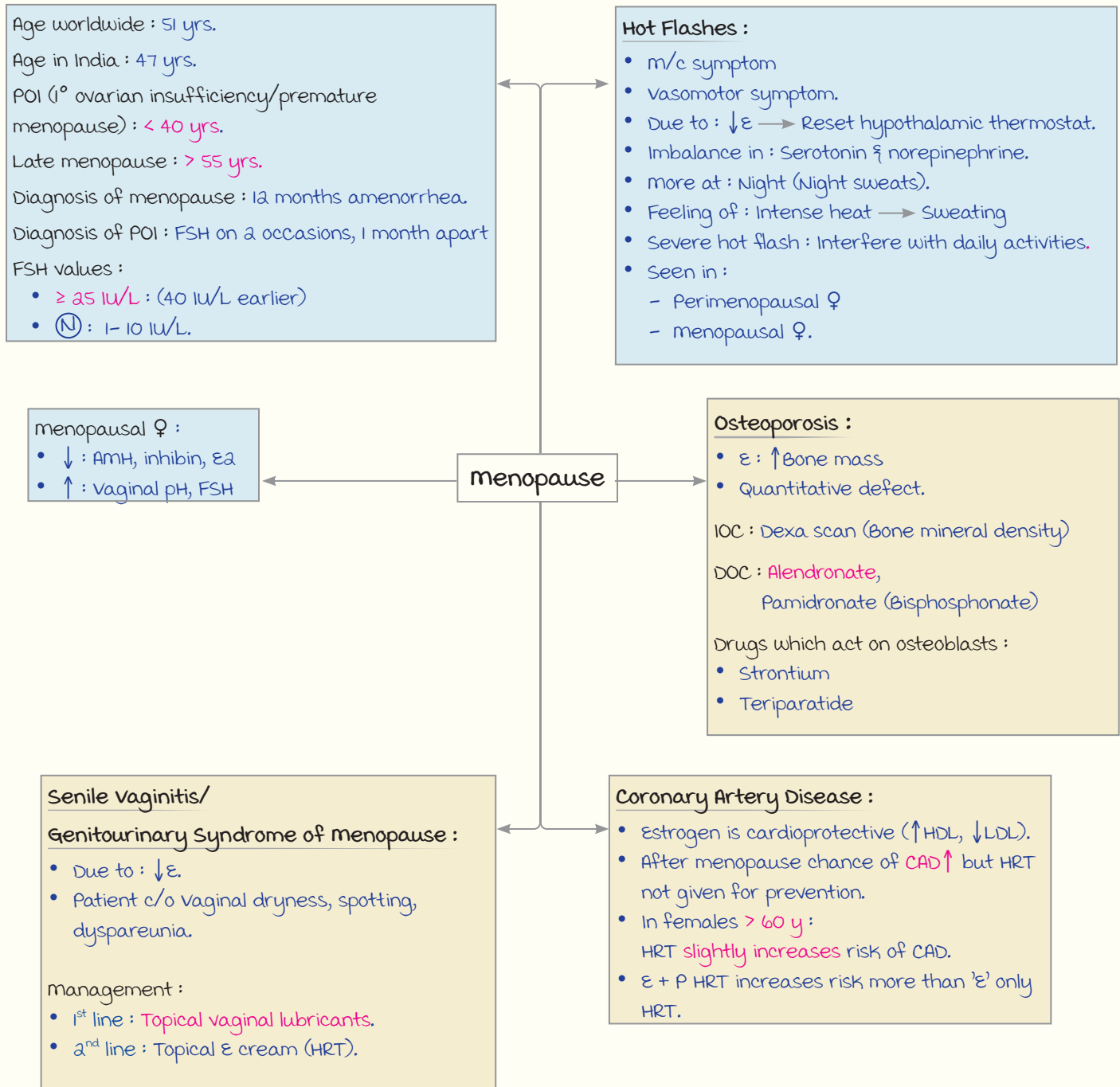
Ways to prevent OHSS :

- Withhold Inj. hCG.
- Replace hCG with GnRH agonist.
- In patients with PCOS : use frozen embryo transfer or pretreatment with metformin.

Menopause

00:36:12

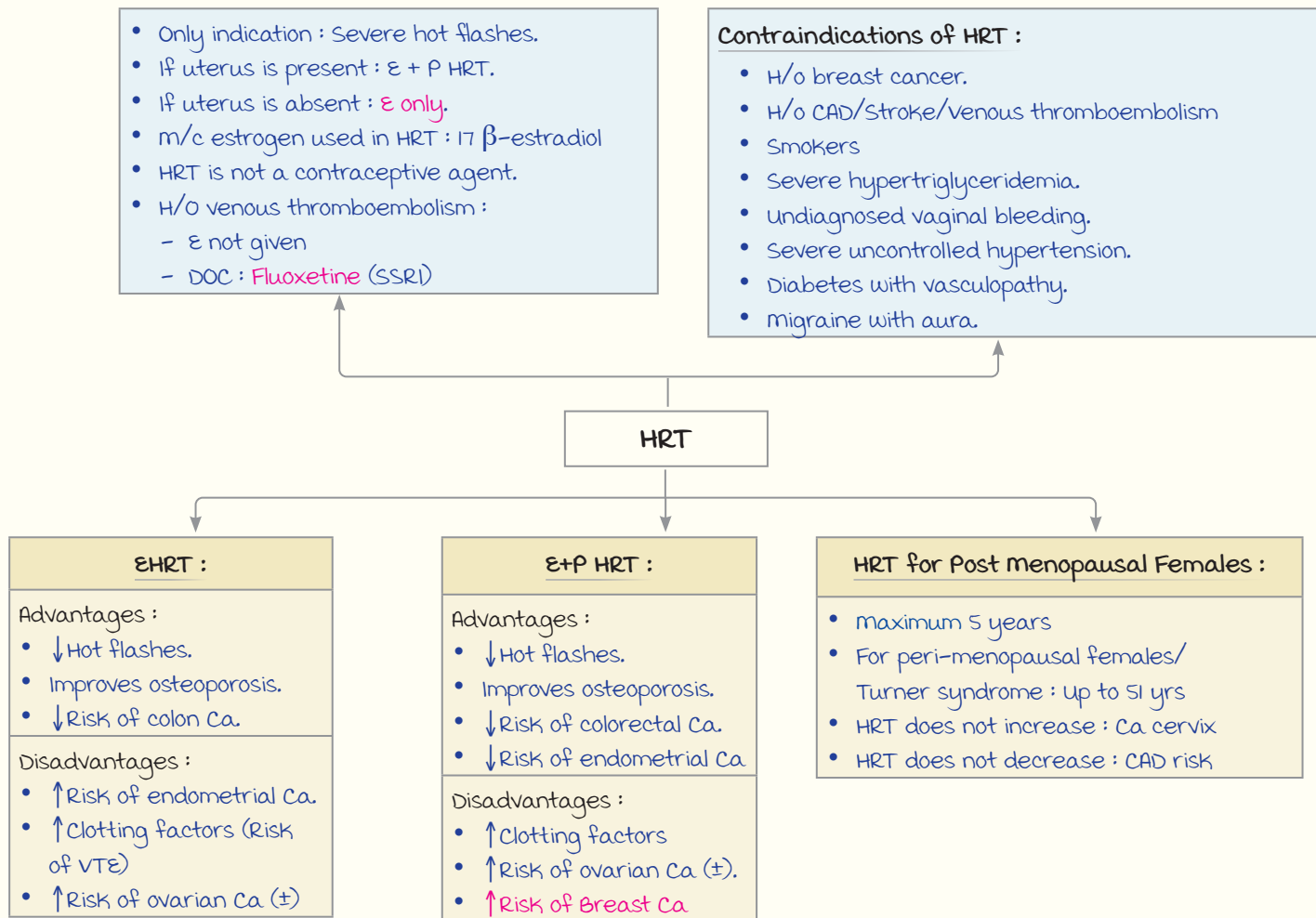
----- Active space -----



----- Active space -----

HRT

00:42:10



AMH, POI & SERMs

00:47:56

----- Active space -----

Anti-mullerian Hormone (AMH) :

- Normal : 1 – 3 ng.
- Secreted by : Granulosa cells of small growing follicles (Preantral & antral).
- POI : ↓ (< 0.5 ng).
- PCOS : ↑ (> 3 ng).

Premature Ovarian Insufficiency (POI) :

- Infertility : Donor egg + IVF.
- metabolic cause : Galactosemia.
- Clinical : 2° amenorrhea + ↑ FSH.
- Hypergonadotropic hypogonadism.
- Chromosomal association :
 - Turners mosaic.
 - Fragile X syndrome.
- Rx for ↓ libido : Testosterone.

Selective Estrogen Receptor Modulators :

Cause	Condition	Rx
Tamoxifen, Clomiphene citrate, Raloxifene	Hot flashes	ε + Bazedoxifene
	Vaginal dryness	Ospemifene
-	Anovulation	Clomiphene citrate (Ovulation induction)
Tamoxifen	Endometrial Ca	-
-	Osteoporosis	Raloxifene (s/e : Hot flashes, avoided)

Key points : Follicle numbers

- maximum : 5th month of IUL (6 – 7 million)
- At birth : 1 – 2 million
- At puberty : 4 – 5 lakh
- Ovulates in lifetime : 500
- Undergo atresia every month : 1000

----- Active space -----

Asherman Syndrome

00:55:24

Pathology :

- Intrauterine adhesions.
- Thin defective endometrium.

m/c cause : Excessive curettage.

maximum risk :

- Postpartum curettage.
- Genital TB.

m/c symptoms :

- Group of symptoms : menstrual symptoms
- Single symptom :
Infertility > 2° amenorrhea > hypomenorrhea

1st investigation : HSG

IOC : Hysteroscopy

Asherman Syndrome

On HSG :



Asherman syndrome :

- multiple filling defects
- Irregular outline (moth eaten appearance).

D/d :



Polyp/fibroid :

- Single
- Smooth, regular outline.

Laboratory investigation (T):

- LH
 - FSH
 - Estrogen
 - Progesterone challenge test
 - E+P challenge test
- Ⓝ
- ve

management :

Hysteroscopic resection of adhesions
+
Insert paediatric Foleys catheter
+
E + P for 3 months

Secondary Amenorrhea

01:02:54

----- Active space -----

Definition :

Absence of menstruation for ≥ 3 months/90 days in previous menstruating females

Cause :

- m/c cause : Pregnancy
- Others :
 - Thyroid dysfunction
 - $\uparrow$ Prolactin
 - Severe anemia

Investigations :

- UPT (R/o pregnancy)
- TSH
- Prolactin
- LH, E_2 , FSH

Diagnosis Based on FSH & LH Levels :

FSH decreased (1 – 3 IU) :

- Hypothalamic causes :
 - Excessive exercise.
 - Bulimia.
 - Anorexia.
 - Stress.
- Pituitary causes :
 - Sheehan's syndrome.
 - Prolactinoma.

FSH : Increased (> 10 IU).

Ovarian : POI.

FSH : Normal (4–10 IU).

- PCOS :
 - Progesterone challenge test +ve.
 - LH : $\uparrow$.
- Asherman syndrome :
 - Progesterone challenge test -ve.
 - $E + P$ challenge test -ve.
 - LH : (N).

Secondary Amenorrhea

Diagnosis Based on Symptoms :

- Secondary amenorrhea with h/o PPH/blood transfusion + Failure to lactate : Sheehan's syndrome ($\uparrow$ PPH $\rightarrow$ Necrosis of anterior pituitary).
- H/o amenorrhea with headache, visual disturbances and galactorrhea : Prolactin secreting pituitary tumor.
- Amenorrhea with $\downarrow$ FSH with :
 - MRI : Empty sella $\rightarrow$ Sheehan's syndrome.
 - MRI : Space occupying mass $\rightarrow$ Prolactinoma.
 - MRI : Normal $\rightarrow$ Hypothalamic cause.

- Secondary amenorrhea with progesterone challenge test +ve : PCOS.
- Secondary amenorrhea with progesterone + estrogen challenge test -ve : Asherman syndrome.

S. Prolactin :

- Normal up to : 20 – 25 ng.
- mild elevation < 50 ng : Repeat prolactin.
- If > 50 ng/mL : MRI.

DOC for hyperprolactinemia : Cabergoline.

DOC for hyperprolactinemia in infertile females : Bromocriptine.

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Note :

menstrual irregularities d/t thyroid dysfunction

Hypothyroidism :

↑ Bleeding.

Hyperthyroidism :

Amenorrhea/oligomenorrhea.

Pituitary Causes :

Sheehan's syndrome :

- m/c symptom : Failure to lactate.
- 2nd m/c symptom : 2° amenorrhea.

Prolactinoma :

- m/c symptom : 2° amenorrhea
- 2nd m/c symptom : Infertility

Pituitary apoplexy (👁️) :

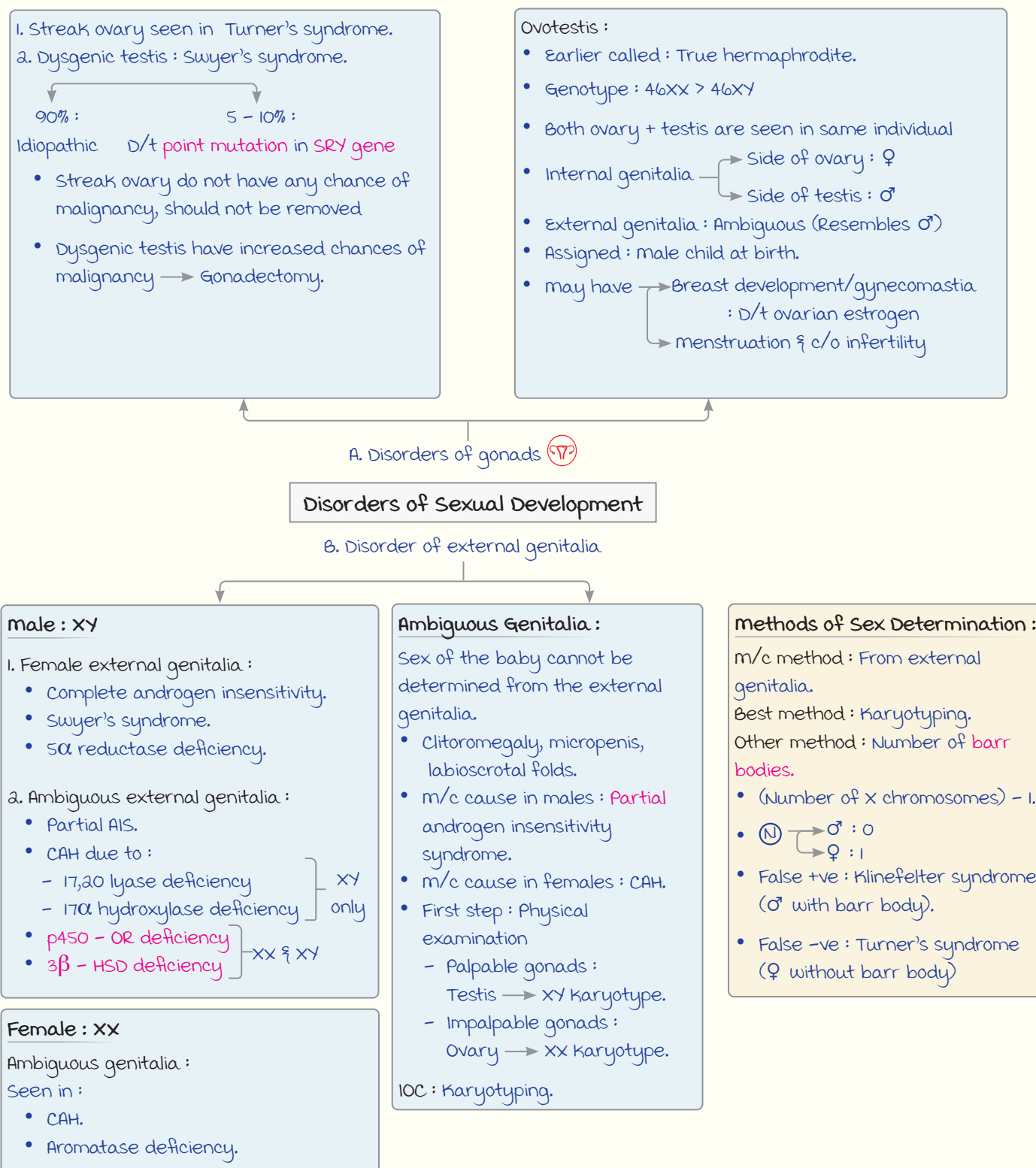
- Pituitary adenoma + PPH.
- Post partum : Shock, hypoglycemia, headache, visual disturbance.
- Rx : Corticosteroids.

GYNAECOLOGY REVISION : PART 5

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Disorders of Sexual Development

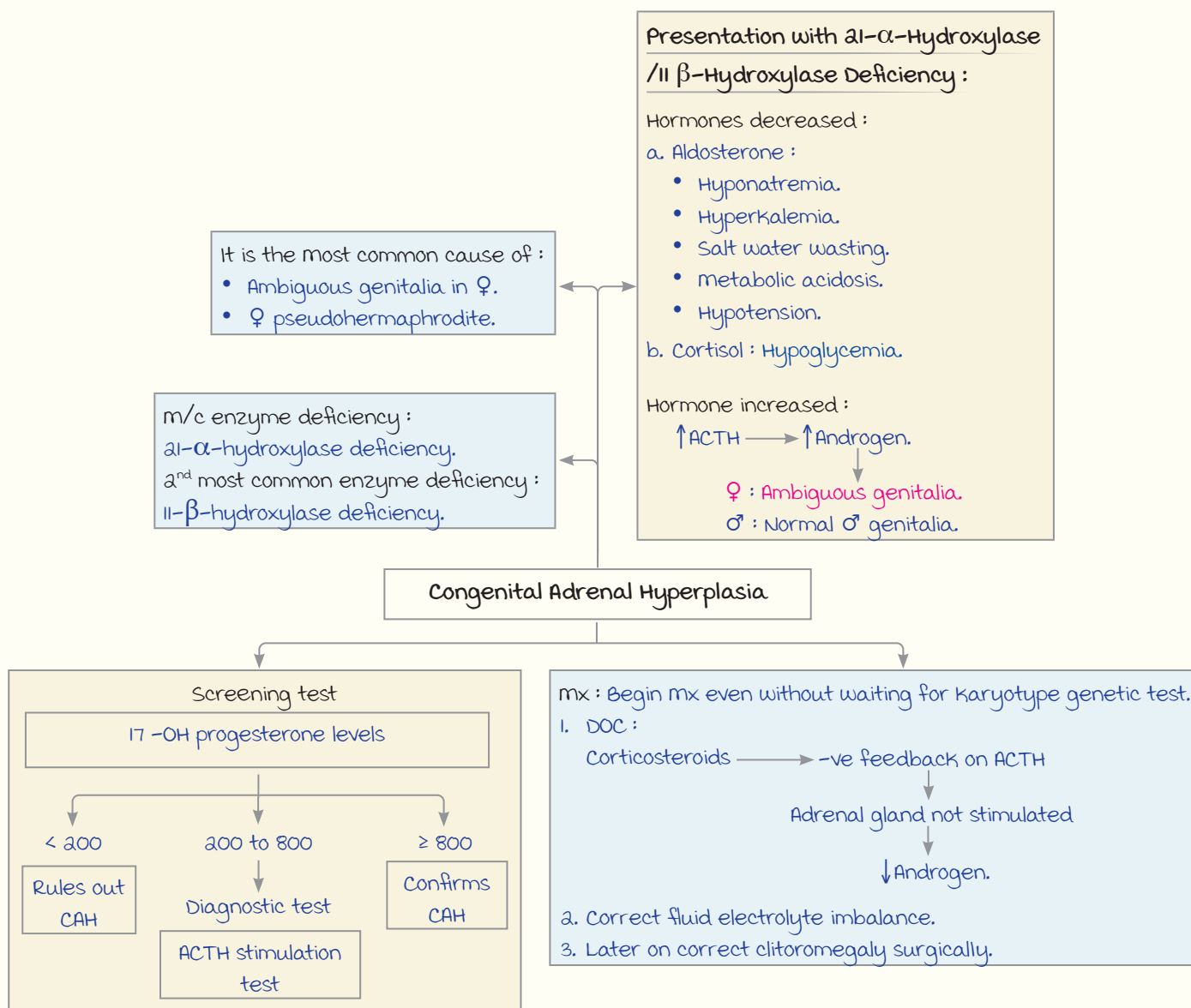
00:01:02



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Congenital Adrenal Hyperplasia (CAH)

00:19:02

**Enzyme Deficiencies Leading to Ambiguous Genitalia (ST) :**

XX	XY	Both
21 α -hydroxylase	17, 20 lyase	3 β -HSD
11 β -hydroxylase	17 α -hydroxylase	p450 OR

Aromatase deficiency :

- maternal androgens $\xrightarrow{\text{Aromatase}}$ Estrogen.
- Ambiguous genitalia in XX + hirsutism in mother during pregnancy.
- 17-OH progesterone : (N).

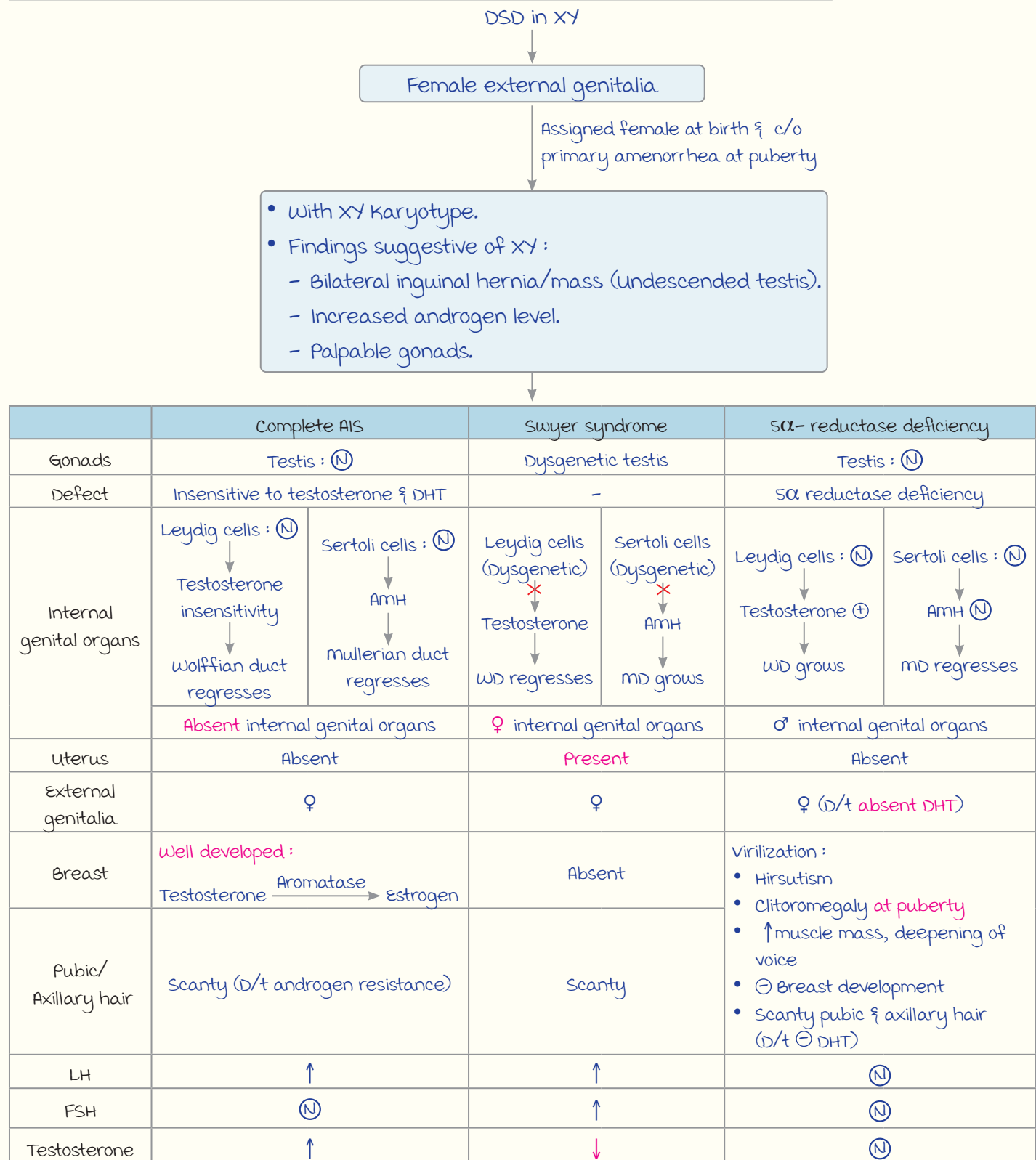
Antley-Bixler syndrome :

- p450-OR deficiency.
- Associated with skeletal anomalies (E.g. : Craniosynostosis).

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Disorders of Development in 46 XY

00:29:44



----- Active space -----

Management :

Androgen insensitivity syndrome (AIS) :

- Female gender-affirming care.
- Gonadectomy :
 - D/t ↑ chances of malignancy.
 - Done at 14 - 16 yrs (After complete breast development).
- Administer 'E' HRT till 51 years.
- Vaginoplasty :
 - Technique → McIndoe technique.
 - Vecchiotti technique (Laparoscopy).
 - Timing : Just before/after marriage.

Swyer's syndrome :

- Female gender-affirming care.
- Gonadectomy :
 - D/t ↑ chance of malignancy.
 - Done immediately after diagnosis.
- Estrogen x 1 yr (For breast development)
 - ↓ F/b
 - Lifelong estrogen + progesterone (uterus ⊕).

Partial AIS vs. 5 α -reductase Deficiency :

Both present as :

- Female with 1° amenorrhea.
- Findings suggestive of 46, XY karyotype : B/L inguinal hernia.
- Ambiguous genitalia.

	Partial AIS/Reifenstein syndrome	5 α reductase deficiency
Breasts	Well developed	Absent
Pubic hair	Scanty	Scanty
Onset of clitoromegaly	At birth	At puberty

Primary Amenorrhoea

00:53:23

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Definition 🚺:

- Absence of menstruation by 13 yrs without breast budding (Tanner stage I).

OR

- Absence of menstruation by 15 yrs with breast budding (Tanner stage 2 or above)

OR

- Absence of menstruation within 3 yrs of thelarche

Causes :

With uterus :

- Kallmann syndrome.
- Gonadal dysgenesis :
 - Turner syndrome (45 XO).
 - Swyer' syndrome (46 XY).
 - Pure gonadal dysgenesis (46 XX).
- Cryptomenorrhoea.

Without uterus :

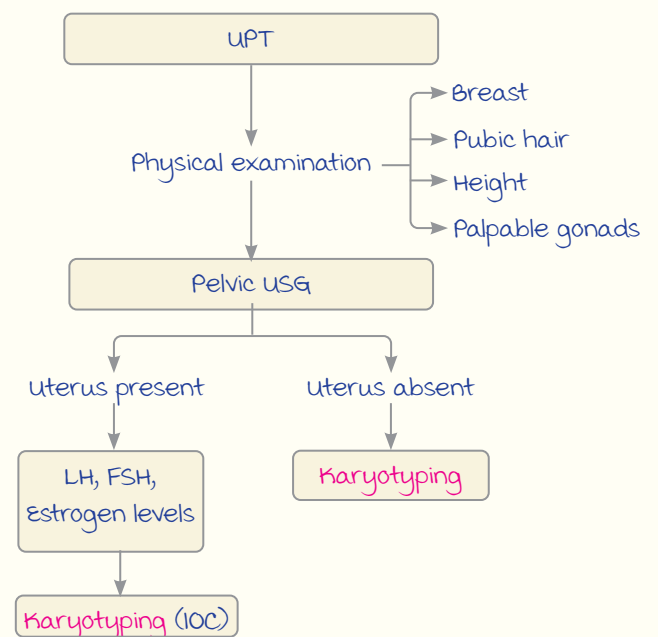
- Mullerian agenesis.
- Complete AIS.

Primary Amenorrhea

Based on LH, FSH & E levels :

- 1° amenorrhea with ↓FSH, ↓LH, ↓E → Hypogonadotropic hypogonadism : Kallmann syndrome.
- 1° amenorrhea with ↑LH, ↑FSH, E↓ → Hypergonadotropic hypogonadism :
 - Turner's syndrome
 - Swyers syndrome
 Gonadal dysgenesis.
- 1° amenorrhea with (N) FSH, ↑LH : AIS.
- 1° amenorrhea with (N) FSH, (N) LH :
 - Mullerian agenesis (uterus ⊖).
 - Cryptomenorrhoea (uterus ⊕, bulky d/t hematometra).

Work up in Primary Amenorrhea :



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Primary Amenorrhea with Uterus Present

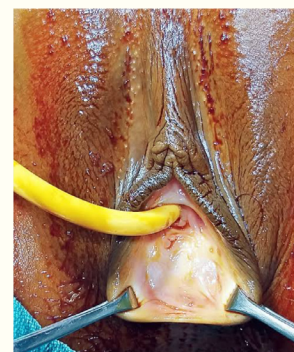
01:02:01

Cryptomenorrhea :

Hidden menstruation.

Clinical features :

- Cyclical pain in abdomen every month.
- Breast development : Present.



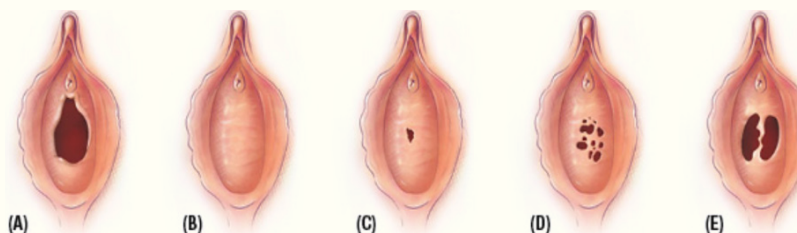
Imperforate hymen

Differential diagnosis :

	Imperforate hymen	Transverse vaginal septum
Local examination	Tense bulging hymen ± bluish colour	Ⓝ
Cough impulse	⊕	Absent
Hematocolpos	more common	Less common
Hematometra	Less common	more common
Endometriosis	Less common	more common
PR examination	Bulge felt on pulp of the finger	Bulge felt on the tip of finger
IOC	-	MRI

mx of imperforate hymen : **Cruciate incision.**Note :

Types of hymen :



A : multiparous/sexually active.

B : Imperforate hymen

C : Virgin ♀.

D : Cribriform hymen.

E : Septate hymen.

Kallmann Syndrome :

Clinical features :

- ♀ complains of 1° amenorrhea with uterus present.
- Breast & pubic hair absent.
- Low FSH & LH.
- **Anosmia.**

Gene involved : **KAL-1** gene.

m/c : 46 XY > 46 XX.

management : **Pulsatile GnRH.**

Gonadal Dysgenesis :

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Turner's syndrome :



- Karyotype : 45 XO
- In some cases, **mosaic** Turner's seen in a cell lines are present : 45 XO/46 XY.
- Gonads : In mosaic
 - One side : 45 XO → Streak ovary.
 - Other side : 46 XY → Dysgenetic testes.
- Risk of malignancy :
 - Streak ovaries : No chances of malignancy.
 - Dysgenetic testes or undescended testes : ↑ chances of malignancy → Gonadectomy.

Turner's Syndrome

Remember in all cases of suspected Turners : Do **Karyotyping**.

Associated features

- Short stature (SHOX gene mutation).
- Webbing of neck
- Low posterior hair line.
- Shield shaped chest.
- Widely spaced nipples.
- SNHL.
- Autoimmune disease :
 - Diabetes.
 - Hashimoto's thyroiditis.
 - Celiac disease.
- Cubitus valgus.
- Heart disease :
Bicuspid aortic valve > Coarctation of aorta.

- m/c tumor seen in dysgenetic gonads : Gonadoblastoma
- m/c malignancy : Dysgerminoma

Ix (🔍) : Karyotyping

- ECHO.
- Renal USG (Associated with horseshoe shaped kidney).
- TSH, T3, T4 (To rule out Hashimoto's thyroiditis).
- CBC, Lipid profile, fasting glucose.
- RFT, LFT.
- Audiometry (To rule out SNHL).
- Anti-endomysial Abs (Risk of Celiac disease).

management :

- Growth hormone (To Rx short stature) : ≤ 8yrs.
 - Estrogen x 1 yr (For breast development)
- ↓ F/b
- Estrogen + Progesterone x lifelong (upto 51 yrs).

----- Active space -----

Turner's syndrome vs. Swyer syndrome :

	Turner's syndrome	Swyer syndrome
uterus	Present	
Breast & pubic hair	Absent	
Karyotype	45 XO	46 XY
Gonads	Streak ovary	Dysgenetic testis
Associated features (Webbing of neck, low posterior hairline etc.)	Present	Absent
Stature	Short	Normal
Gonadectomy	- Not required - Done only in mosaic variant	Required

Primary Amenorrhea with Uterus Absent

01:19:02

	mullerian agenesis	AIS
AKA	mayer-Rokitansky-Kuster-Hauser (MRKH) syndrome	Testicular feminizing syndrome
Karyotype	46 XX	46 XY
External genitalia	♀	♀
Gonads	Ovary	Testes
uterus	Absent	Absent
Breast	Present	Well developed (Tanner stage 4/5)
Pubic hair	Present	Scanty
Serum testosterone	Ⓝ ♀ levels	♂ testosterone levels
LH	Ⓝ	↑
FSH	Ⓝ	Ⓝ
Other features	-	- B/L inguinal hernia - worst reproductive outcome

- management of mullerian agenesis :
vaginoplasty (Just before/after marriage).
- management of infertility in mullerian agenesis : IVF f/b surrogacy.
- mullerian agenesis is associated with :
 - Renal anomalies.
 - Skeletal anomalies.

GYNAECOLOGY REVISION : PART 6

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Endometriosis & Dysmenorrhea

00:00:46

General Points :

m/c site : **Ovary** (Chocolate cyst/endometrioma) > **POD**
> broad ligament (BL) > uterosacral (US) ligament.

L/c site : **Spleen** > **CNS**.

m/c accepted theory : **Sampson's theory** of implantation/retrograde menstruation (For ovary and POD).

For umbilicus : **Halban's** theory of lymphatic spread.

For distant sites like lungs : **Ivanoff's** theory of coelomic metaplasia.

Scar endometriosis : **Direct** spread.

- **Common after** : Episiotomy, cesarean section, hysterotomy, myomectomy.
- **Least likely** : **After hysterectomy**.

Symptoms :

Age : **Reproductive** age (25-35 years)

m/c : **Nulliparous ♀**.

L/c in : **Pubertal ♀**.

Note :

If endometriosis occurs in pubertal females → Suspect **mullerian malformations**.

m/c symptom : Pain (2^o **Dysmenorrhea** > chronic pelvic pain > dyspareunia).

2nd m/c symptom : Infertility.

3rd m/c symptom : Adnexal mass (Chocolate cyst).

Note :

Female c/o **dysmenorrhea + dyspareunia**

↓
Suspect **endometriosis**.

Endometriosis : Ectopic Endometrium

Examination Findings :

P/A exam : **Nothing** significant.

P/V exam :

uterus :

- **Retroverted, fixed** (Cannot be corrected manually).
- **(N) size**.
- **Non tender**.

Adnexa :

- **B/L adnexal mass** (Chocolate cyst) + adnexal tenderness.
- **US ligament** : Nodularity ⊕.

IOC : **Laparoscopy**.

Gold standard : **HPE**.

CA-125 levels : **↑** (>100 IU).

On MRI :

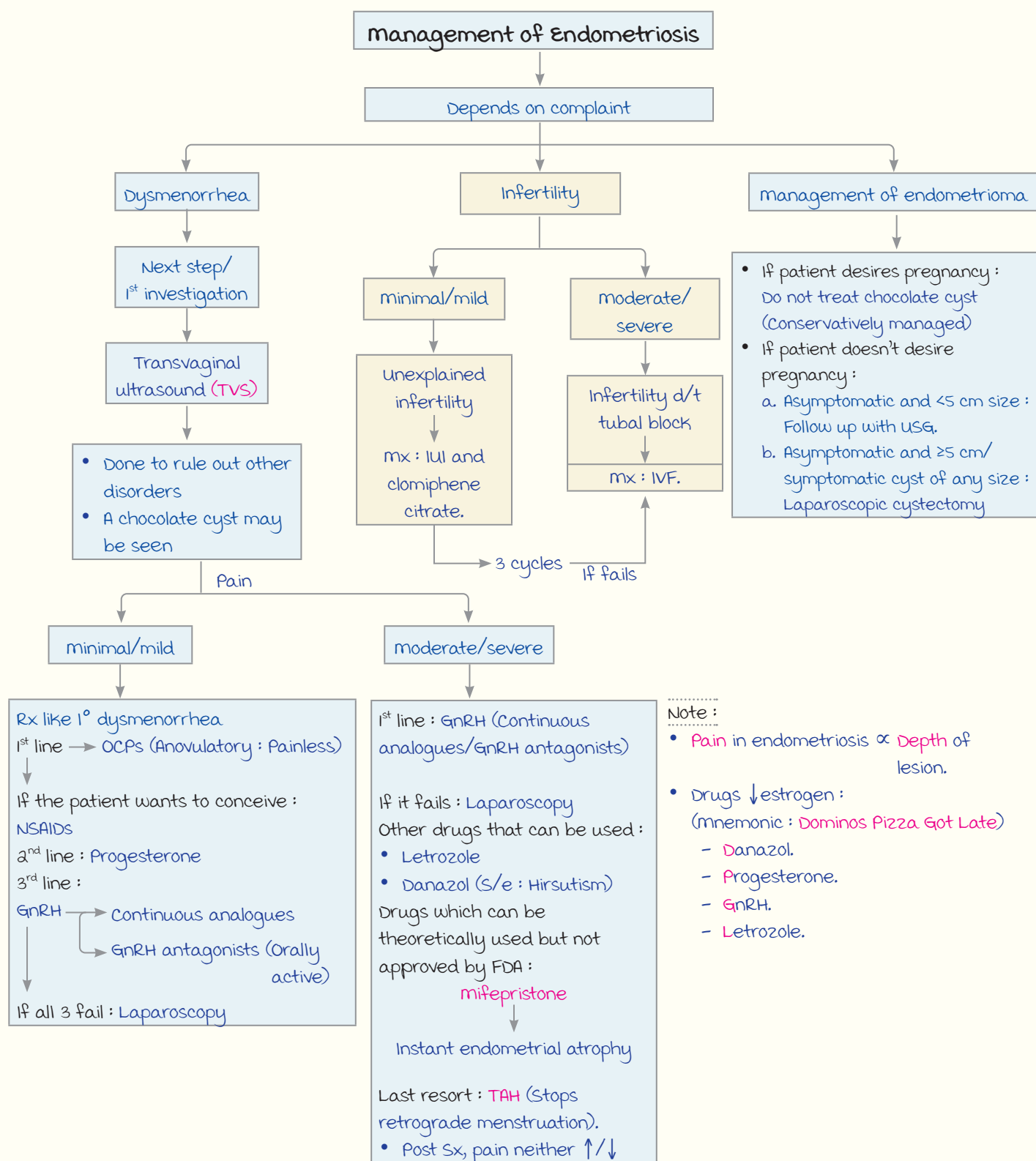
- Done in bladder & bowel endometriosis.
- In deep rectosigmoid endometriosis : **mushroom cap sign (MRI)**.

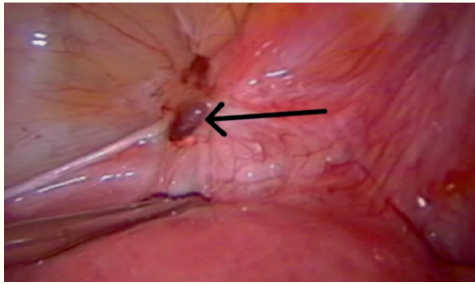


♀ with pain at Sx site,
↑ during periods

↓
Diagnosis : **Scar endometriosis**.

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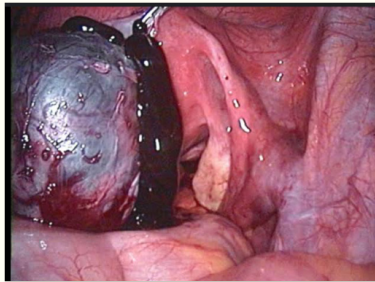




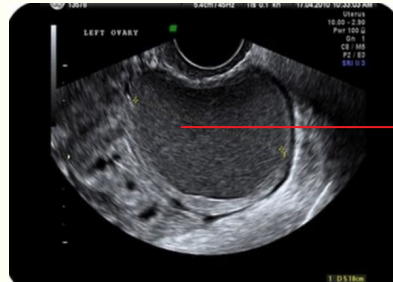
Red flame lesion : Recent lesion

Gunshot/Powder burn
appearance : Older lesions

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Chocolate cyst



USG chocolate cyst

Homogenous
internal
echoes

Presence of chocolate cyst/endometrioma → Severe endometriosis.

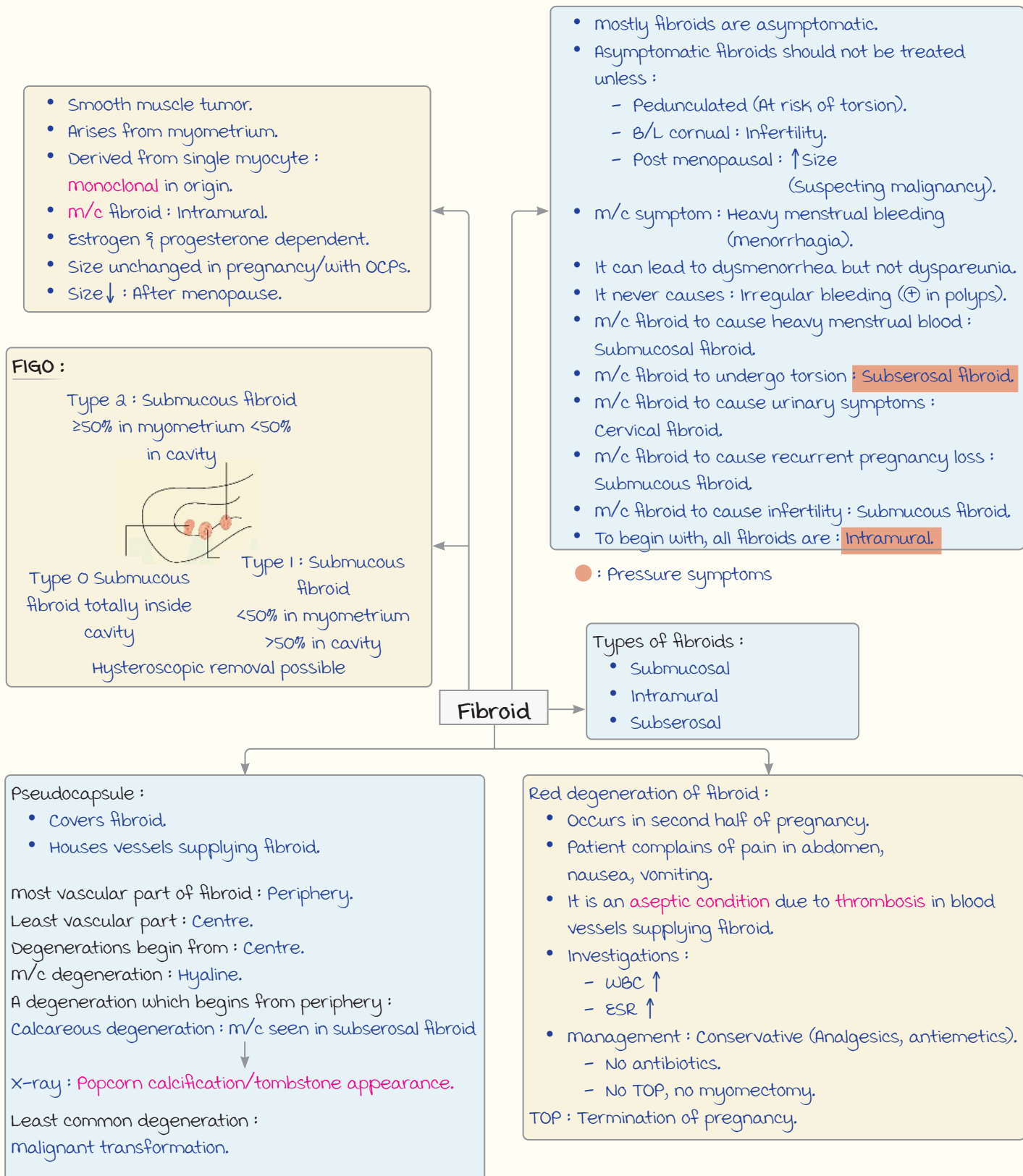
Dysmenorrhea :

	Primary dysmenorrhea	Secondary dysmenorrhea
Cause	Not pathological : Progesterone withdrawal ↓ Vasoconstriction → $\text{PGF-}\alpha$	Pelvic pathology : • Endometriosis (m/c) • PID • Adenomyosis • Fibroid
Pain	menstruation DI → <72 hrs after onset	• Pre-menses → Post-menses • Progressively ↑
Seen in	In young girls, since menarche (Except initial cycles : Anovulatory)	• Reproductive age • New onset dysmenorrhea (Ⓝ prev. cycles)
p/v	Ⓝ	Finding correspond to pathology
Relieved on	• Physical exercise • After marriage • After childbirth	As per pathology
Rx	• OCP : DOC • NSAIDs (Desires to conceive)	

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Fibroid

00:23:35



In case of submucous fibroid leading to heavy menstrual bleeding/infertility : Sx

- Type 0, 1 : Hysteroscopic myomectomy.
- Type 2 : Laparoscopic myomectomy.

- In case of intramural fibroid/subserosal fibroid causing heavy menstrual bleeding

1st try medical management :

1st line drugs (Do not affect size) :

- Tranexamic acid.
- OCP.
- Progesterone.

↓ Fails

2nd line drugs (↓ Size & bleeding) :

- ↓ Estrogen :
 - Danazol.
 - GnRH analogue (Continuously).
 - GnRH antagonist.
 - Letrozole.
- ↓ Progesterone :
 - mifepristone/RU 486 (Progesterone antagonist).
 - ulipristal (Selective progesterone receptor modulator).

management of Fibroid (Only Symptomatic)

If medical management fails in subserosal/intramural fibroid :

↓ Surgery

- Family not complete : Laparoscopic myomectomy.
- Family complete :
 - UAE : uterine artery embolization.
 - HIFU : High intensity focused USG.
 - MRgFUS : MR guided focused USG.
 - Hysterectomy.

In case of pressure symptoms :

D/t only :

- Intramural fibroid.
- Subserosal fibroid.

↓ management

medical Rx : Drugs ↓ E, ↓ P.

myomectomy : Can be done hysteroscopically/laparoscopically.

Hysteroscopically :

1. Type 0 } Submucosal
2. Type 1 } fibroid

Laparoscopically :

1. Type 2 submucosal fibroid
2. Subserosal fibroid
3. Intramural fibroid

After myomectomy, if patient conceives :

Delivery is by LSCS.

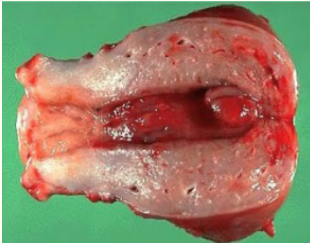
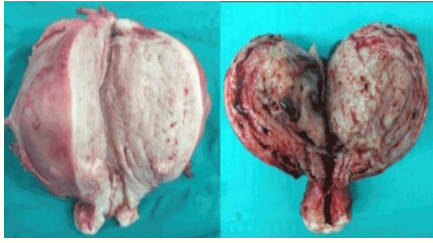
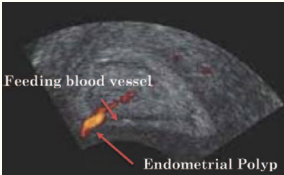
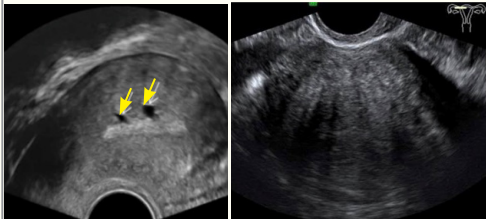
UAE :

- Done via femoral artery using polyvinyl alcohol.
- C/I :
 - ♀ wanting future pregnancy.
 - menopausal ♀.

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Fibroid vs. Polyp vs. Adenomyosis

00:39:40

	Fibroid	Polyp	Adenomyosis
Pathology	Smooth muscle tumor	mucosal outgrowth	Endometrial tissue in myometrium
Incidence	m/c in reproductive age ♀ (25-35 years)	- Seen : All age groups ↑ Age : ↑ Polyp chances	≥ 40 years ♀, multiparous ♀ (m/c)
Symptoms	Heavy menstrual bleeding (HMB)/menorrhagia : m/c symptom	- Reproductive age : Irregular bleeding - menopausal ♀ : m/c cause of postmenopausal bleed	- m/c symptom : menorrhagia + dysmenorrhea - uterine tenderness ⊕
Gross appearance	 - Irregular whorled appearance - Off-white color - Pseudocapsule : ⊕ - Arises from broad base	 - Red, fleshy outgrowth - Narrow base - Supplied by single feeder vessel	 myometrial hypertrophy : - Globular, symmetrically enlarged - Cut section : multiple hemorrhages (b/t menses of endometrial tissue)
Per abdominal (P/A) examination	Abdominopelvic lump : - Unable to converge fingers below lump. - mobile, firm, midline mass - Size : Irregularly enlarged (upto 20 weeks pregnant)	-	-
Per vaginal (P/V) examination	Irregularly enlarged, non-tender uterus ~ upto 20 wks	Normal	- uterus : - Globular, symmetrically enlarged - Tender (Halban sign) - Size : 10-12 weeks pregnant uterus - Adnexal tenderness ⊖
Ultrasound findings following UPT	- TVS : IOC - On doppler : ↑ vascularity around periphery	On doppler :  Feeder vessel sign (m/c seen in endometrial polyp; IOC : Hysteroscopy)	 myometrial cyst (IOC : MRI) venetian blind appearance (Alternating dark & light bands)

Polyp :

c/o metrorrhagia (Never in fibroid).

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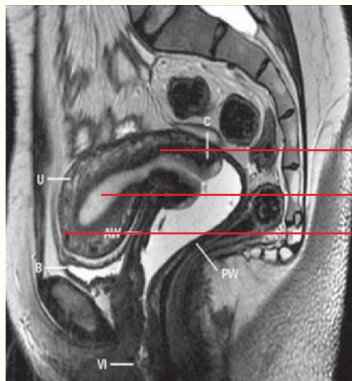
mx :

- Endometrial polyp : Hysteroscopic polypectomy.
- Endocervical polyp : Polypectomy hook.

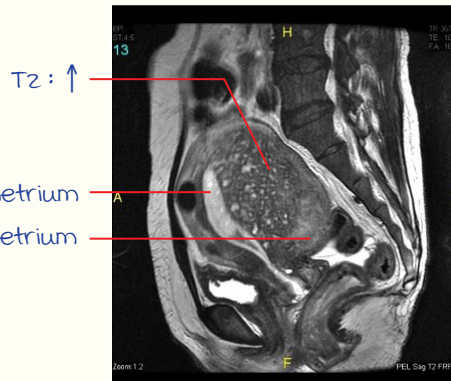
Adenomyosis :

Dx : Clinical.

IOC : MRI.



Ⓝ Tz (Transition zone)



- ≥ 12 mm : Adenomyosis
- < 8 mm : No adenomyosis

mx :

Total abdominal hysterectomy $\rightarrow$ HPE : Gold standard Ix.

Fibroid :

FIGO classification :

FIGO type	Fibroid
0, 1, 2	Submucosal
3, 4	Intramural
5, 6	Subserosal
7	Pedunculated subserosal
8	Extrauterine

Broad ligament fibroid :

True	Pseudo
Arises from BL	Subserous fibroid, growing towards BL
ureter is medial	ureter is lateral

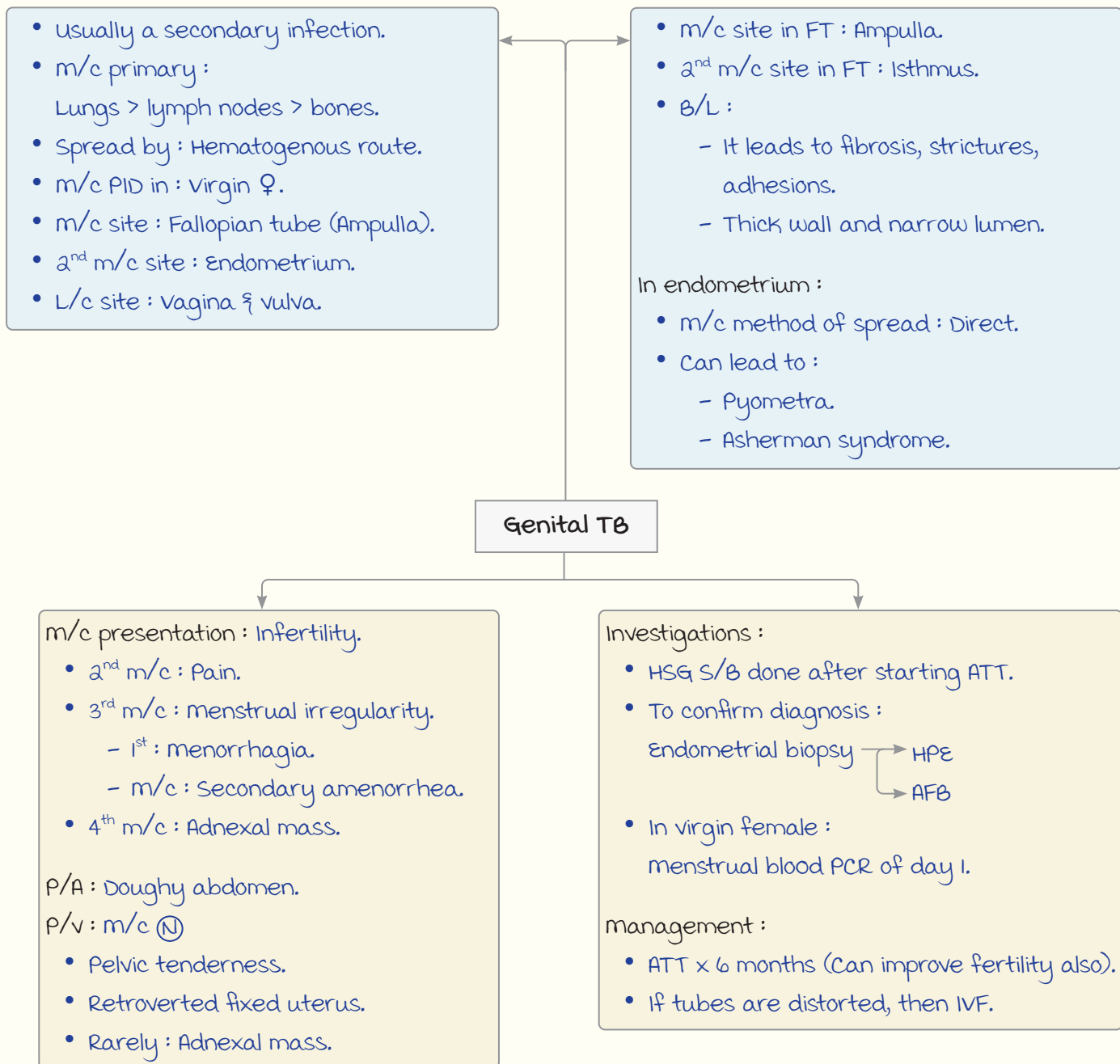
Miscellaneous USG Findings :

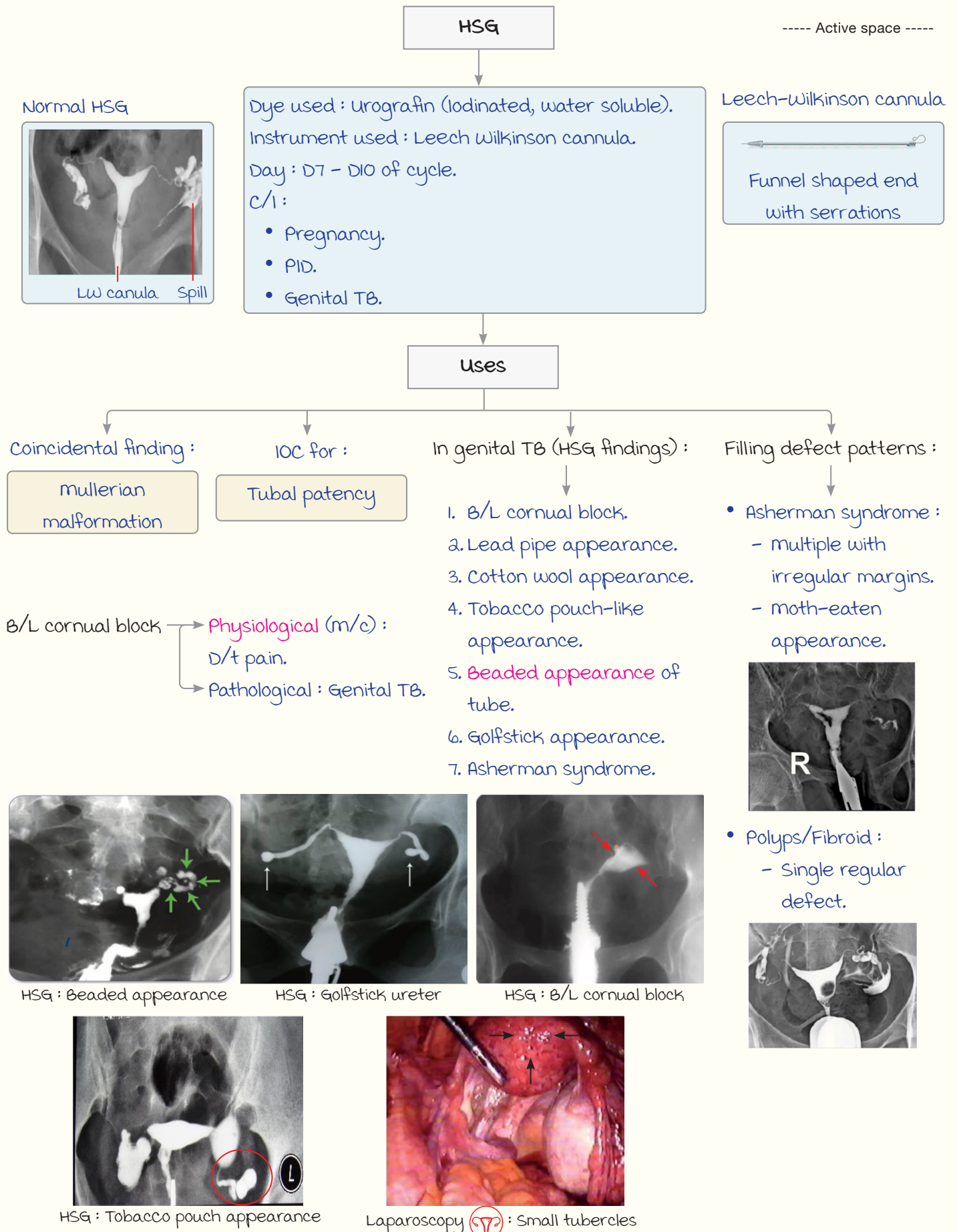
Condition	USG
Ovarian abscess	Heterogeneous internal echoes
Haemorrhagic cyst	Fishnet/reticular appearance
Hydrosalpinx	Retort shaped tube

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Genital TB & Hysterosalpingograms (HSGs)

00:58:17





----- Active space -----

Vaginitis

01:08:22

Physiological discharge : Leucorrhea.

Characteristic :

1. Colorless.
2. Odourless.
3. No pruritis.

Vaginal discharge : From cervix.

Bacterial vaginosis :

- Alteration in vaginal flora
Doderlein bacilli replaced by gardnerella, mobiluncus.
- Discharge :
 - Colour : Greyish white.
 - pH : ≥ 4.5 .
 - Burning, pruritis : $\ominus$.
 - Odour : Fishy.
- Whiff test :
 - Add 10% KOH.
 - Positive : Amine like odour.
- IOC : Saline microscopy.
 - **Clue cells.**
- Gold standard :
- **Gram stain** : Take out Nugent score (≥ 7).
- management : metronidazole (DOC).
- Partner treatment : Not required.

Vaginitis :

IOC : Saline microscopy.

Gold standard : Culture.

Except : Bacterial vaginitis $\rightarrow$ Gram stain.pH of discharge : Always > 4.5 , except Candida < 4.5 .**Trichomonas vaginitis :**

- Agent : Trichomonas vaginalis.
- Discharge :
 - Colour : Frothy, yellowish green.
 - pH : ≥ 4.5 .
- Pruritus : $\oplus$.
- Additional feature : Dysuria.
- P/S exam : Strawberry cervix.
- Whiff test : $+/-$.
- IOC : Saline microscopy (motility $\oplus$).
- Gold standard : Culture (**Stuart media**).
- DOC : metronidazole.
- Partner Rx : Required (STD).

Candidiasis :

- Discharge : Curdy white/cottage cheese.
- **pH : < 4.5 .**
- main complaint : Pruritis.
- A/w : Splash dysuria.
- P/S : white flakes.
- IOC : Saline microscopy (Pseudohyphae).
- Gold standard : Culture.
- DOC : Fluconazole.
- Partner Rx : Only if symptomatic.

Other Features :

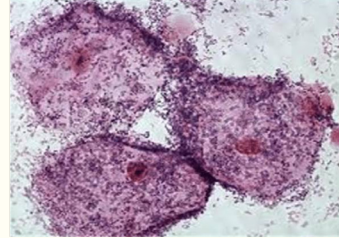
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Epithelial cell → Polymorphic neutrophil ratio :

- Trichomonas
 - Candidiasis
 - Bacterial vaginosis : > 1.
- } < 1.

Bacterial vaginosis (BV) :

- Gardnerella : Gram variable coccobacilli.
 - Diagnostic criteria : **Amsel's criteria**.
 - Greyish white, foul smelling discharge
 - pH ≥ 4.5
 - Clue cell : $\geq 20\%$
 - Whiff test +ve
- } Any 3.



Clue cell : Bacteria adhered to epithelial cells

Candidiasis :

- DOC in pregnancy :
 - Topical imidazoles (Clotrimazole).
 - Oral azoles C/I.
- m/c vaginitis in :
 - Pregnancy (Acidic pH).
 - Diabetics.
 - Antibiotics, steroid, OCP usage.
 - Immunocompromised.



Strawberry cervix

Infection of upper ♀ genital tract : uterus + ovary + tubes + peritoneum.

- m/c cause of PID : multi-organism.
- 2nd m/c cause : Chlamydia.
- m/c cause of **acute** PID : Gonorrhea.
- m/c cause of PID in **virgin** female : Genital TB.
- m/c cause of PID in **IUCD** users : Actinomyces.
- m/c cause of PID in OCP users : Chlamydia.
- DOC for chlamydia :
 - Azithromycin 2 g single dose
 - OR
 - Doxycycline 100 mg BD × 7 days.
 - Treat symptomatic partne
- DOC for gonorrhea :
 - Inj. Ceftriaxone 500 mg IM single dose
 - OR T. Cefixime 800 mg single dose.

CDC Criteria for Diagnosis of PID :

Minimum criteria :

Lower abdominal pain with any of following :

- uterine tenderness.
- Adnexal tenderness.
- Cervical movement tenderness.

Additional criteria :

- Fever.
- mucopurulent discharge.
- microscopy of discharge : Shows abundant WBC.
- Raised ESR.
- Raised CRP.
- Lab test for chlamydia/gonorrhea.

Specific criteria :

- Endometrial biopsy : Endometritis.
- TVS/MRI.
- Laparoscopy.

PID

Signs on USG :

- Adhesions inside tube.
- Adhesions in peritoneal cavity.
- Hydrosalpinx : Retort shaped tube.
- **Beads on string appearance.** } Chronic PID
- **Waist sign** }
- **Cogwheel sign** : Acute PID.

Laparoscopy in PID :

- **Gold standard** investigation.
- With laparoscopy tubes, adnexa can be visualized directly.
- Scoring for conception can be done : **Boer-meisel score.**
 - Not included : Type of organism.
- Specimen can be collected.

Staging of PID :

Gainesville staging.

Stage 1 : No peritonitis.

Stage 2 : Peritonitis present.

Stage 3 : Tubo ovarian mass/abscess.

Stage 4 : Ruptured tubo ovarian mass.

Fitz Hugh Curtis syndrome :

Adhesions formed between liver and anterior abdomen :

Violin string appearance.

- m/c in **chlamydia** > gonorrhea.
- D/t perihepatitis.

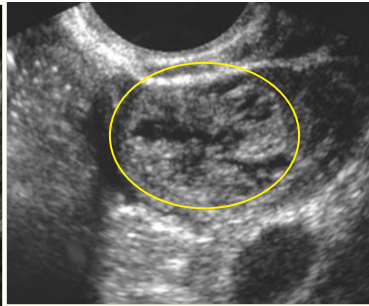
Long-term Consequences :

- **Infertility** (m/c).
- Ectopic pregnancy.
- Hydrosalpinx.
- Chronic pelvic pain.
- Recurrent PID.

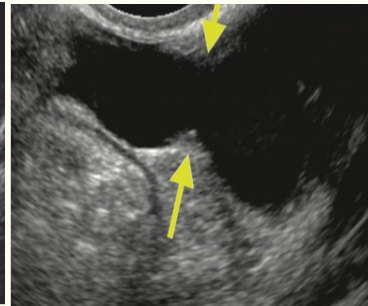
USG :



Beaded appearance of tube



Cogwheel sign



Waist sign

----- Active space -----

Laparoscopy :



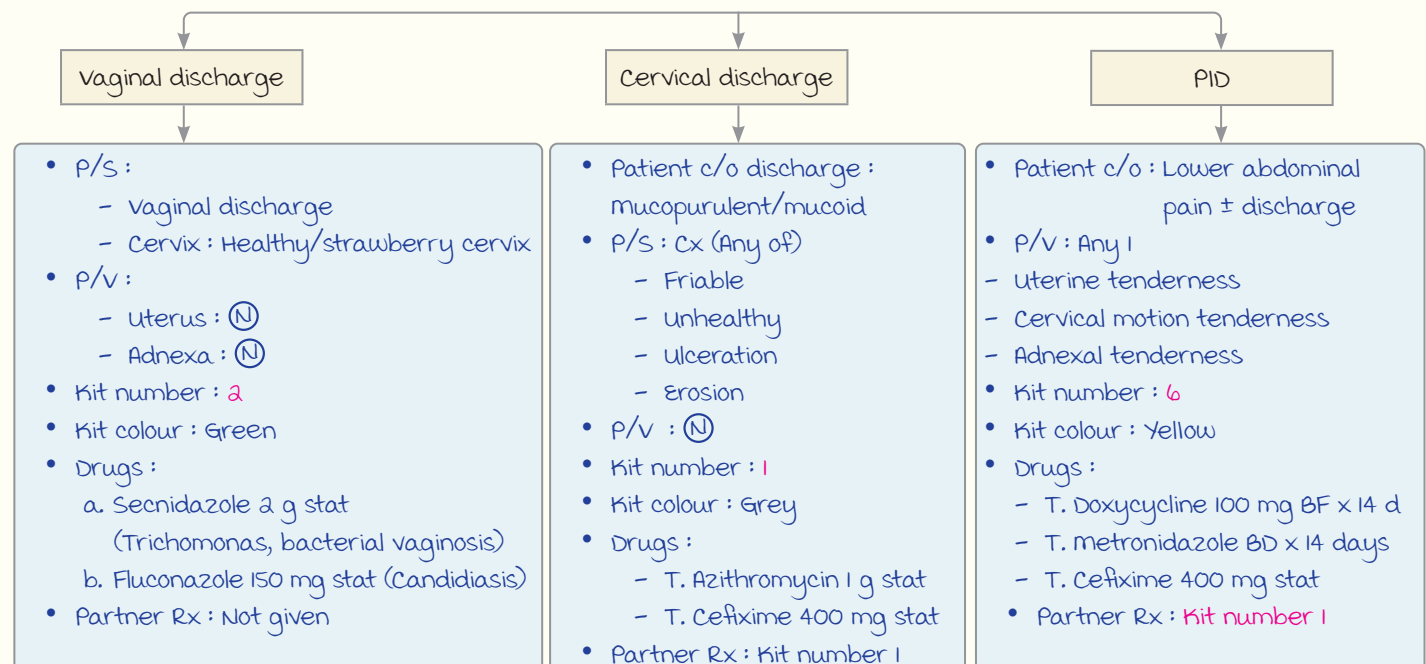
Fitz-Hugh-Curtis syndrome

Note :

Complications uncommon with PID : Pyometra, malignancy.

Syndromic Mx of Infections

01:26:00



- If tested positive for trichomonas : Partner treatment given.

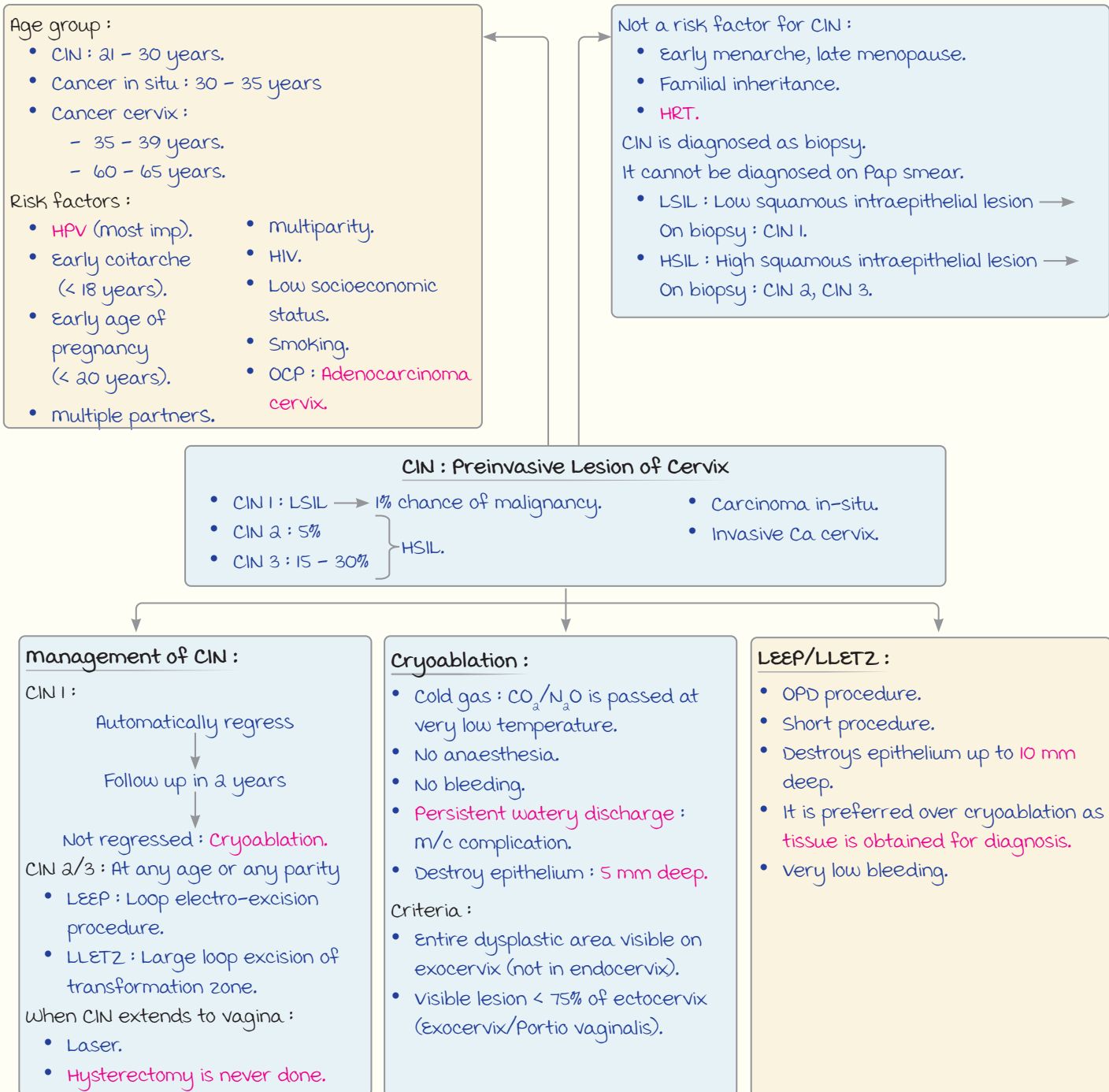
Note : (TV)

IUCD + PID : → Routine : Antibiotics without IUCD removal.
 → Persistent PID/patient demand : IUCD removal + antibiotics.

GYNAECOLOGY REVISION : PART 7

Carcinoma in-situ

00:00:20



HPV

00:11:24

----- Active space -----

- High-risk subtypes :
 - HPV 16 } Responsible for 70% of cancer cervix
 - HPV 18 }
 - m/c HPV leading to squamous cell cancer : HPV 16.
 - m/c HPV leading to adenocarcinoma of cervix : HPV 18.
 - Other high-risk subtypes : HPV 31, 33, 45, 52, 58.
- HPV leads to cancers :
 - In females :
 - a. Cervix.
 - b. Vagina.
 - c. Vulva.
 - In males :
 - a. Penis.
 - b. Oral.
 - c. Anal.
- Low-risk strains : HPV 6, HPV 11.

Leads to :

 - Genital warts.
 - Condyloma acuminata.
 - Laryngeal papilloma in newborn.

HPV proteins needed for :

- Viral replication : E1, E2.
- Malignant transformation :
 - E6 : p53 gene.
 - E7 : RB gene.
- For making HPV vaccine : L1 capsid.

HPV vaccine recommendations :

- Age group : 9 - 26 years.
- High-risk females : 27 - 45 years.
- Ideal age : 11 - 20 years.
- Can be given to :
 - Boys.
 - Sexually active females.
 - HPV positive females.
 - HIV positive females.
 - Cannot be given to pregnant females.
 - Cannot treat CIN I.

HPV :

- It is epitheliotropic.
- Typically infects parabasal cells.

HPV vaccines :

After vaccination, screening should not be stopped.

Bivalent :

- Name : Cervarix.
- Active against HPV 16, 18.
- Protects against CIN, Ca cervix.

Dosage :

- < 15 yrs : 2 doses (0 m, 6 m).
- > 15 years : 3 doses (0 m, 2 m, 6 m).

SAGE recommendation (57) :

- < 21 years : 1 or 2 doses.
- ≥ 21 years : 2 doses.
- HIV positive : 3 doses.

Nonavalent :

- Name : Gardasil-9.
- Active against HPV 6, 11, 16, 18, 31, 33, 45, 52.
- It is approved up to 45 years of age.
- Protects against :
 - CIN/Ca cervix.
 - Genital warts.
 - Vulval Ca.
 - Vaginal Ca.

Quadrivalent :

- Active against HPV 6, 11, 16, 18.
- Gardasil.
- Cervavac.
- Protects against :
 - Genital warts.
 - CIN, Ca cervix.

Note :

Cervavac :

- Recommended from 9 - 26 yrs.
- India's 1st HPV vaccine : Serum institute of India, Pune.

WHO Protocol :

Recommended for resource limited settings.

- Screening begins : 30 years.
- Screening ends : 50 years.
- Screening protocol :
 - See & Treat approach
 - HPV DNA +ve → Treat without biopsy.
 - VIA +ve → Treat without biopsy.
 - See, triage & treat approach
 - HPV DNA +ve (1° screening) → VIA +ve (2° screening) → Treat.
- method recommended by WHO : HPV DNA + VIA > HPV DNA > VIA (visual inspection with acetic acid).
- If HPV DNA is -ve → It is repeated
 - General population : 5 - 10 years.
 - HIV +ve : 3 years.
- If VIA is -ve → It is repeated after : 3 years.
- If HPV is +ve and VIA is -ve → Repeat HPV DNA after 1 - 2 years.

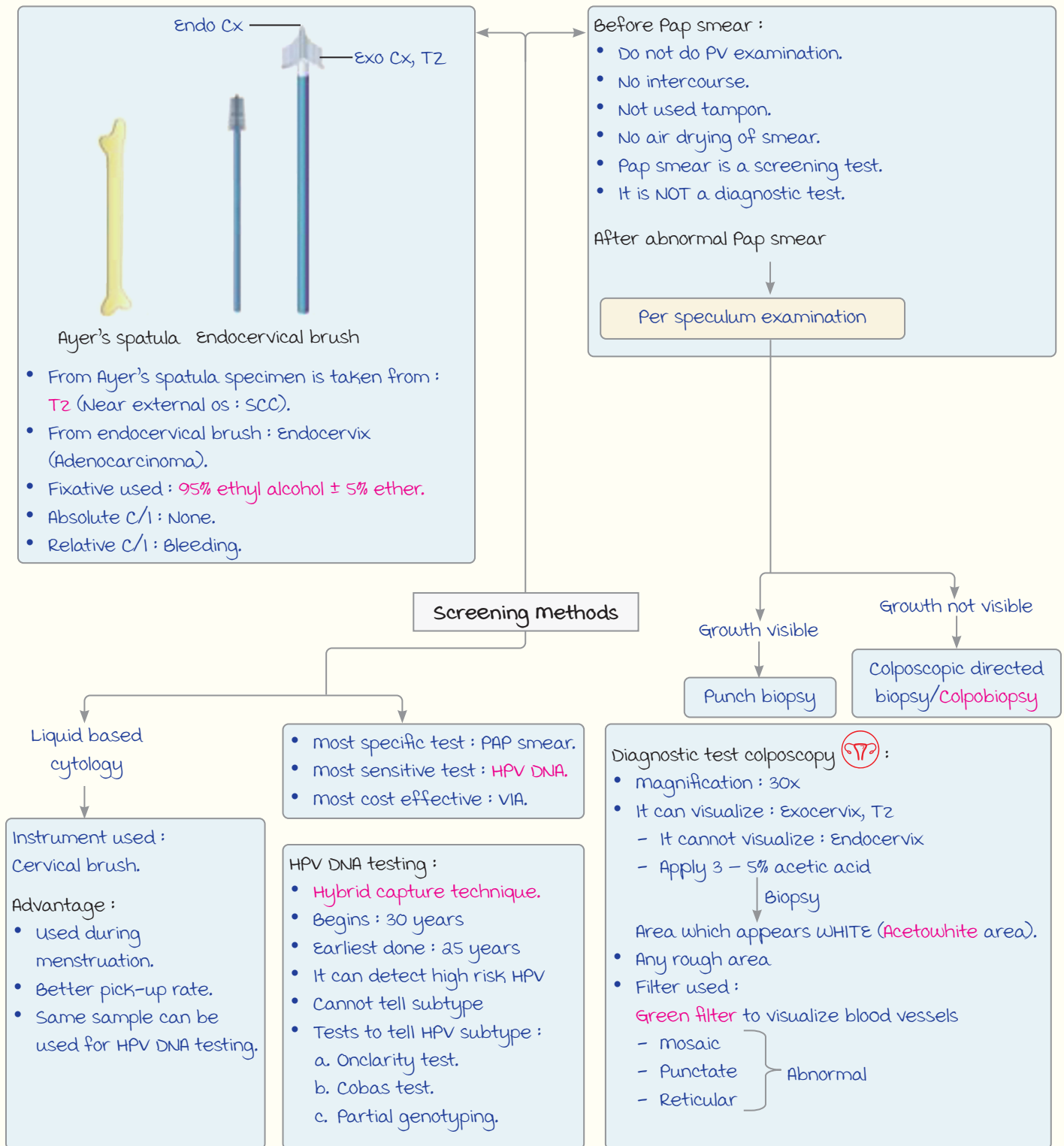
Treatment of choice : LEEP/LLETZ.

ACOG Protocol :

Recommended for resourceful settings.

- Screening begins : 21 years
- method : Pap smear
 - ↓ -ve/Ⓝ
 - Repeated every 3 years
 - ↓
 - Till 30 years.
- From 30 years onwards : HPV DNA + Pap smear (Called as co-test).
- Repeated after 5 years.
- Till age of : 65 years.
- If there is abnormal Pap smear report between 55 - 65 years → Pap smear should be done till 75 years.
- After hysterectomy : Screening stopped.
- After HPV vaccine : Screening not stopped.

----- Active space -----



----- Active space -----

Ca Cervix

00:32:24

- most common variety : **SCC**.
- 2nd most common variety : Adenocarcinoma.
- most common site :
T2 > Squamocolumnar junction.
- most common site for adenocarcinoma : Endocervix.
- most common age — $\begin{cases} 35 - 39 \text{ yrs.} \\ 60 - 65 \text{ yrs.} \end{cases}$
- most common symptom : AUB-m.
- most specific symptom : **Post coital bleed.**
- most common cause of death : Renal failure.
- most common route of spread : Direct > Lymphatic.

Staging (57?) :

- Parametrium involved : Stage II B
- Pelvic side wall is involved : Stage III B
 - ureter/kidney involved : Stage III B
 - Bladder involved : Stage IV A
 - Cystoscopy shows **bullous edema + CT shows hydroureter** : Stage III B
- Pelvic LN involved : Stage III C_i
- Para-aortic LN : stage III C_a.
- Superficial inguinal LN involved : Stage IV B.

m/c Ca :

- In pregnancy : Ca cervix
- In female : Breast ca.
- In female reproductive tract : Ca cervix

Cancer Cervix**IOC :**

- For lymph nodes : **PET-CT**
- For parametrium : **MRI**
- For renal/ureter involvement : **CT**

management :**Early stages :**

- I A_i
 - I A_a
 - I B_i
 - I B_a
- } surgery.

Locally advanced stages :

- I B₃
 - II
 - III
 - IV
- } Chemoradiation.

In a female who c/o postcoital bleeding

Next step : Per speculum examination.

- On P/S examination :
 - If cervix is normal : Pap smear.
 - If cervix is barrel-shaped, indurated, thick, abnormal : Colposcopy + biopsy.
 - If bleeding persists : Endometrial biopsy (R/o endometrial Ca).
 - Growth seen : Punch biopsy.

Management of Cancer Cervix

00:40:04

----- Active space -----

Early stages :

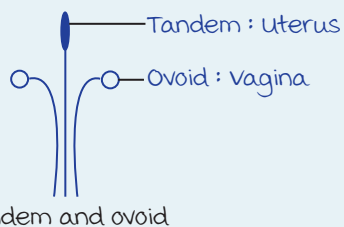
TOC → **Surgery.**

Young age : 35 - 39 yrs	Stage	Old age : 60 - 65 years	
Conization	Stage I A _i	Type 1	No LND
Radical trachelectomy	Stage I A _a	hysterectomy	+ LND
	Stage I B _i		Type 2 hysterectomy + LND
Type 2/3 hysterectomy + LND	Stage I B _a	Type 3 hysterectomy + LND	
Radical trachelectomy : • Remove cervix + 80% parametrium • Stitch uterus to vagina ↓ Conception : Delivery by C-section	-	LND : Pelvic + Paraaortic removed	

management of Cancer Cervix

Locally advanced stages :

- main treatment : **Radiotherapy**
- To increase sensitivity of cells
 - ↓
 - Radiosensitizer (Chemotherapy drug)**
- Drugs used as radiosensitizer :
 - Till date : **Cisplatin.**
 - New drugs added ± **Pembrolizumab.**
 - These drugs should be given : Once weekly.



Radiotherapy :

- Brachytherapy : Source is kept inside body.
 - Intracavitary :
 - Targeting tumor + Paracervical nodes.
 - Use : Tandem + Ovoid.
 - Interstitial therapy : Vaginal mets.
- External beam radiotherapy :
 - Source outside body.
 - Target : Tumor + Pelvic LN ± Paraaortic LN.
 - Given as intensity modulated RT (**IMRT**).
 - Total dose : **45 - 50 Gy.**

----- Active space -----

Pap Smear :

	< 25 years	≥ 25 years
HSIL + Colposcopy (N)	Cone biopsy : • Suspecting endocervical Ca (Not visualized by colposcopy) • In OT under GA	
HSIL	Colposcopic directed biopsy + endocervical curettage	
ASCUS	Repeat Pap smear in 1 yr	HPV DNA (Reflex HPV testing)
LSIL	Repeat Pap smear in 6 m - 1 yr	Colposcopic biopsy + endocervical curettage
AGCUS	• Colposcopy + endocervical curettage • Endometrial biopsy	

ASCUS : Atypical squamous cells of unknown significance

AGCUS : Atypical glandular cells of unknown significance

Pap smear can pick up infections.

Infections not detected :

- Chlamydia
- Gonorrhea

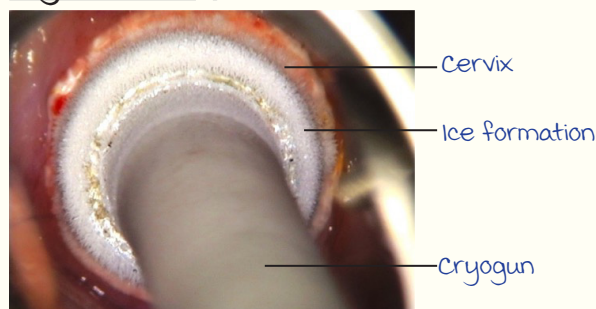
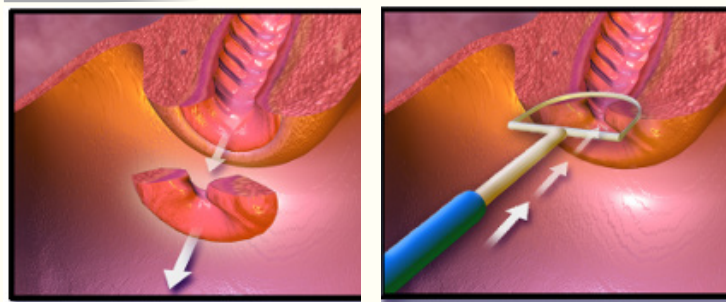
VIA (Visual Inspection With Acetic Acid) : WHO Goals for Cancer Cervix :

With 3 - 5% acetic acid

Epithelium	Color of epithelium
Normal epithelium	Pink
Dysplastic epithelium	White

90/70/90 for 2030 :

- 90% girls : Should be vaccinated for Ca cervix (HPV vaccines).
- 70% : Reproductive age (30 - 50 years) should be screened for Ca cervix.
- 90% : Females with CIN/Ca cervix should get treatment.

Cryoablation :LLETZ/LEEP :Hysterectomy :

Pivot-Rutledge classification :

Hysterectomy	Procedure	
Type 1/simple/total abdominal hysterectomy	• m/c • Remove uterus + cervix • m/c to lead to ureteric injury	
Type 2/Wertheim's/modified radical hysterectomy	Done in malignancies	max. risk of ureteric injury
Type 3/radical hysterectomy		-

Adnexal Mass

00:58:05

----- Active space -----

On P/V :

Benign mass :

Seen in :

- Painful
- U/L
- Cystic
- Tender
- Reproductive age

malignant mass :

Seen in :

- Painless
- B/L
- Solid/firm
- Non tender
- Extremes of age

Hemorrhagic cyst : Reticular cyst/fish net appearance

Chocolate cyst : Cyst with homogenous internal echoes

Hydrosalpinx : Retort shaped tube

Dermoid cyst : Dot & dash/tip of iceberg appearance

On USG :

Signs of malignancy :

- Solid mass/mixed texture
- Papilla
- Thick septa
- Ascites
- Enlarged lymph nodes
- matted bowel loops (Peritoneal caking)
- ↑ vascularity on doppler
- ≥ 10 cm in size

ORADS Scoring :

ORADS 1 :

- Follicular cyst/corpus luteal cyst
- No chances of malignancy

ORADS 2 :

- Risk of malignancy < 1%
 - Endometrioma
 - Dermoid cyst
 - Haemorrhagic cyst
 - Hydrosalpinx
- All 4 : < 10 cm

ORADS 3 :

- Risk of malignancy 1-10%
- Any of above ≥ 10 cm

ORADS 4/5 :

- Risk of malignancy 10-50%
- Signs of malignancy on USG

Adnexal mass :

IOC : TVS

On USG = No signs of malignancy

Reproductive age (17)

Not done : Ca 125

- Raised in benign conditions
- > 200 IU : malignancy

Done : Size of mass

< 7 cm :

- OCP

- Follow up of USG

≥ 7 cm :

- ↑ Risk of torsion

- Surgery.

Adolescent female :

↑ Risk of germ cell tumor

- AFP
- hCG
- LDH

menopausal female :

Epithelial cell tumor

Ca 125 : ≥ 35 IU

Indicates malignancy.

----- Active space -----

Ovarian Cancer : Types & Risk Factors

01:06:45

WHO Classification of Types of Ovarian Cancer :

Epithelial ovarian tumor	Germ cell tumor	Sex cord stromal tumor	metastatic tumor
- m/c = 90% - m/c = Serous tumor - mucinous tumor - Brenner tumor - Endometrioid tumor - Clear cell carcinoma	- m/c = Teratoma - Benign = mature cystic teratoma (Dermoid cyst) - malignant immature teratoma - Dysgerminoma - Yolk sac tumor - Embryonal cancer - Choriocarcinoma	- Estrogen secreting : Granulosa cell tumor - Androgen secreting : Androblastoma or arrhenoblastoma - Sertoli cell - Leydig cell - Hilus cell tumor - Stromal tumors : - Fibroma - Thecoma (Can also secrete estrogen) - Fibrothecoma	Krukenberg tumor

Risk Factors of Ovarian Cancer :

	Proven risk factors	Controversial risk factors
Excessive estrogen	- Early menarche - Late menopause - Obesity - Endometriosis	- PCOS - HRT
Excessive ovulation (Theory of incessant ovulation)	Nulliparity	Infertility and ovulation inducing drugs
Genetic syndrome	- Lynch syndrome - BRCA 1 - BRCA 2	-
Exposure to carcinogens	-	Smoking : - mucinous adenocarcinoma - Protective for others

Protective Factors of Ovarian Cancer :

Related to estrogen	Related to ovulation	Certain surgeries
Physical exercise	Anovulation : multiparity, OCP (most important), breastfeeding	- Hysterectomy - Tubal ligation - Salpingectomy

----- Active space -----

m/c variety of ovarian cancer : Accounts for 90% cases

Types :

- Sometimes serous :
 - Cystadenoma
 - Cystadenocarcinoma
- Maths : mucinous
- Can : Clear cell
- Be : Brenner's
- Easy : Endometrioid

m/c variety : **Serous variety**

- Age : 60-70 years
- mostly : B/L
- Pt. presents as : Non-specific symptoms
- Prognosis : Not good
- Tm marker : **CA 125**
- In mucinous variety :
 - CA 19-9
 - CEA
- HPE : mutation a/w
 - Low grade Tm : pTEN mutation
 - High grade Tm : p53 mutation

Epithelial Cell Tumors

m/c ovarian tumor a/w endometriosis :

- Endometrioid variety.
- Clear cell Ca.

m/c ovarian tumor a/w

endometrial cancer :

- Clear cell Ca.
- Granulosa cell tumor.

Pseudomyxoma peritonei :

- Seen with mucinous variety.
- mucinous material in peritoneum.

Note :

m/c cancer leading to pseudomyxoma peritonei : Appendix Ca.

- most common benign ovarian Tm : Serous cystadenoma.

- m/c malignant ovarian Tm : Serous cystadenocarcinoma.

- m/c ovarian Tm in reproductive age : Dermoid cyst.

- m/c ovarian Tm in pregnancy : Dermoid cyst.

- m/c ovarian cancer in pregnancy : Dysgerminoma.

- m/c cancer in pregnant females : Ca cervix.

----- Active space -----

Second m/c variety :

- Germ cell tumor.
- m/c age : 10-30 yrs.
- mostly : Adolescent female.
- Prognosis : Good.
- usually w/L.

Includes :

- Extremely : Embryonal.
- Difficult : Dysgerminoma.
- To Crack : Teratoma, choriocarcinoma.
- usually : Yolk sac tumor.

- m/c germ cell tumor : Teratoma (Dermoid cyst)
- m/c malignant GCT : Immature teratoma.
- GCT with best prognosis : Dysgerminoma.
- GCT with worst prognosis : **Yolk sac tumor.**
- GCT with **rapid progression** : Yolk sac tumor.
- m/c ovarian tumor in dysgenetic gonads : Gonadoblastoma.
- m/c ovarian cancer in dysgenetic gonads : Dysgerminoma.

Germ Cell Tumors & Tumor markers (57)

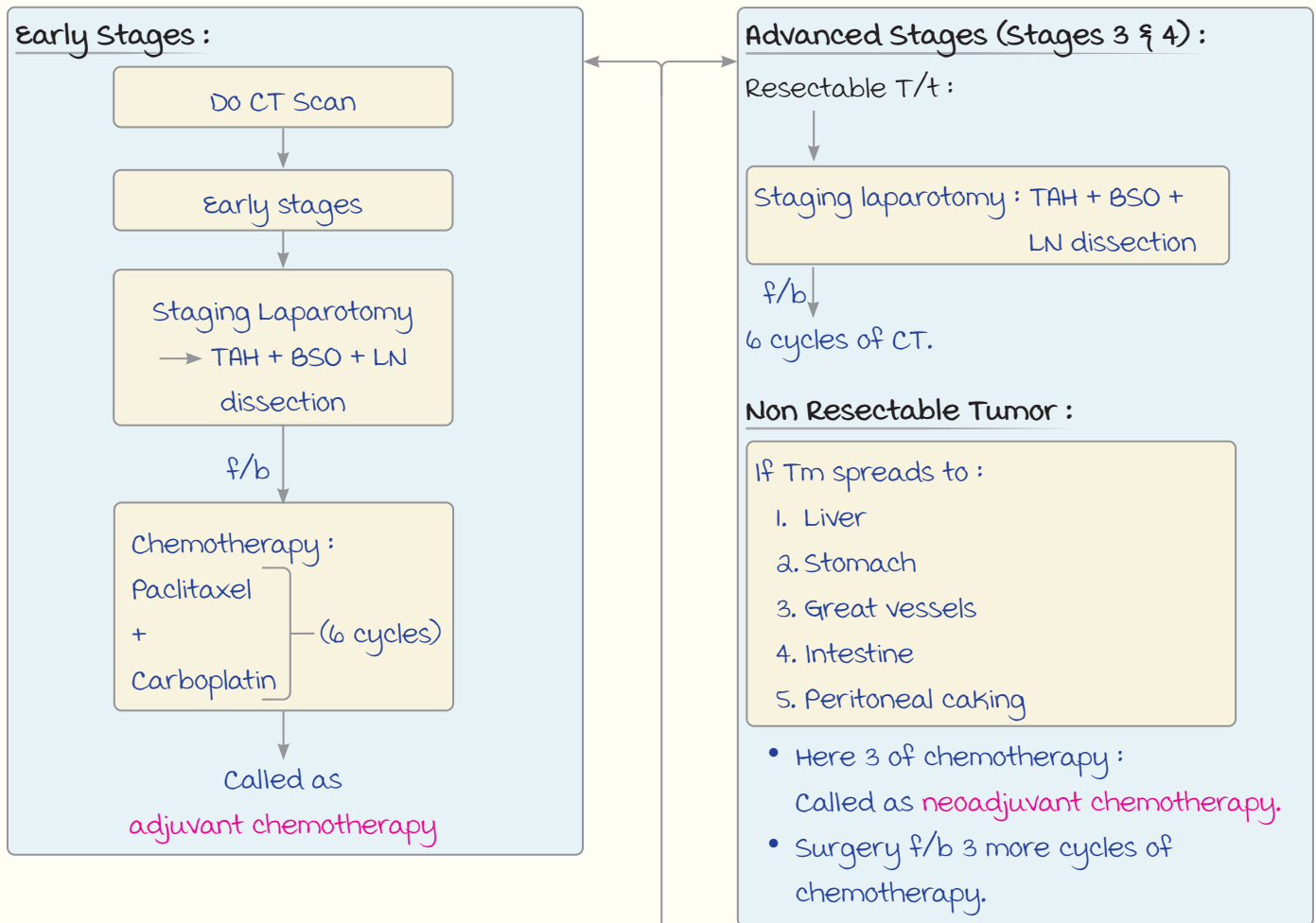
- m/c radiosensitive pelvic organ : Ovary.
- All ovarian Tms are : Radioresistant.
- Only radiosensitive ovarian Tm : Dysgerminoma.
- % of B/L dermoid : 10-15% (~20%).
- % of B/L dysgerminoma : 15-20% (~30%).
- 100% w/L GCT : Yolk sac tumor.

Ovarian tumor markers	Tm markers		
Epithelial cell	CA 125		
Mucinous Tm	CA 19-9, CEA		
Dysgerminoma	m/c ↓ LDH	L/c ↓ hCG	Not seen ↓ AFP
Yolk sac Tm	m/c ↓ AFP	L/c ↓ LDH	Not seen ↓ hCG
Embryonal Tm	α-FP, hCG		
Choriocarcinoma	hCG		
Granulosa cell Tm	AMH, inhibin B		

Ovarian Cancer : Management

01:22:00

----- Active space -----



management of Ovarian Cancer

- Ix to be done before staging laparotomy to know early/advanced stage : CT scan.
- If recurrence of ovarian tumor occurs within 6 months of chemotherapy :
 - **Platinum resistant relapse.**
 - DOC : **Paclitaxel + bevacizumab.**
- If recurrence occurs after 6 months of chemotherapy :
 - **Platinum sensitive relapse.**
 - DOC : Carboplatin + paclitaxel.
- If CA-125 ↑ during treatment :
 - Platinum refractory.
 - DOC : PLD (**Pegylated liposomal doxorubicin**).

Familial ovarian Tm :

- m/c seen with **BRCA 1** > BRCA 2 > Lynch syndrome.
- Age 50-60 yrs (10 yrs younger).
- If a female has a Lynch syndrome/BRCA gene mutation

To prevent ovarian Ca :

- Best : TAH + BSO After completing family by 40 years of age.
 - 2nd best : Endometrial biopsy + annual CA 125 levels.
 - If a female has FIRST degree relative with BRCA mutation and died d/t ovarian Ca
- BRCA -ve
- Endometrial biopsy + Annual **CA 125** levels (**Surveillance tool**).

----- Active space -----

Staging Laparotomy :

Staging of ovarian and endometrial Ca.

Steps ①② :

- Ascitic fluid/NS wash collection.
- TAH + BSO.
- LN dissection : Pelvic & para-aortic.
- Peritoneal biopsy.
- Infracolic omentectomy.
- Appendicectomy : Only in mucinous variety.

Maintenance Chemotherapy :

- BRCA -ve : Olaparib + bevacizumab.
- BRCA +ve : Only olaparib (PARP inhibitor).

Key points :

- Granulosa cell tumor
Brenner's tumor
Thecoma
Fibroma
- + @ pleural effusion + ascites → Meigs' syndrome
- Any other condition + @ pleural effusion + ascites → Pseudomeigs' syndrome

Ovarian Cancers : Appearance & Findings

01:32:12

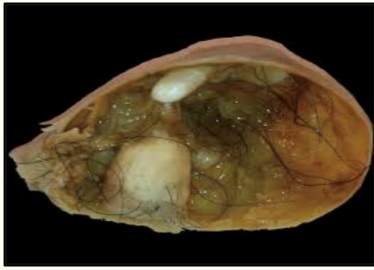
Characteristic Findings :

Findings	Cancer
• Tip of iceberg appearance • Dot & dash appearance • Rokitansky protuberance	Dermoid cyst
Hobnail nucleus	Clear cell Ca
Schiller-Duval body	Yolk sac tumor
Call-Exner body	Granulosa cell tumor
Signet ring cell	Krukenberg tumor
Nests of cell separated by fibrous septa	Dysgerminoma
Reinke's crystal	Hilus cell tumor
Walthard cell nests	Brenner's tumor

Dermoid Cyst :

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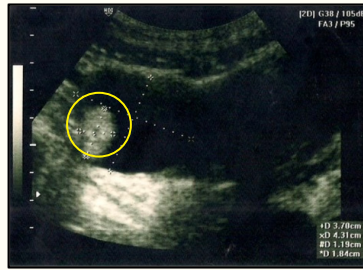
Gross :



Teeth, hair present

- m/c germ cell tumor.
- malignancy :
 - Risk : 0.2-2.1%.
 - m/c site :
Rokitansky protuberance.
 - m/c type :
Squamous cell carcinoma.

USG :

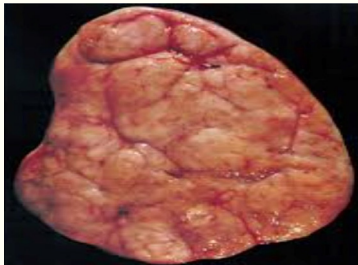


Rokitansky protuberance

- mx :
 - Cystectomy
(max risk for torsion).
 - Oophorectomy (Family complete).
- m/c component : Ectodermal.
- m/c time for torsion :
2nd trimester of pregnancy.

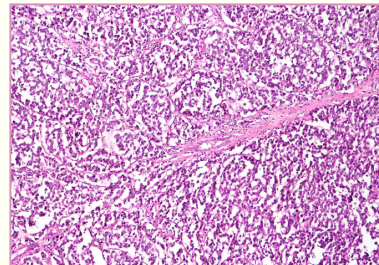
Dysgerminoma :

Gross :

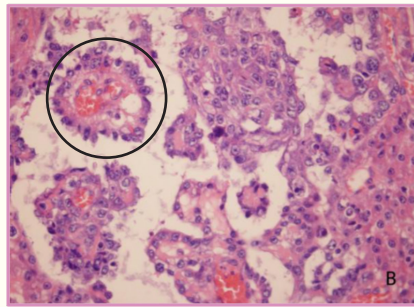


Tan, lobulated, fleshy mass

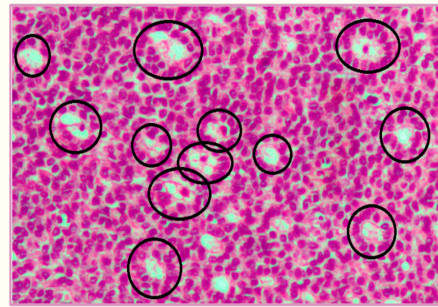
microscopy :

Nests of cells separated by thick septa
(Lymphocytic infiltration)

----- Active space -----

Yolk Sac Tumor :

Schiller-Duval body (~Glomerulus)

Granulosa Cell Tumor :

Call-Exner bodies

- Pubertal age : Precocious puberty.
- Reproductive age : AUB.
- Menopausal age : Post menopausal bleeding.
- ↑ Risk of endometrial Ca → Endometrial biopsy.

Krukenberg Tumor :

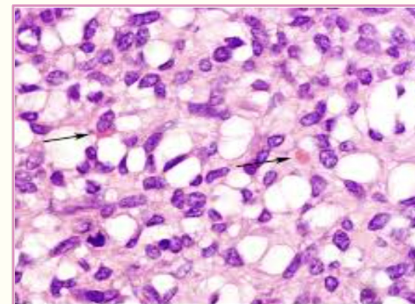
- m/c primary : Pyloric end of stomach Ca. > Breast Ca., Colon Ca.
- Spread : Retrograde lymphatics.

Gross :



Bilateral tumor

microscopy :



Signet ring cell (Nucleus pushed aside)

- Waxy consistency.
- Shape of ovary retained.
- Capsulated.

GYNAECOLOGY REVISION : PART 8

----- Active space -----

Female Infertility

00:00:22

Infertility :

Inability to conceive after 1 year of unprotected intercourse.
Evaluation of couple for infertility :

Condition	Initiation of evaluation (After unprotected intercourse)
Female < 35 yrs	After 1 year
Female : 35 - 40 yrs	After 6 months
- Females ≥ 40 yrs - K/c/o uterine/tubal pathology - Female with previous h/o chemo/radio therapy - H/o Oligomenorrhea/amenorrhea	Less than 6 months

Basic Investigations :

To be done in all couples who complain of infertility :

1. Semen analysis.
2. S. progesterone (Test for ovulation) on day 22.
3. HSG (Test for tubal patency).

In all patients, do :

- A. TSH.
- B. S. Prolactin.

If age of female ≥ 35 years : Also do test for ovarian reserve (AMH).

Infertility

Causes :

- Female infertility : 40 - 50% of infertility
- male infertility : 20 - 30% of infertility
- unexplained : 15 - 30%

Female Causes of Infertility

Ovarian causes (m/c)

Class I :

- Hypogonadotropic hypogonadism.
- E.g. : Kallmann syndrome.

Class II (m/c) :

- Normogonadotropic
- E.g. : Anovulation (PCOS) → most easily treatable.

Class III :

- Hypergonadotropic hypogonadism.
- E.g. : POI (Primary ovarian insufficiency).

Note :

WHO does not include increased prolactin in any class and keeps it as a separate cause.

Tubal causes

- Salpingitis.
- PID.
- Endometriosis.

uterine causes

- Submucosal fibroid.
- Asherman syndrome.
- müllerian malformation (m/c : Septate uterus).

Cervical and unexplained causes

In cervix :
Anti-sperm antibody

Immunological cause of infertility

Now included in unexplained infertility

- Post coital/Huhner/Sims test (Day 12 - 14).
- management : Clomiphene citrate + IUI.

1. Serum progesterone levels :
≥ 3 ng (m/c used).
2. Cervical mucus study :
 - If female has ovulated : Ferning absent.
 - Not ovulated : Ferning (+).
3. Vaginal epithelial study :
 - If female has ovulated : Intermediate cells.
 - Not ovulated : Superficial cells.
4. Endometrial biopsy : Day 26 of cycle
 - If female has ovulated : Corkscrew/sawtooth glands.
 - Not ovulated : Simple tubular glands.
5. Basal body temperature :
Biphasic cycle Indicates → Ovulation.

Tests for Anovulation

Done on : Day 22 of cycle (1 week before menstruation)

Tests which Predict Ovulation :

1. Follicular monitoring :
 - Done from : Day 10.
 - Done by : TVS.
 - Signs of ovulation :
 - a. ↓ Size of follicle
 - b. Fluid in pouch of Douglas.
 - c. Trilaminar endometrium.
2. Urinary LH surge :
Ovulation occurs 24 hours later.

Note :

Time gap b/w blood LH surge & ovulation :
32 - 36 hours.

Endometrial Biopsy :

- Pre-procedure advice :
 - Barrier method of contraception to be used.
 - OCPs to be avoided.
- Indication : Genital TB.
- 2 samples

→ HPE	→ Formalin.
→ AFB	→ Saline.

Rx for Anovulation :

- Overall DOC in anovulation : Clomiphene citrate (CC).
- DOC for anovulation in PCOS : Letrozole.
- DOC for anovulation in pituitary ablation : hMG.
- DOC for anovulation in Kallmann syndrome : Pulsatile GnRH.

Other Causes of Female Infertility

00:10:45

----- Active space -----

Decreased Ovarian Reserve :

↓ No. of follicles.

Tests :

1. ↑ S. FSH (≥ 25 IU $\rightarrow$ POI).
2. ↓ S. Inhibin B (< 45 pg).
3. ↓ Antral follicle count.
4. ↓ AMH.

- Outdated tests : Clomiphene citrate challenge test.
- All tests done on : Day 3.

Except : AMH $\rightarrow$ Done on any day (Best test).

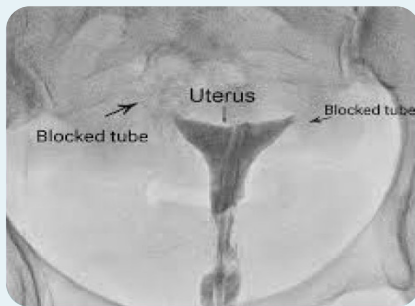
management : IVF with donor eggs

Note :menopausal females : S. FSH ≥ 40 IU.**Tubal Cause :**

- Tubal causes lead to tubal block.
- IOC : HSG (Day 7 – 10)
- Gold standard : Laparoscopic chromopertubation
 - Process : methylene blue/indigo carmine passed into uterus $\rightarrow$ Laparoscope inserted in abdominal cavity to see free spillage of dye.
- In females with block on HSG :
 - U/L block : IUI + CC.
 - B/L block : management depends on site of block.
- Overall management of tubal infertility : IVF

AMH :

- Normal level 1 – 3 IU.
- < 1 IU : Suggestive of POI.
- < 0.5 IU : Diagnostic of POI.

Other Causes of Infertility

B/L cornual block

Bilateral Cornual Block :

m/c cause : Physiological spasm.

m/c pathological cause : Genital TB.

Next step : Hysteroscopic cannulation (Pass a guidewire under hysteroscopic guidance) $\rightarrow$ Laparoscopic chromopertubation (Relieves spasm/releases mucosal plug)

↓ Block persists
IVF.



HSG with B/L distal block

Bilateral Distal Block :

Next step : Laparoscopic chromopertubation.

If bilateral mild block : Fimbrioplasty.

If bilateral severe block : IVF.

If hydrosalpinx (+) :

- Before IVF is done, salpingectomy is needed as fluid present in hydrosalpinx is fetotoxic.
- Hydrosalpinx indicates : Severe block.

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Management of Infertility

00:22:46

Clomiphene :

MOA :

SERM $\rightarrow$ Blocks estrogen receptors in brain $\rightarrow$ $\uparrow$ LH, $\uparrow$ FSH $\rightarrow$ Folliculogenesis
 $\downarrow$
 Ovulation.

Enclomiphene : Isomer mediating anti-estrogenic action.

Ovulation trigger : Inj. hCG.

Dose :

- Starting dose : 50 mg.
- maximum dose : 150 mg.

Ovulation rate : 80%.

Pregnancy rate : 10 - 15%

(D/t blockage of estrogen receptors in cervix & endometrium $\rightarrow$ unsuitable for implantation).

S/E :

- m/c : Hot flashes > Ovarian cyst formation.
- Vaginal dryness.
- Twin pregnancy (5 - 8%).
- OHSS (1%).
- Visual disturbances : Indication for immediate discontinuation.

Pre-requisite : Intact HPO axis.

Letrozole :

- Peripheral action + No action on endometrium/cervix.
- Live birth rate } Comparatively better
- Pregnancy rate } than clomiphene.
- No side effects of hot flashes.
- Pre-requisite : Intact HPO axis.

Timing of Tests :

Test	Time to perform
Test for ovulation	Day 22 of cycle
Test for ovarian reserve	Day 2 - 4 of cycle (Day 3)
AMH	Any day
HSG	Day 7 - 10 of cycle
Post-coital test	Day 12-14 of cycle

.....
Note :
.....

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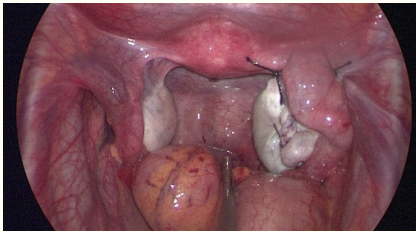
Not a test for ovarian reserve : $S. progesterone > S. estrogen$

(Tested along with s. FSH)

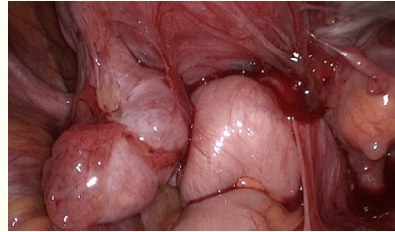


Test for ovulation.

Laparoscopic Imaging (97) :



Genital TB



Frozen pelvis :

1. Genital TB
2. Endometriosis
3. Severe PID

Tubo ovarian abscess

- Seen in $stage\ 3\ PID$ (Gainesville staging)
- Features :
 - Poor general condition
 - $\uparrow TLC$
 - Local tenderness
- USG :
 - mass in ovary with heterogeneous echoes
 - $\uparrow$ vascularity on Doppler



Tubo ovarian abscess

----- Active space -----

Male Infertility

00:32:35

Semen Analysis :

- Done after an abstinence of 2 - 7 days.
- Sample should reach lab within 1 hour.
- Analysis is done on liquefied semen.
- Liquefaction time : 20 - 30 mins.
- WHO semen analysis gives **minimum criteria** needed for conception (Not average values).

Semen analysis parameters	New values (2022)
Semen volume	1.4 mL
Sperm concentration	16 million/mL
Total sperm number	39 million/ejaculate
Total motility	42%
Progressive motility	30%
Vitality	54%
Normal forms/morphology (Strict Tygerberg criteria)	4%

Terminology for Abnormal Semen Analysis :

- Aspermia : Absent semen.
- Oligospermia : Sperm concentration < 16 million/mL.
- Severe oligospermia : Sperm concentration < 5 million/mL.
- Azoospermia : No sperm in semen.
- Teratozoospermia : Abnormal sperm morphology.
- Asthenospermia : Decreased sperm motility.
- Necrozoospermia : Increased nonviable sperm.

most important parameter of semen analysis :
morphology > motility > Concentration.

2 important causes of azoospermia :

1. **Cystic fibrosis** → B/L absence of seminal vesicles & vas deferens.
2. **Klinefelter's syndrome** → Hypoplastic testes.

OATS on semen analysis :

1. Oligospermia
 2. Asthenospermia
 3. Teratozoospermia
- Risk factors :
Smoker/varicocele.

male Infertility

If on semen analysis :
Report is Azoospermia

↓

Repeat semen analysis :
Time gap between semen analysis
1 week (min.) - 1 month (max.)

↓

Azoospermia confirmed

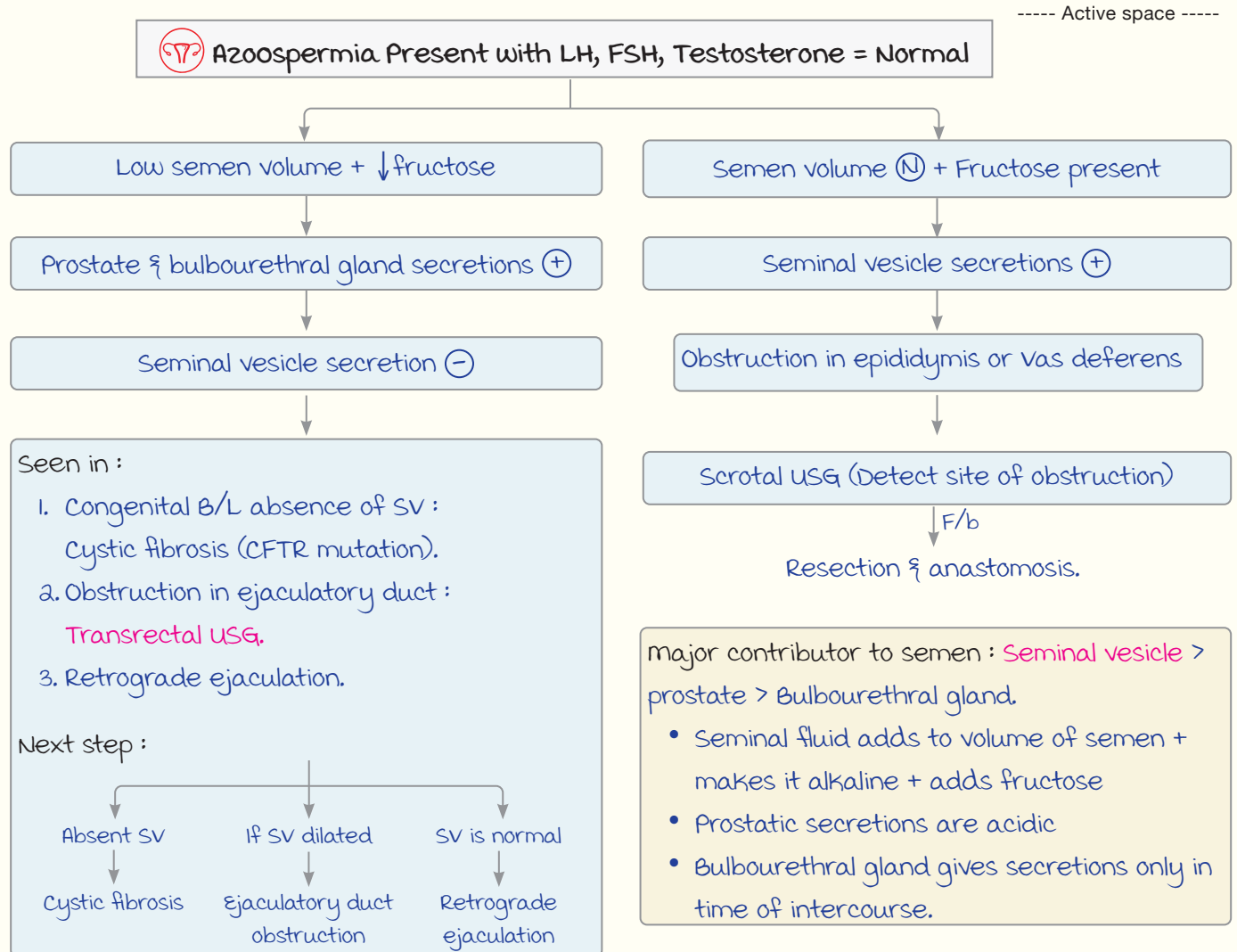
↓

Hormone analysis :
FSH, LH, testosterone.

Single best answer : FSH

- m/c cause of male infertility : Testicular cause.
- m/c reversible cause of male infertility : Varicocele.

	Pretesticular azoospermia	Testicular azoospermia	Obstructive azoospermia
Hormonal profile	Hypogonadotropic hypogonadism : ↓ FSH, ↓ LH, ↓ Testosterone	Hypergonadotropic hypogonadism : ↑ FSH, ↑ LH, ↓ Testosterone	• FSH • LH • Testosterone
Organs affected	Hypothalamus/ Pituitary	Testes	Sperm pathway : Seminal vesicle/ epididymis/ejaculatory duct (ED)/vas deferens (VD)
Causes	• Kallmann syndrome • ↑ Prolactin levels	• mumps orchitis • Klinefelter's syndrome	• Varicocele • Cystic fibrosis • Vasectomy • Obstruction in ED/VD



Klinefelter Syndrome :

- Genotype = 47 XXY.
- Example of **false positive Barr body**.

Phenotype :

- Tall men.
- Long legs.
- Gynecomastia

Testis : Small and hypoplastic.

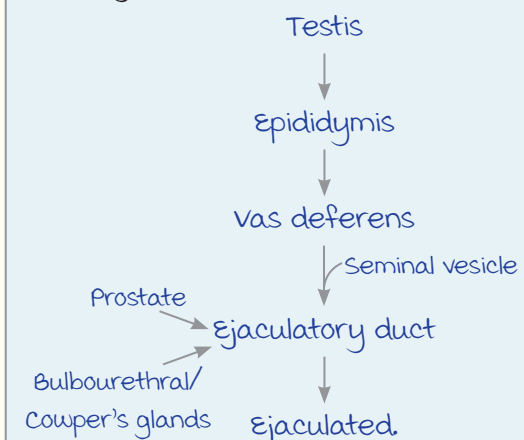
IQ : Decreased.

Life span : Decreased.

Note :

- To measure size of testis : **Prader orchidometer** used.
- False negative Barr body : Turner syndrome.

Pathway for sperms :



----- Active space -----

Assisted Reproductive Technique (ART)

00:47:20

IUI (Not Included Under ART) :

Can be done with : Husband sperm OR donor sperm.

Prerequisite :

- Sperm concentration of : 10 million sperm/mL.
- Sperm morphology : 14% should have normal morphology.
- OPD procedure.
- Processed fertilized sperm (0.4 - 0.8 mL) inserted in uterine cavity with the help of IUI catheter.

Indications :

1. Female-fertility factor infertility.
2. unexplained infertility (Clomiphene citrate + IUI).
3. males with ejaculatory problems (Hypospadias, Epispadias).
4. Oligospermia with sperm conc. : 10 - 16 million/mL.
5. Vaginismus.

In Vitro Fertilization (IVF) :

- To female partner : Give hmg injection so that they hyperovulate
- From D₀ follicular monitoring initiated
- When follicle is 18 - 20 mm (≥ 17 mm) and 2 follicles → Give injection hCG (Ovulation trigger).
- After 32 - 36 hours : Egg retrieval under USG guidance.
- The oocytes picked are incubated with sperms each oocyte → **50,000 to 1 lakh sperm.**
- On day 3 (Cleavage embryo transfer) or day 5 (Blastocyst), embryo transfer is done 2 cm below fundus.

Success rate : 20 - 30%.

Prerequisite : At least 5 million sperm/mL.

Indications :

- Tubal block
- müllerian agenesis (IVF + surrogacy).
- Decreased ovarian reserve (IVF with donor eggs).
- Oligospermia in males with sperm conc. 5 - 10 million/mL.

Cannot be done for :

- Severe oligospermia (< 5 million/mL).
- Asthenospermia.
- Azoospermia.

ART**Intracytoplasmic Sperm Injection (ICSI) :**

Procedure : Same as IVF.

Except : In each oocyte, a single sperm is injected.

Indications of ICSI :

- Testicular azoospermia (After sperms are surgically extracted from testis).
- Asthenospermia.
- Severe oligospermia (Sperm conc. < 5 million/mL).

Remember : Azoospermia means absence of sperms in semen, not in testis.

Surgical methods of sperm extraction :

- TESE : Testicular sperm extraction.
- TESA : Testicular sperm aspiration.

management of male infertility :

Cause	management
Retrograde ejaculation	IUI
Premature ejaculation	SSRI
Ejaculatory problem Hypospermia	IUI
Erectile dysfunction	Sildenafil
Oligospermia sperm count : 10 - 15 million/mL	IUI
Oligospermia sperm count : 5 - 10 million/mL	IVF
Severe oligospermia : < 5 million/mL	ICSI
Nonobstructive azoospermia	Surgically take out sperms from testis or epididymis f/b ICSI

Note :

- **S. FSH** : Best to differentiate b/w obstructive & non-obstructive azoospermia.
- Epididymis
 - Proximal part : Site of sperm maturity.
 - Distal part : Site of sperm motility.

Contraception : Part 1

00:54:46

----- Active space -----

Absolute Contraindications for OCPs

(WHO Class IV) :

- **B**anks : H/o **B**reast cancer.
- **H**ave : Severe uncontrolled **H**ypertension.
- **V**arious : undiagnosed **v**aginal bleeding.
- **S**cheme : **S**moker ≥ 35 yrs.
- **T**o : H/o **T**hrombosis/DVT/MI/Stroke.
- **P**rovide : **P**regnant or h/o cholestasis in pregnancy.
- **H**ome : **H**ypertriglyceridemia.
- **L**oan : Acute **L**iver disease.
- **D**uring : **D**iabetes with **v**asculopathy.
- **m**ay : **m**igraine with **a**ura.
- Postpartum < 21 days.

Absolute Contraindications for IUCD :

For all IUCDs :

1. Undiagnosed vaginal bleeding.
2. Distorted uterine cavity.
3. Pregnancy.
4. Active PID.

For mirena :

- Absolute : H/o breast cancer.
- Relative : Liver disease.

For CuT :

- Absolute : Wilson disease.
- Relative : molar pregnancy.

Note :

- Previous history of ectopic pregnancy/previous history of PID
 - Nulliparity
 - HIV positive
- } **Not c/i** for IUCD

Important Points on Contraception

Important mechanism of Action of All IUCDs

Contraceptive	main mechanism	Others
OCPs (E + P)	Anovulation ($\downarrow$ FSH $>$ LH)	• Thicker cervical mucus • Thinning of endometrium
POP	makes cervical mucus thick	-
Prog implant/ Prog injection	Anovulation	make cervical mucus thick
Cu IUCD	Prevents fertilization (Copper released induces an inflammatory reaction)	Inhibits implantation
mirena	Prevents implantation	-
Centchroman	makes endometrium out of phase and inhibits implantation	-

OCPs & Cancers

 $\uparrow$ Risk of :

- Hepatic adenomas
- Breast cancer in perimenopausal females
- **Cervical cancer** (Reversible risk)

 $\downarrow$ Risk of :

- Endometrial cancer
- Ovarian cancer
- Ovarian cysts
- Benign breast disease
- Colorectal cancer

No effect :

Liver cancer

Note :

HRTs : **No effect** on risk of cervical & ovarian ca.

----- Active space -----

Missed Pills & DMPA Injection

00:01:40

missed Pills :

A. If 1 or 2 pills are missed :

1. Take one pill immediately.
2. Continue schedule.
3. Backup contraception not required.

B. If 3 or more pills are missed in the 1st or 2nd week :

1. Take one pill immediately.
2. Emergency contraception if unprotected intercourse in the last 72 hours.
3. Backup contraception (Barrier method) × 7 days.
4. Continue schedule.

C. If 3 or more pills are missed in the 3rd week :

1. Take one pill immediately.
2. Emergency contraception if unprotected intercourse in the last 72 hours.
3. Backup contraception (Barrier method) × 7 days.
4. Continue schedule.
5. Start next pack as soon as current one is completed.

DMPA Injection :

- Included in NFP by name of : **Antara**
- Dose : 150 mg.
- Route : Im.
- Repeated every : 3 months.
- Window period : 4 weeks.

method of action : Anovulation.

Advantages :

- Decrease seizure frequency.
- Decrease sickling in sickle cell anemia.

Disadvantages :

- Delayed return of fertility.
- Decreased bone mineral density.
- Irregular bleeding.

Scheduling :A female had injection DMPA on 1st July :A. Came on 1st Oct (3 months later) :

1. DMPA inj. given
2. No UPT needed
3. No backup contraceptive required

B. She comes b/w 15th Sept to 28th Oct (**2 wks before & 4 wks after** scheduled date) :

1. DMPA inj. given.
2. No UPT needed.
3. No backup contraceptive required.

C. She comes on 5th November

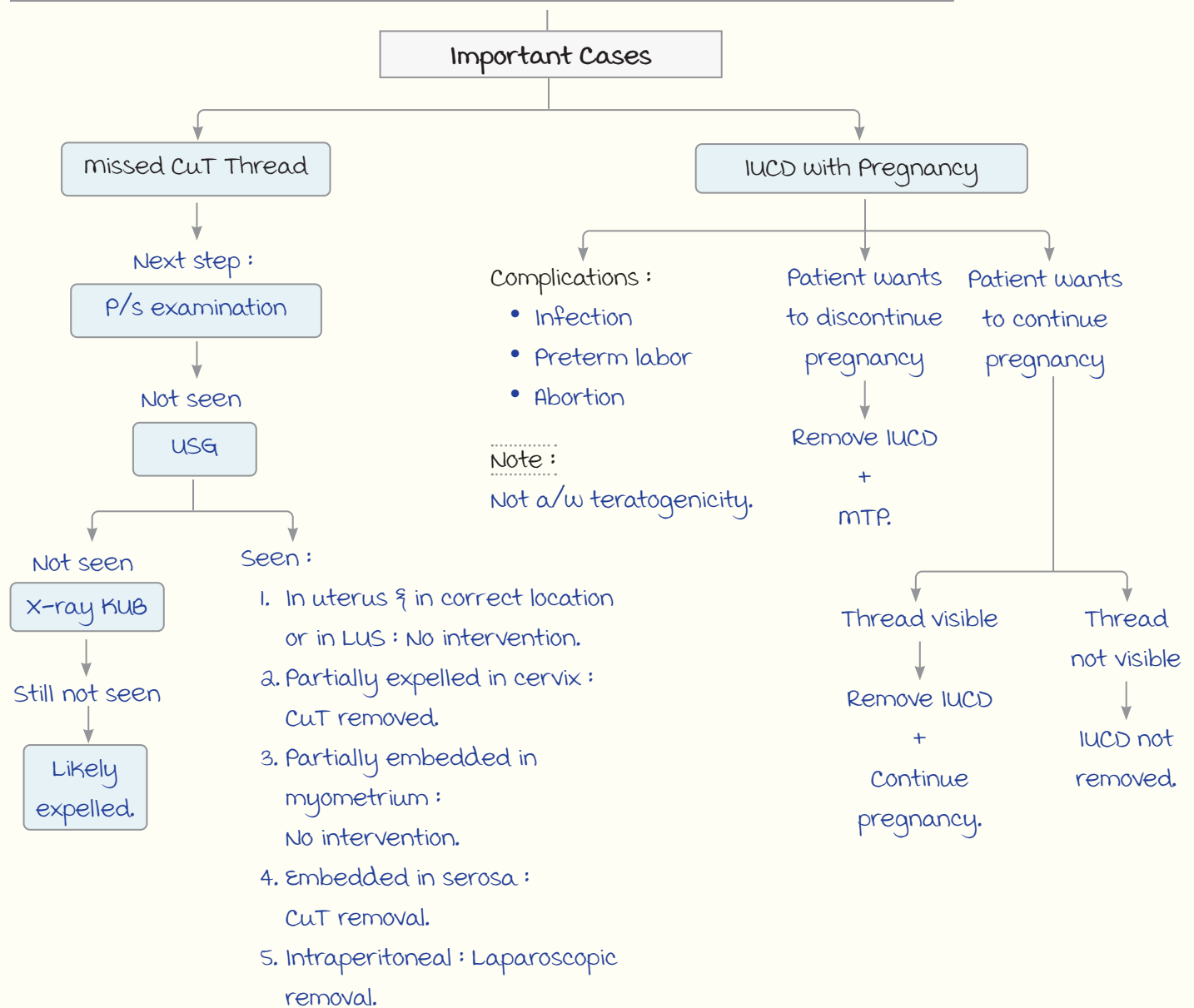
(After 4 months/window period) :

1. Rule out pregnancy.
2. DMPA inj. given.
3. Backup contraceptive × 7 days.

Missing CuT Thread & IUCD with Pregnancy

01:07:33

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Contraception : Part 2

01:10:04

OCPs :

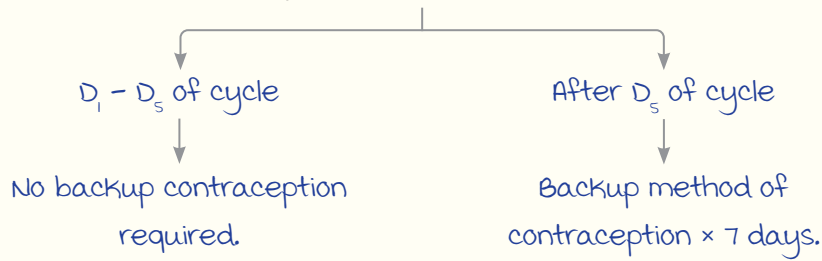
Low dose OCPs : < 50 mcg estrogen (20 – 35 mcg).

3rd generation OCPs :

- Contain 3rd generation progesterone
 - Gestodene.
 - Norgestimate.
 - Desogestrel.
- ↓ Androgenic side effects.
- ↑ Risk of venous thromboembolism.

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Initiation of OCP therapy :



OCPs & PID :

- OCPs ↓ risk of PID.
- m/c PID a/w OCP : Chlamydia.
- m/c vaginitis a/w OCP : Candidiasis.

OCPs for postpartum period :

- Not breastfeeding → Day 21 after delivery.
- High risk of thrombosis → 6 weeks after delivery.
- Breastfeeding female → 6 months after delivery.

Drug interactions :

Avoided with enzyme inducers :

- Anti-epileptics $\xrightarrow{\text{exception}}$ Can be given with :
 - Lamotrigine (↓ efficacy).
 - Levetiracetam.
 - Valproate.
- Rifampicin.
- Griseofulvin.
- Anti-retrovirals.

Transdermal Patches :

- Estrogen + Progesterone contraceptive.
- MOA : Anovulation.
- Absolute C/i : Same as OCPs.
- Reversible contraception.
- No protection against STDs.

method : New patch applied every week × 3 wks $\xrightarrow{F/b}$ Patch free interval × 1 week.

Need for backup contraception :

Patch application	Backup contraception
Within 48 hours of removal of patch	Not needed
After 48 hours of removal	Required × 7 days

CuT vs. Mirena :

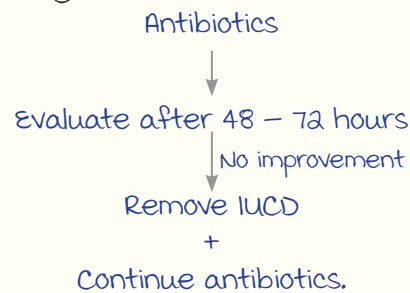
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	CuT	Mirena
MOA	Prevents fertilization	Prevents implantation
Risk of PID	↑ at the time of insertion	↓ (Progesterone → Thick cervical mucus → Impermeable to sperms)
Blood loss	↑	↓
Use as emergency contraceptive	⊕	⊖

Non-contraceptive advantages of Mirena :

- ↓ Endometrial hyperplasia.
- ↓ Blood loss → management of AUB.
- ↓ Risk of endometrial Ca.

Management of PID following CuT insertion :

Long Acting Reversible Contraceptive (LARC) methods :

Components :

- Implants.
- IUCDs.
- Injections.

Note :

Tubal ligation : Permanent method of contraception (Not LARC).

Progesterone Containing Contraceptives :

Use :

- Breastfeeding females (No effect on breast milk).
- H/o thrombosis.

Minipill :

- Component : Norethindrone (m/c).
- MOA : Thickening of cervical mucus.
- Desogestrel containing minipill → Anovulation.

----- Active space -----

POP :

- used when estrogen is contraindicated.
- Contraceptive of choice in **breastfeeding females**.
- High failure rate.
- S/e : **Irregular bleeding**.
- **No pill free period**.
- Norethindrone POP :
 - 1 tablet to be taken at the **same time** daily.
 - Safe period : 3 hours.
 - missed pill > 3 hours → Backup method x 2 days.
 - Pack of 28 tablets.

Emergency Contraception

01:19:58

Drugs :

L : Levonorgestrel (1.5 mg single dose)/
Plan B (**m/c** used).

O : OCPs 2 tablets $\xrightarrow{\text{After 12 hrs}}$ 2 tablets
(Yuzpe's method).

C : Cu IUCD.

C : Centchroman 2 tablets $\xrightarrow{\text{After 12 hrs}}$ 2 tablets.
(60 mg)

u : ulipristal (30 mg).

m : mifepristone (10 - 50 mg).

Not an emergency contraceptive :

1. **misoprostol**.
2. **minipill/POP**.
3. **mirena**.

EC :

- AKA Post-coital pills/Interceptives.
- Best given within 72 hours of unprotected intercourse.
- Can be given up to : 120 hours.
- most effective EC : **CuT**.
- most effective hormonal EC : **ulipristal**.
- Overall efficacy from maximum to least :
CuT > ulipristal > mifepristone > LNG.
- mechanism of action of EC : **Delay ovulation**.
- CuT Action : Prevents fertilization.

Emergency Contraception

- FR of EC : 1%.
- EC cannot protect from STDs.
- EC is not an abortifacient.
- They are not for regular use.
- most women after EC will have bleeding in **3 weeks**.

Cases :

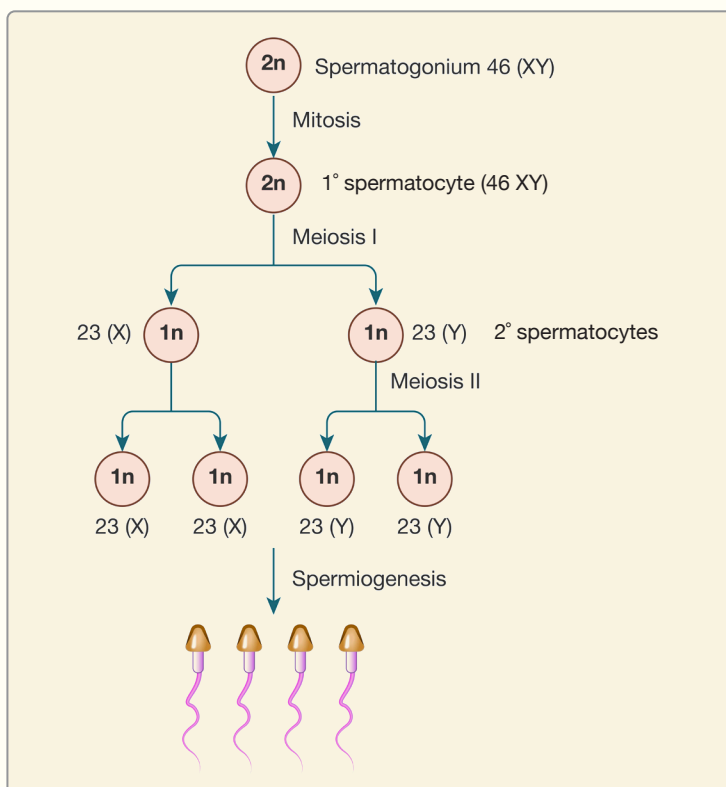
- If bleeding does not occur after 3 weeks of EC :
R/o pregnancy (UPT).
- A female comes on 5th day of unprotected intercourse :
 - Best EC : **CuT**.
 - 2nd best EC : **ulipristal**.
- EC to be avoided in :
 - Breastfeeding female : **ulipristal**.
 - With enzyme inducers :
Hormonal EC (use CuT).
 - Obese females : Avoid **LNG** (use ulipristal/IUCD).

OBSTETRICS REVISION : PART 1

----- Active space -----

Spermatogenesis

00:00:31



Important Steps in Spermiogenesis :

Spermatid	Sperm
Golgi apparatus	Acrosome cap
Nucleus	Head
mitochondrion	middle piece
Centriole	Axial filament/tail

Capacitation :

- Process by which sperms attain ability to fertilize ova
- Site : Female reproductive tract.
- Begins in cervix.
- maximum : Fallopian tube.
- Time taken : **7 hours** (6 - 8 hours).
- **No genetic changes.**
- Changes :
 - Alteration in membrane permeability.
 - Influx of Ca^{2+} & HCO_3^- .
 - Renders sperms **hypermotile**.

Important One-Liners :

Spermatogenesis :

- Begins at puberty.
- Spermatogenesis takes place in seminiferous tubules.
- Time taken for spermatogenesis : **72** days (70 - 74 days)
- Spermiogenesis : Transformation and maturation of spermatids to sperm. There is no mitosis or meiosis.
- Size of sperm : 50 - 60 microns.
- Fertilizable life span : 48 - **72** hours.
- Sperms attain maturity : Proximal part of epididymis.
- Sperms attain motility : Distal part of epididymis.
- Time for spermiogenesis : 10 - 14 days.

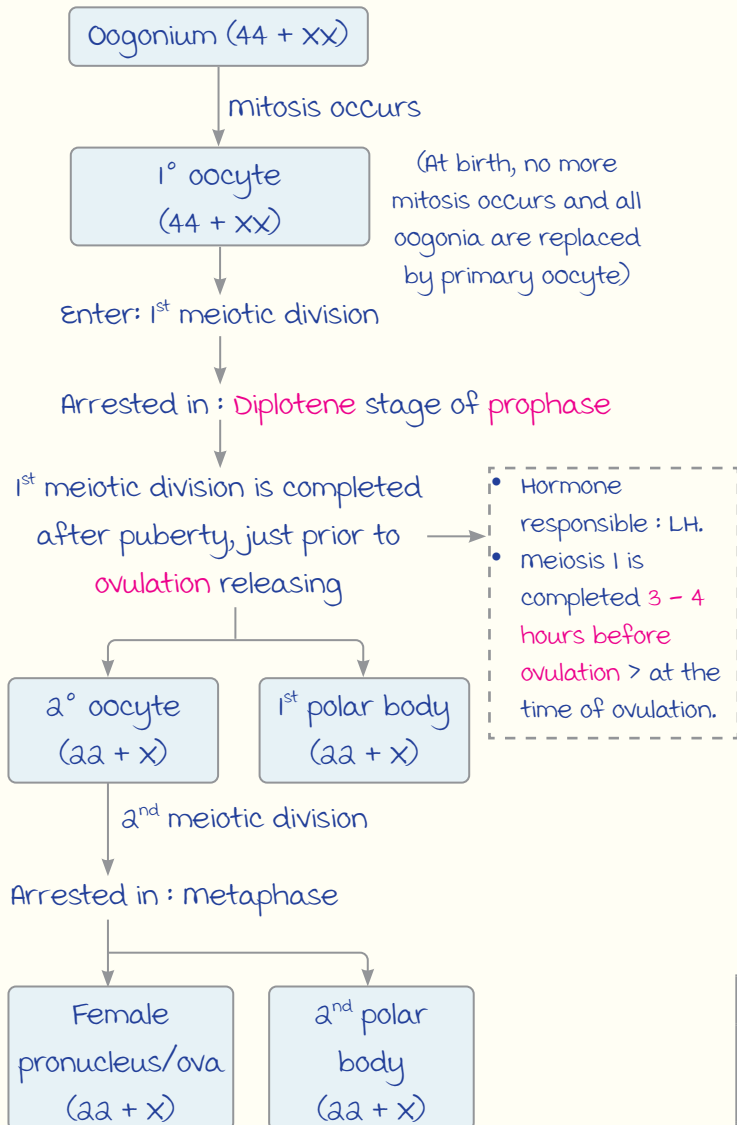
Note :

- Sperms lack **endoplasmic reticulum**.
- Sertoli cells :
 - Temperature sensitive cells of testis.
 - Secrete in intrauterine life :
 - AMH (mullerian inhibiting substance).
 - Inhibin B.
 - Testosterone binding protein.

----- Active space -----

Oogenesis

00:04:08



Important One-Liners :

- Begins in **intrauterine life**.
 - 1st meiotic division is arrested in : Diplotene stage of prophase.
 - 2nd meiotic division is arrested in : metaphase.
 - 1st meiotic division completed at : Puberty.
 - 2nd meiotic division completed at : Time of fertilization.
 - Size of mature ovum : **120 - 130 μm** . (Largest cell in the body).
 - Follicle : 1° oocyte surrounded by follicular cells of ovary.
 - Size of mature follicle : 18 - 20 mm.
 - Follicle = Primary oocyte.
 - Ovum = Secondary oocyte arrested in 2nd meiotic division.
 - Graafian follicle : Secondary oocyte.
 - Fertilizable life span of ova : 12 - 24 hours.
- Ovulation : Conversion of 1° oocyte to 2° oocyte.
- Follicles undergo programmed cell death called as **apoptosis**.
 - Rate of follicular atresia : 1000/month.
 - No. of follicles undergoing ovulation in a lifetime : 400.

Time	Number of follicles
1. 5th month of intrauterine life (20 weeks)	6 - 7 million (maximum)
2. At Birth	1 - 2 million
3. At Puberty	4 lakhs - 5 lakhs

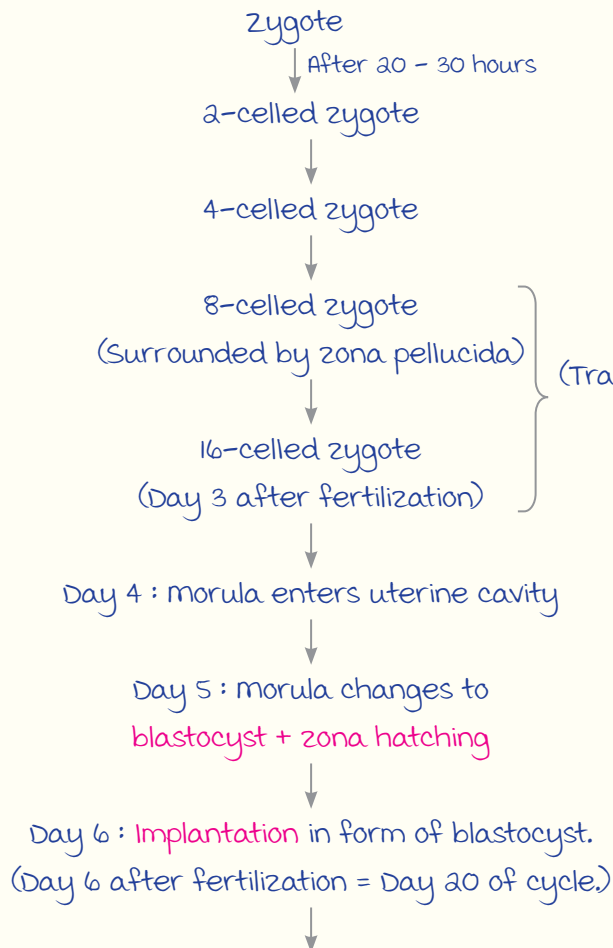
Important One-Liners :

- Events prior to fertilization :
 - Capacitation.
 - Binding to zona pellucida.
 - Acrosomal reaction.
 - Cortical reaction (**Ca²⁺** required).
 - Zona reaction.
- Capacitation provides ability to sperm for acrosomal reaction and to bind to zona pellucida.
- On zona pellucida : Receptor for sperm is zona pellucida glycoproteins (ZPGP) mainly **ZPGP3**.
- Zona pellucida : Prevents polyspermy.

Fertilization & Implantation

00:11:54

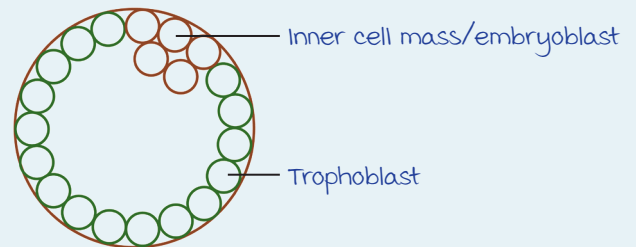
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- Fertilization occurs at **ampulla** of fallopian tube.
- Fertilization occurs on **day 14** of cycle.
- Time taken by sperms to reach site of fertilization : **30 mins**.

morula
(Transport : **Peristalsis in tube**
>
movement of cilia)

Parts of blastocyst (🔗):



Implantation (🔗):

- Site : upper posterior wall of uterus.
- **Implantation window** : uterus is most receptive to conceptus = Day 20 - Day 24 of cycle (**6 - 10 days** after fertilization).
- Thickness of endometrium at time of implantation : 10 - 12 mm (Endometrium has **pinopods** and highest level of **IL-6**).
- Implantation ends : Day 10 - 11 after fertilization (Day 24 - 25 of cycle).

Pre-implantation genetic diagnosis (PGD) :

- Best material : Trophoectoderm (Trophoblast).
- Best time : Day 3 embryo.
- Indications : Detection of
 - Sickle cell anemia.
 - Thalassaemia.
 - Cystic fibrosis.
 - Aneuploidy.
- **Not done for sex determination.**
- Embryo kept in **Ca²⁺ & Mg²⁺ free medium** → Softens zona pellucida.

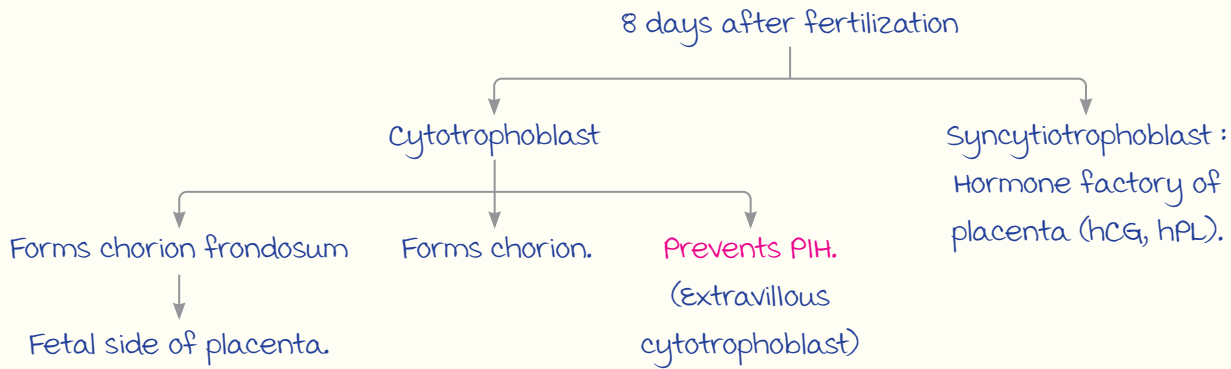
Factors required for implantation (🔗):

1. Apposition : Selectin.
2. Adhesion : **Integrin** (most important).
3. Invasion : matrix metalloproteinases.

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Post Implantation Changes

00:20:46

Trophoblast :**Decidua :**

- Endometrium after implantation.
- **Progesterone** favours decidualization.
- Decidua basalis → maternal side of placenta.
- Fusion of decidua capsularis & parietalis : **14 - 16 weeks.**
(Single uterine cavity formed)

Arias Stella reaction :

- Hypersecretory changes on HPE during implantation.
- D/t progesterone.
- Also seen in :
 - Ectopic pregnancy.
 - OCP usage.

Note :

Amnion : Formed by ectoderm.

Signs in Pregnancy

00:25:40

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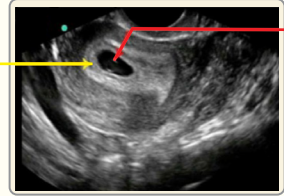
Presumptive signs : Signs felt by patient herself	Probable signs : Signs seen by doctor	Positive signs (Definitive)
Amenorrhoea	Goodell's sign : Softening of cervix (1 st sign to become positive at 6 weeks).	Fetus seen on ultrasound
Nausea/vomiting	Hegar's sign : • Softening of uterine isthmus at 6 - 10 weeks. • On bimanual palpation, vaginal and abdominal fingers touch each other.	Hearing fetal heart sounds : • Hand held doppler (10 w) • Stethoscope (20 w)
Fatigue	Chadwick's sign : Blue/violet discoloration of vagina and vulva (Jacquemier's sign).	Palpation of fetal parts
Increased urinary frequency (T1 & T3)	Osiander's sign : Pulsations felt in lateral fornix of vagina.	Fetal skeleton seen on X-ray
Breast changes	Palmer's sign : • Rhythmic painless uterine contractions. • Not felt by patient.	-
Quickening	Piskacek's : Unequal growth of uterus	
	Positive pregnancy test	
	Increased pigmentation	

Signs in Pregnancy

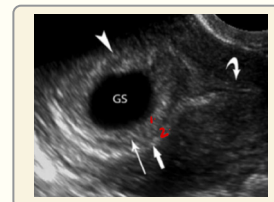
USG signs (97) :

Also know

- Hartmann's sign/placental sign : Bleeding at the time of implantation.
- Quickening.
 - Perception of first fetal movement by mother.
 - Primi : 18 - 20 weeks.
 - Multi : 16 - 18 weeks.
- Internal ballotement : At 16 - 20 weeks.
- External ballotement : At 20 - 24 weeks.

Endometrium  Gestational sac

Intradecidual sign : 1st sign of pregnancy on USG indicating interstitial implantation



Double decidual sac sign :

- 1 : Inner ring → Decidua capsularis
2 : Outer ring → Decidua parietalis



Double bleb sign

Yolk sac and amniotic sac are the two blebs

Lightening :

- At 36 weeks : Height of uterus is at level of xiphisternum. Hence, patient has respiratory discomfort.
- After 36 weeks : Head of fetus goes into pelvis so height of uterus decreases. Hence, breathlessness is relieved called **welcome sign**/lightening (Seen in late T3).

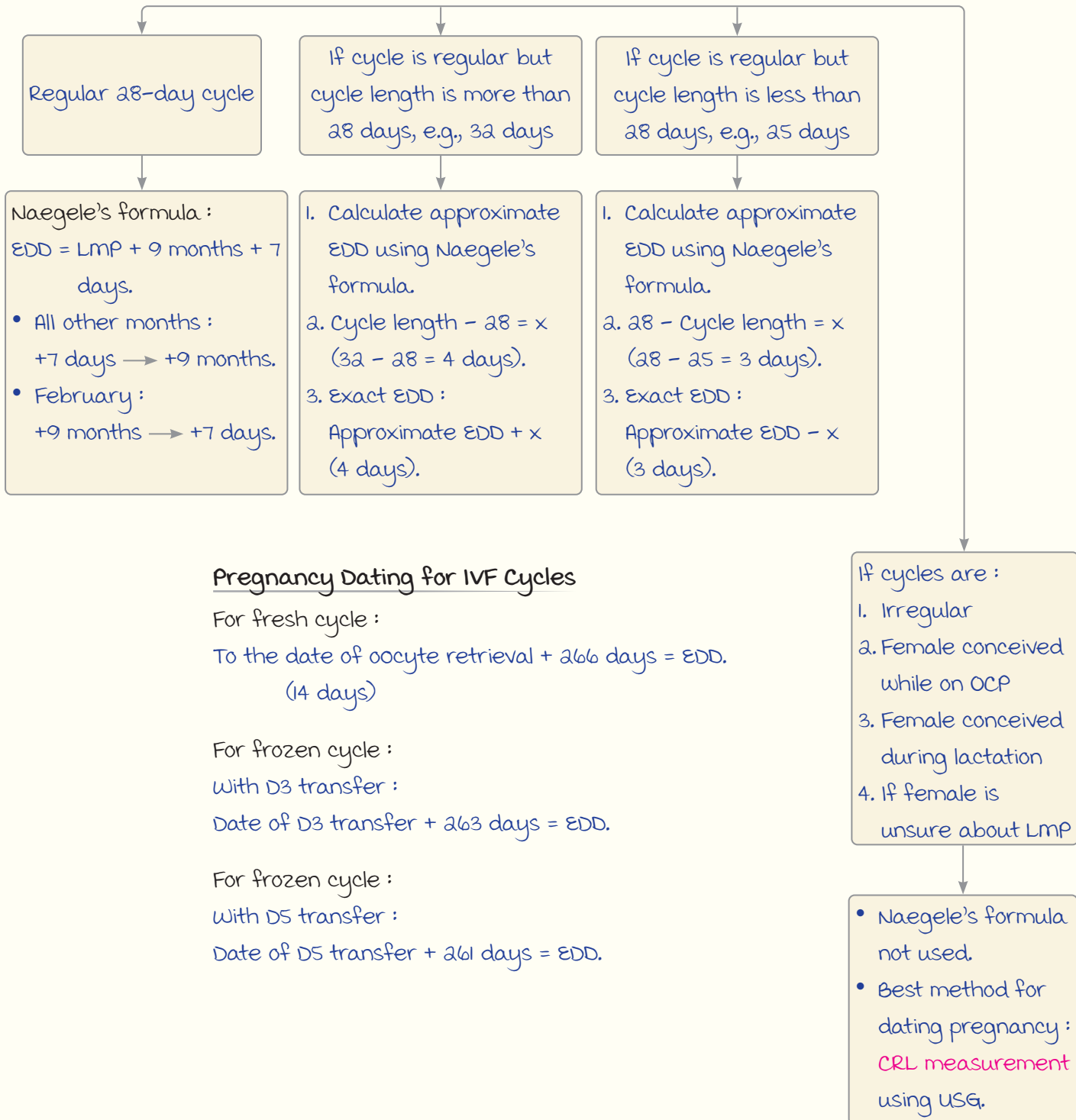
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Pregnancy Dating

00:34:10

Total duration of pregnancy : 9 months + 7 days (40 weeks/280 days).

Pregnancy Dating for Natural Conception :



Pregnancy Dating for IVF Cycles

For fresh cycle :

To the date of oocyte retrieval + 266 days = EDD.
(14 days)

For frozen cycle :

With D3 transfer :

Date of D3 transfer + 263 days = EDD.

For frozen cycle :

With D5 transfer :

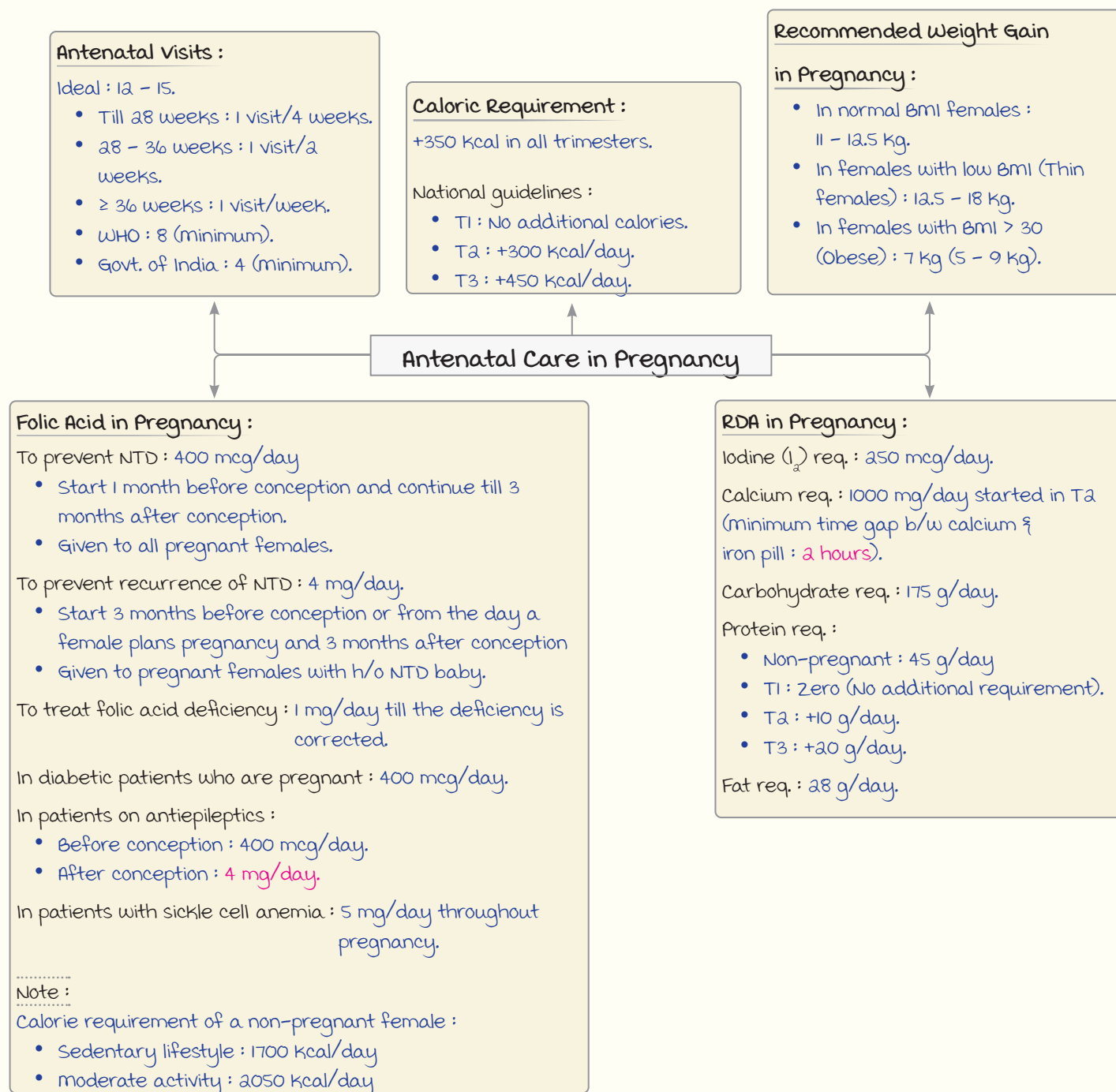
Date of D5 transfer + 261 days = EDD.

OBSTETRICS REVISION : PART 2

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Antenatal Care in Pregnancy

00:00:08



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Antenatal Care : Vaccines and Tests

00:05:36

Vaccines :

vaccines which are safe	vaccines to be given in special circumstances	vaccines which are absolutely contraindicated	
All dead vaccines can be given e.g., • Hepatitis A • meningococcus • Hepatitis B • Rabies • Pneumococcus	• Yellow fever • Polio • Typhoid	• mumps • measles • Rubella • Smallpox	• Chicken pox • BCG • HPV vaccine

Concept 1 :

- All killed vaccines are safe in pregnancy.
- All live vaccines are C/I (Pregnancy C/I for 1 month after live vaccine).

Concept 2 :

Exceptions to absolute contraindication :

1. Yellow fever.
2. Polio.
3. Typhoid.

Antenatal Care in Pregnancy : Part 2**Tetanus Vaccine :**

Td (Tetanus diphtheria) vaccine :

- All pregnant females should be given Td vaccine.
- Number of doses : 2.
- Time : 1st dose at 1st antenatal visit and 2nd dose after a gap of 4 weeks.
- If pregnant female was immunized in past 3 years and had received 2 doses ; then in current pregnancy only Td booster given

Tdap vaccine :

- Tetanus toxoid + diphtheria toxoid + acellular pertussis is also recommended during pregnancy.
- ACOG : At least 1 Tdap vaccine should be given between 27 - 36 weeks to all pregnant females to protect newborn from pertussis.

At the First visit :

- ABO, Rh.
- Hemoglobin.
- HIV testing (Opt out approach).
- VDRL.
- HBsAg.
- Blood sugar : **DIPSI test**
 - 75 g glucose (Irrespective of meals) → Blood sugar checked after 2 hours
 - Done at 1st visit & repeated b/w 24 - 28 wks.
- Urine routine microscopy (Repeated every trimester).
- TSH.
- Aneuploidy screening. } If patient can afford

At 35 - 37 weeks :

- Screening for group B Streptococci (Rectovaginal swab).

Repeat CBC : 24 - 28 weeks (ACOG).

GOL : Hb done at least 4 times in pregnancy.

.....
Note : TORCH test is not mandatory in Indian guidelines

USG in Pregnancy : Part 1

00:10:49

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USGs in Pregnancy :

- Dating/Viability scan : 6 - 8 weeks.
- Nuchal translucency scan : 11 - 13 weeks + 6 days.
- Anomaly/Target scan/Booking scan/Level 2 scan : 18 - 20 weeks (upto 22 weeks).
- Growth scan : 32 & 34 weeks.

Important Points :

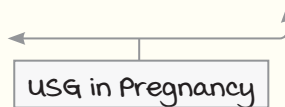
- In all pregnant females, target scan should be done (18 - 22 weeks) → Gives detailed anatomy to look for gross congenital anomalies.
- If during entire pregnancy only one USG has to be done, it should be a level 2 scan, i.e., **anomaly scan**.
- If fetal echo is needed, it is done between 22 - 24 weeks.
- Amniotic fluid can be assessed by T2 & T3 USG.
- For placental localisation, best time to do USG is at 32 weeks → Repeat at 36 weeks.
- In all pregnant females with previous h/o preterm labor or T2 abortion ACOG recommendation : TVS for cervical length measurement b/w 16 - 24 weeks.

Estimation of Gestational Age by USG :

Trimester	Parameter
T1	CRL > MSD
T2	BPD > HC
T3	Combination of all parameters > FL

CRL : Crown - rump length MSD : mean sac diameter
BPD : Biparietal diameter HC : Head circumference
FL : Femur length

- Best overall parameter for gestational age estimation : **CRL**.
- Abdominal circumference is best for assessing growth of fetus, i.e., in case of **macrosomia and IUGR**.
- CRL can be used till 13 w + 6 days (CRL < 84 mm).
- most accurate gestational age can be determined by CRL between 7 and 9 weeks.
- CRL (mm) + 42 = Gestational age in days.
- Smallest CRL which can be measured = 5 mm.
- mean sac diameter in mm + 30 = Gestational age in days.



Structures seen in Pregnancy on USG :

USG Sign	Seen earliest at TVS
1. Gestational Sac (Earlier)	4.5 - 5 weeks
2. Yolk sac	5 - 5.5 weeks
3. Embryo/fetal pole	5.5 weeks (CRL measurable)
4. Cardiac activity (Seen on m-mode)	6 weeks

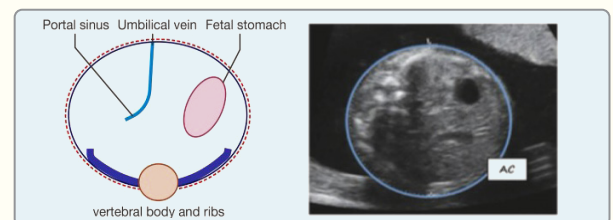
All structures are seen 1 week earlier in TVS than on TAS.

Cut off :

- Cut off of MSD to get fetal pole : 25 mm.
- Cut off of CRL to get cardiac activity : 7 mm.
- Cut off of hCG to get G sac on TVS : 2000 IU.

For Growth of Fetus :

Parameter used : **Fetal abdominal circumference**.



Abdominal circumference

- Abdominal circumference (AC) measurement on USG → AC should be measured in a plane with :
 - P : Portal sinus
 - U : Umbilical vein
 - S : Stomach
 } **Hockey stick sign**
- Clinically : Fetal weight can be estimated using **Johnson's formula**.
- On USG : Best method to estimate fetal weight is by combination of HC, AC, FL, and BPD using **Hadlock/Shepard formula**.
- AC ≥ 350 mm indicates macrosomia.

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Neural Tube Defects (NTD)

00:20:32

Neural Tube Defects :1st congenital anomaly detected on USG**Anencephaly** (🐭):

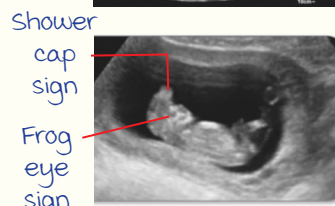
- m/c NTD (Cranial defect).
- Brain tissue herniates out without covering of scalp, meninges and hair.
- It can be detected earliest on USG at 10 weeks.
- Best diagnosed at 14 weeks.
- Polyhydramnios : D/t absent fetal swallowing.
- Higher association with female fetal sex.
- Post term labor > Preterm labor.
- Recurrence risk after 1 baby with NTD = 4%.
- Recurrence risk after 2 babies with NTD = 10%.



Anencephaly



Mickey mouse sign



Shower cap sign

Frog eye sign

USG Signs of Anencephaly :

1. Mickey mouse sign : Triangular face
2. Frog eye sign : Bulging eye sign
3. Shower cap sign : Irregular appearance of head region due to herniation of brain tissue

Encephalocele :

Cranial defect : Brain tissue herniates out covered by meninges, scalp & hair.



Encephalocele

Important USG Images of Anomalies :

For congenital anomalies :

- Best USG : Level 2/Target scan.
- Level 1 scan can detect some anomalies like NTD.

For gross congenital anomalies :

- Screening test : Level 1 USG.
- Diagnostic test : Level 2 USG.



Spina bifida

- A. Banana sign : Downward displacement of cerebellum
B. Lemon sign : Frontal bossing

Seen in :

- Spina bifida
- Arnold Chiari malformation

Note :

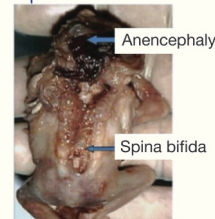
IOC for chromosomal anomalies : Karyotyping.

Spina Bifida :

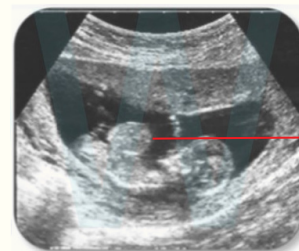
Caudal defect.

Craniorachischisis :

- Cranial and caudal defect seen together.
- Fetus has both anencephaly and spina bifida.



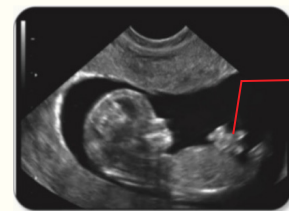
Craniorachischisis



Smooth, regular sac

Omphalocele

- Abdominal wall content herniates out covered in a sac.
- Associated with anencephaly (Indication for karyotyping).



Cauliflower like app.

Gastroschisis

- Abdominal wall content herniates out w/o any covering
- Not a/w anencephaly

Aneuploidy Screening

00:26:22

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First Trimester :

11 - 13 weeks + 6 days.

Biochemical test (Dual test) :

- PAPP-A.
- hCG ($\uparrow$ in Down's syndrome).

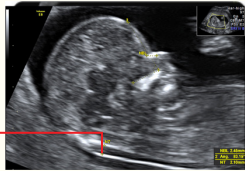
USG (11 - 13 weeks + 6 days) :

Nuchal translucency :

- Fluid collection below neck.
- Normal : < 3 mm.
- Causes of $\uparrow$ NT (≥ 3 mm) :
 - a. Aneuploidies : Down syndrome $>$ Turner syndrome $>$ Trisomy 18/13.
 - b. Congenital heart disease
 - c. TTTs.
 - d. Genetic syndromes :
 - DiGeorge syndrome.
 - Noonan syndrome.

Next step :
karyotyping

Fluid collection below the neck



Nuchal translucency

If NT on USG ≥ 3 mm and nuchal karyotype is normal
 $\downarrow$
Fetal echo.

Screening test

Second Trimester :

16 - 18 weeks.

Biochemical test :

1. Triple test/Bart test/Kettering test :

- Alpha-fetoprotein
- hCG
- unconjugated E3 (Estriol)

2. Quadruple test : Triple test + Inhibin A ($\uparrow$ in Down's syndrome)

USG (Soft tissue markers) :

- Nuchal fold thickness ≥ 6 mm.
- Absent nasal bone.
- Short femur/humerus.
- Sandal gap.
- Duodenal atresia (Double bubble sign)
- Simian crease.
- Echogenic intracardiac focus.
- Echogenic bowel.
- Choroid plexus cyst.

$\geq 2 \oplus$
 $\downarrow$
+ve test.

Combined test : Dual test + Nuchal translucency.
Integrated test : PAPP-A in T1 + NT in USG in T1 + Quadruple test in T2.

Aneuploidy Screening :

Done in all pregnant females irrespective of age

NIPT
(Non-invasive prenatal testing)

2° screening test

Cell free fetal DNA :

- Disadvantages :
 - 2° screening test.
 - Expensive.
- Sample : maternal blood.
- Source : Placental cell apoptosis.
- Can be done anytime ≥ 10 wks.
- Highest sensitivity. (99%)

-ve

- No further testing needed (Reassure the patient)
- Best screening test

+ve

Karyotyping

If screening is positive

Karyotyping (Diagnostic test)

Tissue for karyotyping obtained by

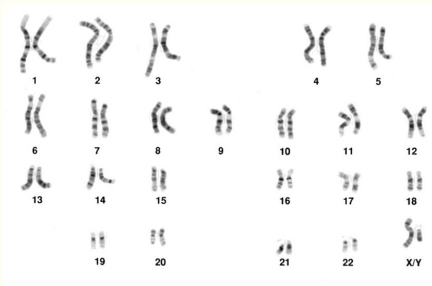
T1

Chorionic villi sampling
(Best time : 11 to 13 weeks + 6 days)

T2

Amniocentesis
(Best time : 16 - 18 wks)

----- Active space -----

Karyotyping :

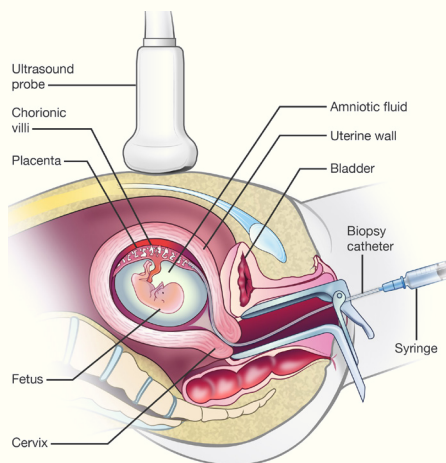
Normal male karyotype



Down syndrome



Turner syndrome



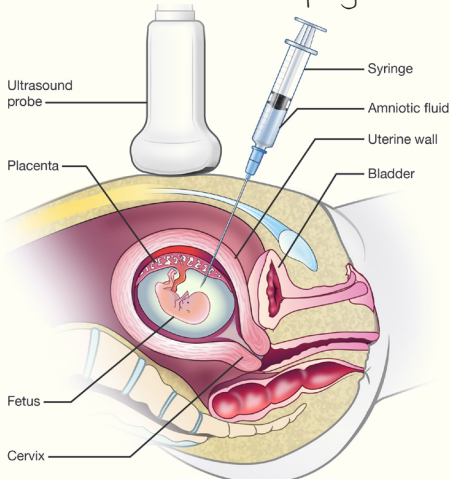
Chorionic villus sampling

Chorionic villus sampling :

- Earliest done at 10 wks.
- < 10 wks → Oromandibular defect + limb defects.
- Risk of fetal loss : 1%.
- 2 cell lines maybe obtained if there is placental **mosaicism**.

Amniocentesis :

- Can be done b/w 15 - 20 weeks.
- Early amniocentesis (12 - 14 weeks) C/I → Fetal loss.
- Risk of fetal loss : 0.5%.



Amniocentesis

Cordocentesis :

- Blood obtained from umbilical vein.
- Risk of fetal loss : 3%.
- Only indication : **Rh isoimmunized pregnancy**.
- Done at ≥ 18 weeks.

USG in Pregnancy : Part 2

00:39:53

1st sign of pregnancy on USG : **Intradecidual sign**.

1st structure seen on USG : **Gestational sac** → True : Intrauterine pregnancy.
 → Pseudo : Ectopic pregnancy.

1st structure that denotes intrauterine pregnancy : **Yolk sac**.Cardiac activity seen on USG at : **6 weeks (TVS)**.Gestational sac visible on TVS : **5 weeks**.

Single best parameter :

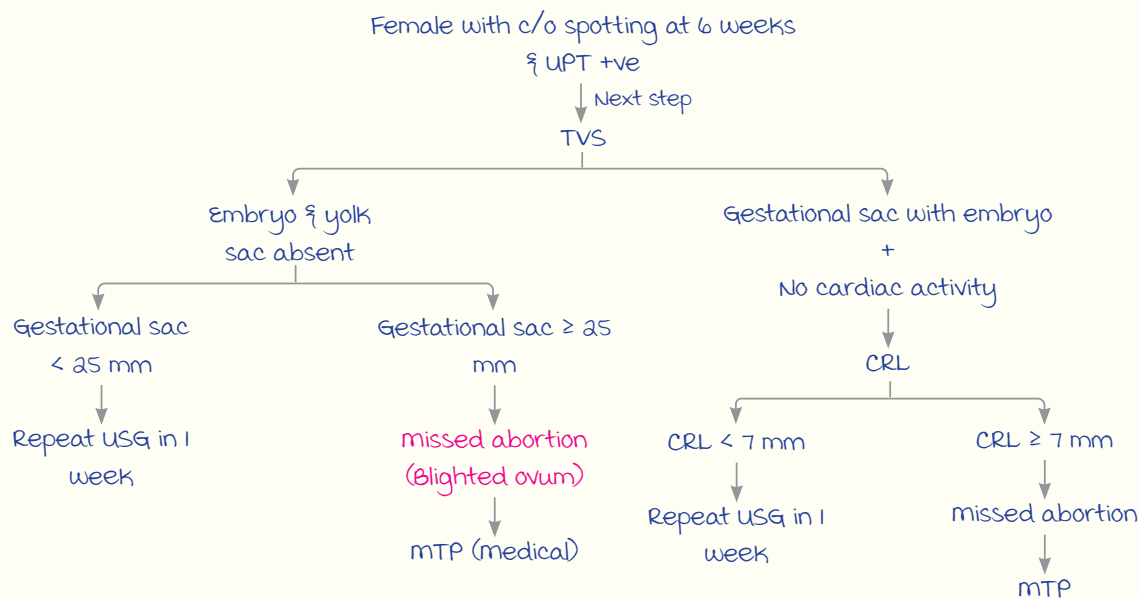
- Gestational age assessment : CRL.
- Fetal growth : Abdominal circumference (AC).

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Planes of measurement :

- BPD measurement : Plane passing through thalami & 3rd ventricle.
- HC measurement :
 - Thalami + 3rd ventricle + Cavum septum pellucidum visualized.
 - Indication : Altered shape of head (E.g. Brachycephaly) in T2.

Missed Absorption on USG :



Diagnosis of missed abortion (57) :

- MSD : ≥ 25 mm + no fetal pole seen (Blighted ovum).
- CRL : ≥ 7 mm + no cardiac activity.

Detection of NTD :

- 1st congenital anomaly detected on USG.
- Biochemical marker :
Acetylcholinesterase (most specific).

hCG Levels to Visualize

Gestational Sac :

- Cut off values
 - TVS : 2000 IU.
 - TAS : 6500 IU.
- minimum level on TVS : 1000 IU

Key Points : AFP

- Synthesized by fetal yolk sac Liver + GIT.
 - Tested in maternal serum b/w 15 - 20 weeks (16 - 18 weeks).
 - ↓AFP :
mnemonic : Diabetic GOAT.
 - Diabetic mother.
 - Gestational trophoblastic disease.
 - Obese mother/Overestimated POG.
 - ↑AFP :
 - NTD
 - Abdominal wall defect.
- Next step → USG

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Teratogens

00:51:00

Radiation Exposure :

- Maximum radiation permissible in pregnancy : 5 rad.
- Exposure ≥ 10 rad : Definitive damage.
- Maximum risk of radiation exposure 8 - 15 wks.
- If a female undergoes X-ray due to road traffic accident and she is 6 weeks pregnant
→ Continue pregnancy.
- Investigations safe in pregnancy :
 - USG.
 - MRI.
- Investigations not done : X-ray & CT.
- Absolutely C/I : **Radioactive iodine.**

Drugs ☹️ :

- Lithium : Causes **Ebstein anomaly**
 - Patient presents with bipolar disorder.
 - Chest x-ray : Box shaped heart.
- Thalidomide : Phocomelia (Proximal limb defect)
- Chloramphenicol : Grey baby syndrome.
- Valproate : NTD (m/c : spina bifida)
- ACE inhibitors : Renal anomaly.
- Isotretinoin : microtia/anotia.
- Indomethacin :
 - Premature closure of ductus arteriosus.
 - C/I in ≥ 32 weeks.
- Warfarin :
 - Leads to chondrodysplasia (Crosses placenta).
 - Abnormal cartilage formation.
 - Absent nasal bone.
 - Stippling of epiphysis.
 - Only indication : mechanical valve replacement.

Teratogens

Alcohol
(Fetal
alc. syn-
drome)

Gaa's = Growth restriction.
Famous = Abnormal facial features (Smooth philtrum, thin vermilion border, small epicanthal folds).
Beer = Abnormal brain development, microcephaly.
Bar = Abnormal behavioural development.

Antibiotics Safe in Pregnancy :

- **C**ephalosporins
- **A**mpicillin/ amoxicillin/ amoxyclav
- **m**etronidazole
- **P**enicillins

Antibiotics C/I in T3 :
Sulfamethoxazole.**Antibiotics C/I in First Trimester :**

- **N**ot = Nitrofurantoin
- **F**irst = Fluconazole
- **T**rimester = Trimethoprim

Antibiotics C/I Throughout**Pregnancy :**

F = Fluoroquinolone → Toxic to cartilage.
A = Aminoglycoside → Ototoxic and nephrotoxic.
T = Tetracycline (If needed, Doxy may be given) → Teeth discoloration.

Growth Periods :

Periods	Definitions	Comment
1. Pre-embryonic	Day of fertilization upto 2 wks after that	All or none law : Teratogenic exposure during this period either leads to abortion or fetus escapes any injury
2. Embryonic period	3 - 8 wks after fertilization	most teratogenic period
3. Fetal period	≥ 9 wks after fertilization (Till delivery)	-

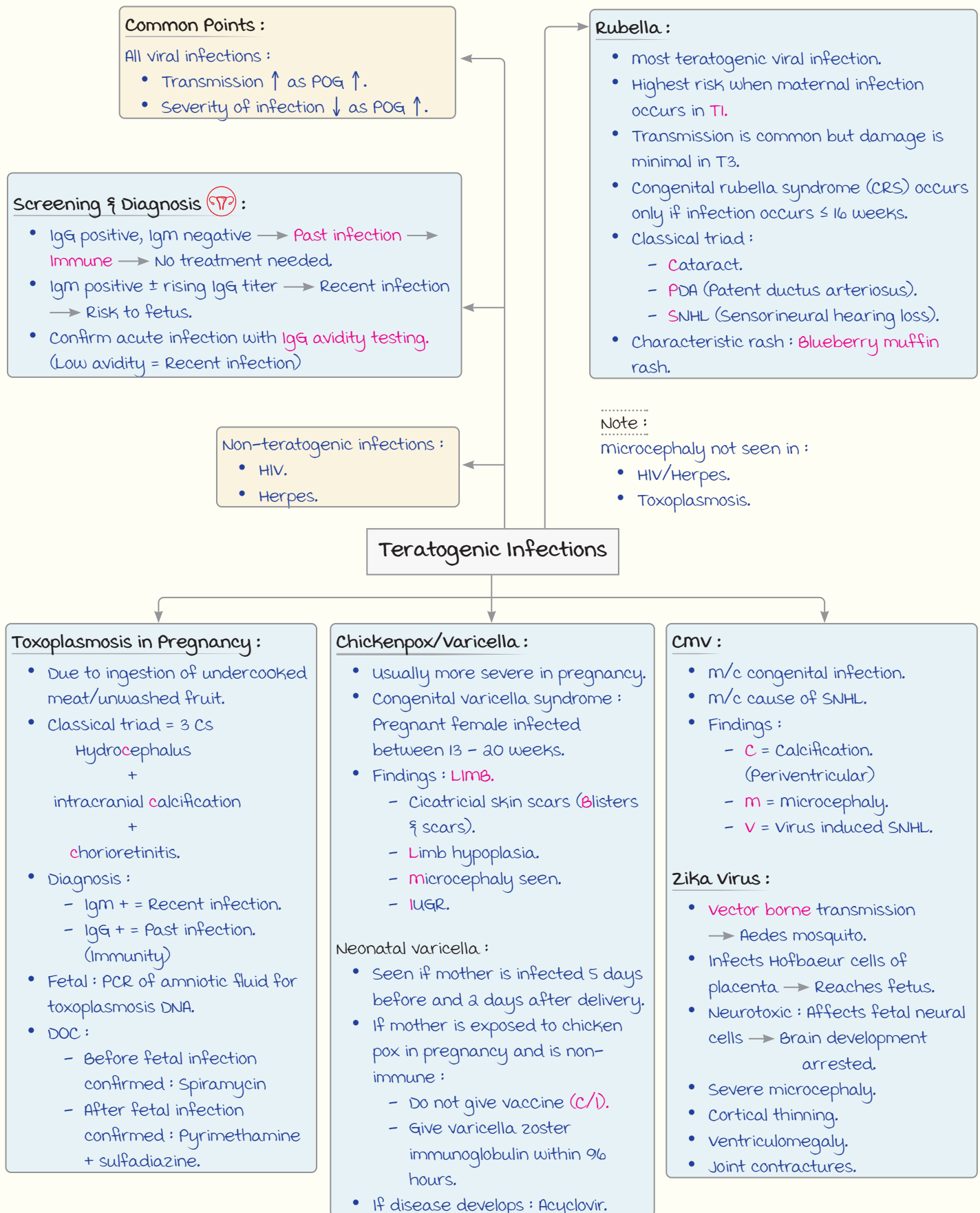
Minimum Time Gap :

- Any live vaccine and pregnancy : **1 month.**
- Between tamoxifen and pregnancy : **3 months** (minimum : 2 months).
- Between isotretinoin & pregnancy : **1 month** (1 - 3 months).
- Between previous LSCS and pregnancy : **1.5 years** (minimum : 6 months).
- Between VVF repair and pregnancy : **1 year.**

Note :**Calculation of :**

- Growth period : Day of fertilization
 - POG/Period of pregnancy : 1st day of LMP.
- } Difference = 2 weeks.

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Drug in Pregnancy

01:07:07

Anticoagulants in Pregnancy :

- Pregnancy : hypercoagulable state (All factors ↑ except II & XIII).
- Warfarin crosses placenta.
- Anti-coagulant of choice in pregnancy : LMWH (Does not cross placenta).
- For DVT in pregnancy : LMWH.
- For APLA syndrome in pregnant female : LMWH + Aspirin.
- For APLA syndrome with previous h/o abortion/thrombosis, in pregnancy : LMWH + Aspirin.
- For APLA syndrome with h/o preterm labor (D/t UPI/PIH) : **Aspirin**.
- For thromboprophylaxis after delivery in patients at risk : LMWH.

Patient on warfarin presents at 40 weeks :

1. Stop warfarin.
2. Plan C-section.
3. Vit K injection to mother & baby after delivery.

Only Indication of Warfarin in**Pregnancy :**

Patient with previous h/o mechanical valve replacement.

In first trimester : At < 12 weeks	LMWH (DOC) or warfarin < 5 mg/day
12 - 36 weeks	warfarin
At 36 weeks	Heparin (warfarin → PPH and ICH in baby)
On day of labor	Stop all anticoagulants
Restart anticoagulation : 6 hours after vaginal delivery/12 hours after C-section	Warfarin + UFH ↓ Stop UFH when INR = 2.5 & continue warfarin.

Drugs in Pregnancy**Antiepileptics in pregnancy :****Safest :**

- Levetiracetam.
- Lamotrigine.

most teratogenic :**valproic acid.**

In pregnancy :
Increased clearance of antiepileptics =
↓ Hence
monitor drug levels every 4 weeks

Note :

Anticoagulant of choice in bioprosthetic valve re-placement : **Aspirin**.

- Case 1 : If a female is on antiepileptics and conceived
↓
Continue same antiepileptic. (except valproate)
- Case 2 : If a female is on polytherapy prior to pregnancy → Shift to monotherapy.
- Case 3 : Patient coming for 1st antenatal visit at 16 weeks, K/c/o epilepsy and on valproic acid → Next step : Continue valproic acid.

Antimalarial in pregnancy (✓) :

- Safe :
 - a. Chloroquine.
 - b. Quinine.
 - c. Artesunate.
 - d. Artemether.
- C/I :
 - a. Primaquine.
 - b. Halofantrine.
 - c. Doxycycline.

DOC for candidiasis in pregnancy :

- Topical imidazole (miconazole).
- Fluconazole C/I.

DOC for bacterial vaginosis in pregnancy : metronidazole (Avoided in 1st Tr).

Thyroid Disorders in Pregnancy

01:15:02

----- Active space -----

Size of thyroid gland	Increases but goiter is pathological
α subunit	α subunit TSH $\approx$ α subunit of hCG
Total T3 and total T4	$\uparrow$
Thyroid binding globulin	$\uparrow$ (D/t $\uparrow$ Estrogen)
Free T3, T4	Normal (Euthyroid)
Normal TSH	0.1 - 2.5
Aim of TSH	TSH < 2.5

Remember :

- Fetal thyroid gland becomes functional at 12 weeks of pregnancy.
- In TI fetus is dependent on maternal thyroid hormone.
- maternal T3 and T4 can cross placenta.
- maternal TSH cannot cross placenta.

Thyroid Disorders in Pregnancy

- m/c cause of hypothyroidism in pregnancy : **Hashimoto's thyroiditis**.
- If a female with hypothyroidism conceives, during pregnancy : $\uparrow$ Dose of thyroxine. (30 - 50%)
- m/c cause of hyperthyroidism in pregnancy : **Graves' disease**.
- DOC for hyperthyroidism in pregnancy : **PTU**.
- DOC for hyperthyroidism in TI : **PTU**.
- DOC for hyperthyroidism in T2/T3 : **Carbimazole/methimazole**.
- DOC for thyroid storm in pregnancy : PTU (Propranolol only for symptomatic relief).
- methimazole and carbimazole are C/I in TI : Leads to **aplasia cutis** in fetus.

Infections in Pregnancy

01:18:18

Viral Infections :

viral infections that do not cause GCA/Hydrops fetalis :

- HIV.
- Herpes.
- Hepatitis.

mode of delivery :

- Active genital herpes at term : C-section.
- HIV : vaginal delivery.

----- Active space -----

Syphilis :

- Complication :
 - Still birth.
 - Does not cause recurrent abortion.
- **Kassowitz law** (Congenital syphilis) :
As no. of pregnancy losses $\uparrow \rightarrow$ Pregnancy outcome improves.

Infections Causing Hydrops :

- Parvovirus.
- Syphilis.

Key points

1. Fetal period : ≥ 11 weeks from LMP till delivery.
2. Prolactin level in pregnancy :
 - Highest prolactin level : Late T3 (**Not** during breastfeeding)
 - Prolactin levels $\downarrow$ after delivery
 - 1st stimulus for breast milk secretion : $\downarrow$ Estrogen & $\downarrow$ Progesterone.
 - Hyperprolactinemia in pregnancy :
 - DOC : **Bromocriptine**.
 - Indication for Rx : **Visual disturbances**.
3. Anticoagulant of choice in bioprosthetic valve replacement in pregnancy : Aspirin

OBSTETRICS REVISION : PART 3

----- Active space -----

Hematological Changes in Pregnancy

00:00:20

Increased	Decreased	Same in pregnancy
- Blood volume - Plasma volume (40%) - RBC volume (20%) Hemo-dilution →	Hematocrit OR packed cell volume (RBC vol/plasma volume)	- Bleeding time - Clotting time
- Hb mass (g) - Reticulocyte count (D/t ↑ EPO)	Hb conc. (g/dL)	mCHC
Plasma protein mass (g)	Plasma protein conc. (g/dL)	
- Globulin - Thyroid-binding globulin - Sex hormone-binding globulin	- Albumin - A:G ratio = N = 1.7:1 - Pregnancy = 1:1	
All clotting factors (Pregnancy is a hypercoagulable state)	Factor II, III	
S. Fibrinogen - WBC count (15,000) - Neutrophilia - ESR - CRP Inflammatory markers	- Fibrinolytic activity : - Protein C - Protein S - Antithrombin - Platelet count (Benign gestational thrombocytopenia) - Eosinopenia	

- Size of spleen = ↑ by 50% in pregnancy.
- Size of pituitary = ↑ by 125-135% in pregnancy.
- BMR ↑ by 10-20%.
- Plasma osmolarity & oncotic pressure : ↓ d/t hemodilution.

Anemia in Pregnancy

00:04:46

Classification :

Physiological anemia (Hb ≥ 11 gm%)	Pathological anemia (Hb < 11 gm%)
- Cause : D/t hemodilution - m/c cause of anemia in pregnancy - Normocytic normochromic anemia	- Iron deficiency anemia (m/c) - microcytic hypochromic anemia

WHO Classification of Anemia in Pregnancy :

- mild : Hb → 10-10.9 gm%.
- moderate : Hb → 7-9.9 gm%.
- Severe : Hb → < 7 gm%.
- very severe anemia : Hb → < 4 gm%.

Note :

Anemia is the m/c indirect cause of maternal mortality in India

----- Active space -----

Anemia Mukht Bharat Program (९७) :Aim : To prevent anemia; **Not to treat** anemia.

Interventions :

1. Digital hemoglobinometer.
2. IFA pill (Iron and folic acid pill).
3. Deworming.
4. Food fortification.
5. Delayed cord clamping
(Prevents neonatal anemia).
6. Addressing other causes.

IFA Pill :

Iron = 60 mg.

Folic acid = 500 mcg

(Prevents megaloblastic anemia).

Color of pill = Red.

Note :

Folic acid → 1st 12 weeks POG :
Prevents NTD's.

→ From 4^m of pregnancy :
Prevents megaloblastic anemia.

Remember :

- To prevent anemia = IFA pill is given from 4th month of pregnancy.
- To treat anemia = IFA pill should be **started** from the time when anemia **detected**.

All reproductive age females

- Given IFA pill once per week.
- Deworming done :
Tab. Albendazole 400 mg x twice a year.

When She Plans Pregnancy :

- Stop IFA pill.
- Start folic acid supplementation (400 mcg/day).

Continue only folic acid till 3 months after conception.

At 4th month of Pregnancy :

Start IFA tablet → 1 pill/day.

Continue throughout pregnancy or for at least
180 days in pregnancy.To be continued till : **180 days after delivery.**

Deworming to be done using :

Tab Albendazole = 400 mg once in second trimester.

Remember :

- Dose of IFA pill to **prevent** anemia = 1 pill/day.
- Dose of IFA pill to **treat** anemia = 2 pills/day.

management of Iron Deficiency Anemia in Pregnancy :

----- Active space -----

methods of Iron supplementation :

1. Oral iron.

2. Parenteral iron (C/I in T₁).

- m/c used : Iron sucrose (iv).

- max dose per day : 200 mg.
- max dose per week : 600 mg (Alternate days).

- Best form : Iron carboxy maltose.

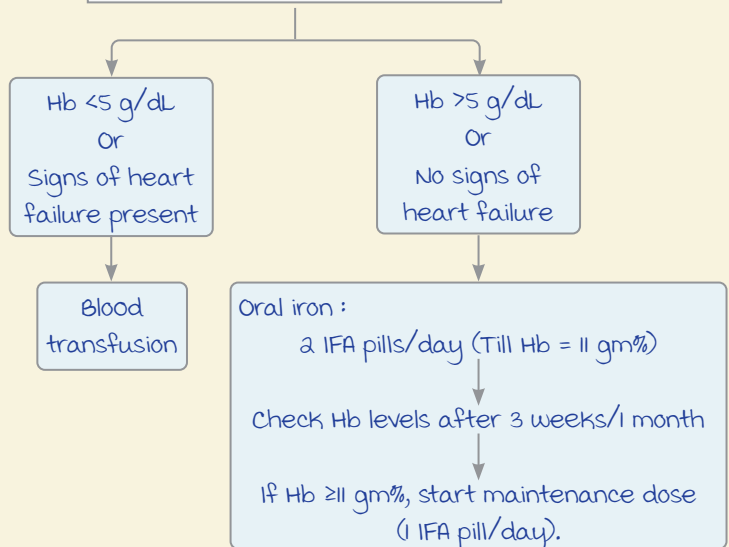
- max dose/day : 1000 mg/day.

3. Blood transfusion.

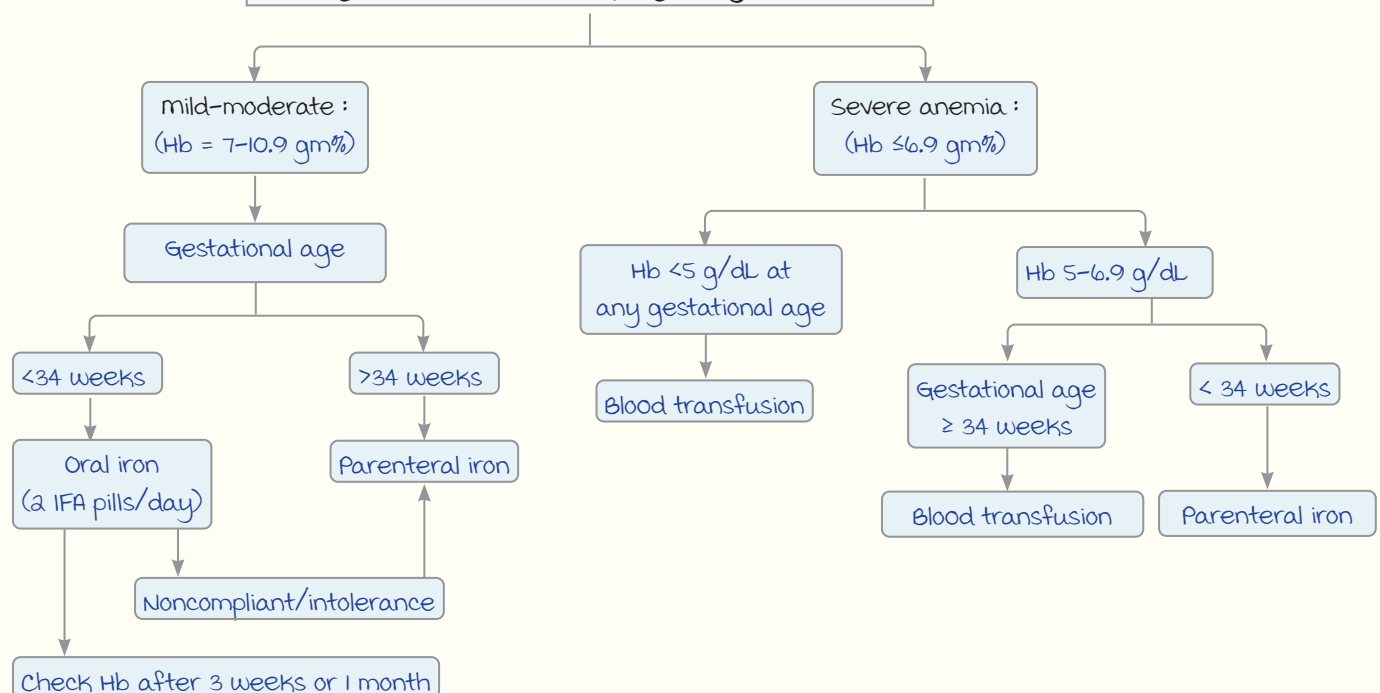
Important PYQs (97%) :

- Rate of increase of Hb after oral iron and parenteral iron is same = Increase in Hb is seen after 3 weeks of starting iron and increase is ≈ 0.7 g/dL per week.
- Calculation of parenteral iron doses in mg = use **GANZONI** formula
 $2.4 \times \text{pre-pregnancy wt of pt} \times \text{Hb deficit}$
 $(14 - \text{Pt Hb}) + 500 \text{ mg}$
 (For replenishing storage)
- Indications for blood transfusion in anemia :
 - unstable vitals (Indicates hemorrhage).
 - Signs of heart failure.
 - Thalassemia (NESTROFT test positive).
 - Hb < 5 g/dL at any gestational age.
- 1 PRBC : $\uparrow$ Hb by 1 gm% in 24 hrs.

management of Anemia in T1



management of anemia in pregnancy in T2 and T3



----- Active space -----

Renal & Respiratory Changes in Pregnancy

00:21:42

Renal Changes in Pregnancy :

Size of kidney	Increases by 1 cm
Hydrourerter	• B/L hydrourerter (D/t smooth muscle relaxant effect of progesterone). • Hydrourerter is more on right side (D/t dextrorotation of uterus).
Renal blood flow	Increases
GFR	Increases
Filtering capacity of kidney	Increases
S: urea, uric acid and creatinine	Decrease during pregnancy
Proteinuria	⊖
Glycosuria	⊕ (Physiological)

Respiratory Changes :

Remember
Transverse diameter chest increases by 2 cm
Diaphragm is pushed up by 4 cm
Circumference of chest increases by 6 cm
Subcostal angle increases from 68° to 103°
Parameters which increase in pregnancy
Inspiratory capacity
Tidal volume
minute ventilation
Parameters which remain unchanged
Inspiratory reserve volume
Respiratory rate
vital capacity
Rest of the parameters decrease

2,3-DPG : ↑.

Tissue O₂ demand : ↑.Arteriovenous O₂ gradient : ↓.maternal Hb O₂ dissociation curve : Shifts to right.

Cardiovascular Changes and Heart Disease in Pregnancy

00:25:38

----- Active space -----

Increased	Decreased	Remain same
Cardiac output	Peripheral vascular resistance d/t progesterone	JVP
Stroke volume	Systolic BP	Left ventricular ejection fraction
Heart rate	Diastolic BP	Pulmonary capillary wedge pressure
Femoral venous pressure (↑risk of varicose veins & hemorrhoids)	MAP	-

Normal Cardiac markers in Pregnancy

(Do Not Indicate Heart Disease)

- PR : ↑.
- BP : ↓.
- JVP = Remains constant.
- S → S1 = Loud + prominent splitting.
→ S2 = (N).
→ S3 = Easily heard.
- ECG :
 - Left axis deviation as heart is pushed upward and outward by the diaphragm.
 - **mild ST depression.**
 - Q wave in lead II/III.
- Chest x-ray :
 - mild cardiomegaly.
 - Straightening of left heart border.
- murmur :
 - Ejection systolic murmur grade I to II is physiological.
 - Continuous (mammary) murmur.

Indicators of Heart Disease in Pregnancy :

- JVP : ↑.
- Heart sounds :
 - S2 : Loud + Prominent splitting.
 - S4 : ⊕.
- murmur :
 - Ejection systolic murmur (≥Grade 3/6)
 - Presence of diastolic murmur (Physiological in 20% cases).
- X-ray : marked cardiomegaly.

Symptoms :

- Hemoptysis.
- Progressive dyspnea.
- Non-dependent edema.
- Orthopnea.
- Paroxysmal nocturnal dyspnea.
- Chest pain.

Heart disease in pregnancy

----- Active space -----

Heart disease in pregnancy

Most Common in Heart Disease in

Pregnancy :

- m/c heart disease in pregnancy :
 - Cause : Rheumatic heart disease.
 - Lesion : mitral stenosis.
- Second m/c heart disease in pregnancy :
 - Cause : Congenital heart disease.
 - Lesion : Atrial septal defect.
- m/c congenital valvular HD in pregnancy : **mitral valve prolapse.**
- m/c cyanotic congenital heart disease in pregnancy : Tetralogy of fallot.
- HD with maximum risk of maternal mortality : **Eisenmenger syndrome :**
 - 2° pulmonary hypertension.
 - mortality rate : 50% (**Pregnancy C/I**).
- m/c time of death in patients with Eisenmenger syndrome : At the time of delivery or within 1 week of delivery.
- m/c maternal mortality is seen with HD in pregnancy : mitral stenosis.

Heart Disease Where

Pregnancy is C/I :

- LV ejection fraction **<35%.**
- Severe mitral stenosis (valve area $<1 \text{ cm}^2$).
- Severe aortic stenosis (valve area $<1 \text{ cm}^2$).
- NYHA grade 3 or 4.
- Marfan syndrome with aortic root dilation ($\geq 4 \text{ cm}$).
- Coarctation of aorta with aortic valve involvement
- Pulmonary hypertension.
 - Primary.
 - Secondary (Eisenmenger syndrome).
- Peripartum cardiomyopathy with residual defects.
- Fontan surgery with residual defect (Done for left hypoplastic heart)

management of HD in Labor :

- Preferred mode of delivery in heart disease patients : vaginal delivery.
- Induction of labor.
- Inducing agents :
 - Cervix is ripe : Oxytocin.
 - Cervix is not ripe : Foley catheter.
 - Other preferred agents : misoprostol.
 - **Relative C/I** : Dinoprostone.
- Delivery position : Semi recumbent position with left lateral tilt (Avoid lithotomy $\rightarrow$ $\uparrow$ venous return).
- For pain relief (**mandatory**) : Epidural analgesia.
- IV fluids during labor : **Restrict** to 1 mL/kg/hr.
- Restrict P/V examination
- Cut short 2nd stage of labor : Forceps > vacuum.
- 3rd stage of labour : Active management of the third stage of labor done :
 - methylergometrine is contraindicated.
 - Oxytocin can be used.
 - In case of mitral stenosis give diuretics.

Supine Hypotension Syndrome :

Clinical features :

- Lightheadedness on lying supine (**~ 32 weeks**).
- Normal on standing.

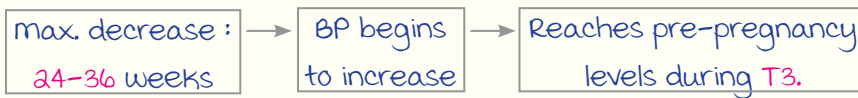
management : Advise patient to lie in **left lateral position**.

Clinical significance :

Position for C-section : Supine position with $\rightarrow$ $\uparrow$ venous
pillow under right hip return.

BP Changes in Pregnancy :

----- Active space -----



m/c time to diagnose pre-eclampsia : T3.

Physiological edema in pregnancy :

- D/t fluid retention during pregnancy (~ 6.5 L).
- Seen in dependent parts of body.
- Disappears after rest.
- Pitting ⊕.

venous pressure in lower limbs : ↑ in pregnancy :

- ↑ risk of varicose veins & hemorrhoids.

CO Changes in Pregnancy :

- Normalizes within 10 days after delivery.

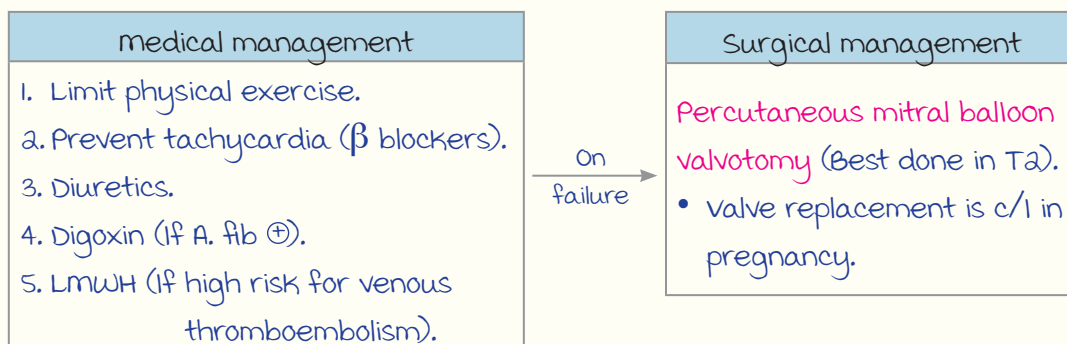
maximum chance of heart failure (Overall maximum CO) :

Immediate postpartum > 2nd stage of labor > Late 1st stage of labor > 28-32 weeks POG.

Mitral Stenosis in Pregnancy :

m/c cause of heart disease in pregnancy.

management :



----- Active space -----

Peripartum Cardiomyopathy :

Heart failure : Last month of gestation or < 5 months post-delivery
+ no h/o heart disease

↑ incidence in :

- Twin pregnancy.
- Pre-eclampsia.
- Obese females.

Etiology :

- viral myocarditis.
- Prolactin.

management :

- Bromocriptine.
- Anticoagulants if LVEF < 35%.

Recurrence :

- High recurrence rate.
- Future pregnancy c/I if LVEF < 50% / residual defect.

IOC : Echocardiography (LVEF < 45%).

Warfarin in Pregnancy :

Pt on warfarin presents at :

Heart Disease Associations :

Condition	Heart disease
mother has pregestational diabetes	VSD
mother has syphilis with anti-Ro/ anti-La antibodies	Congenital heart block
mother took lithium in pregnancy	Ebstein anomaly
Congenital rubella syndrome	Patent ductus arteriosus
Turner's syndrome	Bicuspid aortic valve
Down's syndrome	Endocardial cushion defect > VSD > ASD

Twin Pregnancy

00:47:31

----- Active space -----

Dizygotic twins :

- 2 ova released at same time fertilized by 2 sperms in same act.
- Incidence : Two-third of twins.
- Sex of twin can be same/different.
- Incidence varies from country to country.
- Identifiable risk factors like maternal family history of twinning, use of clomiphene (5-10% chances of twins), use of HMG (30% chances of multiple pregnancy) & IVF.

monozygotic twins :

- Single ova fertilized by a single sperm and then it divides into 2.
- Incidence : 1/3rd of twins (1 in 250 pregnancies).
- Sex of twins : Same.
- Risk factor : IVF.
- These twins have :
 - Same blood group.
 - Same HLA typing.
 - Same karyotype.
 - Different fingerprints.

Twin Pregnancy

Types of twins :

- Dizygotic twins : Are always DCDA (m/c).
- monozygotic twins :
 - If division occurs at ≤72 hours (morula stage) : DCDA.
 - If division occurs between 4 and 8 days (Blastocyst stage) : MCDA (m/c monozygotic).
 - If division occurs between 9 and 12 days (Embryonic disk) : MCMA.
 - If division occurs at >12 days : Conjoined twins.

Imp PYqs :

- All conjoined twins are MCMA.
- In monochorionic monoamniotic and conjoined twins delivery should be between 32-34 weeks by C-section (b/t risk of cord entanglement).
- m/c variety of conjoined twins is Parapagus (Joined at lower abdomen and pelvis) followed by : Thoracopagus.
- Least common variety of conjoined twins is rachipagus (Joined at vertebral column) followed by : Craniopagus.
- IOC for knowing chorionicity = USG (TVS)
- Time : 10-14 weeks.
- Twin not possible = MADC.

IOC for chorionicity : TVS.

Prognosis : DC > MC.

Best time to do USG to know chorionicity : 10-14 weeks.

USG of DCDA Twins



Twin peak sign

USG of MCDA Twins



T-sign

markers of Dichorionicity on USG :

- 4 layers of membrane between twins.
- Thickness of membranes ≥ 2 mm.
- Different sexes of fetus.
- 2 different placentas.
- Twin peak sign/Lambda sign positive.

Complications of Twin Pregnancy :

Fetal Complications of Twin Pregnancy :

- All complications are more in monozygotic than dizygotic twins.
- m/c fetal complication of twin pregnancy : IUGR
- Only complication which is more in DZ twins : Cumulative risk of anomaly
- Complications specific to monochorionic twins :
 - Twin to twin transfusion syndrome (TTTS).
 - Twin anemia polycythemia sequence (TAPS).
 - Twin reversed arterial perfusion (TRAP).
 - Selective IUGR.
- Complications specific to monoamniotic twins :
 - Cord entanglement.
 - Conjoined twins.

TTTS :

- Seen m/c in : mCDA.
- L/c seen in : mCMA.
- USG diagnosis of TTTS :
Following two should be present :
 - mCDA twins.
 - One twin with **oligohydramnios** and other with **polyhydramnios**.
- A vascular connection between twins :
m/c : Deep vascular connection.
- One twin : Donor twin → will have oligohydramnios.
- Other twin : Recipient twin → will have polyhydramnios

Prognosis in Twins :

DCDA : Best prognosis
monochorionic twins : Poor prognosis

Staging of TTTS (Quintero staging) :

- Stage 1 : On USG, bladder of oligo-hydramnios twin is seen
- Stage 2 : Bladder of oligo-hydramnios not seen (Anuria).
- Stage 3 : **Doppler changes in MCA.**
- Stage 4 : Hydrops in one/both twins.
- Stage 5 : Death of one or both twins.

Others

Twin Anemia-Polycythemia Sequence (TAPS) :

Variant of TTTS :

One twin has anemia (no oligohydramnios), and the other polycythemia (no polyhydramnios).

Twin Reversed Arterial Perfusion (TRAP) :

- Rare complication.
- One twin has no heart.
- Donor twin perfuses deoxygenated blood to the acardiac twin through umbilical artery".
- In the acardiac twin, lower extremities may develop, giving rise to **acardiac acephalus** or no part may be identifiable ; then it is called **acardius amorphous**.

- Prognosis of TTTS depends on Recipient twin.

management :

- upto 28 weeks : Fetoscopic laser ablation.
- >28 weeks : Amniocentesis from polyhydramniotic twin sac.



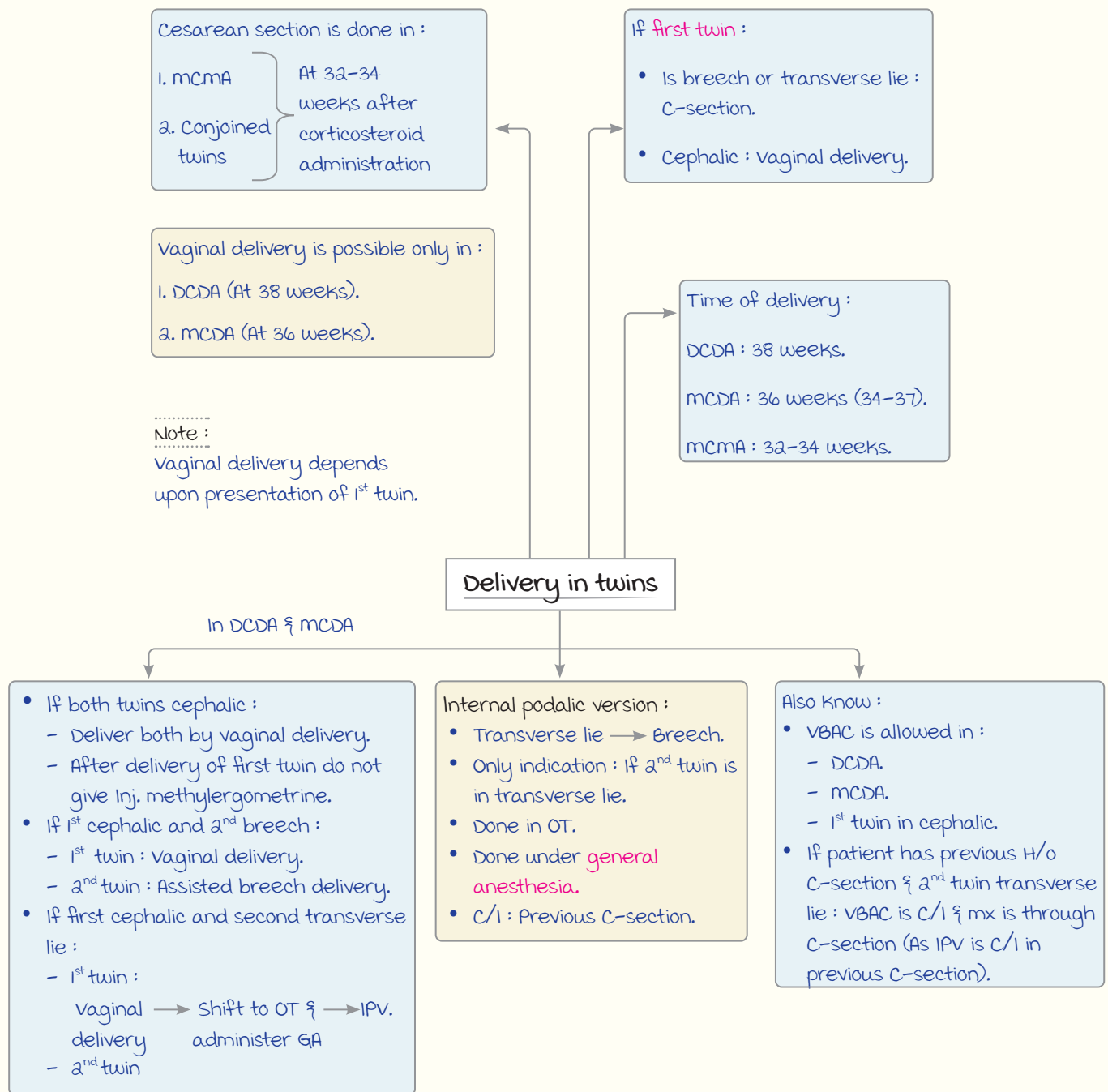
Acardiac acephalus



Acardiac amorphous

Delivery in Twins (57):

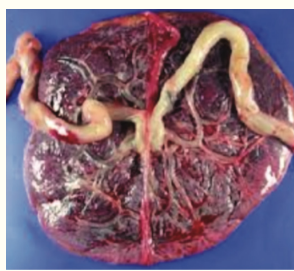
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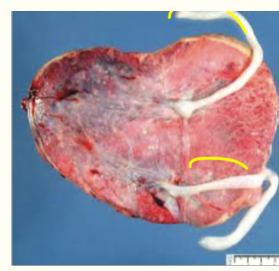
Placenta of Twin Pregnancy :



DCDA
(Thick membranes)



mcda
(Thin membranes)



mcma
(No membrane)

----- Active space -----

Hyperemesis Gravidarum

01:11:33

A patient with twin pregnancy having excessive nausea & vomiting, leading to weight loss $\geq 5\%$ or ketonuria.

Causes :

- $\uparrow$ HCG $\rightarrow$ Twin pregnancies.
 $\rightarrow$ molar pregnancies
- H. pylori.

Complications :

- Hypochloremia.
- Hypokalemia.
- metabolic alkalosis.
- Vitamin B₁ deficiency $\rightarrow$ Wernicke's encephalopathy.
- Vitamin K deficiency.

management :

Investigation : USG.

1st step in treatment : Dietary modifications.

- Small, frequent meals.
- Avoid spicy & fatty food.
- Avoid lying down immediately after meals.

1st line drug : Pyridoxine $\xrightarrow{\text{Fails}}$ Pyridoxine + Doxylamine.

DOC :

Admission & Nil per oral (NPO) + IV fluids + Ondansetron.

Superfecundation vs. Superfetation

00:14:05

Superfecundation	Superfetation
2 ova released at different times in the same cycle $\downarrow$ Fertilized by two different acts of coitus	2 ova released and fertilized in two different cycle
Can be seen in humans (Rare)	• Not possible in humans : $\text{D/t } \uparrow \text{ Estrogen \& progesterone}$ $\downarrow$ $\text{LH \& FSH} \downarrow$ $\downarrow$ Anovulation • Theoretically possible till 14-16 weeks.

Hellin's rule :

If incidence of twins in a country is 1 in 80 pregnancies $\rightarrow$ Incidence of triplets : 1 in 80² pregnancies.
 $\rightarrow$ Incidence of quadruples : 1 in 80³ pregnancies.

OBSTETRICS REVISION : PART 4

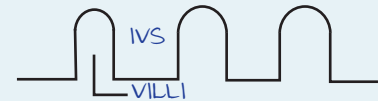
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Placenta

00:00:31

- Described as
 - Discoidal.
 - Deciduate.
 - Hemochorial.
- Weight : 500 gm.
- Weight of placenta : Fetus = 1 : 6.
- Gestational age at which weight of placenta = wt. of fetus : **17 weeks.**
- Diameter : **15 - 22 cm.**
- Thickness : **2.5 - 4 cm.**
- Placentomegaly : Thickness ≥ 4 cm.
- Seen in :
 - Twin pregnancy.
 - Rh incompatibility.
 - Diabetes.
 - Small placenta seen in :
 - PIH
 - IUGR.
 - molar pregnancy.
 - Infections : **Parvovirus & syphilis.**
 - uteroplacental insufficiency.

- maternal side of placenta (1/5th of placenta) formed by decidua basalis.
- Fetal side formed by chorion frondosum (4/5th of placenta).
- Cord attachment : Fetal side of placenta (Center).



- Volume of placenta (500 mL) :
 - villi has 350 mL of fetal blood.
 - IVS has 150 mL of maternal blood.
- Pressure in IVS : 10 mmHg.
- Placental circulation :
 - Fetoplacental circulation is via umbilical artery & vein and established by $D_{IT} - D_{al}$.
 - uteroplacental circulation is via spiral arteries (Open into IVS) established by D_{IS} after fertilization.
- Normal location : upper uterine segment.
- Functional unit : Cotyledon.

hCG :

- Glycoprotein.
- α & β subunit.
- Produced by syncytiotrophoblast.
- α -subunit similar to LH, FSH, TSH.
- β subunit : Specific.
- Function : maintains the corpus luteum of pregnancy (Similar to LH).
- Doubling time of hCG : 48 hours.
- $T_{1/2}$: 36 hours.
- hCG appears in maternal blood **8 days** after fertilization, i.e., Day 22 of cycle/5-6 days before missed period.
- Levels during pregnancy :
 - Peaks at 10 weeks then drops
 - Plateaus at 16 weeks
 - Remains in blood at low level throughout pregnancy
 - Disappears from blood at 2 weeks after delivery

Placenta

Placental hormones (Proteins) formed by syncytiotrophoblast (Steroids)

hPL :

- AKA human chorionic somatomammotropin (hCS).
- Produced by syncytiotrophoblast.
- Similar to GH and prolactin.
- Detected earliest at 3 weeks of pregnancy.
- Maximum production : **36 weeks.**
- Responsible for insulin resistance in pregnancy.
- marker of placental function.

Progesterone :

- Produced by corpus luteum till 6-7 weeks.
- After 8 weeks, produced by placenta.
- Substrate : **maternal LDL cholesterol.**
- Functions :
 - Prepares the endometrium for implantation.
 - Decreases myometrial contractility.

Estrogen :

- most specific in pregnancy : **E3.**
- most common in pregnancy : **E2.**
- Placenta cannot synthesize estrogen alone unless it gets precursors from fetus (Fetal adrenals give DHEA-S which is used by placenta to form estrogen).

IVS : Intervillous space
hPL : Human placental lactogen

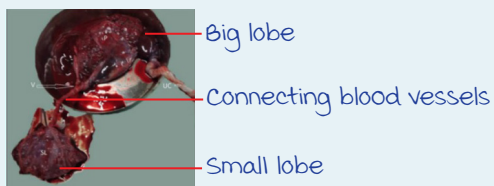
Note :

Prolactin formed by anterior pituitary & decidua

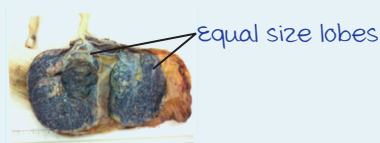
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Placenta with 2 lobes :

- ↑ Risk of APH/PPH.



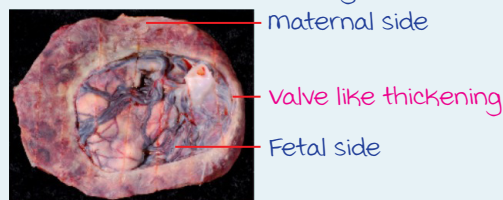
Placenta succenturiata



Placenta bilobata

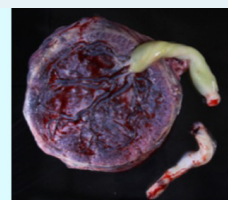
maternal side > fetal side of placenta :

Surrounds fetal side in the form of ring



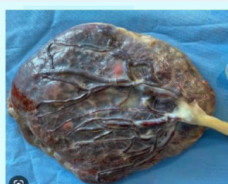
Circumvallate placenta :

- Associated with :
 - Antepartum hemorrhage.
 - Pre-term labor.
 - Low birth weight

Circummarginate placenta
(Not significant)

Placental Anomalies

Cord attached to the margins of placenta



Battledore placenta

Cord first attached to membranes then vessels separate to attach to placenta (D/t loss of Wharton's jelly)

velamentous
insertion of cord

Vasa Previa :

- m/c seen with **velamentous insertion of cord**.
- Leads to perinatal mortality
- Cord is attached to membranes over internal os

membrane
rupture
↓
Cord ruptures

↓
Fetal blood loss

↓
Severe fetal distress (Out of proportion to blood loss)

Types :

- 1 (m/c) : D/t velamentous cord insertion.
- 2 : D/t placenta succenturiata.

IOC : USG Doppler.

mx : Emergency C-section.

C/i : Artificial rupture of membranes.

Note :

Cord vessels :

- Right umbilical artery
- Left umbilical artery
- Left umbilical vein

Placental localization (USG) :

- T3 at 32 weeks.
- Repeat at 36 weeks.

Blood flow :

- uterus at term : 750 mL/min (~10% of cardiac output at term).
- utero-placental (at term) : 500 – 750 mL/min.
- IVS : Blood replaced 3 times/minute.

D/D for Fetal Distress Post ARM :

- Cord prolapse (Bleeding $\ominus$).
- vasa previa (Bleeding $\oplus$).

Placenta Previa :

- Placenta located in lower uterine segment.
- Leads to maternal mortality due to maternal blood loss.

Singer's Alkali Denaturation Test :

- AKA APT test.
- To differentiate fetal from maternal blood.

Principle : HbF in fetal blood is resistant to acids & alkali.

Reagent : KOH/NaOH.

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Hypertension in Pregnancy

00:19:04

Chronic HTN in pregnancy :

- A hypertensive patient conceives, increase in BP seen **before 20 weeks** without proteinuria and without end organ damage.
- BP does not come back to normal even after 12 weeks of delivery.

Pregnancy Induced HTN (PIH) :

- A normotensive patient conceive during pregnancy, the blood pressure increased **after 20 weeks of pregnancy**.
- BP becomes normal within 12 weeks of delivery.

Pre-eclampsia :PIH with **proteinuria** OR signs of **end organ damage**.**Gestational HTN :**

PIH without Proteinuria OR signs of end organ damage.

Chronic HTN with superimposed Pre-eclampsia :

- A chronic hypertensive patient develops :
- Sudden onset proteinuria.
 - End organ damage.
 - Uncontrolled BP.
 - At **≥ 20 weeks**.

Proteinuria :urine dipstick **≥ +2 (Screening test)**

OR

Excretion of protein **≥ 300 mg** in 24 hrs (**Diagnostic test**)

OR

Protein - Creatinine ratio **≥ 0.3**.**Signs of end-organ damage :**

Any one of the following :

- Platelet count **< 1 lakh**.
- Liver enzymes raised **≥ 2 times** its **(N)** value.
- S.creatinine **≥ 1.1 mg/dL**.
- Pulmonary edema.
- Visual symptoms/Headache.

Eclampsia :

Generalized tonic clonic seizure in a patient with severe preeclampsia.

Early preeclampsia (⚠) :PE seen **before 34 weeks** of pregnancy.

Elevated BP

Urine dipstick test

↓ **≥ +2**

24hr protein excretion.

Hypertension in PregnancyBP **≥ 140/90 mm Hg** on **2 occasions 4 hours apart**.If BP **≥ 160/110 mmHg** **15 mins** → Repeat BP **≥ 160/110 mmHg** → Start anti-HTN drug.**Mild PE :**

- BP **≥ 140/90 mmHg** but **< 160/110**.
- No signs of end organ damage.
- No signs of impending eclampsia.

Criteria that do not differentiate between mild & severe PE :

- Proteinuria.
- Oliguria.
- IUGR.

Severe PE :

- BP **≥ 160/110 mmHg**
- OR
- Signs of end organ damage ⊕
- OR
- Signs of impending eclampsia ⊕

Symptoms of impending eclampsia in severe PE (⚠) :

H : Headache (Severe).**E : Epigastric pain.****V : Visual disturbances.**

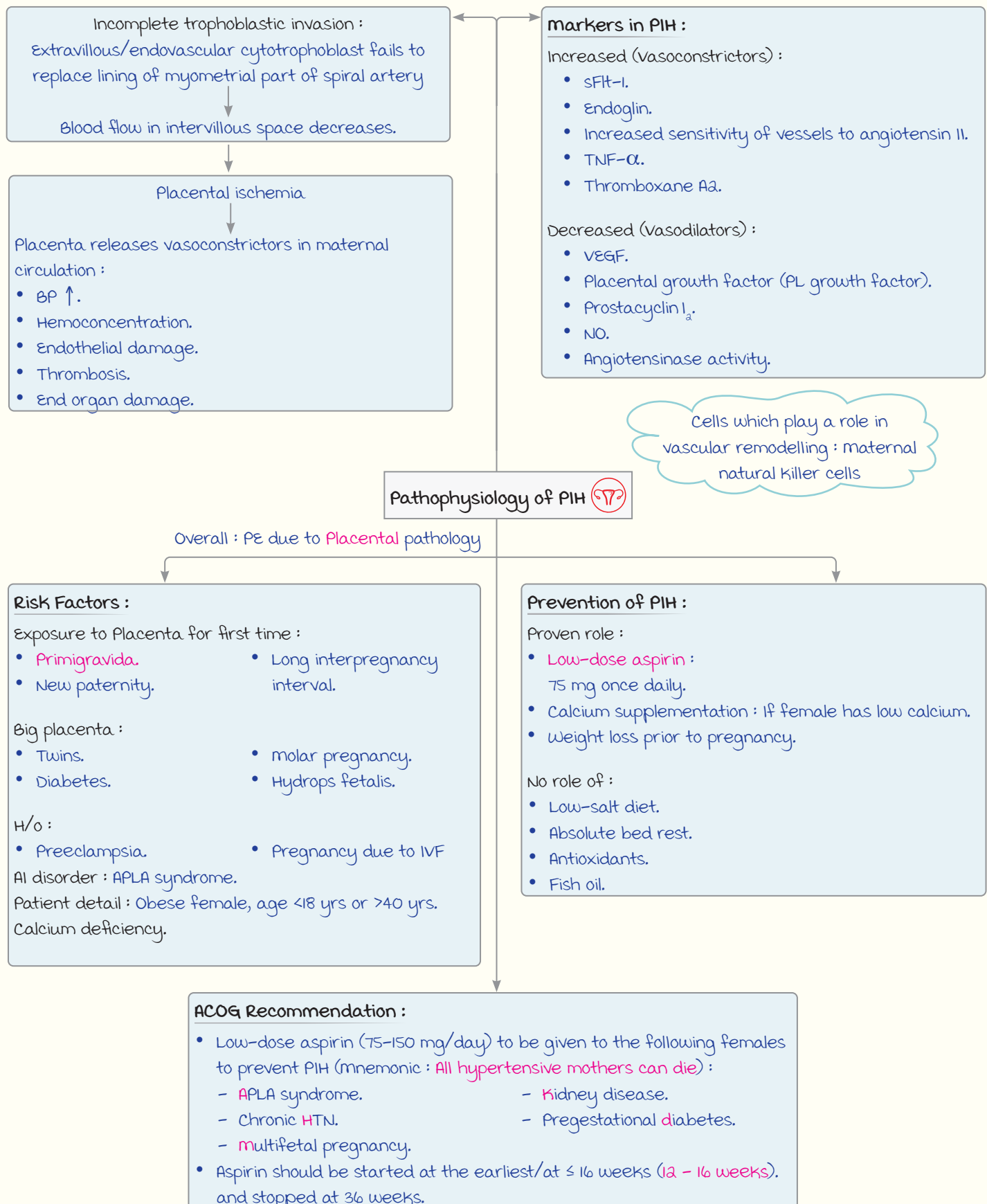
- Fundal examination : **Silver wire appearance.**

All patients of PE should be admitted & following Ix to be done :

1. Platelet count
2. LFT
3. S. creatinine
4. Fundal examination

To differentiate between mild and severe PE

----- Active space -----



----- Active space -----

Common Points :

1. Fetal monitoring to be done in all patients of PIH.

Started at 32 weeks.

	mild	Severe
- DFMC - NST - BPS - USG for growth - umbilical artery doppler	Daily Weekly Weekly 3 weeks	Daily 2 weeks

2. Definitive management in all patients of PIH :
Termination of pregnancy (TOP) by induction of labor.

- Best mode of delivery : **vaginal**.
 - mild PE : 37 weeks.
 - Severe PE : 34 weeks
 - Impending Eclampsia
 - Eclampsia
 - HELLP syndrome
 - Abruptio placenta
 - Fetal distress
- } Immediate TOP

management of mild PE :

- Treat as outpatient.
- Advise :
 - BP measurement : Twice weekly.
 - Fetal monitoring from 32 weeks.
 - Inform about signs of impending eclampsia.
 - Antihypertensive : $\pm$.
- NO role of :
 - Aspirin.
 - Bed rest.
 - Salt restriction.
 - MgSO_4 .
- Definitive mx : TOP at 37 wks by vaginal delivery.

management**management of Severe PE :**

To all patients :

- MgSO_4 (DOC) : To prevent eclampsia.
- Antihypertensives : To prevent intracranial hemorrhage.
- Corticosteroids (First dose given).

Do immediate TOP without giving 2nd dose of corticosteroids in :

- Impending eclampsia.
- Eclampsia.
- HELLP syndrome.
- Abruptio placenta.
- Fetal distress.
- DIC.

In case of uncomplicated pregnancy with severe PIH

IOL at 34 weeks

For any other complication

2nd dose of corticosteroid given

Induction of Labor (IOL)

management of impending eclampsia**management of eclampsia :**

- 1st step :
 - Prevent patient from falling.
 - Secure airway.
- 2nd step : MgSO_4 .
- 3rd step : IV anti-hypertensives.
- Definitive management : Immediate TOP.
- mode of delivery : vaginal.

- 1st step : MgSO_4 .
- 2nd step : Anti-hypertensives.
- Definitive management : Immediate TOP.
- Time of intervention : After 1st dose of corticosteroid.

Note :

umbilical artery doppler done in :

- PIH.
- utero placental insufficiency.
- IUGR.

Antihypertensives in pregnancy :

- Indication : BP $\geq 160/110$ mmHg.
- NICE guidelines : Persistently elevated BP $\geq 150/100$ mmHg.

Target BP :

- Systolic : 120 – 130 mmHg.
- Diastolic : 80 – 90 mmHg.

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Drugs used in severe pre-eclampsia	Drugs used for chronic hypertension	Absolutely C/i
1st line : • IV Labetalol (maximum : 220 mg). • IV Hydralazine (maximum : 30 mg). • Oral Nifedipine.	1st line : • Oral Labetalol. • Oral Nifedipine. • Oral methyldopa.	• ACE inhibitor. • Angiotensin receptor blocker (Losartan). • Diazoxide.

key points :

- Safest antihypertensive : α -methyldopa
- Diuretics :
 - C/i in Pre-eclampsia.
 - Not c/i in chronic HTN.
- Labetalol is C/i in :
 - Asthma.
 - Heart failure.

Last resort :

Sodium nitroprusside : s/e $\rightarrow$ Cyanide poisoning.Anti Hypertensives & MgSO_4 Important points on MgSO_4 :

- DOC for prevention and treatment of seizures in PIH.
- MOA : Centrally acting (Blocks NMDA receptors).
- Not an antihypertensive.
- Low therapeutic range : 4–7 mEq/L.
- 1 ampoule of MgSO_4 : 2 mL of MgSO_4 = 1 g of MgSO_4 (50% conc.).

Pritchard Regimen :

- Loading dose (14 ampoules) :
 - IM : 10 g of 50% MgSO_4 (5 g in each buttock).
 - IV : 4 g of 20% MgSO_4 .
- No renal function test is needed for loading dose.
- maintenance dose :
 - IM : 5 g/4 hours of 50% MgSO_4 .
 - Given for :
 - a. 24 hrs after last seizure
 - or
 - b. 24 hrs after delivery

Sibai regime (IV only) :

- Recommended by ACOG.
- Loading dose : 4g MgSO_4 .
- maintenance dose : 2 g MgSO_4 .

Before giving maintenance dose check :

- Deep tendon reflexes (knee jerk) $\oplus$.
- Resp rate ≥ 12 breaths/min.
- urine output ≥ 100 mL/4 hrs or ≥ 30 mL/hr

Signs and symptoms of MgSO_4 toxicity :

- At 10 mEq/L :
 - 1st sign : Loss of knee jerk.
 - Diaphoresis.
 - Slurring of speech.
- At 12 mEq/L : Respiratory paralysis & arrest.
- At 15 mEq/L : Cardiac conduction defect & cardiac arrhythmias.
- At 20 mEq/L : Cardiac arrest.
- Oliguria **not a sign** of MgSO_4 toxicity.
- Serum mg levels are **NOT** routinely monitored.
- Antidote : Calcium gluconate.

Other :

Important points on MgSO_4 :

- Neuroprotective agent : All babies born at < 32 weeks.
- $\downarrow$ Incidence of abruptio placenta.
- It can prevent + Treat seizures in pregnant females with high BP.
- It has tocolytic action (Occurs at a concentration at which it's toxicity is seen).
- Normally causes $\downarrow$ FHR.
- Persistent $\downarrow$ FHR in Pre-eclampsia : Abruptio.

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Key points (📌):

- First sign of pre-eclampsia : **High BP**.
- First organ affected in PIH : **Kidney**.
 - HPE : **Glomeruloendotheliosis**.
- Classification for hypertensive retinopathy : **Keith-Wagner** classification.

Eclampsia (📌):

- Cause of seizure : **Cerebral hypoxia + edema**.
- Types :
 - **Antepartum** (m/c & worst prognosis).
 - **Postpartum** (Within 48 hrs of delivery).
 - **Intrapartum**.
- m/c MRI finding : **PRES** (Edema in posterior cerebral hemisphere).

Choice of anesthesia for C-section in PIH :

- Pre-eclampsia/eclampsia with controlled seizure : **Neuraxial** > Epidural.
- Eclampsia with seizures : General anesthesia.

Jaundice in Pregnancy

00:50:00

Differences for jaundice in T3

D/D	Imp points	Lab finding	mx
HELLP syndrome : • RR (Recurrence rate) : 4-7 % • Complication of severe PE (80% pts will have high BP) • C/o pain abdomen in T3	Tennessee criteria for dx : • H : Hemolysis + • EL : Elevated liver enzymes (>2 times the normal value) + • LP : Low platelet count (<1 lakh)	Any 2 of below 4 should be present to dx hemolysis : 1. Peripheral smear : Schistocytes/Burr cells. 2. S.bilirubin ↑ 3. LDH ↑; Haptoglobin ↓ 4. Severe anemia without evidence of blood loss	Similar to severe PE : 1. MgSO₄ given 2. Antihypertensives 3. Definitive mx : Immediate TOP (After 1st dose of corticosteroids)
Cholestasis in pregnancy : • m/c in : - Twin pregnancy - Female fetus • RR : 70% • Risk factor : ↑estrogen	• C/o itching in palm & sole. • No pain in abdomen • Fetal complications : - Stillbirth - Cardiac arrhythmia - meconium aspiration syndrome	↑Bile acids & salts	• DOC : ursodeoxycholic acid - ↓itching - Improves fetal outcome • Fetal monitoring : From 32 weeks of pregnancy • TOP : 37-38 weeks
Acute fatty liver of pregnancy : • Pregnant female : Heterozygous deficiency of LCHAD enzyme. • Fetus : Homozygous deficiency of LCHAD enzyme.	• Pain abdomen in T3 • Altered sensorium • Hepatorenal failure • Complications : - Hypoglycemia - DIC - Pancreatitis - Renal failure	↑NH ₃	Immediate TOP

Doppler in Pregnancy

00:55:16

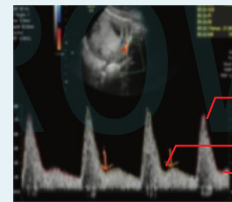
----- Active space -----

middle cerebral artery doppler :

- Checks : Peak systolic velocity (PSV).
- Detects : Anemia.
- Interpretation :
 - PSV ≥ 1.5 mom : Fetal anemia.
 - PSV < 0.8 mom : Fetal polycythemia.
- Indications :
 - TTTS (Twin to twin transfusion syndrome).
 - Rh incompatibility.
 - Vasa previa.

uterine artery doppler :

- Between 22 - 24 weeks :
 - Normally done to predict PIH.
 - Look for : Diastolic notch (DN).
 - Interpretation :
 1. DN disappears : Normal.
 2. Diastolic notch $\oplus$ beyond 24 wks : $\uparrow$ Risk of PIH.
- Between 11 - 13 weeks :
 - Look for : Pulsatility index (PI).
 - Interpretation :
 - $\uparrow$ PI $\rightarrow$ $\uparrow$ Risk of early onset pre-eclampsia.



uterine artery doppler

Role of Doppler In Pregnancy

Umbilical Artery Doppler

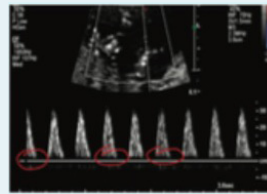
- POG $\uparrow \rightarrow$ $\uparrow$ Blood flow in diastole.
- S/D ratio :
 - $\downarrow$: Normal pregnancy.
 - ≥ 3 : mild PIH $\rightarrow$ TOP at 37 weeks

Note :

Reversal of flow in

Ductus venosus $\rightarrow$ Immediate TOP

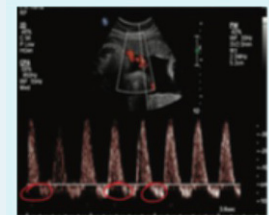
Absent diastolic flow :



TOP : At 34 weeks

- If pregnancy is < 34 wks
 - Doppler
 - NST
 - BPP
 To be done.
- TOP at 34 weeks
- mode of delivery : vaginal.

Reversed diastolic flow :



TOP : Between 31 - 32 weeks

- If pregnancy $< 31 - 32$ weeks
 - Doppler
 - NST
 - BPP
 To be done.
- TOP at 31 - 32 weeks.
- mode of delivery : C-section.
- Reversed end diastolic flow
 - + Late deceleration on CTG
 Immediate TOP.

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Intrauterine Growth Restriction (IUGR)

01:01:59

Weight of fetus (EFW/AC) < 10th percentile for gestational age.

Classification :

Based on weight of fetus :

moderate IUGR : 3rd - 10th percentile.

Severe IUGR : < 3rd percentile.

Based on weeks of gestation :

Early onset IUGR :
IUGR at < 32 wks.

Late onset IUGR :
IUGR at ≥ 32 wks.

Diagnostic Criteria for IUGR

ISUOG : International society of ultrasound in obstetrics & gynaecology

Early onset IUGR :

- EFW or AC < 3rd percentile
 - OR
 - EFW or AC 3rd - 10th percentile with any 1 of the following :
 - umbilical artery
 - OR
 - uterine artery
- PI > 95th percentile

Late onset IUGR : Any two of following ⊕.

- EFW < 10th percentile.
- AC < 10th percentile.
- Low growth velocity : Growth velocity < 2 quartiles for particular gestational age.
- Abnormal doppler :
 - Cerebro-placental ratio < 5th percentile
 - MCA PI < 5th percentile (D/t brain sparing effect).
 - umbilical artery PI > 95th percentile (D/t ↓ blood flow in umbilical artery).

Note :

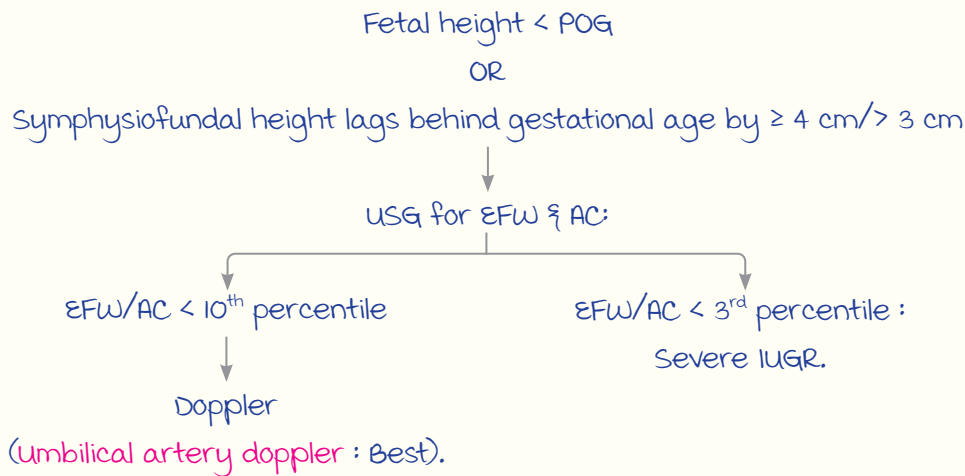
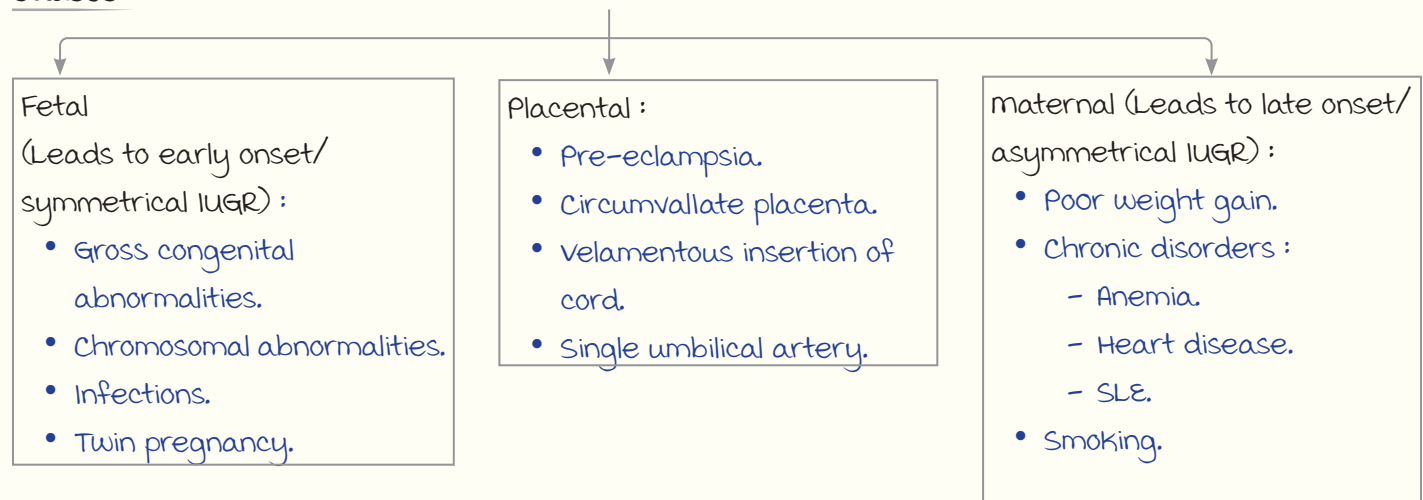
uterine artery doppler may be normal in late onset IUGR.

SGA vs. IUGR :

	Small for gestational age	IUGR/FGR (Fetal growth restriction)
EFW/AC	< 10 th percentile	
Doppler	Normal	Abnormal
Growth velocity	Normal	Abnormal

Clinical Diagnosis of IUGR :

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Causes :Follow up of High Risk Patients :

Antenatal fetal monitoring :

Ideal at 32 weeks (Earliest at 28 weeks) :

USG for EFW/AC :

- Every 4 weeks
- Severely IUGR (EFW/AC < 3rd percentile) : Every 2 weeks

Doppler :

- Normal : Repeat weekly
- Abnormal : Repeat twice weekly
- AEDF/REDF : Repeat thrice weekly
 - NST : 6 - 12 hourly
 - BPP : Every day

----- Active space -----

Termination of Pregnancy :

- Normal doppler with :
 - EFW < 3rd percentile : 37 weeks.
 - EFW > 3rd percentile : 39 weeks.
- uterine artery doppler :
 - PI > 95 : 37 weeks.
- Associated oligohydramnios : 34 weeks.

Preterm Labor (PTL)

01:12:00

Classification Based on Gestational Age :

Early preterm labor :
At < 34 weeks

Late preterm labor :
At ≥ 34 weeks

Risk factors :

- Previous h/o preterm birth (PTB).
- Uterine overdistension, E.g. : twins, polyhydramnios.
- Infection (Asymptomatic bacteriuria, bacterial vaginosis).
- Short cervical length (< 25 mm).

2 conditions where preterm labor can be prevented :

1. Short cervix (≤ 2.5 cm).
2. Previous h/o PTL.

ACOG recommendation :

- Pregnant females with previous h/o PTL

TVS for cervical length
(Done around 16-24 weeks).

- IOC : TVS.

Findings on TVS to predict PTL :

1. Length of cervix < 2.5 cm.
2. U-shaped cervix.

- Other predictors for PTL :

Fibronectin protein : ≥ 50 ng/mL in cervicovaginal secretions.

Preterm Labor : Prevention

uterine contraction ⊖

Progesterone to prevent PTL :

- If any 1 of following is present :
 - Cervical length ≤ 2.5 cm.
 - H/o PTL/T2 abortion.
 - Cervical cerclage.
- Started at 12 weeks.
- Stopped at 36 weeks.

Cervical cerclage :

Done if all of the 3 conditions are fulfilled :

- Singleton pregnancy.
- H/o PTL/T2 abortion.
- TVS : Cervical length ≤ 2.5 cm.

Case-based approach :

- Previous h/o PTL ⊕ :
 - Length of cervix < 2.5 cm : Cervical cerclage + progesterone.
 - Length of cervix ≥ 2.5 cm : Progesterone.
- Primigravida with cervical length < 2.5 cm : Progesterone.

----- Active space -----

Established PTL :

- uterine contractions $\oplus$ at < 37 weeks.
- ACOG guidelines :
4 in 20 minutes OR 8 in 60 minutes.
- FOGSI guidelines :
1 in 10 minutes OR 6 in 60 minutes.

+

- Cervical changes : Dilatation of cervix/ $\downarrow$ length of cervix.

- Dilatation :

2 - 4 cm :

Suspected PTL

Rx based on cervical length.

> 4 cm :

Established PTL

Rx irrespective of cervical length.

Pregnant woman at < 37 wks : uterine contraction $\oplus$ & Cervical dilatation > 4 cm**Established PTL**

POG < 34 weeks :

- Steroids.
- Tocolytics (48 hrs).
- mgSO_4 (POG < 32 wks).
- No role of :
 - Antibiotics.
 - IOL.

POG $\geq$ 34 weeks :

- Corticosteroids.
- No role of :
 - Tocolytics.
 - mgSO_4 .
 - Progesterone.
 - Antibiotics.
 - IOL.

management of PTLuterine contraction &
Cervical dilatation of 2 - 4 cm**Suspected PTL**

Check POG

POG > 30 wks

Next step

TVS

Cut-off : 1.5 cm

Cx length
< 1.5 cm

Rx of PTL.

Cx length
 $\geq$ 1.5 cm

Normal dilatation.

POG < 30 wks

Next step

TVS

Cut-off : 2.5 cm

Cx length
< 2.5 cm

Rx of PTL.

Corticosteroids :

- DOC for fetal lung maturity.
- m/c :
 - India : Dexamethasone.
 - Overall : Betamethasone.

- Betamethasone : 2 injections 12 mg each, 24 hours apart (im).
- Dexamethasone : 4 injection 6 mg each, 12 hours apart (im).
 - C/i : Chorioamnionitis.
- Role : It accelerates fetal lung maturity.
- Corticosteroids action begins after 48 hours of first dose.

Tocolytics :

- use : To buy time for corticosteroid actions.
- used for 48 hours only.
- Never given after 34 weeks of pregnancy.

 mgSO_4 :

Role : used as a neuroprotective agent In all PTL at < 32 weeks.

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Note :

- Low birth weight : weight < 2.5 kg.
- Very low birth weight : weight < 1.5 kg.

Tocolytics :

ACOG recommendations :

- Till 32 weeks : Indomethacin.
- 32 - 34 weeks : Nifedipine (Tocolytic of choice in India).

Indomethacin :

- Side effects :
 - Premature closure of ductus arteriosus (If given beyond 32 weeks).
 - Oligohydramnios.

Atosiban :

- Tocolytic of choice in heart disease.
- MOA : Oxytocin antagonist

mgSO₄ : Both tocolytic action & toxicity seen at 9 - 10 mEq/L (Hence not used).

Contraindications :

Absolute C/i :

- PTL ≥ 34 weeks.
- Intrauterine fetal death.
- Fetal anomaly.
- Chorioamnionitis.
- maternal hemodynamic instability.

Relative C/i :

Dilation of cervix ≥ 3 cm.

PROM vs PPRM

01:27:20

	PROM	PPROM	
membrane rupture	37 weeks – onset of labor	< 37 weeks	
mx	• Sterile P/S exam (PV is c/i) • IOL • Group B strep. prophylaxis if : - membrane rupture ≥ 18 hours - Fever ≥ 4 hours during labor • No role of : Corticosteroids/ tocolytics	At > 34 weeks	At < 34 weeks
		• Steroids • Antibiotics • Fetal monitoring • maternal monitoring • TOP : At 37 weeks	• Steroids • Antibiotics • Tocolytics • mgSO₄ (If < 32 weeks) • Fetal & maternal monitoring

Prolonged rupture of membranes : membrane ruptured for ≥ 18 hours.

Cord Prolapse

01:30:19

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Cord prolapse :

Cord lies below presenting part + Outside introitus : Seen after ROM.

Occult prolapse :

- Cord lies alongside the presenting part after ROM but is **not visible/not palpated** outside.
- Diagnosed based on abnormal FHR.

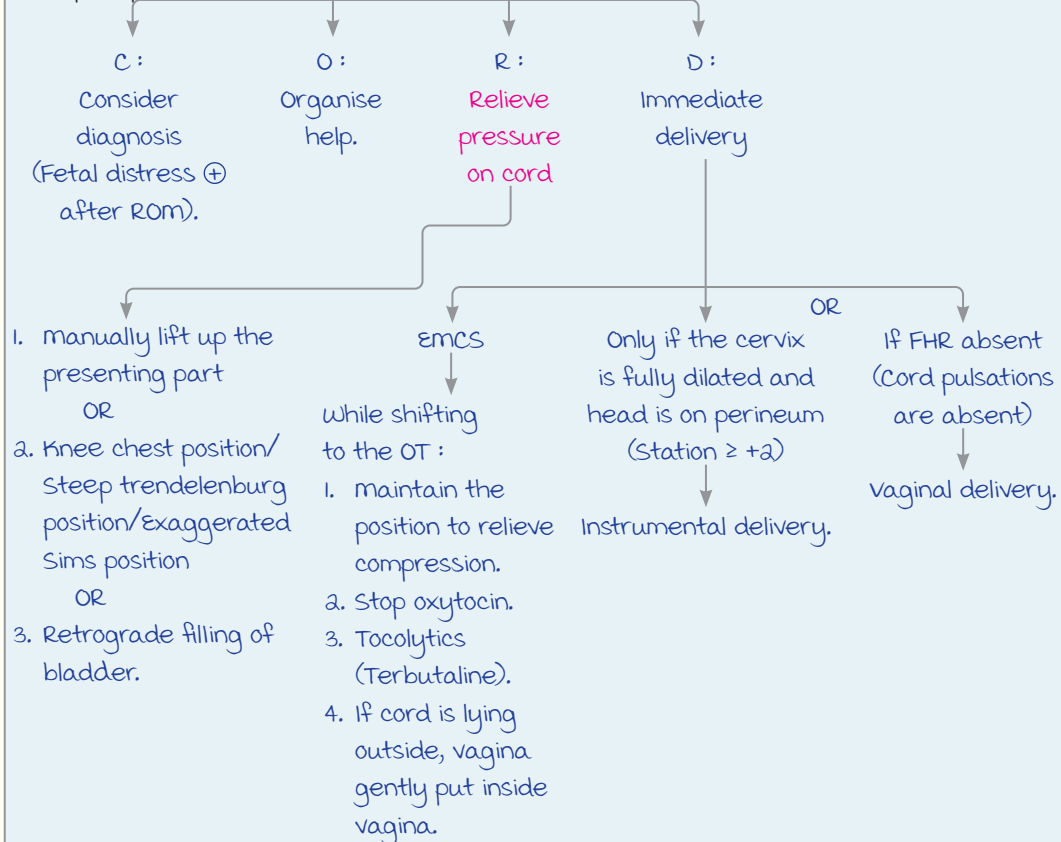
Funic Presentation :

Cord lies between presenting part & cervix with intact membranes.

Cord Prolapse

management

Cord prolapse :



Funic presentation :

- On PV : Feels like bag of worms.
- Emergency C-section done.
- Rupture membranes c/i.

OBSTETRICS REVISION : PART 5

Amniotic Fluid Disorders

00:00:28

Volume of amniotic fluid (97) :

Gestational age	Volume
32 - 34 weeks (maximum)	1000 mL
At term/40 weeks	800 mL
≥ 42 weeks	200 mL

Sources of amniotic fluid :

Gestational age	Main contributor
1 st trimester	maternal plasma
12 - 20 weeks	Fetal skin
≥ 20 weeks	Fetal urine

Color of amniotic fluid :

Color	Seen in
Straw coloured/Turbid	Term
Green (D/t meconium : Biliverdin ⊕)	- Fetal distress - Breech - Transverse lie
Golden (D/t bilirubin)	Rh incompatibility
Tobacco juice/Brown	Intrauterine death of fetus
Saffron color/Yellowish green	Post-term pregnancy
Dark red	Concealed abruption

Oligohydramnios vs. Polyhydramnios :

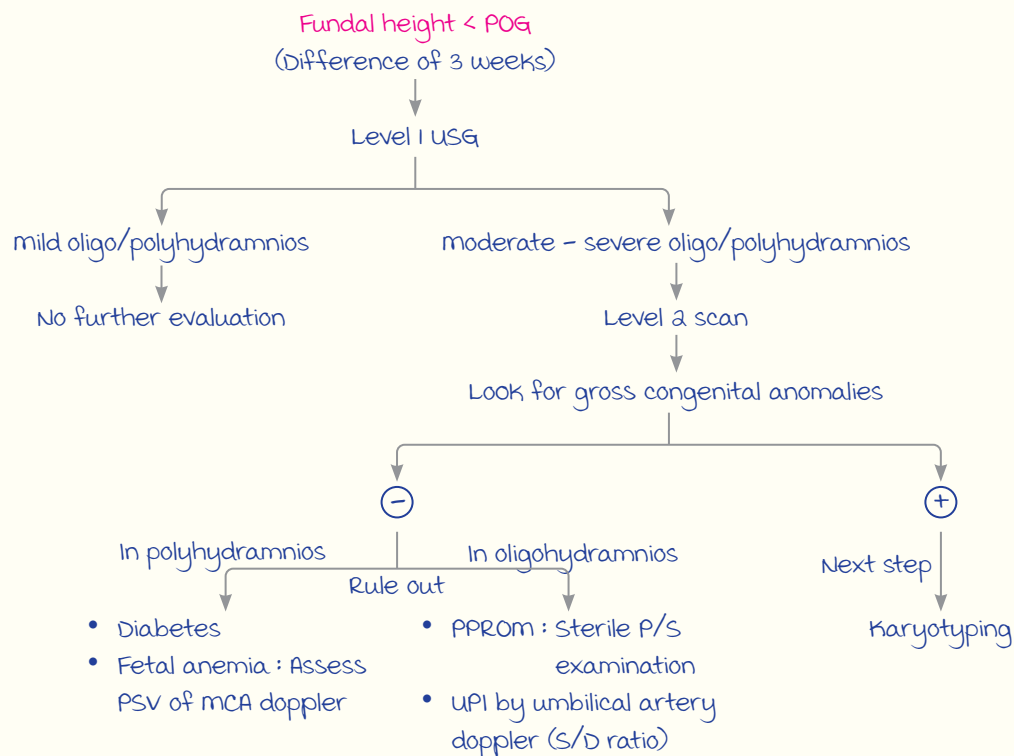
	Oligohydramnios	Polyhydramnios
Volume	< 200 mL	≥ 2 L
AFI	< 5 cm	≥ 25 cm
SVP	< 2 cm	≥ 8 cm
m/c cause	mild	Idiopathic
	Severe	- Renal anomalies - Posterior urethral valve : key hole sign - Congenital GI anomalies : - Oesophageal/duodenal atresia - Intestinal obstruction - Cleft lip/palate

	Oligohydramnios	Polyhydramnios
Other causes	a) ↓ Transudation across placenta (Small placenta) : • UPI • PIH • IUGR	a) ↑ Transudation across placenta (Placentomegaly) : • Diabetes • Twin pregnancy • Rh -ve pregnancy
	b) ↓ Urine output : • Drugs : Indomethacin/ ACE inhibitors	b) ↑ Urine output : • Twin pregnancy • Diabetes • Bartter syndrome (Polyuria) • Fetal anemia d/t : - Thalassemia - Parvovirus infection - Rh incompatibility
	c) ↓ volume of amniotic fluid : • PROM • Post-term pregnancy • Leak following amniocentesis	c) ↑ Transudation from fetal skin : • NTD • Abdominal wall defect
	d) Chromosomal anomaly : Triploidy	d) Chromosomal anomalies : Trisomies
	• Twin to twin transfusion syndrome • TORCH infections	

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SVP : Single largest vertical pocket.

management :



----- Active space -----

Termination of pregnancy :

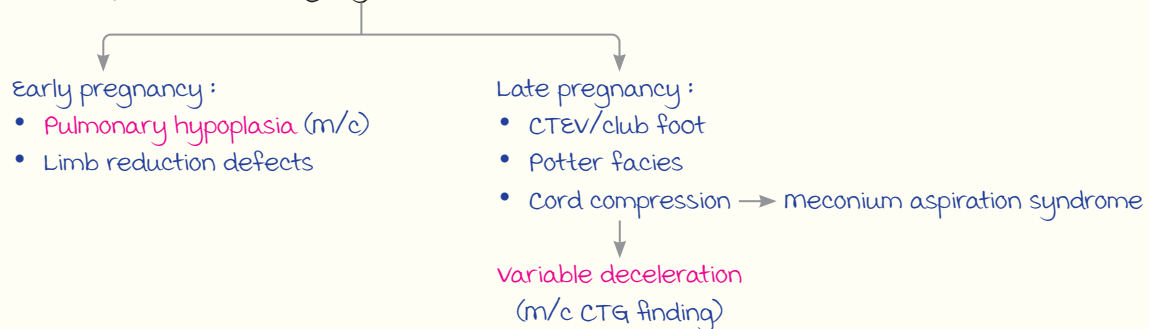
- mild - moderate oligo/polyhydramnios : 39 weeks.
- Severe oligo/polyhydramnios : 37 weeks.

Note :

- Tests to differentiate amniotic fluid from vaginal discharge :
 - a. Nitrazine test.
 - b. Fern test.
- Water component of amniotic fluid : 98 - 99%.

Oligohydramnios :

Complications of oligohydramnios :



Indication for amnioinfusion : Persistent variable deceleration on CTG.

Amniotic band/Streeter's syndrome (ASP) :

PPROM → Severe oligohydramnios → Constriction band →
(membrane tightly wrapped around the fetus)



Distal digit amputation

Note :

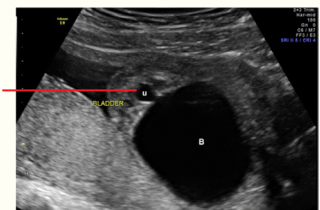
1. Phocomelia :

- Proximal limb amputation.
- s/e of thalidomide.

2. Indomethacin :

- used to treat polyhydramnios.
- C/I ≥ 32 weeks → Premature closure of ductus arteriosus.

Key hole sign



Posterior urethral valve

Diabetes in Pregnancy

00:16:36

----- Active space -----

Pregestational diabetes (Type non-A diabetes) :

- Diabetic female conceives.
- Blood sugar levels raised from **day 1** of pregnancy.
- Hyperglycemia → Free radical formation → Congenital malformation.
- Does not resolve after delivery.

Gestational diabetes :

(Priscilla White A diabetes)

- Nondiabetic female conceives but becomes diabetic during pregnancy d/t insulin resistance.
- Insulin resistance is d/t **hpl** (No role of hCG) and increases as pregnancy advances.
- Significant IR develops between 24 - 28 weeks pregnancy.
- Gestational diabetes develops between 24 - 28 weeks (Organogenesis is complete).
- Does not lead to congenital malformation.

Diagnosis of pregestational diabetes :

- FBS : ≥ 126 mg/dL
- RBS : ≥ 200 mg/dL
- 2-hour PP : ≥ 200 mg/dL
- HbA1C : $\geq 6.5\%$

Diabetes in Pregnancy

Type A diabetes can be :

- A1 = Gestational diabetes controlled on diet.
- A2 = Gestational diabetes controlled on drugs (metformin/insulin).

Congenital malformation in diabetes :

- m/c system involved : **CVS** > CNS.
- m/c congenital malformation : **VSD** > NTD.
- most **specific** : **Caudal regression syndrome/ Sacral agenesis.**



Caudal regression syndrome

- m/c CVS anomaly : **VSD.**
- most specific CVS anomaly (Does not resolve after delivery) : **TGA.**
- m/c CVS finding (Reversible after delivery) : **HOCm.**

Diagnosis of Gestational Diabetes :

In India, DIPSI guidelines are followed :

- Test at 1st antenatal visit + repeated at 24 - 28 weeks of pregnancy (All pregnant females).
- Fasting : Not required.

Procedure :

Give 75 g of glucose to patient mixed in 300 mL of water irrespective of previous meals (To be drunk in 5 minutes).

Result :

- If 2-hour PP < 140 mg/dL : Normal (Repeat test at 24 - 28 weeks).
- If 2-hour PP ≥ 140 mg/dL : Gestational diabetes.
- If 2-hour PP ≥ 200 mg/dL : Pre-gestational diabetes.

Important points :

- minimum time gap between 2 tests : 4 weeks.
- If patient comes for first time after 28 weeks : Only tested once.

----- Active space -----

Management of Diabetes in Pregnancy

00:22:08

Ix to be done In pre-gestational and gestational diabetes :

1. DIPSI test.
2. NT scan.
3. Level-2 USG.
4. Urine $\left\{ \begin{array}{l} \rightarrow \text{Routine} \\ \rightarrow \text{micro} \end{array} \right\}$ Increased chances of asymptomatic bacteriuria
5. Screening for PIH :
BP/dipstick in every visit.
6. Antenatal fetal monitoring from 32 weeks :
 - a. NST } weekly
 - b. BPP }
 - c. USG for fetal growth x every 3 weeks.
(If 3rd weekly not possible :
At least 2 USG for fetal growth)
7. Only in pre-gestational :
 - a. HbA1C.
 - b. Fetal ECHO.

Number of antenatal visits :

- Well-controlled GDM/no complications :
Routine antenatal visit.
- Not well-controlled GDM/complications seen :
 - T2 : Every 2 weeks.
 - T3 : Every week.

metabolic goals in diabetes (mNT or insulin) :

- FBS : 95 mg/dL.
- 1-hour PP : 140 mg/dL.
- 2-hour PP : 120 mg/dL.
- HbA1c : < 6% (6.5% max).
- Average capillary glucose < 100 mg/dL.

Obstetric management (Termination) :

- Gestational diabetes :
 - Well-controlled on diet (Type A₁)
 - Type A₂ well-controlled on insulin/drugs.
 - Type A₂ not controlled on insulin :
37 - 38 weeks + 6 days.
- Pre-gestational well-controlled ≥ 39 weeks.

management of Diabetes in Pregnancy

management of pregestational diabetes :

1. medical nutrition therapy (mNT) :
Diet modification + weight counselling.
2. Insulin.
3. Low dose aspirin : To prevent PIH.
(Started < 16 weeks POG).

management of gestational diabetes :

Give medical nutrition therapy (Diet modification) alone for 2 weeks.

After 2 weeks

Check 2 hr PP glucose levels

value < 120 mg/dL

Continue mNT

Since insulin resistance increases during pregnancy, check 2-hour PP levels :

- T2 : Every 2 weeks
- T3 : Every week

value ≥ 120 mg/dL

mNT + insulin

Initially :
Check 2-hour post-prandial levels every third day to adjust dose of insulin

Complications of Diabetes in Pregnancy

00:29:07

----- Active space -----

Fetal complications :

- Fetal hyperglycemia.
- **macrosomia** (m/c).
- Shoulder dystocia.
- ↑ Chances of abortion, IUD, stillbirth.

Note :

If there is diabetic vasculopathy :
Patient may have oligohydramnios and IUGR.

Neonatal complications :

- **Neonatal hypoglycemia**
- Hypocalcemia.
- Hypokalemia.
- Hypomagnesemia.
- RDS (Insulin : ↓ Surfactant production).
- Necrotizing enterocolitis
- Hyperviscosity
- Hyperbilirubinemia
- **Polycythemia**

Never seen in babies of diabetic mothers :

- **Anemia**
- mental retardation

Macrosomia :

Birth weight ≥ 4 kg.

Risk factors :

- Post-term pregnancy.
- Diabetic mother.
- male fetus.
- maternal obesity.

Diagnosis :

Abdominal circumference (Best parameter for fetal growth) : ≥ 35 cms on USG.

mode of delivery :

Vaginal delivery.

Indications for C-section :

- Diabetic female, if weight of fetus ≥ 4.5 kg.
- Nondiabetic patient if weight of fetus ≥ 5 kg.
- If patient has macrosomic baby in this pregnancy and h/o cesarean section (VBAC is C/I).

Complications of Diabetes

Shoulder Dystocia

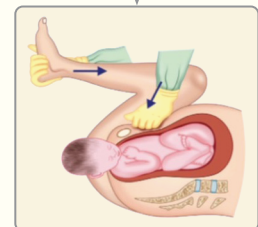
m/c fetal complication of shoulder dystocia :
Erb's palsy
(Porter's tip hand $\rightarrow$ C₅ - C₆ involvement)

Shoulder Dystocia (T) :

- Delay in delivery of shoulder by ≥ 1 minute after delivery of fetal head.
- Obstetric emergency.
- Sign : **Turtle sign**.

Algorithm for management of shoulder dystocia (T) :

- **H** : Call for help.
- **E** : Give liberal episiotomy ; Empty the bladder.
- **L** : Legs $\rightarrow$ **McRoberts maneuver**.
 - Flex the thighs and abduct them
 - Leads to straightening of sacrum and increases available space.
- **P** : Suprapubic pressure (Fundal pressure C/I).
- **E** : Enter hand in pelvis and rotate shoulder (Rubin II maneuver/Woods corkscrew maneuver).
- **R** : Remove posterior arm (Deliver posterior arm) $\rightarrow$ Jacquemier maneuver.
- **R** : Roll over patient on all 4 limbs (Gaskin maneuver or all-fours maneuver).
- Last step in mgt of shoulder dystocia : Push the head of baby back into uterus, followed by cesarean $\rightarrow$ **Zavanelli maneuver**.
- Fundal pressure is C/I in shoulder dystocia.
- **IGF** : Hormone for fetal growth (No role of growth hormone).



McRoberts maneuver



All-fours maneuver

Note :

m/c nerve injured during McRoberts maneuver : **Lateral cutaneous nerve** of thigh.

----- Active space -----

Note :

Pregnancy → Fasting hypoglycemia.
 → Post-prandial hyperglycemia.

Considerations in management :

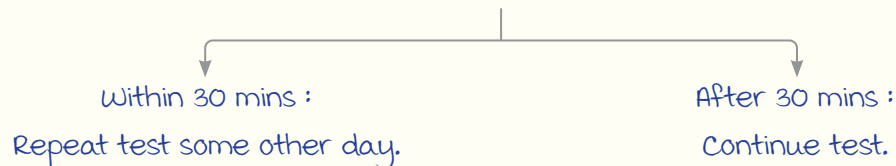
Insulin in pregnancy :

- Insulin secretion ↑.
- Insulin resistance ↑.
- Dose of insulin :
 - ↑es as pregnancy advances.
 - ↓es during labor (D/t nil oral).

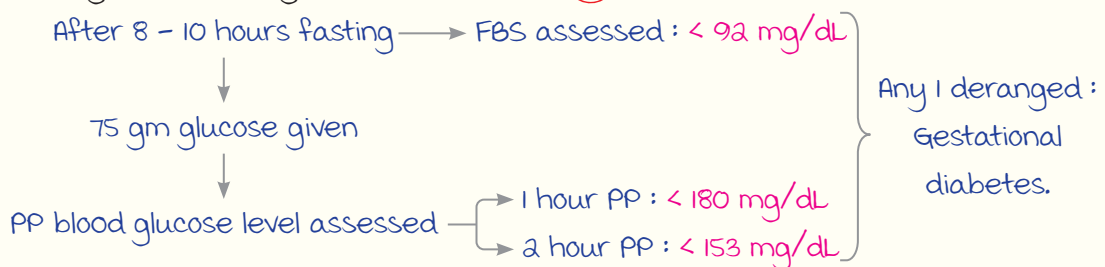
Glycosuria :

- Physiological during pregnancy.
- Indication to screen for diabetes.

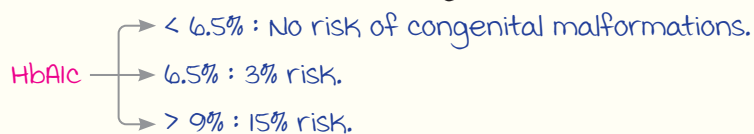
If vomiting occurs during DIPSI :



IADPSG guidelines for gestational diabetes ⁽⁹⁷⁾ :



Risk assessment for overt/pre-gestational diabetes :

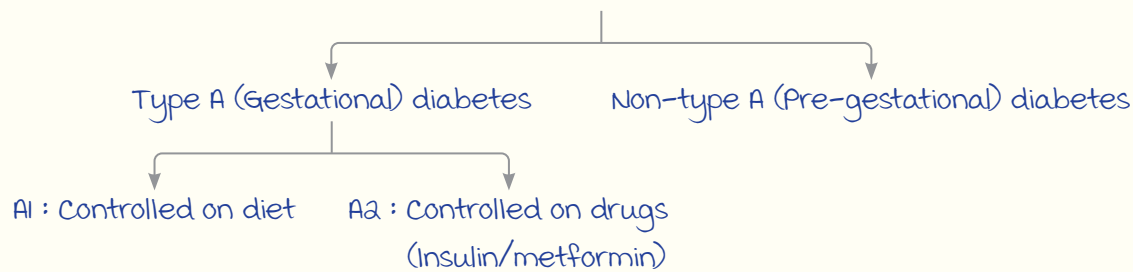


Best method to ↓ risk of congenital malformations : Strict glucose control.

Dose of folic acid in diabetic patients : 400 mcg/day.

Priscilla-White classification (57) :

----- Active space -----



OHAs in pregnancy :

- C/I as it crosses the placenta → Fetal hypoglycemia.
- Safe in pregnancy :
 - metformin : Indication → Non-compliance to insulin.
 - Glyburide.

DOC for diabetes in pregnancy : Insulin.

- Route : SC.
- m/c used : Human premix insulin.
- 1 vial : 40 IU.
- Syringe can be used maximum upto 14 times.
- Stored at 4 - 8°C.

Indications for umbilical artery doppler :

- Diabetics with vasculopathy.
- Diabetics with PIH.

Estimation of fetal weight:

- Clinical : Johnson formula.
- USG : Hadlock/Shepard formula.

Fetal lung maturity :

- Tests :
 - Phosphatidyl glycerol (Best).
 - Lecithin/sphingomyelin ratio $\geq 2 : 1$ → mature lungs.
- Surfactant :
 - Produced by Type II pneumocyte at 20 wks (In amniotic fluid by 28 wks).

Complications of diabetic pregnancy with UPI/vasculopathy :

1. Oligohydramnios.
2. IUGR.

OGTT in GDM patients : Done in all 6 weeks after delivery.

----- Active space -----

Rh Negative Pregnancy

00:47:51

Pathophysiology :

mother is Rh-negative
Fetus is Rh-positive

If fetal blood enters maternal blood, Rh antigen in fetal blood stimulates maternal immune system

IgG antibodies formed against fetal Rh antigen
(Can cross the placenta)
Leading to

- | | |
|--|--|
| <ul style="list-style-type: none"> • Fetal hemolysis • Fetal anemia • Hyperbilirubinemia • Hepatosplenomegaly • Placentomegaly • PIH • Polyhydramnios | <p>Fetal</p> <p>></p> <p>maternal</p> |
|--|--|

Algorithm when Rh negative female visits**at 1st antenatal visit :**

1st step : Check husband ABO/Rh.

If husband is Rh +ve = 50% chances fetus will be Rh +ve → High risk pregnancy

Next

DO : Indirect coomb test (ICT)

To check for Rh antibodies

ICT -ve

Rh non-
isoimmunised
pregnancy

ICT +ve

Rh
isoimmunised
pregnancy

Rh Negative Pregnancy**management of Rh non-iso immunized pregnancy :**

If ICT -ve at 1st visit and 28 wks

Give anti-D to mother at 28 wks

Called as antepartum prophylaxis

Plan delivery at 39 - 40 wks

- Don't give methyl ergometrine
- Do early cord clamping

After delivery :

Check fetal Rh status + DCT
Rh +ve & DCT -ve

Antepartum prophylaxis :

- To Rh -ve pregnant female, if ICT is +ve at 28 weeks.
- Dose : 300 mcg (1500 IU).
- Route : Im.
- Aim : Protection of **current** pregnancy.

Postpartum prophylaxis :

- Given if : Fetus is
 1. Rh +ve
 2. DCT -ve
- Ideally should be given : within 72 hours.
- Can be given : upto 28 days after delivery.
- Dose : 300 mcg.
- Route : Im.
- Aim : To protect **next** pregnancy

Give anti-D to mother
(Post-partum prophylaxis)

Other Indications for Anti-D :

1 st trimester	2 nd and 3 rd trimester
• Abortion • molar pregnancy • Ectopic pregnancy • Chorionic villus sampling • mTP	• Amniocentesis • Trauma • APH • manual removal of placenta • External cephalic version • Fetal death
Dose : 50 mcg	Dose : 300 mcg

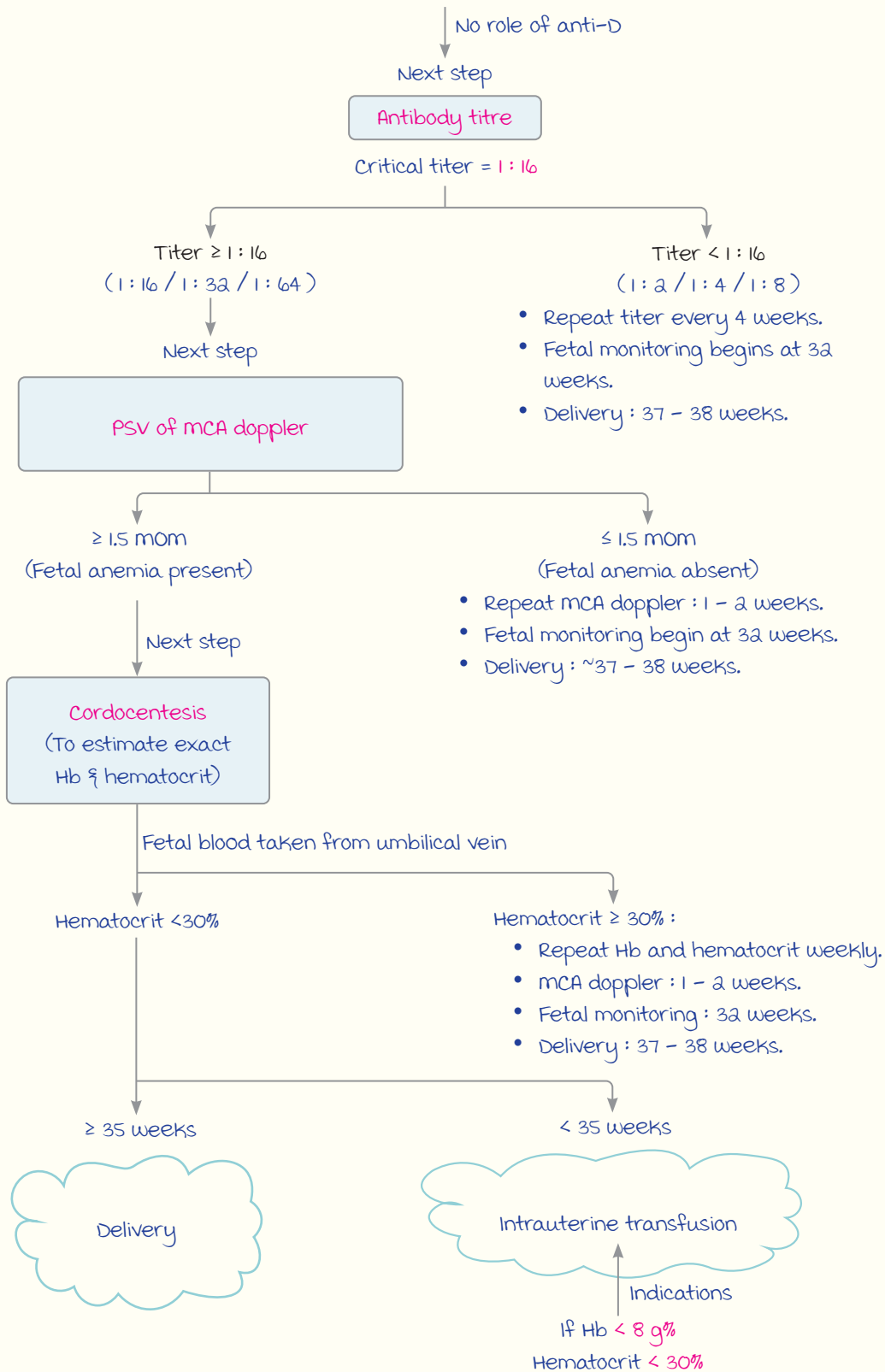
Note :

Cordocentesis : Not an indication for anti-D.

management of Rh Isoimmunized Pregnancy (STP) :

----- Active space -----

If ICT is +ve either at first antenatal visit or at 28 weeks,
such a pregnancy is called **Rh isoimmunized pregnancy**



----- Active space -----

Anti-D :

c/i :

- Rh -ve husband (Rh -ve fetus).
- ICT +ve/DCT +ve.

Dose :

- TI : 50 mcg.
 - Ta/T3 :
 - After delivery :
- 300 mcg $\xrightarrow{\text{Neutralizes}}$ $\left\{ \begin{array}{l} 15 \text{ mL of fetal RBC.} \\ 30 \text{ mL of fetomaternal hemorrhage.} \end{array} \right.$

Rh Antigen :

Appears on fetal RBCs 38 days after fertilization.

D antigen :

- main Rh antigen located on chromosome 1.
- Autosomal dominant inheritance.

mode of Delivery :

- vaginal delivery.
- Cord clamping : ICT $\left\{ \begin{array}{l} \rightarrow +ve \text{ (Anemia)} \rightarrow \text{Delayed cord clamping (1 - 3 min).} \\ \rightarrow -ve \rightarrow \text{Early cord clamping (< 1 min).} \end{array} \right.$

Kleihauer - Betke Test :

- AKA Acid elution test.
- Quantitative test used to calculate dose of anti-D.

Principle : HbF resistant to acid/alkali.

Reagent : Citric acid phosphate buffer.

Hydrops Fetalis :

Occurs d/t heart failure when Hb < 5 gm % hematocrit < 15%.

USG diagnosis :

1. Diagnostic findings :

- Pleural effusion.
- Pericardial effusion.
- Ascites.
- Skin edema (Buddha sign : Subcutaneous scalp edema).

Any ≥ 2 $\oplus$

2. Other findings :

- a. Placentomegaly.
- b. Polyhydramnios.

----- Active space -----

Types :

- Immune : Rh incompatibility > ABO incompatibility.
- Non-immune :
 - CVS disease (m/c).
 - Anemia.
 - Chromosomal disorder
 - Parvovirus infection.
 - Syphilis.
 - TORCH infection

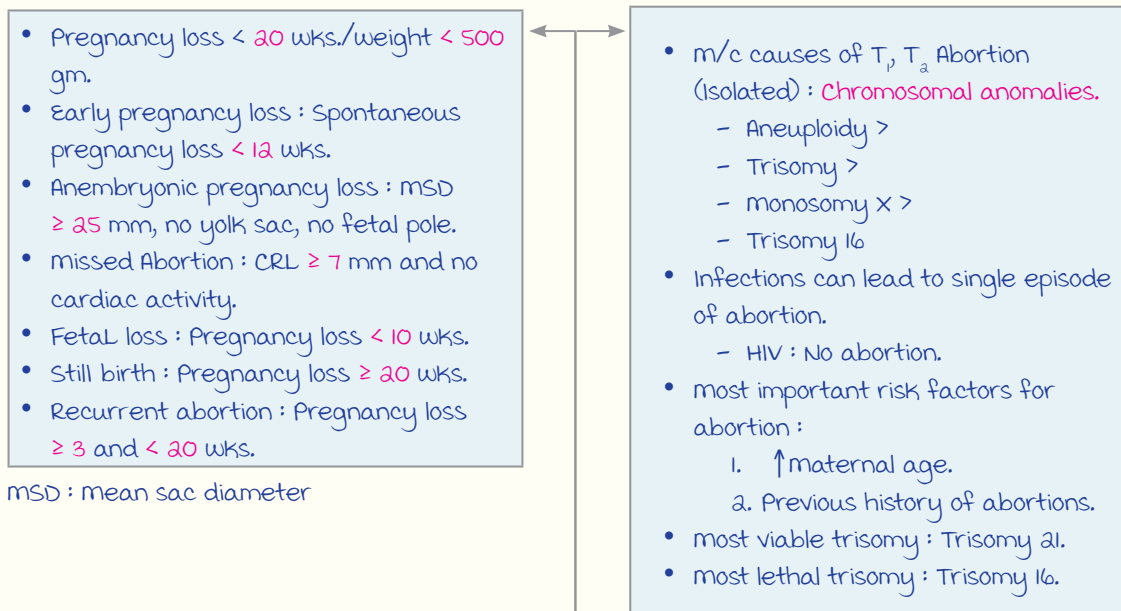
Note :

- Infections not causing hydrops :
 - Herpes.
 - Hepatitis.
 - HIV.
- Intrauterine death of fetus :
 - Robert's sign : Gas in great vessels (1st sign).
 - Spalding sign : Overlapping of skull bones.
 - Ball sign : Hyperflexion/hyperextension of fetal spine.

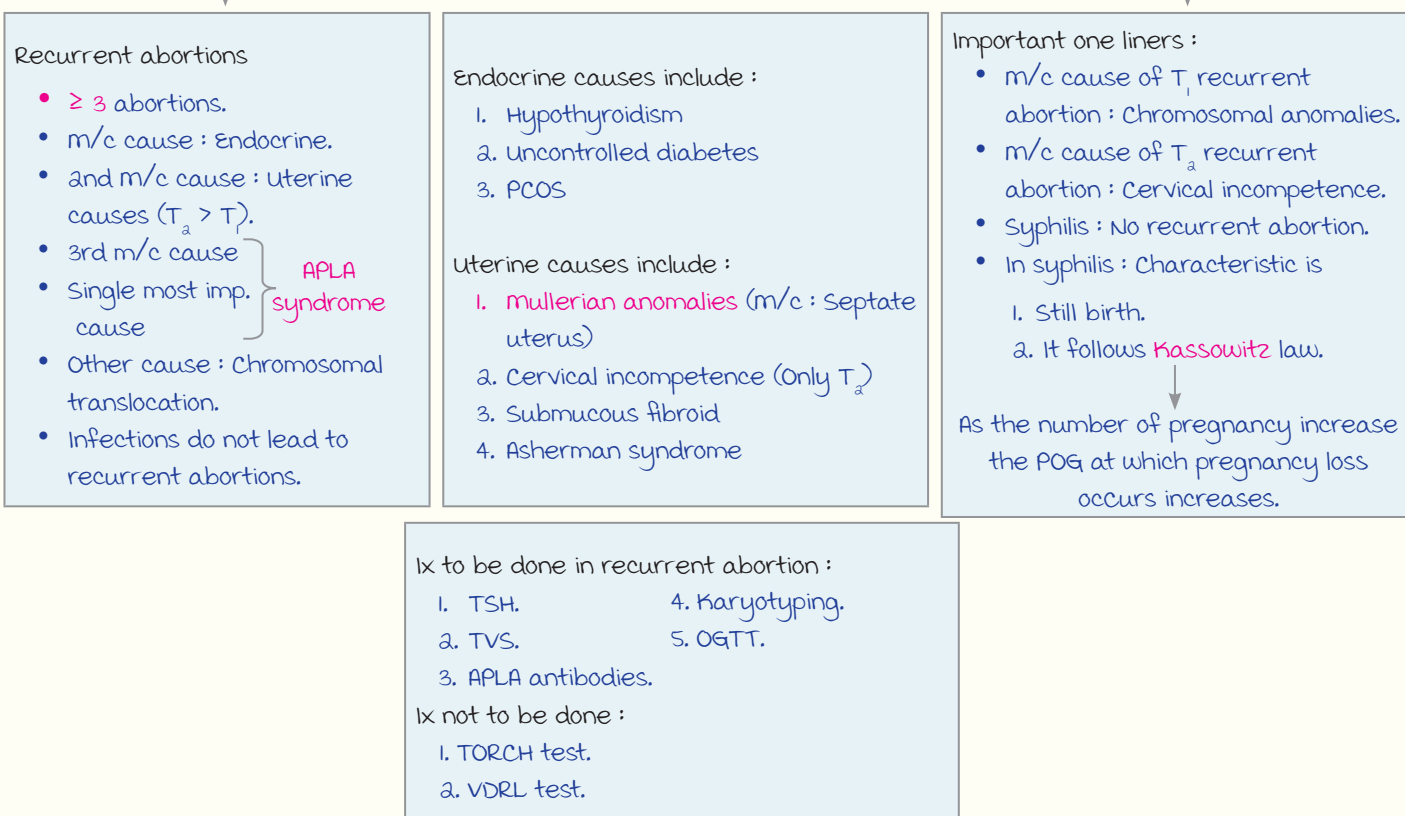
OBSTETRICS REVISION : PART 6

Abortions

00:00:27



Abortions



Cervical Insufficiency/Incompetent Os

00:07:10

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Spontaneous dilatation of cervix due to shortening of cervix (in T_a).
IOC : USG

Diagnosis :

History based diagnosis :

≥ 3 abortions of **second trimester** confirm cervical insufficiency (No need for USG).

USG-based diagnosis :

History of one or two second trimester abortion or ≥ 1 PTB or cervical surgery. TVS done in present pregnancy shows length of cervix ≤ 2.5 cm.

On TVS : Cervical Insufficiency.

- Length of cervix : ≤ 2.5 cm.
- Internal Os : Dilated.
- Shape of cervix : 'U' shaped.

management of cervical insufficiency :

- Cervical cerclage (if indicated)
- Progesterone

Next steps :

- Case 1 : Pregnant female has history of three T_a abortions.
Next step : Cervical cerclage + progesterone.
- Case 2 : Pregnant female has history of two T_a abortion.
Next step : Cervical length on TVS ≤ 2.5 cm : Cerclage + progesterone
 > 2.5 cm : Only progesterone
- Case 3 : Pregnant female with history of one PTL or T_a abortion + length of cervix ≤ 2.5 cm.
Next step : Cervical cerclage + progesterone.
- Case 4 : Pregnant female with no History of abortion but cervical length < 2.5 cm.
Next step : Only progesterone.

Cervical Insufficiency

Cervical Cerclage :

It can be done in pregnant females and non-pregnant females.

Pregnant females :

1. Transvaginally (m/c) :

- most common : McDonald cerclage (Purse string suture applied)
- Shirodkar cerclage (using mersilene tape).
- Wurm's cerclage : Done as emergency/rescue cerclage.
- 2 sutures applied : AP, transverse.

2. Transabdominally :

- Methods :
 - Benson and Durfee cerclage (Outdated).
 - Laparoscopic cerclage.
- It can be done during **pregnancy** and before pregnancy.
- Indication : Failed transvaginal cerclage.
- The suture should be removed only after female has completed her family.
- Delivery should be done by C-section.

Non-pregnant females :

- Preferred method of cerclage : Laproscopic cerclage.
- methods to know in non-pregnant female that cervix is incompetent :
 - If **Hegar dilation No. 8** can be passed without resistance.
 - Foley's catheter No. 16** Can be removed without deflating bulb.

Concepts on Cerclage (TV) :

McDonald's cerclage (m/c) :

- Suture : Nonabsorbable monofilament purse string sutures applied on : **Portio vaginalis.**
- Time : **12 - 24 weeks.**
- C/I :
 - Ruptured membranes (Chorioamnionitis).
 - Bleeding.
 - uterine contractions.
 - Fetal demise/gross congenital anomalies.
 - Current pelvic infection.
- Relative C/I : **Placenta previa.**
- Time to remove stitch : **37 weeks.**
 - Can be removed earlier in pre term labour.

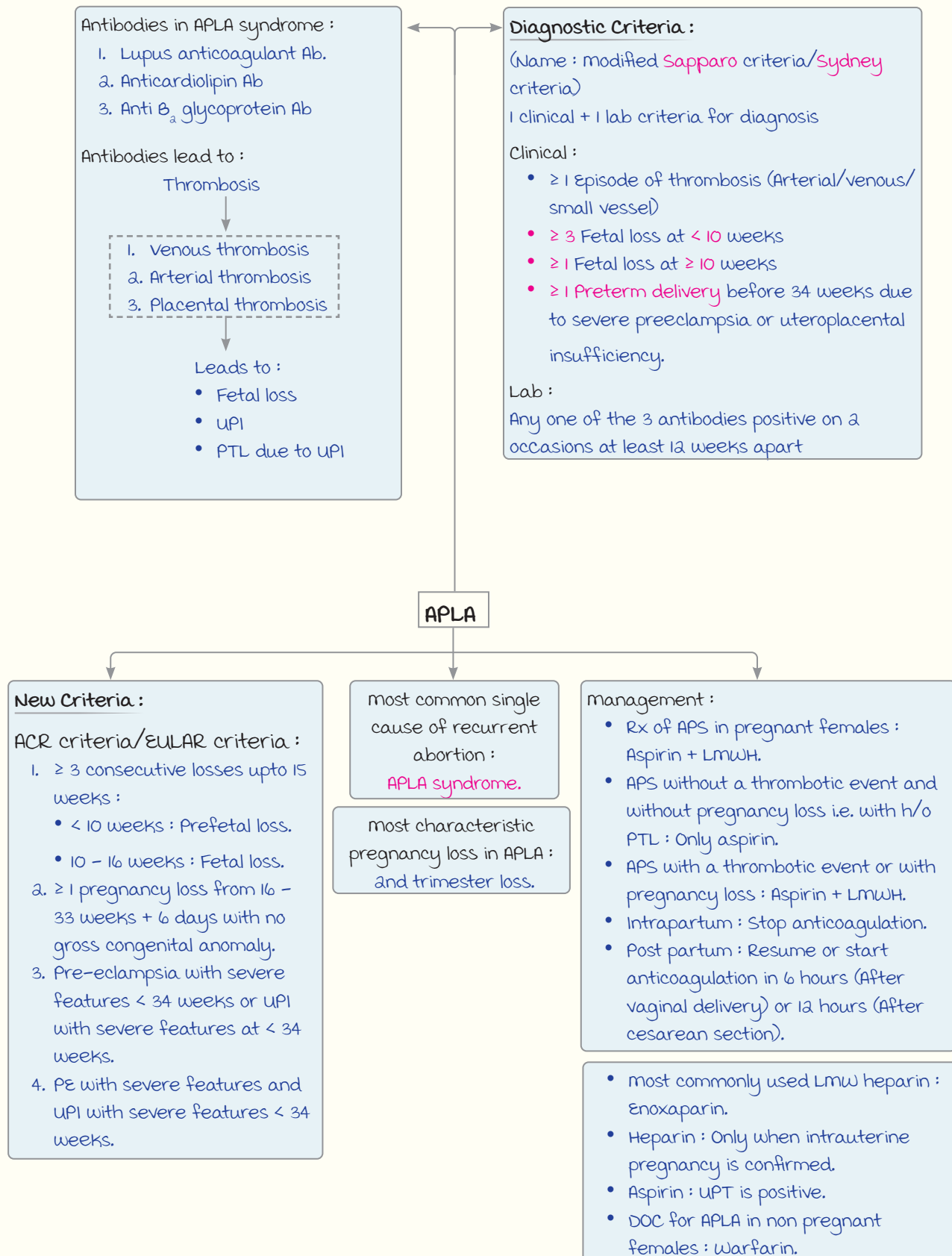
Shape of cervix on USG as it dilates (TV) :
Normal $\rightarrow$ T shaped $\rightarrow$ Y shaped $\rightarrow$ V shaped $\rightarrow$ U shaped.

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Antiphospholipid Antibody (APLA) syndrome



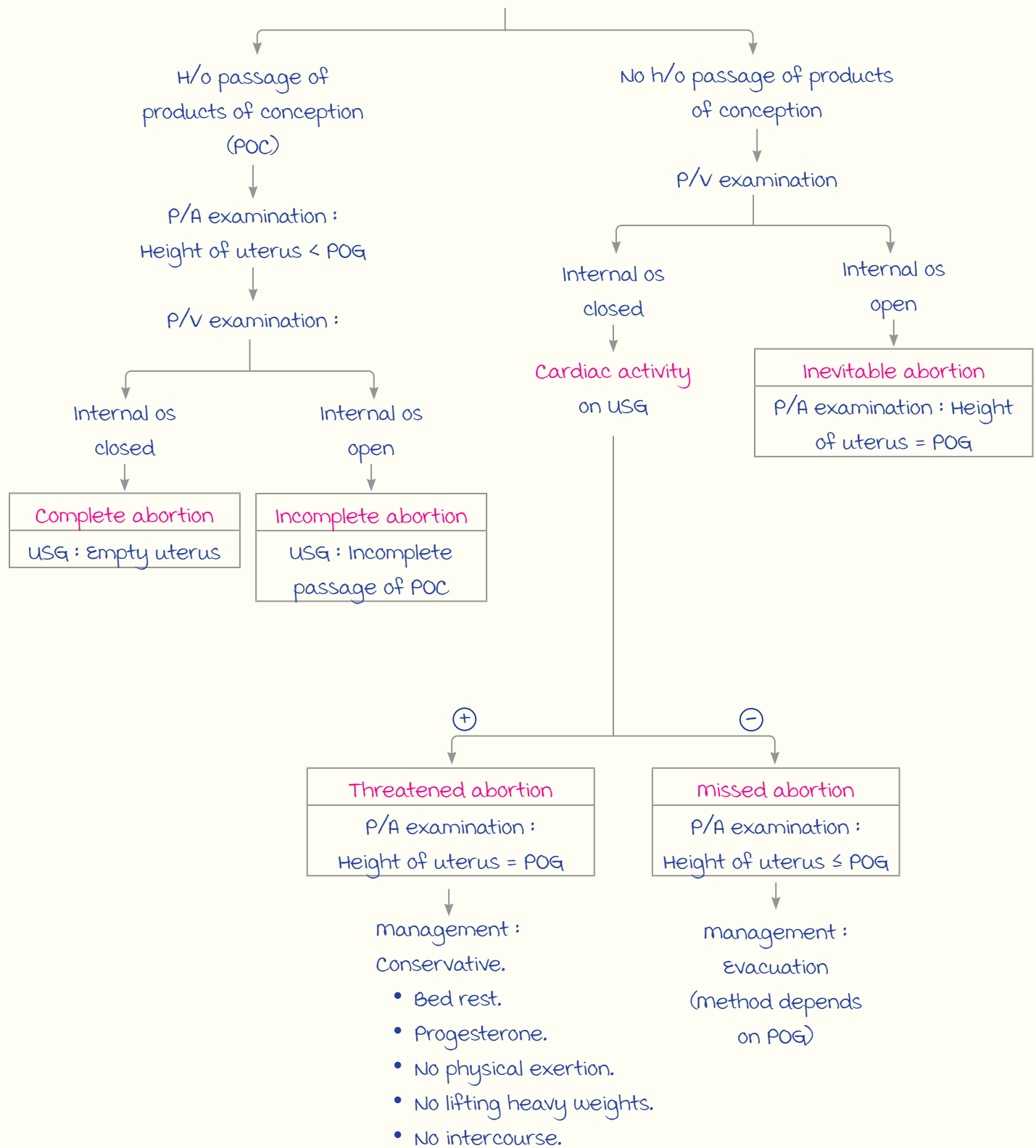
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Diagnostic Flow Chart for Abortions

00:26:46

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----- Active space -----

Medical Termination of Pregnancy

00:30:10

MTP Amendment 2022 :

MTP can be done till 24 weeks.

- In case of contraceptive failure : MTP is done till 20 weeks.
- In severe congenital anomalies of fetus : No upper limit (if permitted by medical board).

Qualification to do MTP :

For doing MTP till 12 weeks :

RMP who has assisted in 25 MTPs (At least 5 should be as primary surgeon).

For doing MTP between 12 and 20 weeks :

- RMP with 6 months of house job in Obs and Gyne or RMP with 1 year of experience in any hospital with all facilities.
- Diploma/degree/DNB in Obs and Gyne.

medical board = Gynae + Pedia + Radiologist

- Single doctor opinion : upto 20 weeks.
- Two doctors opinion needed : 20 - 24 wks.

MTP ⚠ :

- Female's consent is needed for MTP if ≥ 18 yrs.
- In minor/mentally retarded patients : Legal guardian's consent required.
- marriage certificate
- FIR report of rape
- All records of MTP should be maintained for 5 years.
- Lower limit of age to do MTP : No lower limit.

MTP

methods to do MTP**T_i :**

- medical abortion
- Suction evacuation
- menstrual regulation
- manual vacuum aspiration

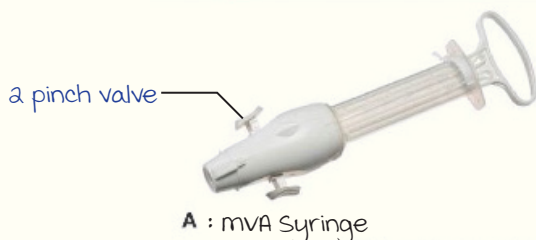
T_a :

- medical abortion using :
 - misoprost
 - Extra amniotic ethacridine
 - Intra amniotic saline
 - Oxytocin
- Surgical
 - Dilatation and evacuation
 - Suction and evacuation
 - Hysterotomy

medical abortion :

GOI and WHO = upto 9 weeks.

- USG not mandatory, done on OPD basis.
- D1 : T. mifepristone : 200 mg oral
- D3 : T. misoprost
800 mcg oral/buccal/p/v 24 hrs
Bleeding
- D15 : Ensure process is complete

**Suction Evacuation**

- Done using Karman cannula.
- Number of Karman cannula corresponds to size of uterus/ 1 size lesser.
- Pressure generated = 600 mmHg.
- Suction evacuation can be done maximum till 16 weeks.
- Suction evacuation is best method of MTP = 10 - 12 weeks.
- In areas where electricity is not available = MVA

Manual Vacuum Aspiration

- Done using MVA syringe (2 pinch valves)
- Capacity of syringe = 60 mL.
- Pressure generated = 660 mmHg > 600mmHg.
- menstrual regulation can be done maximum till 6 weeks.
- MVA can be done upto 12 weeks.

Suction & Evacuation :

----- Active space -----

Anesthesia given : Paracervical block

Signs of complete evacuation :

- ↓ Bleeding.
- Bubbles seen in suction tube.
- Grating sensation.
- Gripping sensation on Karman's cannula.

uterine perforation during suction evacuation :

With Hegar's dilator	With Karman cannula
• Wait & watch • monitor patient	• Stop procedure • Do not remove cannula • Resuscitation & laparotomy

Second Trimester Abortions :

medical abortion (m/c) :

- Done on IPD basis.
- misoprostol **400 mcg** P/V → Expulsion of POC.
4 - 5 hourly maximum 5 doses (mini labor)

Dilatation & evacuation :

- using ovum forceps.
- Followed by blunt curettage.
(except in : molar pregnancy → Sharp curettage)

Best method of MTP :

POG	method
≤ 9 weeks	medical abortion
10-12 weeks	Suction evacuation
> 12 weeks	Abortion using misoprostol (IP)

Indications for MTP :Social : Contraceptive failure.Humanitarian : Rape victim.Therapeutic : Pregnancy causes risk to mother.Eugenic : Gross congenital anomalies in fetus.Medical Abortion Kit :

Tab methotrexate :

- Approved for T₁ abortions by WHO
- Not approved by GOI.

----- Active space -----

Ectopic Pregnancy

00:40:40

m/c site : **Fallopian tube** (Ampulla)
 m/c nontubal site : Ovary.
 Least common site : C-section scar/
 Cervical ectopic.
 Heterotopic pregnancy : Twin
 pregnancy; one is intrauterine, other
 ectopic - mx : **Always surgical.**

Triad of symptoms :

- Amenorrhea (6 - 10 weeks), pain in lower abdomen and bleeding P/V.
- most specific symptom : Pain in abdomen

Symptoms :

To suspect ruptured ectopic :

1. Orthostatic hypotension.
2. Shoulder tip pain (D/t hemoperitoneum)
3. Urge to defecate.

Named criteria :

Site	Criteria
Cervical	Palman, Rubin
Abdominal	Studdiford
Ovarian	Spiegelberg

Signs : Ruptured ectopic

1. Shock ($\uparrow$ PR, $\downarrow$ BP).
 2. P/A : Abdominal distension, rebound tenderness.
 3. Localizing signs — Guarding.
Rigidity.
- } + uPT $\oplus$

Ectopic Pregnancy**Risk Factors :**

- Maximum risk :
 - Previous h/o ectopic pregnancy.
 - Previous h/o tubal surgery.
- m/c : PID/salpingitis.
- Contraceptives :
 - **Decrease absolute risk** of ectopic but if failure of contraception occurs, then ectopic pregnancy risk is increased.
 - m/c : Tubal sterilization > mirena > IUCD > POP.

P/V Finding of Ruptured Ectopic :

- most important : Adnexal mass.
- Cervical movement tenderness (D/t peritonitis)
 - Also seen in PID.
- uterus is soft and enlarged but size is less than POG.

Management of Ruptured Ectopic :

- Always surgical.
 - No role of conservative management or medical management.
- Route of surgery :**
- If vitals stable : Laparoscopy.
 - If unstable vitals : Laparotomy.
- Surgery of choice : u/L salpingectomy.**
- Surgeries never done for ruptured ectopic :**
- Salpingo-oophorectomy.
 - Linear salpingostomy (Done for unruptured ectopic).

Case :

Female with 6 - 8 weeks amenorrhea, pain in abdomen, spotting, vitals unstable, uPT+ve

Next step :

- If on P/A, guarding and rigidity present : Ruptured ectopic $\rightarrow$ Sx management.
- If guarding and rigidity are not present : FAST $\rightarrow$ Confirm if ruptured.

Investigations Done in Unruptured Ectopic :

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1. TVS : IOC

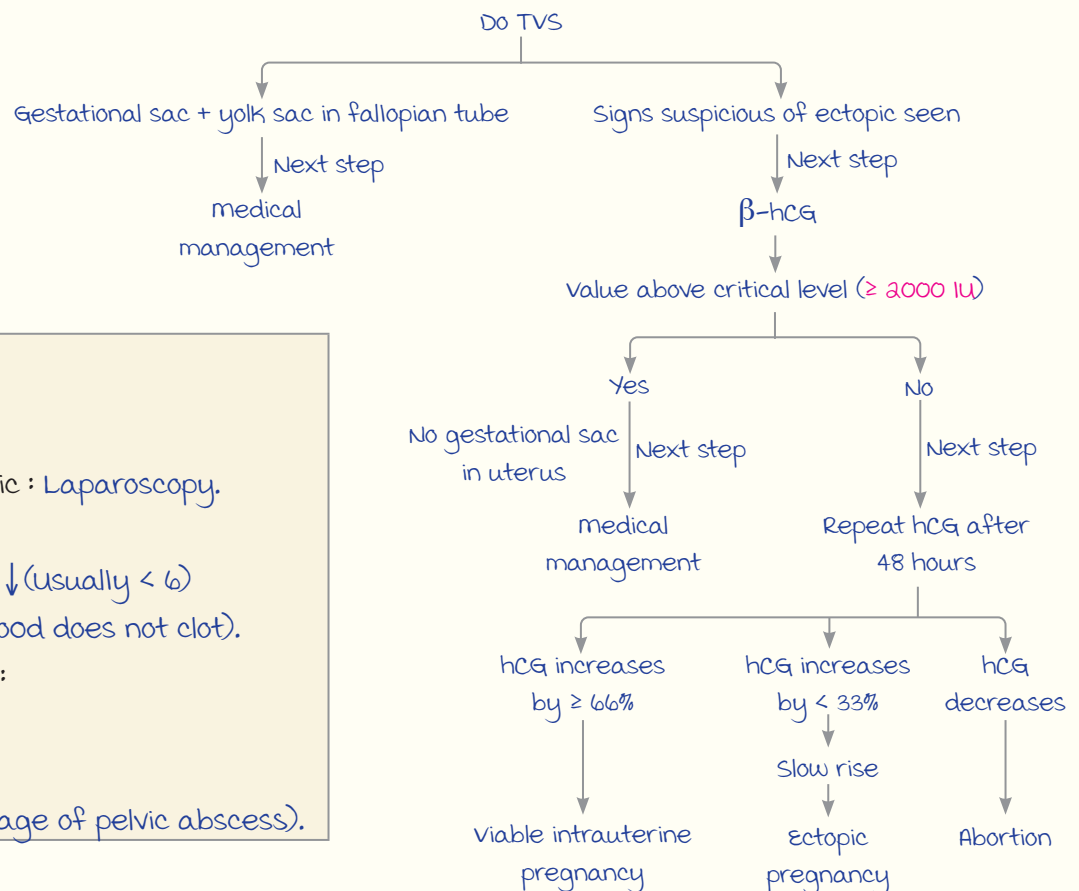
- Confirmed sign of ectopic pregnancy : Gestational sac + yolk sac seen in fallopian tube.
- Suspicion of ectopic if :
 - Only gestational sac in fallopian tube.
 - Bagel sign +ve.
 - Doppler : **Ring of fire appearance.**
 - Adnexal mass.
 - Empty uterus.

2. Beta-hCG :

Critical value of hCG is that value of hCG above which in all intrauterine pregnancies, gestational sac is visible inside uterus.

- TVS : 2000 IU.
- TAS : 6500 IU.

Algorithm for diagnosis of unruptured ectopic :



mx  :

IOC in ectopic : TVS.

Gold standard Ix in ectopic : Laparoscopy.

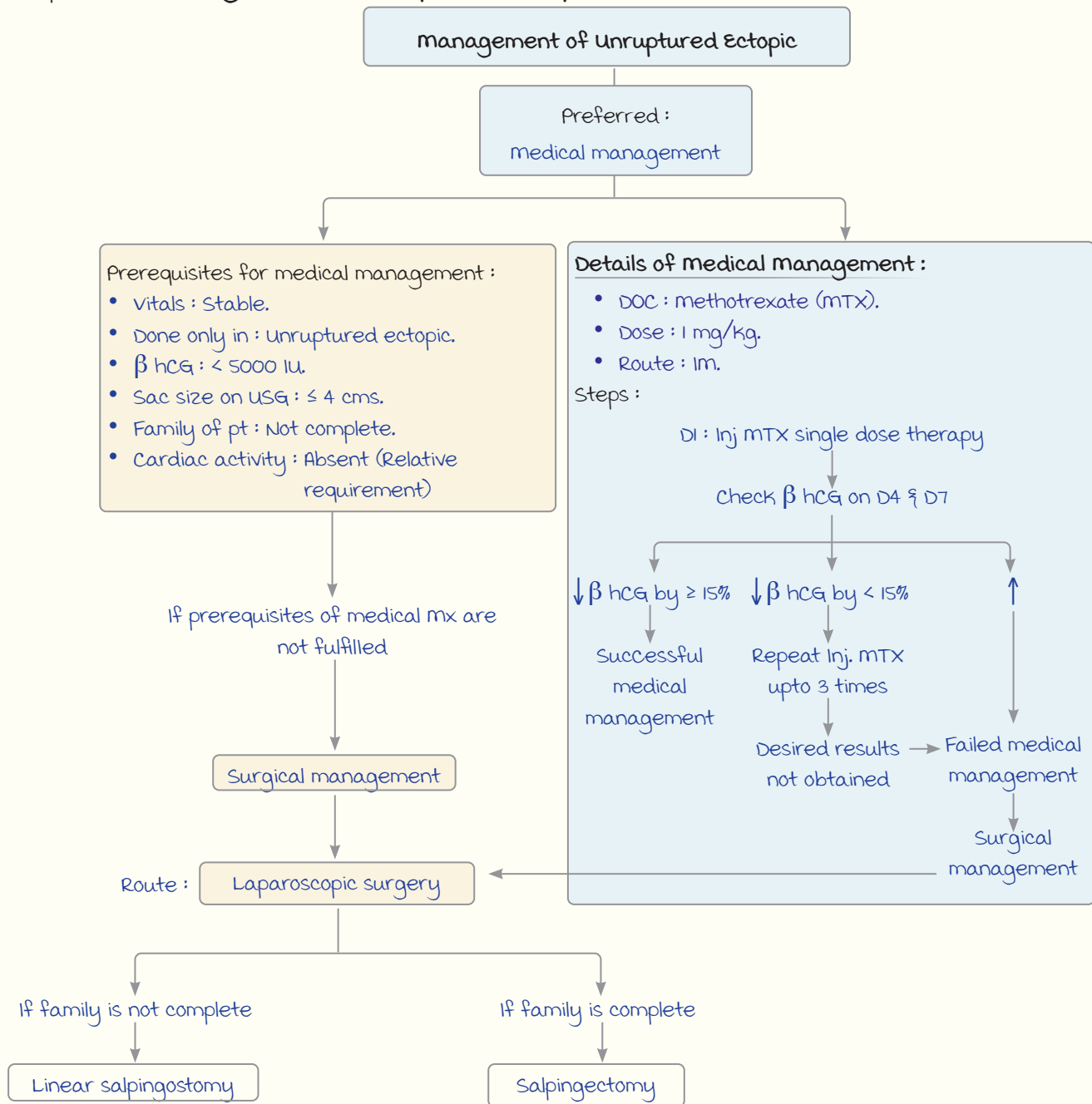
Ix which may be done :

- s. Progesterone : ↓ (usually < 6)
- Culdocentesis (Blood does not clot).

Investigation never done :

- Hysteroscopy.
- HSG.
- Colpotomy (Drainage of pelvic abscess).

----- Active space ----- **management of Unruptured Ectopic :**



Expectant mx :

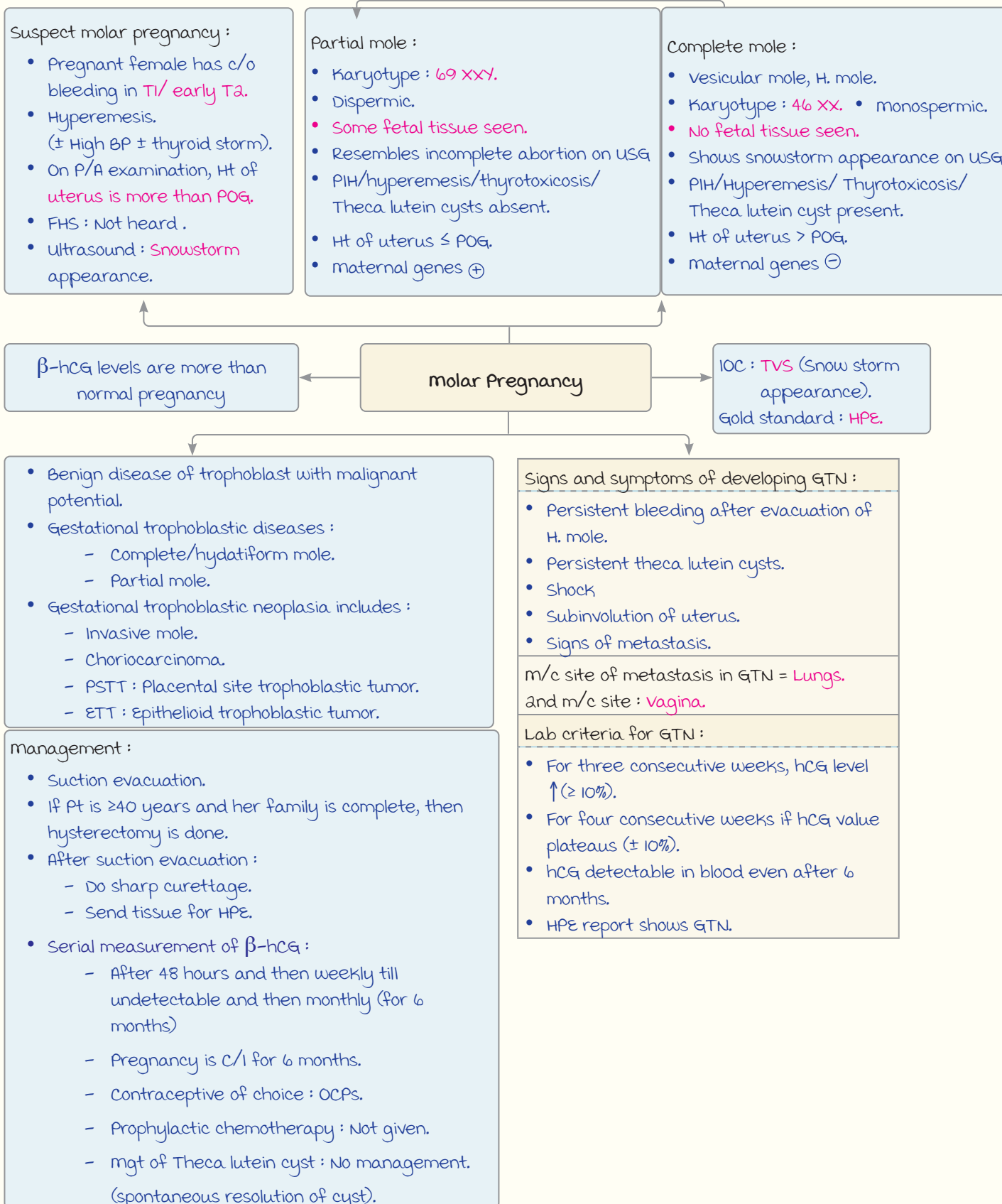
- Rarely done.
- hCG : < 1000 IU.
- Best results are seen with hCG < 200 IU.
- Falling trend of hCG.
- No cardiac activity.

----- Active space -----

Molar Pregnancy

00:58:39

Gestational trophoblastic diseases



----- Active space -----

GTN – Choriocarcinoma and invasive mole (97%) :

- m/c GTN after molar pregnancy : Invasive mole.
- Choriocarcinoma m/c occurs after : H. mole
- m/c GTN after full term delivery : Choriocarcinoma.
- PSTT occurs m/c after : Full term pregnancy.

FIGO staging of GTN :

- Stage I : Limited to uterus (Low risk).
- Stage II : Spread to vagina (m/c site : Just below urethra).
- Stage III : Lung metastasis.
- Stage IV : Liver/Brain metastasis.

Gestational Trophoblastic Neoplasia

WHO risk score for GTN

Low risk	High risk
• GTN develops after molar pregnancy. • metastasis : lungs/ vagina. • hCG values $< 10^3$ IU. • Age of patient < 40 years. • Number of metastasis < 4.	• GTN develops after full term pregnancy. • metastasis : Brain/ Liver. • hCG $\geq 10^5$ IU. • Age ≥ 40 years. • Number of metastasis ≥ 8.
Total score ≤ 6	Total score ≥ 7

Single drug used in :

- Low risk GTN : methotrexate.
 - Given as : **multidose therapy** (D1, D3, D5, D7).
 - Alternating with Folinic acid (D2, D4, D6, D8).
- multiagent chemotherapy : **emACO**/Bagshawe regimen.
 - E : Etoposide.
 - m : methotrexate.
 - A : Actinomycin.
 - C : Cyclophosphamide.
 - O : Oncovin.

management of GTN

	WHO risk score	management
Stage I : Low risk	-	methotrexate
Stage II :	≤ 6 : Low risk	
Stage III :	≥ 7 : High risk	• emACO • metastasis to brain :
Stage IV : High risk	-	Radiotherapy

----- Active space -----

Key Points on molar Pregnancy :

Risk factor : vitamin A deficiency.

HPE :
 → Complete & partial mole : villi seen.
 → Choriocarcinoma: villi absent.

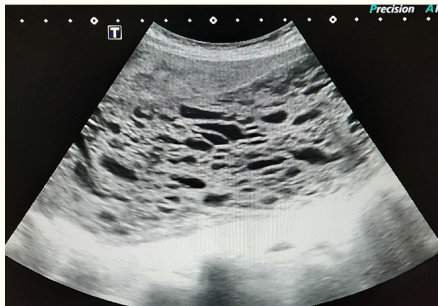
Risk of conversion to GTN :

	% chances of GTN	% chances of choriocarcinoma
Partial mole	4%	1%
Complete mole	20%	2 - 4%

Tumor markers :

- PSTT : hPL.
- Choriocarcinoma/GTN : hCG.

Complete mole :



Snowstorm appearance on USG

Lung metastasis :


 Cannon ball appearance (m/c)
 (and m/c : Snowstorm appearance)

OBSTETRICS REVISION : PART 7

----- Active space -----

Placenta Previa : Classification & Clinical Features

00:00:22

Antepartum hemorrhage : Bleeding at ≥ 28 weeks (Period of viability) till delivery of placenta.

Battledore placenta : Does not cause APH.

Older Classification of Placenta Previa :

Type I (Lateral) :

Lower edge in the lower uterine segment within 2 cm of internal os.

Type II (marginal) :

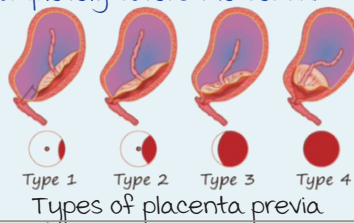
Lower edge reaching the internal os.

Type III (Incomplete) :

Placenta partially covers internal os.

Type IV (Complete) :

Placenta completely covers the cervix.



New Classification of Placenta Previa :

Low lying placenta : Placental edge ≤ 2 cm from internal os (Type I).

Placenta previa : Placental edge either reaching the OS or covering it (Type 2, 3, 4).

Placenta previa viewed in T1/T2 on USG $\rightarrow$ undergoes placental migration during T3 in 90% cases
Hence $\downarrow$
USG diagnosis done at 32 wks f/b 36 wks.

Note :

- In posterior variety of placenta previa : **Stallworthy sign** is positive.
- Type 2** (Old classification) : **Dangerous variety** of placenta previa.

Clinical Features :

History :

- Bleeding in T3 (Bright red) :
 - Painless.
 - Causeless.
 - Recurrent.
 - Warning hemorrhage present.
- No pain in abdomen.

P/A exam :

- Ht. of uterus = POG.
- uterus : **Soft, relaxed, non-tender.**
- FHS : Easily heard.
- Fetal parts : Easily palpable.
- malpresentation : **Transverse lie > Breech**

P/V exam and P/S exam : C/I.

Investigation :

- Screening Ix : TAS.
- IOC : TVS.

Placenta Previa

Significance of USG :

- To confirm diagnosis.
- To detect malpresentation.
- To rule out placenta accreta spectrum.

Best time to do USG to rule out previa :

- 32 weeks f/b 36 weeks.
- Placenta previa with $\rightarrow$ myometrial invasion of villi previous h/o LSCS $\downarrow$ Prone to placenta accreta spectrum (PAS).

Note :

m/c cause of transverse lie :

- Overall : **Prematurity.**
- At term : **Placenta previa.**

----- Active space -----

Placenta Previa : Risk Factors & Management

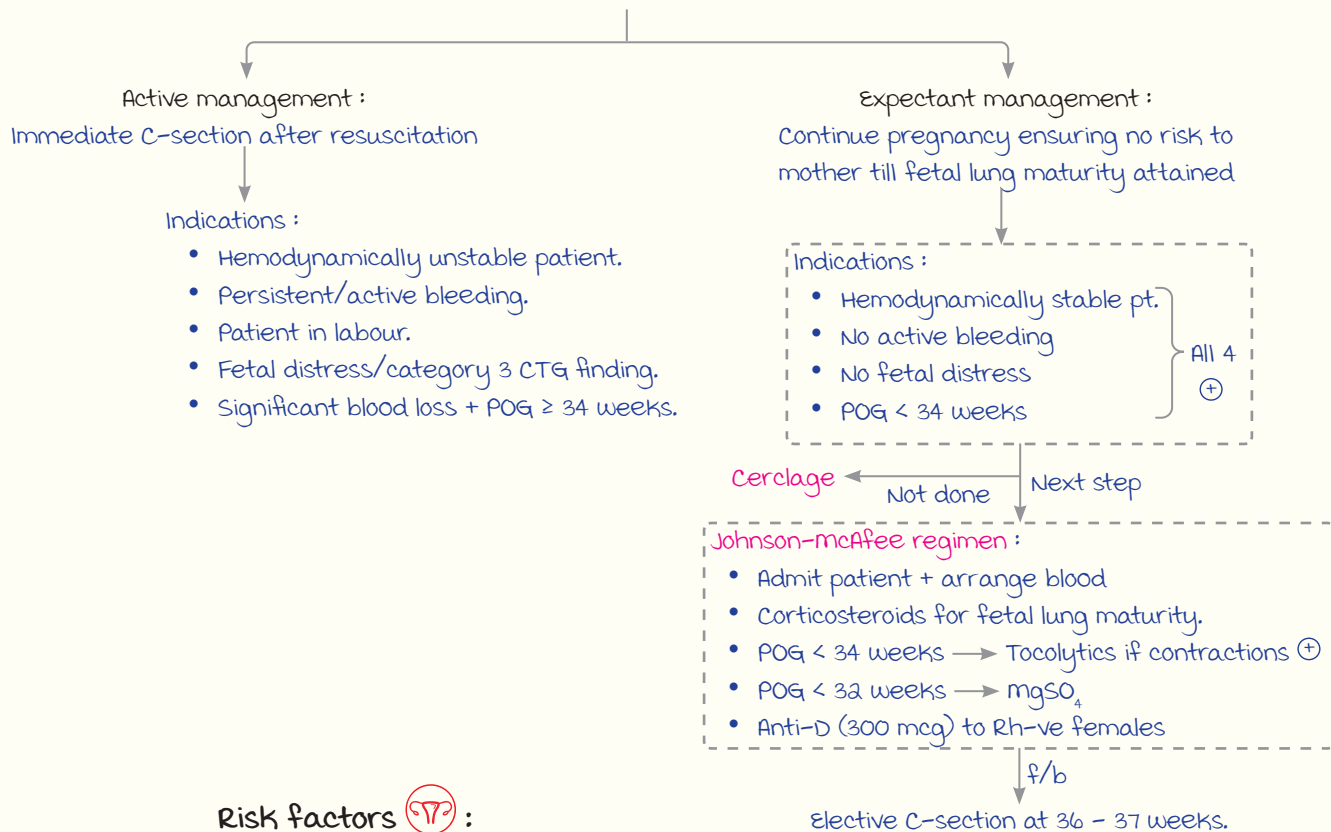
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1. Incidental Detection of Placenta Previa on Level 2 USG :

Repeat USG at 32 weeks & 36 weeks
 ↓
 Persistence of placenta previa/low lying placenta
 ↓
 Elective LSCS at 36 - 37 wks + 6 days.

Advice to be given at 20 weeks :

- Avoid intercourse.
- Avoid standing for ≥ 4 hours.
- Avoid medium to severe exercise.
- Immediately report if bleeding $\oplus$.
- Avoid lifting heavy weights.

2. Known Case of Placenta Previa with Bleeding ⚠ :**Risk factors ⚠ :**

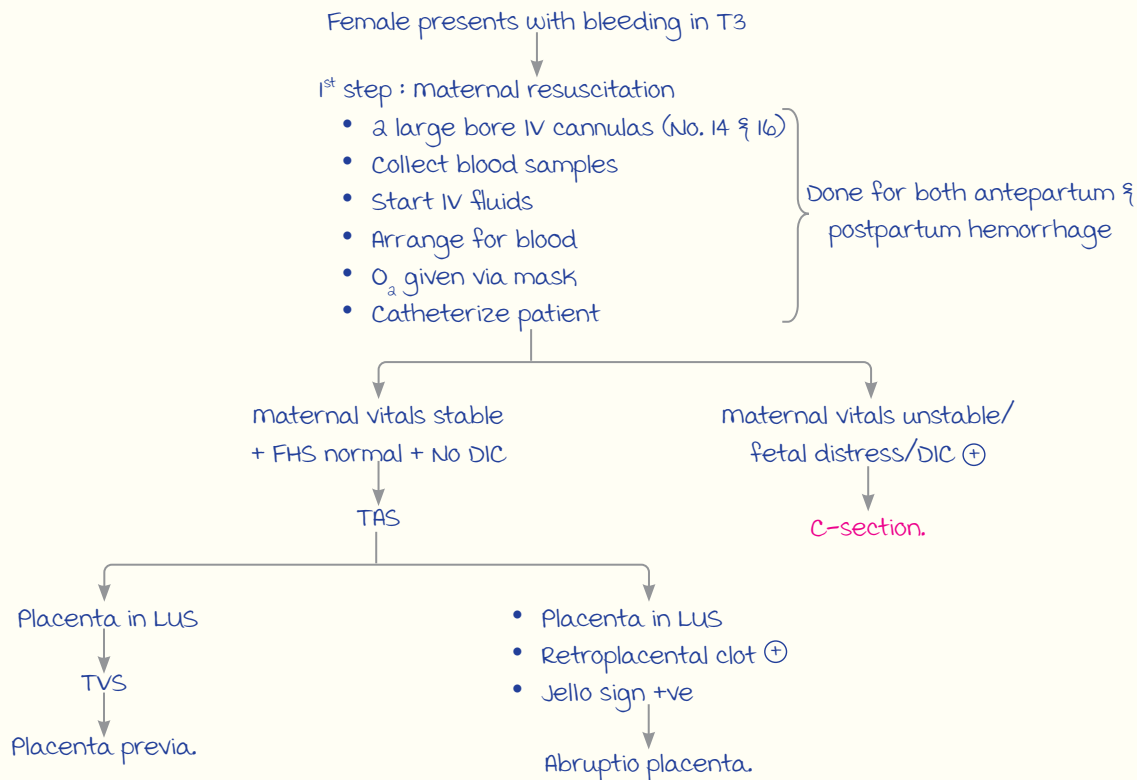
- ↑maternal age.
- ↑maternal parity.
- Big placenta :
 - Twin pregnancy.
 - Succenturiate lobe.
 - Diabetes.
 - Placenta bilobata.

- Defective endometrium :
 - Endometritis.
 - H/o C-section/myomectomy.
 - H/o IVF (↑maternal age, twins).
- Smoking.

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APH : Diagnostic Algorithm

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Abruptio Placenta

00:15:10

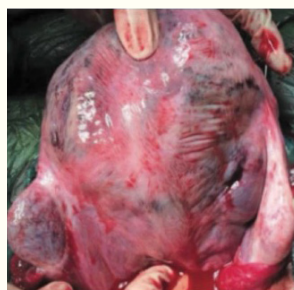
Abruptio is premature separation of normally situated placenta.

Types :

- Revealed variety.
- Concealed type : Blood collects in uterus behind placenta giving rise to couvelaire uterus (Bruised uterus).
- mixed type (m/c).

Note :

Placental separation releases **thromboplastin** which initiates contraction + leads to DIC



Couvelaire uterus

Presentation :

- Patient gives h/o trauma/high BP (PIH) :
- Pain in abdomen + bleeding P/V in T3 :
 - Blood dark red in color.
 - Not recurrent.
 - No warning hemorrhage.

Page classification :

- Grade 0 : Retrograde (Retroplacental clot seen after delivery).
- Grade 1 : Pain + bleeding + FHS normal.
- Grade 2 : Fetal distress ⊕.
- Grade 3 : Fetal death/maternal shock ± DIC seen.

Abruptio Placenta AKA Accidental Hemorrhage/Uterine Apoplexy

P/A finding :

- Ht. of uterus > POG.
- Uterus : Tense, tender and rigid.
- FHS : Not heard easily.
- Fetal part : Not palpable.

P/v examination :

Not C/I in abruptio but should be done after ruling out previa.

Investigation :

- Abruptio is a clinical diagnosis.
- On TVS (Not diagnostic) :
 - Placenta in upper segment
 - Retroplacental clot
 - Jello sign : Shimmering of placenta on maternal/transducer movement

Principle for mx :

- Aim : Deliver the patient ASAP (↑ Time interval between onset of abruptio & delivery → ↑ Chances of DIC)

management :

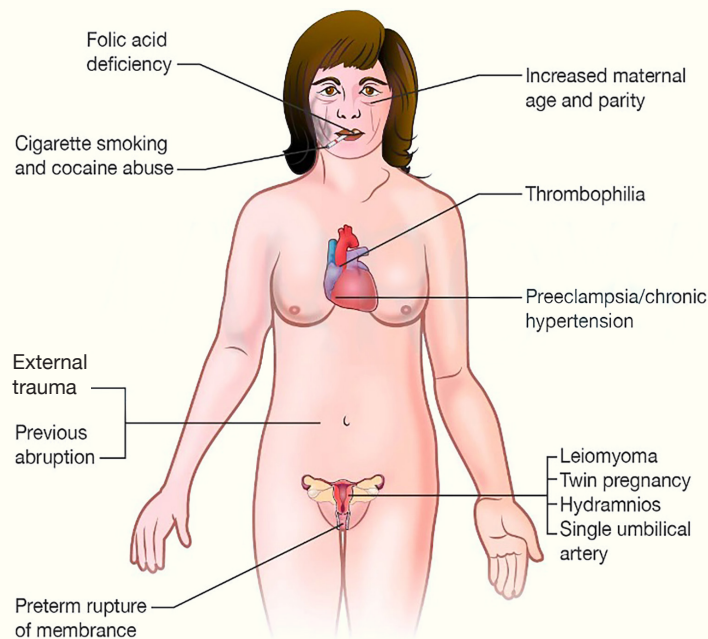
- Abruptio + fetal distress/maternal condition unstable or DIC : C-section.
- Abruptio + fetal death + mother unstable or DIC : C-section.
- Abruptio + fetal death + mother stable : vaginal delivery.
- Abruptio + DIC : Correct DIC $\xrightarrow{f/b}$ C-section.
- Abruptio + No emergency condition : Continue pregnancy upto 34 wks.
- If gestational age ≥ 34 wks + FHS normal + maternal vitals stable + imminent vaginal delivery : vaginal delivery.

Significance of ARM :

- It releases prostaglandins & expedites delivery.
- It temporarily decreases bleeding.
- It helps in making a diagnosis with blood stained liquor.

Risk Factors :

----- Active space -----

D/D of Bleeding ≥ 28 weeks :

1. Antepartum hemorrhage.

2. uterine rupture :

- H/o previous uterine Sx : C-section/classical caesarian/myomectomy.
- C/o abdominal pain + bleeding.
- P/A :
 - Palpable gap over uterus.
 - Fetal parts felt superficially.
- FHS : Bradycardia → Absent FHS.
- P/v :
 - Loss of fetal station OR
 - migration of station (E.g. : Initial +3 → +2)

3. Preterm labor :

- Blood mixed mucus discharge (Show).
- Progressive dilatation of cervix.

Couvelaire uterus :

- Concealed abruptio → Retroplacental blood → Invades myometrium → Bruised appearance of uterus.
- ↑ Risk of PPH.
- Not an indication for hysterectomy.

----- Active space -----

Obstetric Formulae

00:27:04

Gravida : Total number of times a female
has conceived (Past + present).

Parity : Total number of past pregnancies that
crossed the period of viability (28 weeks in India)

Twins/triplets :
Counted as 1.

Abortion : Any pregnancy loss < 20 weeks.

methods of Representation :

$G_a P_b$:

- a : Gravida.
- b : Parity.

$G_a P_{b+c}$:

- a : Gravida.
- b : Parity.
- c : Abortion.

GTPAL system : $G_a P_{f+g+h+i}$

- a : Gravida.
- f : No. of term deliveries (≥ 37 weeks).
- g : No. of preterm deliveries (20 - 36 weeks + 6 days).
- h : No. of abortions.
- i : No. of living children at present (Twins/triplets counted as 2 & 3 respectively).

Note :

- multipara : ≥ 2 previous births.
- Grand multipara : ≥ 5 previous births.

Antepartum Fetal Monitoring : NST & BPP

00:31:40

----- Active space -----

Case :

A pregnant female at 34 weeks c/o decreased fetal movement

Next step :

- Screening test : **modified BPS** > Non-stress test.
- Diagnostic test : Biophysical score.

Indications of NST :

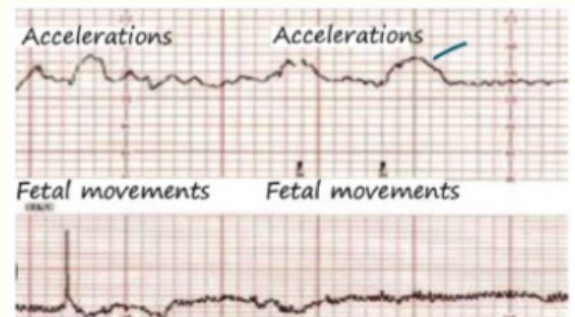
1. In pregnant females who c/o decreased fetal movements.
2. In all post term pregnancies.
3. In all high risk pregnancies beginning from 32 weeks.

Procedure :

- Patient lies in the left lateral position for 20 minutes.
- Fetal heart rate is continuously monitored and fetal heart accelerations are noted with respect to fetal movements.



Nonstress test (NST)



Antepartum Fetal monitoring

Components of NST :

- FHR : Normal range between 110 - 160 bpm
- FHR variability :
 - FHR shows a variability of 5 - 25 bpm.
 - If FHR is fixed, it has bad prognosis.
- FHR accelerations :
 - At < 32 weeks : ↑ in FHR by **> 10 beats/min** lasting for 10 seconds with fetal movement.
 - At ≥ 32 weeks : ↑ in FHR by ≥ 15 beats/min lasting for ≥ 15 seconds with fetal movement.

False positive rates of NST are high.

Next step

Biophysical/manning score (Diagnostic)

Inference :

A. In 20 mins : If there are ≥ 2 accelerations

Reactive/negative NST (Normal)

False negative rates of NST are very low :
No further testing required.

B. If in 20 mins there are < 2 accelerations

Repeat NST x 20 mins

Total = 40 mins
Still if < 2 accelerations

Non-reactive/positive NST

----- Active space -----

Biophysical Profile (manning Score) :

Done with help of USG for a period of 30 mins.

Five components :

1. **T** : Fetal **tone**.
2. **B** :
Fetal **breathing** movement.
3. **meningitis** :
Gross body **movement**.
4. **Always** :
Amniotic fluid (SVP).
5. **Notorious** : NST.

Criteria for biophysical profile :

- Nonstress test : 2 points if reactive (Atleast 2 episodes of FHR accelerations of at least 15 bpm & at least 15 s duration).
- Fetal breathing movements : Two points if one or more episodes of rhythmic breathing movements of > 30 s within a 30 min observation period.
- Fetal tone : Two points if one or more episodes of extension of a fetal extremity or fetal spine with return to flexion.
- Amniotic fluid volume : Two points if a single deepest vertical pocket ≥ 2 cm (Horizontal dimension should be atleast 1 cm).
- Fetal movement : Two points if ≥ 3 discrete body or limb movements within 30 mins of observation.

Interpretation & management :

Score (Out of 10)	management
10 or 8 with AFV normal	• Normal fetus : Repeat BPS weekly • Normal pH of fetus
6 with normal AFV	Equivocal $\left\{ \begin{array}{l} \geq 37 \text{ weeks : Deliver} \\ < 37 \text{ weeks : Repeat test in 24 hrs} \end{array} \right.$
8 or 6 with decreased AFV	Deliver ≥ 36 weeks (In these cases, chronic asphyxia is suspected)
4	Deliver ≥ 32 weeks
2 or 0	Immediate delivery (Certain of fetal asphyxia and significant fetal acidaemia)

} Risk of fetal acidemia

modified BPS :

Screening test that includes 2 parameters :

1. NST
2. AFI

Intrapartum Fetal Monitoring : CTG

00:40:29

----- Active space -----

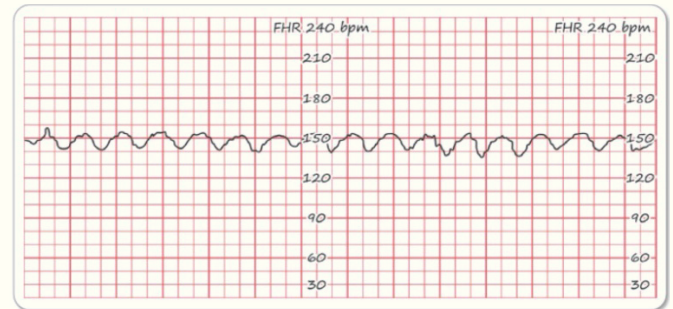
NST done during labour.

Graphs :

- upper graph : FHR.
- Lower graph : uterine contractions.

Fetal Heart Rate :

- Normal range : 110 – 160 bpm.
- < 110 bpm : Bradycardia.
- **Persistent fetal bradycardia** → Bad prognosis.



Sinusoidal heart rate pattern

Variability :

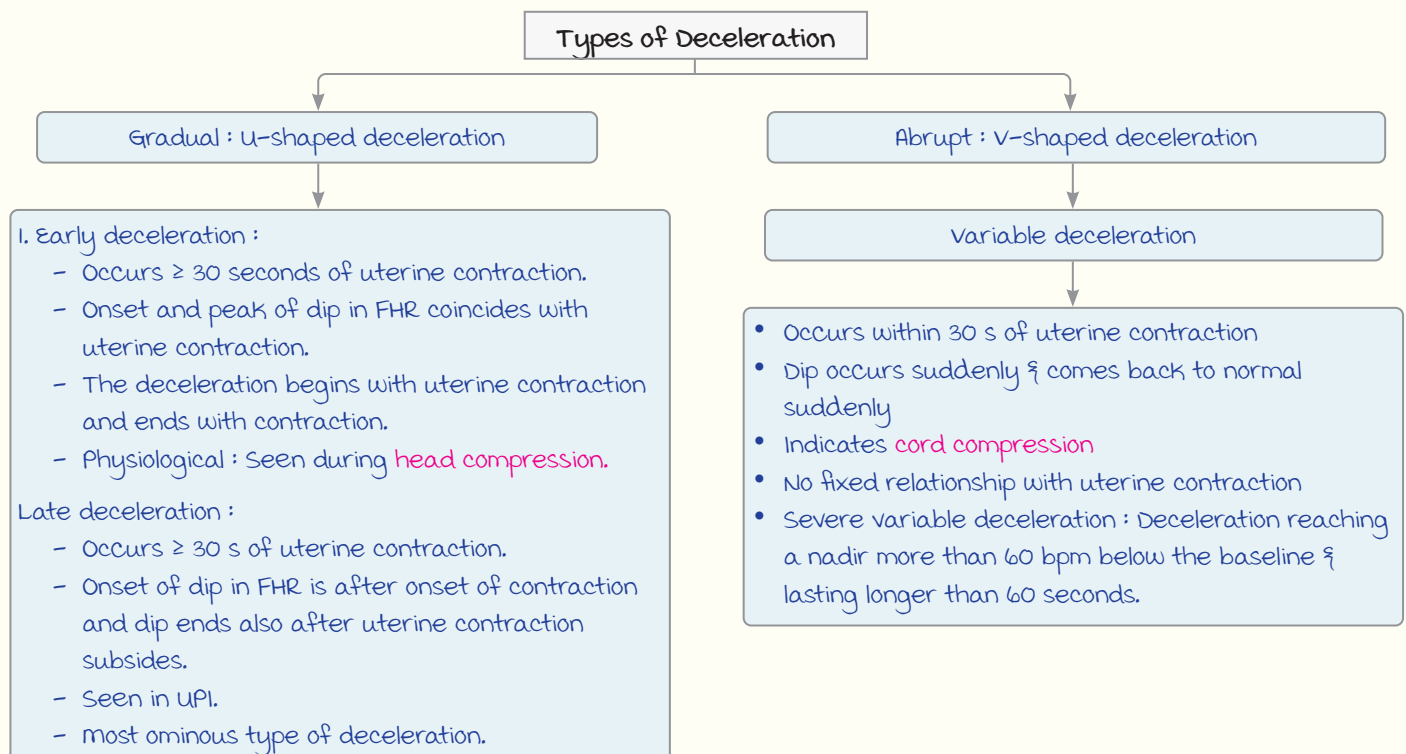
- Normal variability : 5 – 25 bpm.
- **Absent variability/sinusoidal pattern** ⊕ → Bad prognosis.

Deceleration :

↓ in FHR by 15 bpm x 15 seconds (may last upto 2 minutes).

Prolonged deceleration :

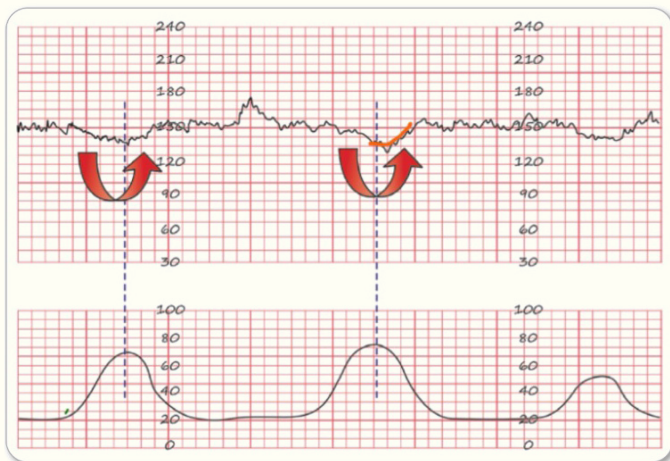
- Deceleration lasting 2 – 10 minutes.
- Cause : Hyperstimulation of uterus (Oxytocin/misoprost).



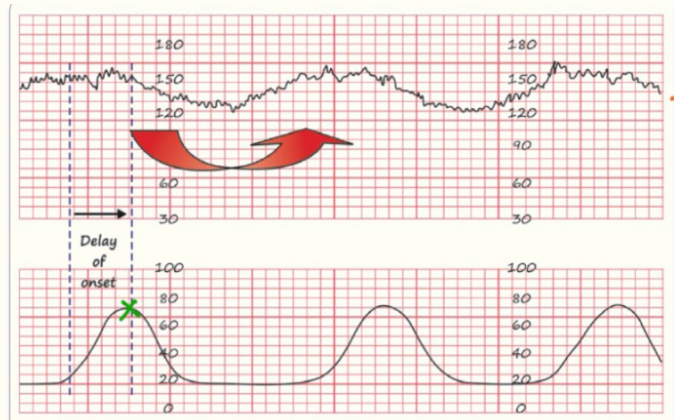
Note :

Deceleration lasting ≥ 10 mins → Change in baseline FHR.

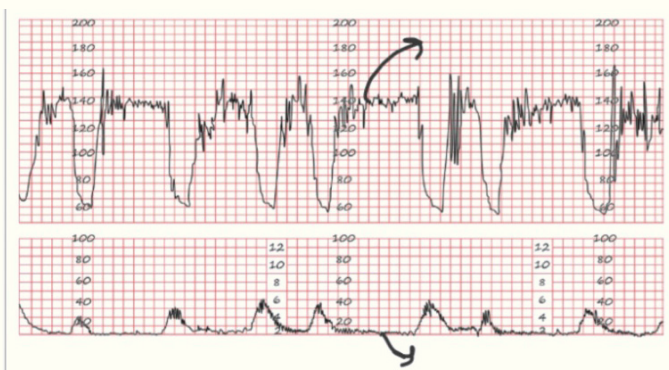
----- Active space -----



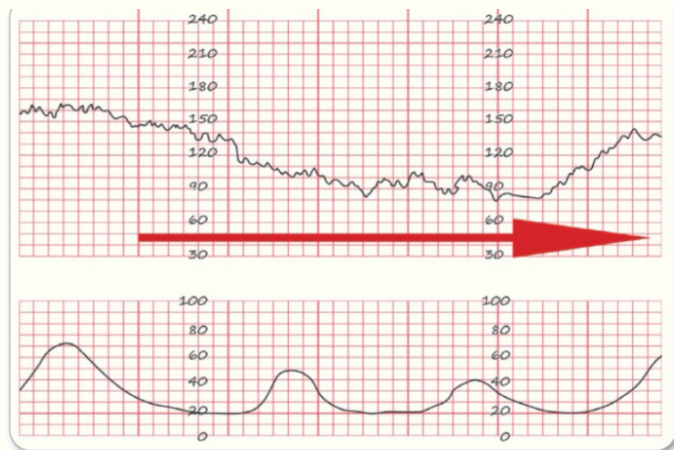
Early deceleration



Late deceleration



Variable deceleration



Prolonged deceleration

Normal CTG (Category I CTG) :

- FHR : 110 - 160 bpm.
 - Variability : 5 - 25 bpm.
 - Acceleration : +/-.
 - Early deceleration : +/-.
 - Late deceleration
 - Variable deceleration
- } Absent.

Category 3 CTG :**1. Sinusoidal heart rate pattern :**

Seen in fetal anemia → Rh incompatibility
 → TTTS
 → Vasa previa

2. Absent variability with any of the following :

- Persistent bradycardia
- Persistent late deceleration
- Persistent variable deceleration

management of category 3 CTG :

In utero resuscitation + emergency C-section

Steps for in-utero resuscitation :

1. Left lateral position.
2. O₂ by mask.
3. Stop oxytocin.

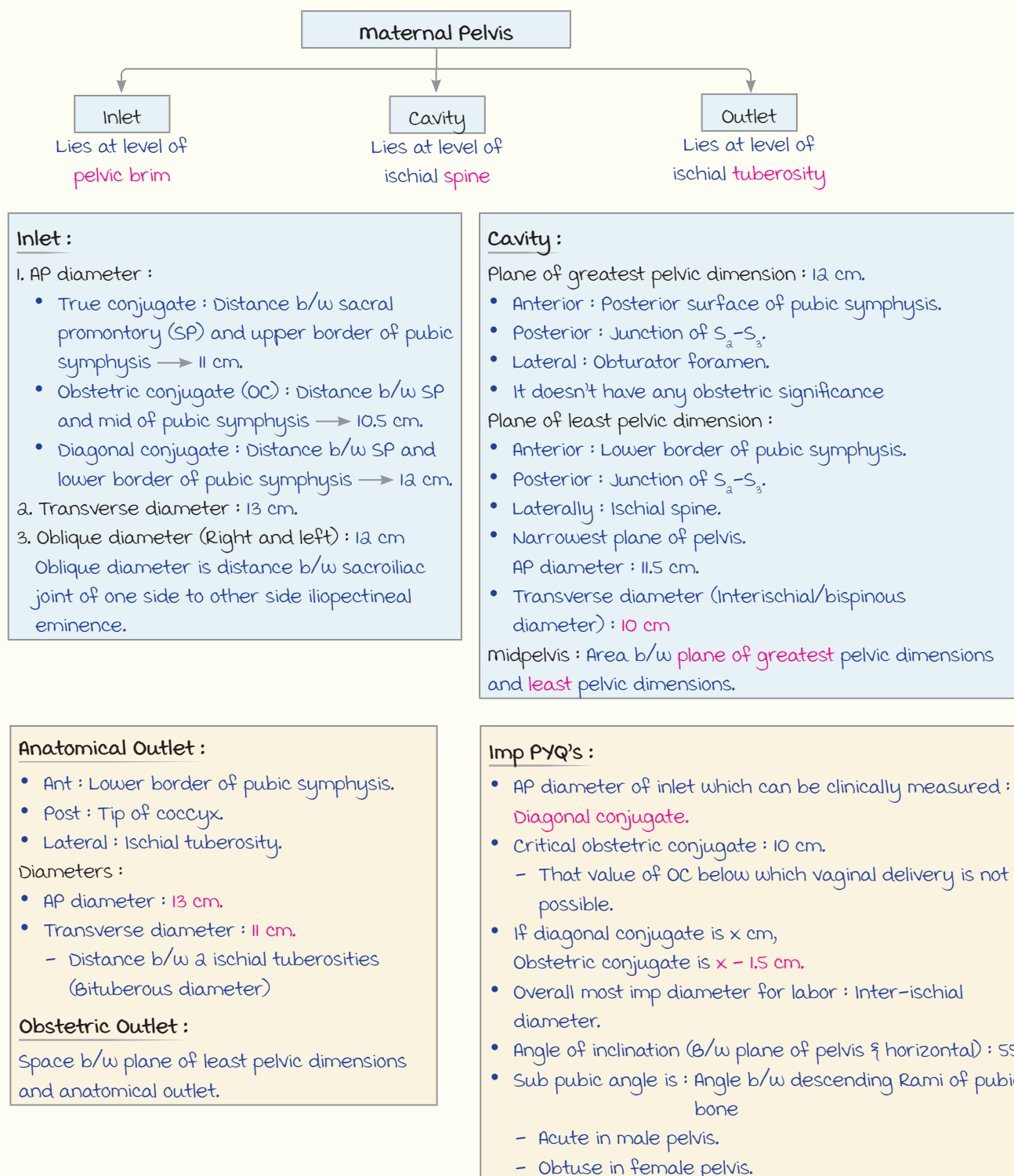
TTTS : Twin to twin transfusion syndrome

OBSTETRICS REVISION : PART 8

----- Active space -----

Maternal Pelvis

00:00:40

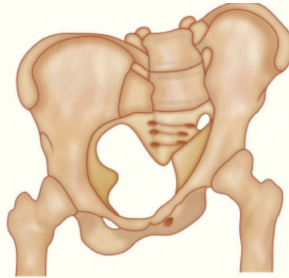


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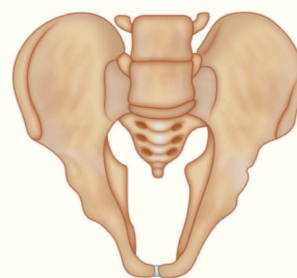
Contracted Pelvis :

If any of major diameters of pelvis is decreased by 1 cm or if :

- Obstetric conjugate < 10 cm : Contracted inlet.
- Interischial diameter < 8 cm : Contracted midpelvis.
- Bituberous diameter < 8 cm : Contracted outlet.
- Naegele's pelvis : Only one Ala is present, i.e., one Ala of sacral bone is absent.
- Robert pelvis : Both Ala are absent.
- mx of contracted pelvis : Always cesarean section.



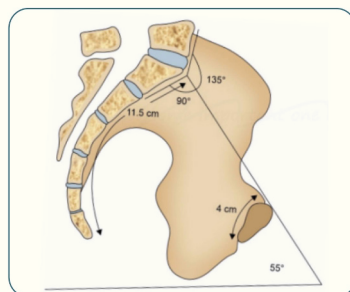
Naegele's pelvis



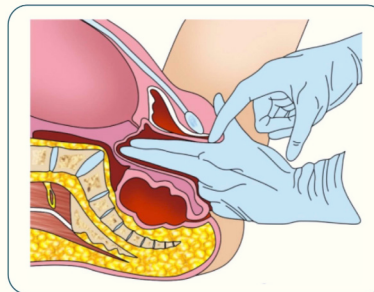
Robert pelvis

Caldwell moly Classification :

	Gynecoid (♀ type)	Android (♂ type)	Anthropoid (Ape like)	Platypelloid (Flat bowl like)
Diagram of inlet	m/c type of pelvis : 50% (m/c) 	20% 	25% 	Least common type of pelvis : 5%
Shape of inlet	Transverse oval	Heart shaped	Anteroposterior oval	Flat bowl like pelvis
Transverse and AP diameter	TD > AP diameter	TD > AP diameter	AP diameter > TD	TD >>> AP diameter
Ischial spine	-	Prominent	-	-
Side walls	Parallel and broad	Convergent	Parallel and narrow	Divergent
Subpubic angle	Obtuse	Acute	-	-



Angle of inclination



Diagonal conjugate being measured

Diagonal conjugate :

- Normal : 12 cm
- Contracted : OC < 10 cm.

Fetal Skull

00:11:52

----- Active space -----

AP Diameters :

- AP > transverse
- Longest AP : mentovertical (14 cm) > $\overbrace{\text{submentovertical} = \text{Occipitofrontal}}^{11.5 \text{ cm}}$
 - mentovertical : Encountered in brow presentation (LSCS only).

Transverse Diameters :

- Always smaller than AP diameters.
- In ascending order, TD of skull are :
 - Miss : Bimastoid $\rightarrow$ 7.5 cm.
 - Tina : Bitemporal $\rightarrow$ 8 cm.
 - So : Super subparietal $\rightarrow$ 8.5 cm.
 - Pretty : Biparietal $\rightarrow$ 9.5 cm
(m/c engaging transverse diameter)
- Diameters which are 9.5 cm :
 - Biparietal diameter.
 - Suboccipitobregmatic diameter.
 - Submentobregmatic diameter.

Parts of Fetal Skull			
Parts	Definition	Position of head	Engaging diameter
Vertex	Part of skull lying b/w anterior fontanelle and posterior fontanelle	Flexed	Suboccipitobregmatic (SOB) diameter : 9.5 cm
		Deflexed (Neutral)	Occipitofrontal diameter (11.5 cm)
		Slightly deflexed	Suboccipitofrontal diameter (10 cm)
Brow	Part of skull b/w anterior fontanelle and root of nose (with supraorbital ridges)	Partially extended	• mentovertical (mv) diameter : 14 cm • Always do cesarean
Face	Part of skull b/w root of nose and chin	Fully extended	Submentobregmatic (SMB) diameter : 9.5 cm or Submentovertical (Smv) diameter: 11.5 cm

moulding :

moulding : It is the alteration of the shape of the skull while passing through the resistant birth passage during labor.

Grading : There are three gradings of moulding.

1. Grade 1 : Skull bones touch but do not overlap.
2. Grade 2 : Skull bones overlap but can easily be separated.
3. Grade 3 : Fixed overlapping of skull bones.

Grade 1 can be seen normally.

Grade 3 :

moulding + slow progress of labor indicates CPD

Lie :

Relationship b/w long axis of centralized uterus and long axis of fetus

Note :

Before commenting on lie - always correct the dextrorotation of uterus.

- m/c lie : Long axis of fetus and long axis of uterus coincide with each other.
- Oblique lie : Long axis of fetus makes an angle with long axis of uterus.
- Transverse lie : Fetal long axis and maternal long axis are 90° to each other.
 - It is the m/c cause of cord prolapse.
 - management of transverse lie is C-section.

Presentation :

It is that part of fetus which lies at lower pole of uterus.

Cephalic
(Normal)

malpresentations :

- Breech (m/c).
- Shoulder presentation :
 - Seen in transverse lie.
 - mx : C-section.

Presenting Part :

The part of presentation which lies directly over internal os and hence, is the part felt first on P/v examination.

In cephalic presentation :

Presenting part could be :

- vertex (m/c) : Seen in fully flexed, deflexed head, partially flexed.
- Brow : Seen in partially extended head.
- Face : Seen in complete extension.

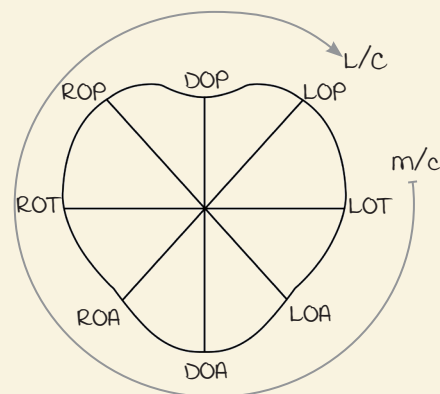
Denominator :

Bony point of reference on the presenting part which comes in relationship to the maternal pelvis.

Presenting Parts	Denominator
Vertex	Occiput
Breech	Sacrum
Brow	Frontal eminence/ bone
Face	mentum (Chin)

Position :

Relationship of denominator to maternal pelvis

**Key Points :**

- m/c position : LOT > LOA.
- m/c occipito-anterior position : LOA.
- m/c occipito-posterior position : ROP.
- m/c position in breech : Left sacro anterior.
- m/c position in face : Left mento anterior.

Unstable lie :

----- Active space -----

Lie of fetus keeps changing beyond 37 weeks.

Causes : Idiopathic > placenta previa > polyhydramnios & abnormal pelvis.

Note :

On P/V examination → Fully flexed head : **Posterior** fontanelle felt.
 → Slightly deflexed head : **Anterior** fontanelle (Sinciput) felt.

Attitude :

Relationship b/w different parts of the fetus.

Normal attitude : Flexion.

Abnormal attitude : Face & brow.

Station :

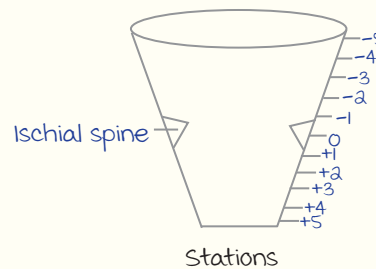
Position of fetal head with respect to **ischial spine**.

'0' station : Head is engaged

'+2' station : Forceps & vacuum can be applied.

Ischial spine : Site for **pudendal nerve** block.

(Ligament pierced : Sacrospinous ligament).

Cephalopelvic Disproportion (CPD) :

Best method for diagnosis : Trial of labour.

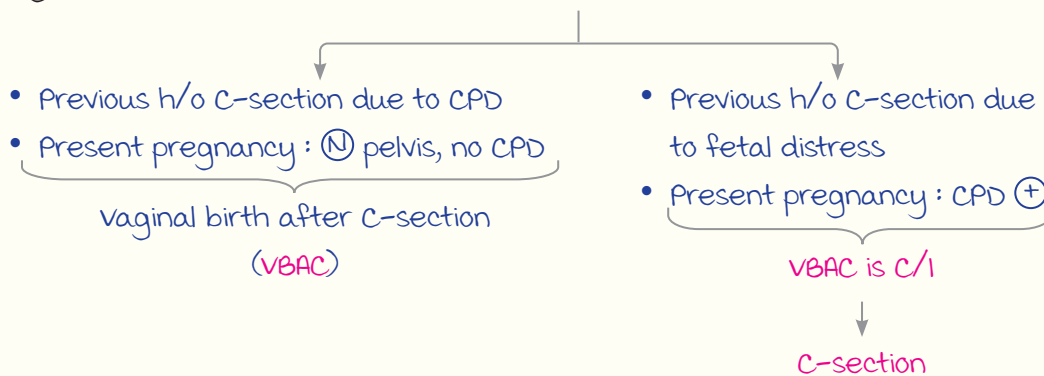
Radiological investigation : **mRI** > CT > X-ray pelvis.

Note :

Pelvic assessment :

- mainly done in primigravida.
- Done at 38-39 weeks.

management :



----- Active space -----

1st & 2nd Stages of Labour

00:26:02

1st stage (Latent Phase) :

Begins : Onset of true labour pains.
 Latent phase is not plotted on partogram.
 mainly concerned WITH : Effacement & dilation of cervix.

Average duration :

- Primi : 12 hrs.
- multi : 8 hrs.

Prolonged latent phase :

- Primi : ≥ 20 hrs.
- multi : ≥ 14 hrs.

mx : **Therapeutic rest** (Never do C-section).

1st stage (Active Phase) :

Begins :

- Old WHO guidelines : 4 cm.
- New guidelines : 5 cm.
- ACOG guidelines : 6 cm.

mainly concerned with :

1. Dilation of cervix.

2. Descent of fetal head.

minimum dilation :

Dilation	Descent
Primi = 1.2 cm/hr	1 cm/hr
multi = 1.5 cm/hr	2 cm/hr

Old WHO guidelines : 1 cm/hr.

Stages of labor = Normal & Abnormal

Slow active phase : **Protracted active phase**

• Rule out :

1. Occipitoposterior position :
 mx $\rightarrow$ Wait & watch.

2. CPD :

mx $\rightarrow$ C-section

• mx : Augmentation of labour
 (ARM + Oxytocin).

• Active phase arrest :

- Dilation

+ No change in Cx dilatation for 4 hrs
 WITH adequate contraction and for 6
 hrs WITHOUT adequate contraction.

+ Ruptured membranes.

- mx : C-section.

When to suspect CPD :

1. Caput +
2. moulding (Grade 3) +
3. Slow progress of labour.

Second Stage of labor :

- Begins : Full dilation of cervix
- Ends : Delivery of baby.
- Avg duration :
 - Primi : 50 mins.
 - multi : 30 mins.
- Prolonged 2nd stage :

	Without epidural	With epidural
Primi	≥ 3 hrs	≥ 4 hrs
multi	≥ 2 hrs	≥ 3 hrs

• mx of prolonged 2nd stage :

Station

Above + 2

At/below + 2

C-section

Instrumental delivery

- Bandl's ring present
 (Indicates obstructed labor)
- moulding/caput present
 with second stage arrest/
 prolonged (Indicate CPD)

C-section

Obstructed Labour

00:32:20

----- Active space -----

Causes : Transverse lie, conjoined twins etc.

Clinical Features :

maternal dehydration, exhaustion, tachypnea, tachycardia.

On P/A : UUS is thick, tender & tonically contracted.

- LUS : Thinned & stretched.
- Bandl's ring (Depression/groove felt b/w UUS & LUS).

On P/V :

- Hot and dry vagina.
- Caput ++ or +++
- moulding ++ or +++
- Hematuria +/-.



Bandl's ring

Mx :

- C-section immediately.
- Do not give oxytocin.

Complications :

1. Rupture of uterus.
2. VVF (After 7 - 10 days).

Augmentation of Labour :

Augmentation means **accelerating** the progress of labor.

Indication : Protracted active phase after ruling out occipitoposterior position and CPD.

Methods :

1. Artificial rupture of membranes done by Kocher forceps.
2. Oxytocin : Given at 6 mU/min increased at interval of 20-40 minutes by 6 mU.

C/I of ARM :

1. maternal HIV infection.
2. maternal genital herpes infection.
3. Intrauterine death of fetus.
4. vasa previa/funic presentation.

Note :

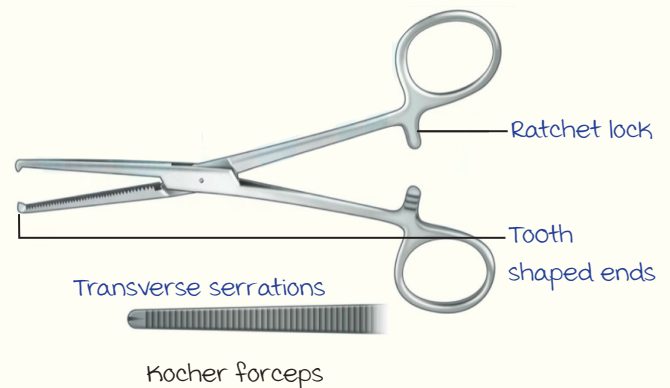
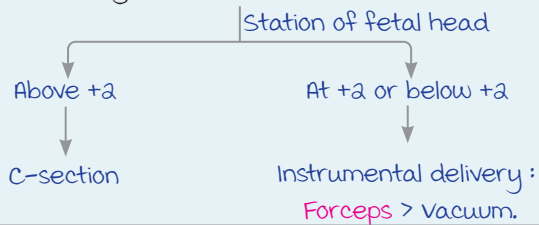
In polyhydramnios → **Controlled** rupture of membranes.

mx of Fetal Distress in Labor :

In first stage of labor : (Dilatation < 10 cms).

mx : C-section.

In second stage of labour (Dilatation is 10 cm) :



In Obs, there are 2 rings :

Bandl's ring	Constriction ring/Schroeder ring
Retraction ring	Constriction ring
Due to obstructed labor	Injudicious use of oxytocin leads to localised spasm of uterus
B/w UUS and LUS	Can be seen anywhere
Seen in 2nd stage of labor	1st or 2nd stage of labor
Felt and seen P/A	Felt P/V
A/w maternal dehydration and fetal distress	mother and fetus are normal
Relieved by C-section	• Relieved by sedation • If ring reappears then do a lower vertical C-section

----- Active space -----

Uterine Contractions

00:36:36

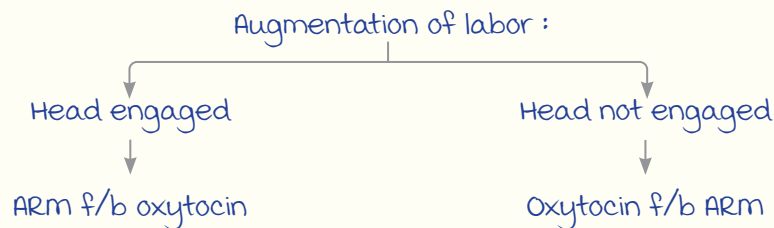
Prerequisite For Labour :

3Ps :
 → Passenger : Fetus.
 → Passage : Pelvis.
 → Push : uterine contractions.

Adequate contractions :

- Frequency : 3 contractions in 10 mins.
- Duration : 40-45 seconds each.
- Pressure generated : 200-250 montevideo units in 10 min.

management of inadequate contractions:



Note :

Interval b/w P/V examination in labor:
 → Without oxytocin : 4 hrs.
 → With oxytocin : 2 hrs.

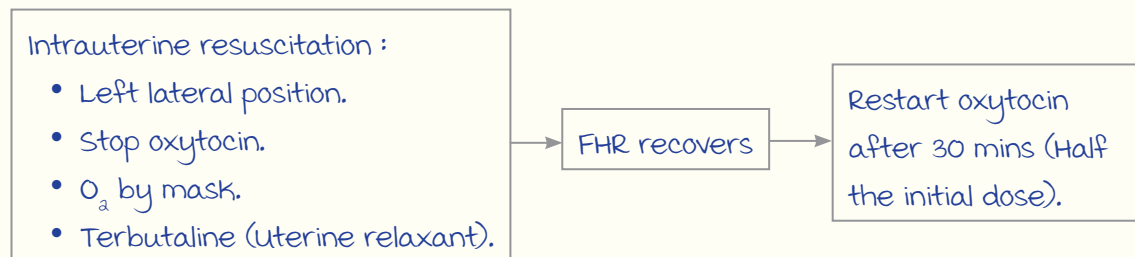
Tachysystole :

≥ 5 contractions in 10 minutes (↓ Blood supply to fetus → Fetal distress).

Cause : Excessive use of oxytocin/misoprostol (iatrogenic).

CTG findings : Prolonged deceleration.

management :



Partogram

00:39:52

modified WHO Partogram :

Based on old WHO recommendations.

Active phase : Begins at 4 cm.

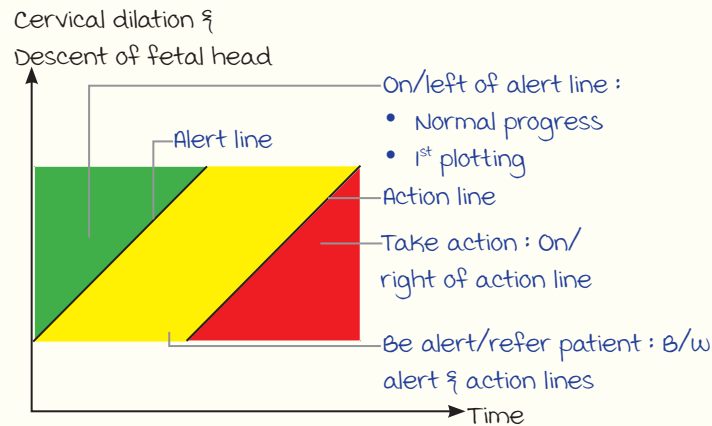
Latent phase & 2nd stage : Not plotted.

Two lines : Given by Philpott.

- Alert line : minimum dilation in active phase $\rightarrow$ 1 cm/hr.
- Action line : Drawn 4 hrs to right of alert line

----- Active space -----

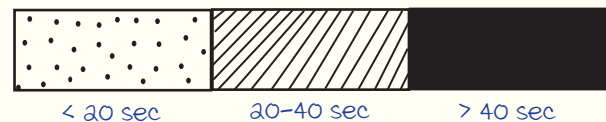
Progress :



Size of boxes :

- Small box : 30 mins.
- Large box : 1 hr.

Representation of duration of contractions :



Symbols on partogram :

- C : Clear amniotic fluid.
- m : meconium stained amniotic fluid.
- B : Blood stained amniotic fluid.
- X : Dilation of cervix.
- O : Descent of fetal head.

Labor Care Guide/Next Generation Partogram :

Based on new WHO guidelines.

Features :

- Active phase & 2nd stage of labor plotted.
- Active phase begins at 5 cm.
- No alert & action lines :
 - Dilation during active phase can be < 1 cm/hr.
- upper limit for every cm dilation defined separately.

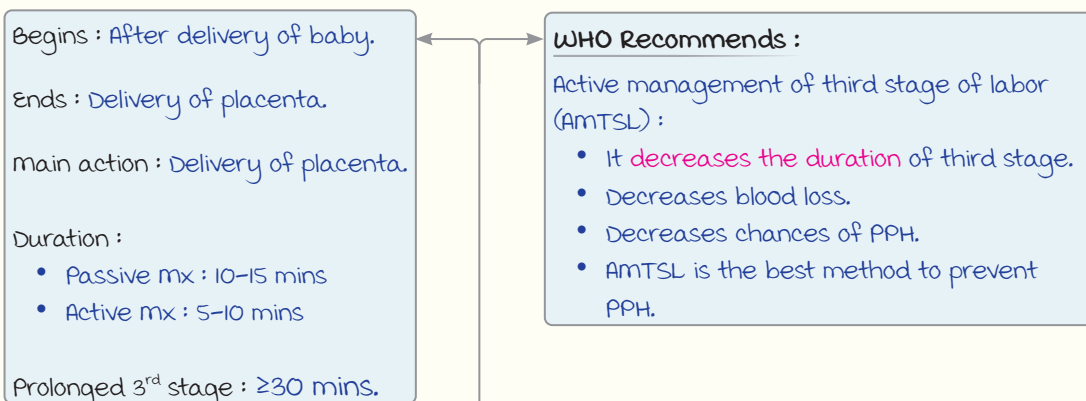
Supportive care section : New addition.

- Companion.
 - Oral fluid.
 - Pain relief.
 - Posture : mobile/supine/left lateral.
- } Yes/No.

----- Active space -----

3rd Stage of Labour

00:51:00

**3rd Stage of Labor****Steps in AMTSL :**

1. Inj. uterotonic within **1 minute** of delivery of baby or immediately after delivery of anterior shoulder of baby (most important step)
2. **Delayed cord clamping** (Clamping cord in 1-3 minutes of delivery)
3. Delivery of placenta by **controlled cord traction** (modified Brandt Andrews technique)
4. Intermittent assessment of uterine tone (Earlier : uterine massage).

Note : Early cord clamping is not a part of AMTSL.

Conditions of Early Cord Clamping :(Clamping within **1 minute** of delivery)

- If neonatal resuscitation is needed (Birth asphyxia) } **Absolute indication**
- K/C/O congenital heart disease } **Relative indication**
- If in RH -ve female indirect Coombs test is negative (Non-isoimmunized pregnancy) }
- In preterm infants } **Delayed cord clamping**
- In COVID 19 + patients }
- In macrosomic babies }
- In post-term cases }
- In HIV positive }



Controlled cord traction

uterotonics :

Recommended by WHO :

Oxytocin : 10 IU IM or IV infusion.

If oxytocin is not available following can be given :

1. methylergometrine : (methergine) **0.2 mg IM**
 - Never given IV as it can lead to hypertension
2. Syntometrine : Fixed dose combination of 5U oxytocin and 0.5 mg methylergometrine.
3. Carbetocin : Synthetic oxytocin.
 - 100 mcg slow IV.
4. misoprost PGE₁ :
 - 600 mcg oral, sublingual or P/v for AMTSL.
 - WHO recommends distribution to all pregnant females during antenatal period.

Note :4th stage of labor :

- 1-2 hrs period after delivery of placenta.
- Physiological chills in this stage (D/t blood loss & misoprost).

Note :

Carboprost : PG F-2α is used for management of PPH.

AMTSL :

----- Active space -----

Best method to prevent PPH.

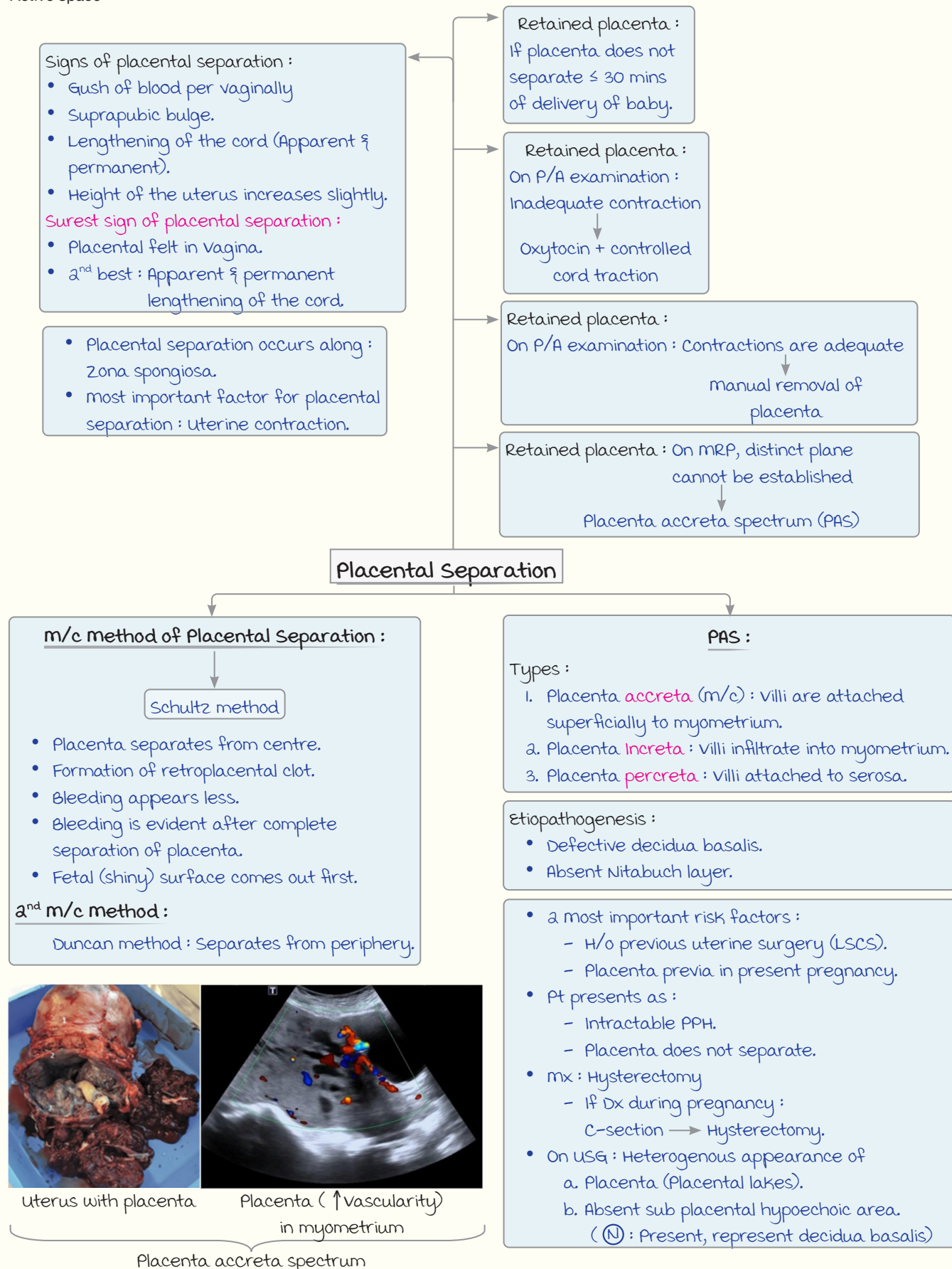
Not included :

- Early cord clamping.
- Balloon tamponade.
- Bimanual compression.
- uterine massage.

Disadvantage : Retained placenta (D/t administration of uterotonic before delivery of placenta).

manual removal of placenta : Done if cord breaks during controlled cord traction.

----- Active space -----



OBSTETRICS REVISION : PART 9

----- Active space -----

Malpresentation and Malposition

00:00:36

Transverse lie :

- Presentation : Shoulder
- Vaginal delivery : Not possible
- maximum risk of : Cord prolapse
- If patient comes in antenatal period with transverse lie : ECV (At ≥ 36 weeks)
- If patient with transverse lie :
 - Comes in latent phase : ECV
 - Comes in active phase : C-section
- m/c/c of transverse lie : Prematurity
- m/c/c of transverse lie in term pregnancy : Placenta previa

Brow :

- Engaging diameter : mentovertical (14 cm)
- mx : C-section
- Anterior fontanelle and supraorbital ridge palpable

Face :

- m/c cause : Anencephaly
- Denominator : mentum
- mentoposterior :
 - mx : C-section
 - vaginal delivery
 - Instrumental delivery
 } Not possible
- mentoanterior :
 - mx : vaginal delivery
 - Engaging diameter: Submentobregmatic diameter (9.5 cm)
 - Head of baby is born by :
 - Forceps : Flexion
 - Vacuum : Not possible
- m/c seen in platypelloid pelvis

malpresentation and malposition

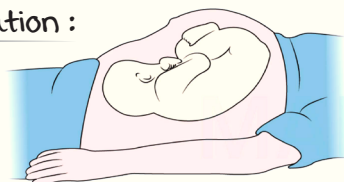
Occipito Posterior Position :

- m/c OP position: ROP
- m/c cause of OP position: Deflexed head
- 2nd m/c cause of OP position: Android pelvis
- m/c outcome in position : Head of baby rotates $3/8^{\text{th}}$ of circle $\rightarrow$ DOA $\rightarrow$ vaginal delivery
- Best management of OP : Wait & watch

Deep Transverse Arrest :

- m/c pelvis : Android (D/t prominent ischial spines)
- m/c level : Ischial spine (Internal rotation occurs)
- Best management : C-section

On P/A Examination :



Occipitoposterior

PYQs :

On P/A Examination in OP position :

- Subumbilical flattening
- Fetal limbs are toward umbilicus
- Fetal back is toward mother's flank
- FHS are best heard in mother's flank area

----- Active space -----

Breech

00:05:13

Types of Breech :

Flexed/complete breech	Frank Breech (m/c)	Knee presentation	Footling presentation (L/c)
- Hip flexed - Knee flexed	- Hip flexed - Knee extended	- Hip extended - Knee flexed	- Hip extended - Knee extended
P/v : Heel + buttocks + genitalia + anal opening	P/v : Buttocks + genitalia + anal Opening	P/v : Knees felt	P/v : Feet is felt
m/c in multigravida	Least chances of cord prolapse. m/c in primigravida	- Chances of cord prolapse increased - C-section is done - ECV not done	maximum chances of cord prolapse ↓ C-section

- Overall maximum chance of cord prolapse : Transverse lie > footling > Knee.
- Breech with extended face is called as Stargazer breech.

Breech

maneuvers in breech

- Best mx of breech : C-section
- Vaginal delivery is possible in breech aka assisted breech delivery
- Absolute indication for cesarean in breech :
 - Footling presentation
 - Knee presentation
 - stargazer breech
 - Previous h/o c-section
 - Breech with macrosomia
 - Preterm breech

Part	maneuver
For delivery of buttocks	Groin traction (For flexed breech)
For delivery of extended legs in frank breech	P inard maneuver (Pressure on p opliteal fossa)
Shoulder	Lovset's maneuver

Delivery of after coming head in breech :

- Head is the last part to be delivered in breech
- m/c cause of death in breech : Intracranial haemorrhage.
- Head is born by movement of : Flexion
- mostly breech is dorso-anterior. In dorso-anterior breech methods of delivery of fetal head are
 - Burn marshall technique
 - malar flexion and shoulder traction (mauriceau-Smellie-Veit technique)
 - Piper's forceps
- In case of dorso-posterior breech : Prague maneuver

External Cephalic Version :

----- Active space -----

Procedure :

- Per-abdominally, baby is rotated from transverse lie or breech and made cephalic
- Done in OPD
- Anesthesia not needed. Tocolytics are used to relax

ix :

Breech/transverse lie in singleton pregnancy.

- In Rh negative females : Anti-D given (300 mcg).
- No routine induction of labor after ECV
- Should be done under continuous fetal monitoring
- Previous cesarean sections : Relative C/I

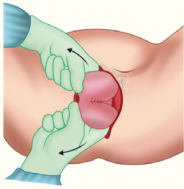
External Cephalic Version

Pre-requisites for ECV	Absolute contraindications for ECV
Period of gestation ≥ 36 weeks : Usually done at 37 weeks (Less chances of reversion)	Oligohydramnios
Liquor should be adequate and membranes should be intact	Ruptured membranes
Can be done in early labor (Latent phase) if membranes are intact	Active phase of labor
Singleton pregnancy.	Twin/multifetal pregnancy
Fetal heart rate should be normal on CTG.	Abnormal fetal heart rate on CTG

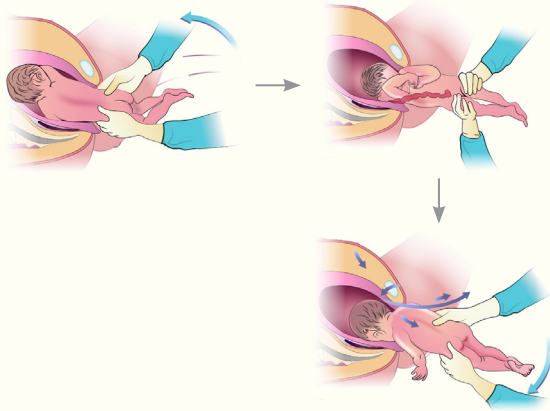
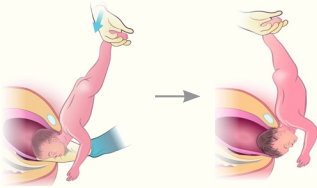
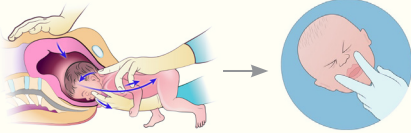
Relative contraindications for ECV :

- PIH
- Heart disease in mother
- IUGR
- Previous LSCS

Maneuvers in Breech :

maneuver/technique	Done in	To Deliver
Groin traction 	Flexed breech	Buttocks & legs

----- Active space -----

maneuver/Technique	Done in	To Deliver
Pinard maneuver 	Frank breech	legs
Lovset's maneuver 	-	shoulder
Burns-marshall technique 	Dorso-anterior breech	After coming head in breech
mauriceau-Smellie-veit technique : malar flexion + shoulder traction 		
Piper's forceps.		
Prague maneuver 	Dorso-posterior breech	

Induction of Labour

00:16:43

----- Active space -----

Bishop Score :

mnemonic	Parameter	0	1	2	3
Delhi	Dilatation of cervix	Closed	1-2 cm	3-4 cm	> 5 cm
Police	Position of cervix	Posterior	mid	Anterior	—
Employed	Effacement of cervix	30%	40-50%	60-70%	≥ 80%
Special	Station of fetal head	Above -2	-2 station	-1, 0 station	Below ischial spine
Commandos	Consistency of cervix	Firm	medium	Soft	—
Score (maximum : 13)	≤ 5 = Poor score : Ripening should be done before IOL ≥ 6 = IOL can be done ≥ 9 = max. success of IOL				

Indication of Labor

= means to initiate process of Labor

modified Bishop Score :

- Effacement of cervix is replaced by length of cervix (measured by TVS) : Length of 2-4 cms
→ Score : 1
- Simplified Bishop score includes only three parameters (DES) :
 - a. Dilatation of cervix
 - b. Effacement of cervix
 - c. Station of fetal head
 } maximum score = 9

C/I for IOL :

Contraindications for induction of labor :

- Severe CPD
- Contracted pelvis
- Transverse lie, Brow presentation, Face (mento-posterior position)
- Fetal distress : Nonreassuring fetal heart rate
- Placenta previa
- Classical cesarean section
- Previous hysterotomy
- Previous myomectomy
- Active genital herpes infection

methods of Inducing Labor :

mechanical methods (Best) :

- Foley's Catheter (Filled with 30-50 mL of NS) or extra amniotic saline infusion. It is best method for IOL in previous cesarean patients
- Stripping of membranes } ↑ PG

medical method :

- misoprostol = PGE1 = Tab. 25 mcg given 4 hourly P/V. (max. 6 doses)
- Dinoprostone = PGE2 = Available in two forms :
 - Cerviprime gel : 0.5 mg given 6 hourly, max. doses = 4
 - Cervidil = Slow release dinoprostone formulation. Contains 10 mg dinoprostone placed in posterior vaginal fornix

IOL in patients with previous C-section : mechanical methods > Dinoprostone



Cervidil

----- Active space -----

Instrumental Delivery

00:24:06

Forceps and Vacuum :**Indications :**

1. Dilatation of cervix : Full dilatation
2. Station of fetal head : $\geq +2$
3. Ruptured membranes
4. Episiotomy given
5. Consent taken from patient
6. Empty bladder

Indications of Instrumental Delivery :

1. Fetal distress in 2nd stage of labour
2. Prolonged 2nd stage
3. Maternal exhaustion

Prophylactic use :

- Heart disease
 - PIH
- } To cut short 2nd stage of labour

Advantages of vacuum over Forceps

1. vacuum can rotate head of fetus
2. vacuum can be applied when cervix ≥ 6 dilated

Vacuum cannot be used :

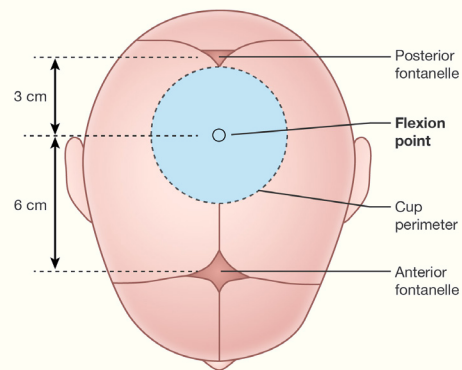
1. Prematurity (ICH)
2. mentoanterior
3. After-coming head of breech

Conditions where forceps are preferred over vacuum :

- Heart disease
- Fetal distress

Absolute C/I of instrumental delivery :

1. CPD/contracted pelvis
2. Obstructed labour
3. Transverse lie
4. Brow
5. mentoposterior face presentation



- vacuum is applied at flexion point.
- Flexion point lies :
 - 3 cm anterior to posterior fontanelle (Preferred)
 - 6 cm posterior to anterior fontanelle
- Initial pressure : 0.2 Kg/cm^2
- max pressure : $0.8 \text{ Kg/cm}^2 \leq 600 \text{ mmHg}$
- vacuum leads to more of fetal injury

Forceps :

- Types of forcep applications :
 - Low forcep : $\geq +2$ station
 - Outlet forcep : $+3, +4, +5$ station
 - Head on perineum
 - Scalp visible at introitus
- (Only outlet forceps : Wrigley's)

Advantage of forcep over vacuum

1. Preterm
2. mentoanterior presentation
3. Delivery of head in breech

Forceps lead to more **maternal** injury**Fetal complications are more with forceps:**

- Facial nerve palsy
- Brachial plexus injury
- Corneal injury
- Intraventricular hemorrhage

Traction force applied in forceps :Primi = maximum **20** KgMulti = maximum **13** Kg

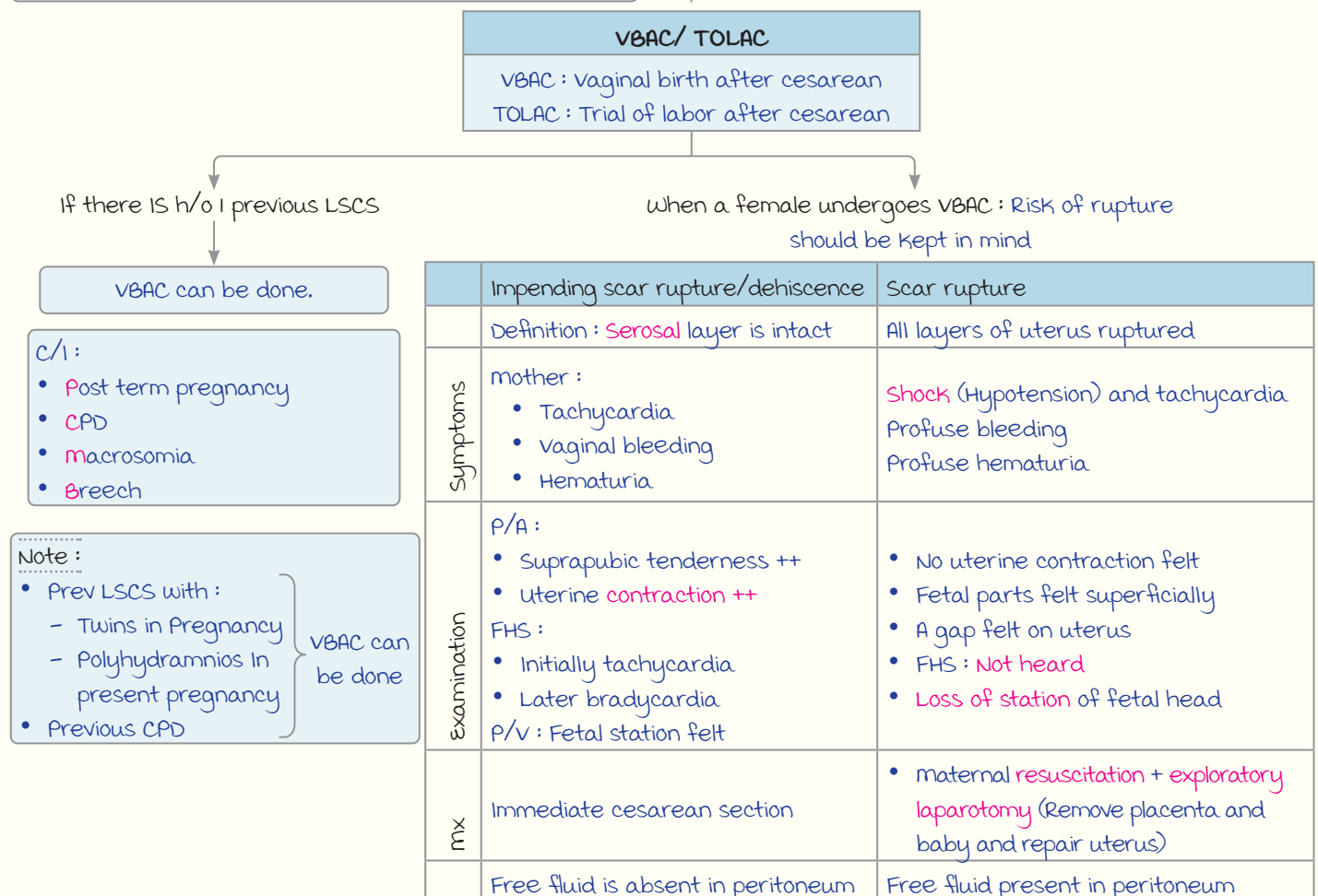
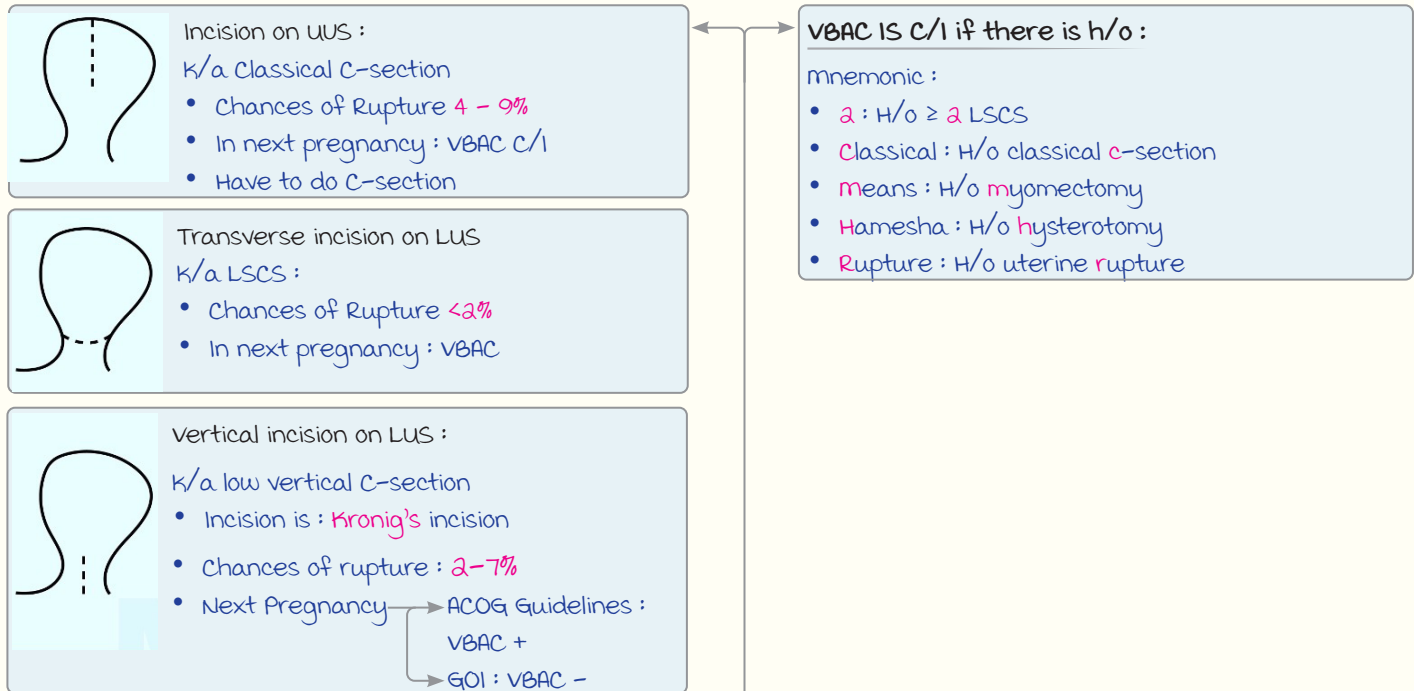
Forceps is applied along : Occipitomenal diameter

VBAC/TOLAC

00:31:40

----- Active space -----

Types of Cesarean Section :



----- Active space -----

Indications for classical Cesarean section :

- **Cancer cervix** : Absolute indication.
- Severely **contracted pelvis**.
- Lower uterine segment adhered to bladder.
- Placenta previa with bleeding.
- Lower uterine segment is not formed : Preterm C-section.
- Perimortem C- section.

Key Points about Cesarean Section :

Level of analgesia : **T4**.

- In epidural analgesia/painless labour : T10.

Position : **Supine** with **15°** left lateral tilt of table/place wedge under the right hip.

Time gap between cesarean section & next pregnancy :

- Ideal : **18** months.
- minimum : **6** months.

Elective cesarean section : Done at **39** weeks (Non-medical reasons).

Postpartum Hemorrhage

00:42:47

----- Active space -----

Definition of PPH :

Blood loss ≥ 500 mL after vaginal delivery.

OR

Blood loss ≥ 1000 mL after cesarean section

within 12 weeks of delivery

ACOG definition :

Blood loss ≥ 1000 mL irrespective of type of delivery

OR

Bleeding associated with signs and symptoms of **hypovolemia**.

Clinical definition :

A fall in **hematocrit** $\geq 10\%$ between admission and postpartum period

OR

Excessive bleeding needs **blood transfusion**

- m/c cause of **maternal mortality** in India $\rightarrow$ PPH
- m/c cause of **shock after delivery** $\rightarrow$ PPH
- Primary PPH : Occurs within **24** hours of delivery.
m/c cause : **Atonic** PPH
- Secondary PPH : Occurs ≥ 24 hours but within 12 weeks of delivery
m/c cause : Retained placental tissue
- Causes of PPH - 4T'S
 - T = Decreased Tone of uterus (m/c)
 - T = Trauma
 - T = Retained Tissue
 - T = Thromboplastin, i.e bleeding disorder

uterotonics used in management of PPH :

1. Oxytocin :

- Dose : 20-40 IU
- Route : **Im or IV infusion**
 - Never give IV bolus $\rightarrow$ Hypotension & cardiac arrest
 - Infusion should not be prepared with 5% D $\rightarrow$ Water intoxication.

2. methylergometrine : If oxytocin is unavailable

- Dose : 0.2 mg Im
- C/I :
 - After delivery of 1st twin
 - Heart disease
 - Pre-eclampsia
 - Eclampsia
 - Rh-ve pregnancy.
 - PPH (D/t $\uparrow$ BP)

3. misoprostol :

- Dose & Route
 - WHO : 800 mcg Sublingual/oral
 - ACOG : 600-1000 mcg oral/sublingual/per rectal
- **Cannot be given IV**
- S/E : Hyperthermia
- C/I : Previous C-section patients
- Asthma not a C/I

4. Carboprost (PGF_{2 α}) :

- AKA Dinoprost/Hemabate
- Dose :
 - 250 mcg Im repeated every 15 - 90 mins
 - maximum 8 doses/2 mg
- C/I : Bronchial asthma

5. Tranexamic acid : Given **within 3 hrs** of hemorrhage

Drugs not used in mx of PPH :

1. Carbetocin : used in AMTSL
2. Dinoprostone (PGE₂) : used in IOL

Initial management of PPH

maternal resuscitation :

- massage uterus** (P/A done)
- Insert **2 large bore IV cannulae** (16 or 14 No.).
- Give **1000 mL crystalloids**.
- Place foley catheter for fluid input-output charting.
- Investigations $\rightarrow$ CBC, blood grouping and Rh typing.
- Arrange for blood and blood products

Perform a per abdominal examination followed by per vaginal examination

Normal uterine tone

Traumatic PPH

P/V examination

To check for

Laceration/
Hematoma

Repair/Appropriate management

Reduced uterine tone/tone **maintained only with massage**

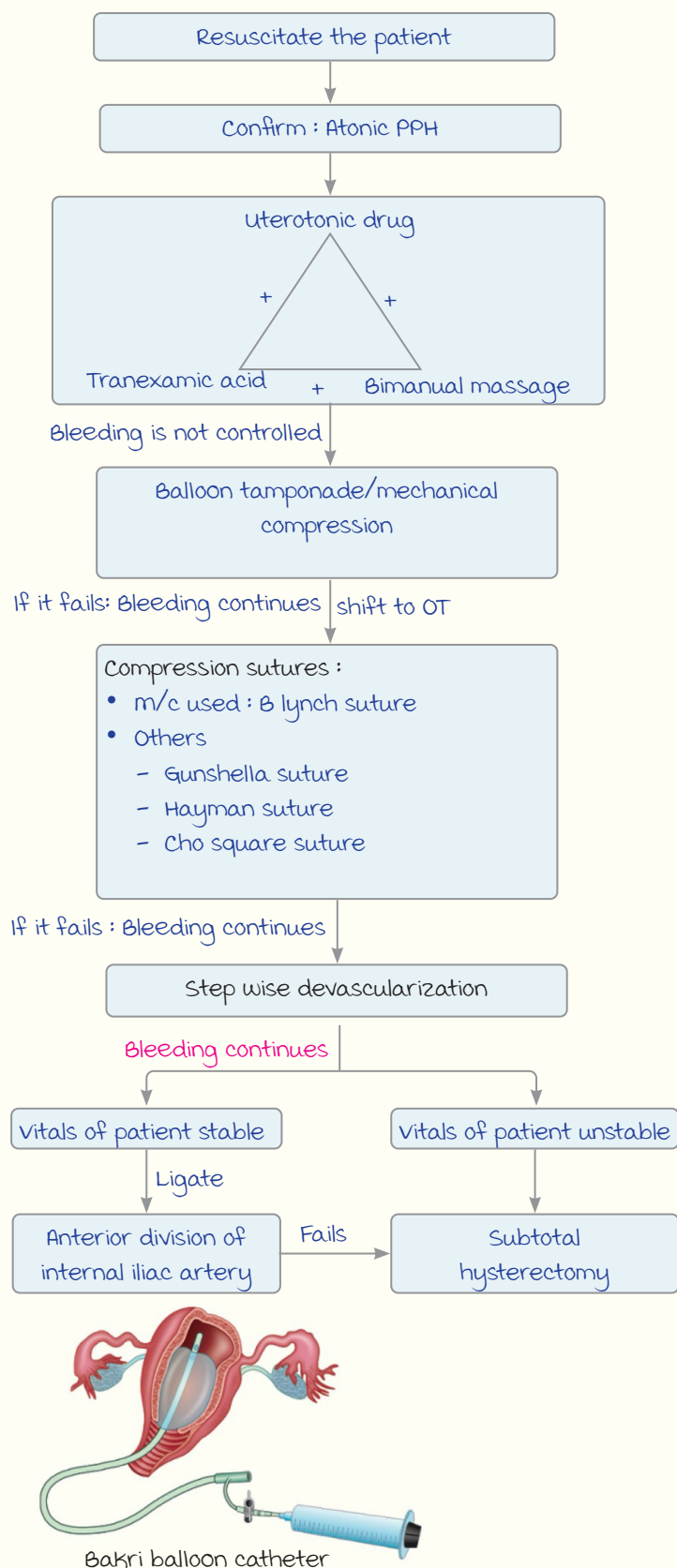
Atonic PPH

P/V examination

Ensure entire placenta and its membranes have been expelled to **confirm atonic PPH**

uterotonics + Tranexamic acid + bimanual massage } 30m

----- Active space -----

Algorithm for management of Atonic PPH :

uterine massage : As per WHO it is not a part of AMTSL but has been shifted to management of PPH

mechanical compression :

- Done by Bakri balloon catheter
- maximum capacity of bakri balloon catheter : 500 mL
- Others which can be used :
 - SB esophageal catheter
 - Condom catheter
- If bleeding stops : Bakri balloon catheter should be removed after 12 hours

In Stepwise Devascularization Arteries which are Ligated :

1. **uterine artery**
Site = As uterine artery turns upward in broad ligament
2. **Ovarian artery : Anastomosing branch**
3. **Anterior division of internal iliac artery**
(As it gives origin to uterine artery)

- TAH/simple hysterectomy : Removal of uterus and cervix
- **Subtotal hysterectomy** : Cervix spared

Note : If after initial resuscitation and oxytocin, bleeding is not controlled and at any point of time patient's vitals become unstable. Do not follow the algorithm and directly proceed to **uterine artery ligation**.

Postpartum Collapse

00:54:31

----- Active space -----

Post Partum Haemorrhage :

- m/c cause of postpartum collapse
- Pt will have excessive bleeding then will go into shock
- P/A = Tone of uterus **decreased**
- P/V = Excessive bleeding

Uterine Inversion :

Pt goes into immediate shock after delivery

Cause : **mismanaged 3rd stage** of labor

Shock : Neurogenic

Hemorrhagic (Cause of death)

On P/A : Cup like depression below umbilicus/uterus may not be felt

On P/V : Globular mass (Fundus) felt in vagina

Postpartum collapse

If Patient after delivery has breathlessness/
resp difficulty/hypoxia + shock

Amniotic fluid embolism

- Initially bleeding is less then pt goes into 2nd phase where she has DIC
- Also called as anaphylactoid syndrome of pregnancy
- It is a clinical diagnosis
- Investigation (may be useful) : Pulmonary artery catheterization
- P/A : (N)
- P/V : (N)

If Pt after delivery has PPH

Visual disturbance

+

Headache

+

Hypoglycemia

+

Shock

Diagnosis : Pituitary apoplexy

D/t :

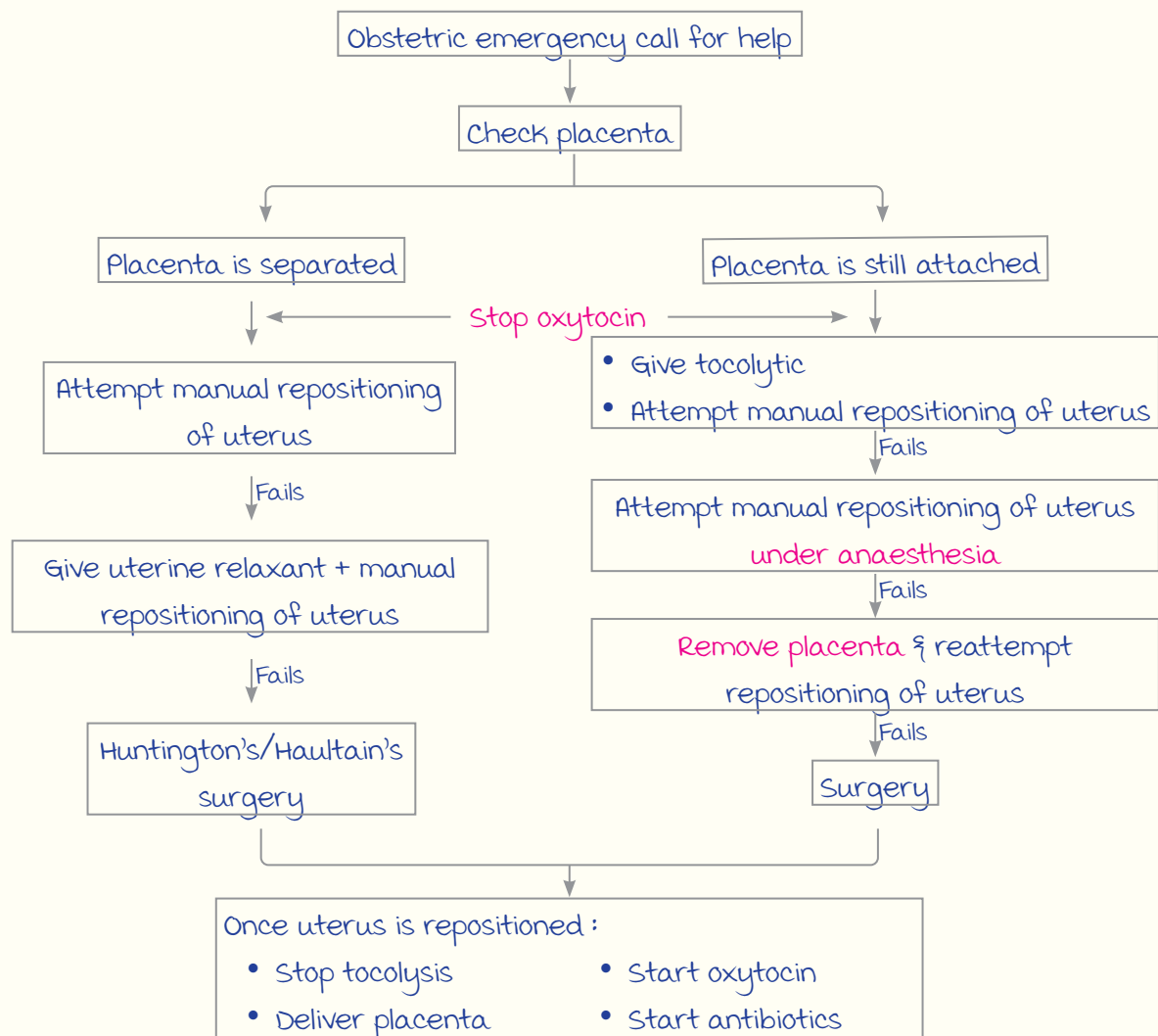
Prolactinoma during pregnancy

PPH after delivery

Sheehan syndrome

mx : Steroids (1st line)

----- Active space ----- management of uterine Inversion :



Vulval Hematoma :

C/F : Difficulty passing urine & severe pelvic pain after delivery.

Artery involved : Pudendal artery

management :

- Check vitals
- Conservative management :
Analgesics & sitz bath.

- Deteriorating vitals
- ↑ Size of hematoma
- Severe unbearable pain

Sx : Drain hematoma.

Episiotomy

01:03:31

----- Active space -----

- It is a surgically planned incision given on perineum
- WHO does not recommend routine episiotomy in all females undergoing vaginal delivery
- Episiotomy can be median/**mediolateral** (60°)

	median	mediolateral
No of muscle fibres cut	↓	↑
Bleeding	↓	↑
Repair & healing	Quick	Slow
Wound gaping	↓	↑
Dyspareunia	↓	↑
Involvement of anal sphincter	↑ Risk	Not involved

To Prevent perineal tear recommended :

- Perineal massage
- Warm compress at perineum
- Controlled delivery of head
- Perineal support
- Head of fetus should be **first flexed** and then **extended** (**Ritgen** maneuver) with one hand and other hand should support perineum

Not Recommended :

- Routine episiotomy
- **Fundal pressure.**

Indications for Episiotomy :

- After coming head of breech
- Instrumental delivery
- macrosomia
- Shoulder dystocia



Episiotomy & Perineal Tear

muscles not cut during episiotomy :

- Ischiococcygeus/coccygeus
- Ischiocavernosus
- Anal sphincter
- Obturator internus

Sequence of episiotomy repair 972

mucosa
↓
muscles
↓
skin

Perineal Tear :

Grade 1 : Tear involving vaginal mucosa

Grade 2 : Tear of perineal muscles

Grade 3 : Involvement of anal sphincter

- 3a : Tear involving **< 50%** of thickness of external anal sphincter.

- 3b : Tear involving **≥ 50%** of thickness of external anal sphincter.

- 3c : Tear involving internal anal sphincter.

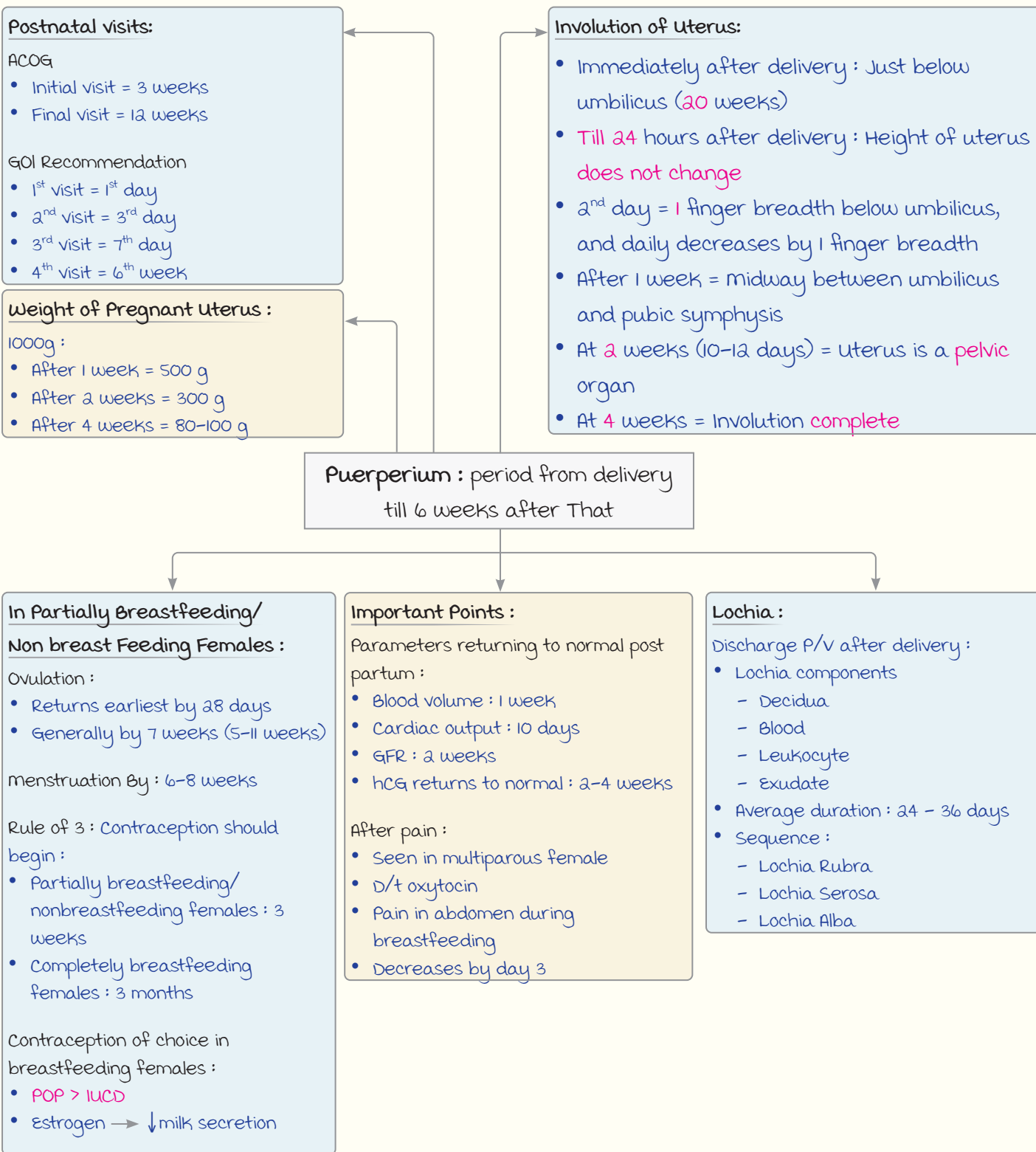
Grade 4 : Tear involving **anal epithelium** or **rectal mucosa.**

Grade 3 and grade 4 : Complete perineal tear (Repair ↓ GA).

----- Active space -----

Puerperium

01:07:58



----- Active space -----

Postpartum Blues :

- Due to **sudden withdrawal** of estrogen and progesterone
- mild depressive symptoms, eg. insomnia, fatigue anxiety
- No Rx needed
- Develops in **2 - 4 days** postpartum that lasts for **< 2 weeks** (usually < 10 days)

Postpartum Depression :

- Seen : 2 weeks-12 months postpartum
- Pt is **unstable**, there is phobia and anxiety
- mx : Antidepressants

Psychosis

Seen in < 0.1% cases

Develops = 2 - 3 days postpartum

- Patient has attention deficit, confusion and altered sensorium
- mx : Antidepressant and antipsychotic

Subinvolution :

Causes :

- Retained placental fragments
- Infection

C/f :

- Excessive bleeding
- Prolonged lochia

On examination :

uterus is bigger than expected

IOC = Pelvic USG

mx : uterotonic **f/b** → Gentle curettage

- Risk of Asherman syndrome ⊕

Puerperal Pathology**Puerperal Pyrexia :**

Increase in temperature of 38.0°C (100.4°F) or higher

Postpartum endometritis (Cause) :

- m/c site : Placental site.
- Risk factor : C-section
 - H/o chorioamnionitis
 - H/o prolonged rupture of membranes

Presentation :

- Fever
- Subinvolution of uterus
- uterus tenderness
- Foul smelling Lochia +/-

mx : Broad spectrum antibiotics

Septic Pelvic Thrombophlebitis :

- It is a **clinical** diagnosis

C/f :

- Suspicion of septic pelvic thrombophlebitis arises when a patient with endometritis, who has fever, does not resolve by giving broad spectrum antibiotics
- m/c vein involved : **Right ovarian vein**

Investigations :

- Blood culture-negative
- CT/MRI : may show thrombosed vein
- P/v : Cord like structure may be felt

mx : Add heparin to broad spectrum antibiotics

